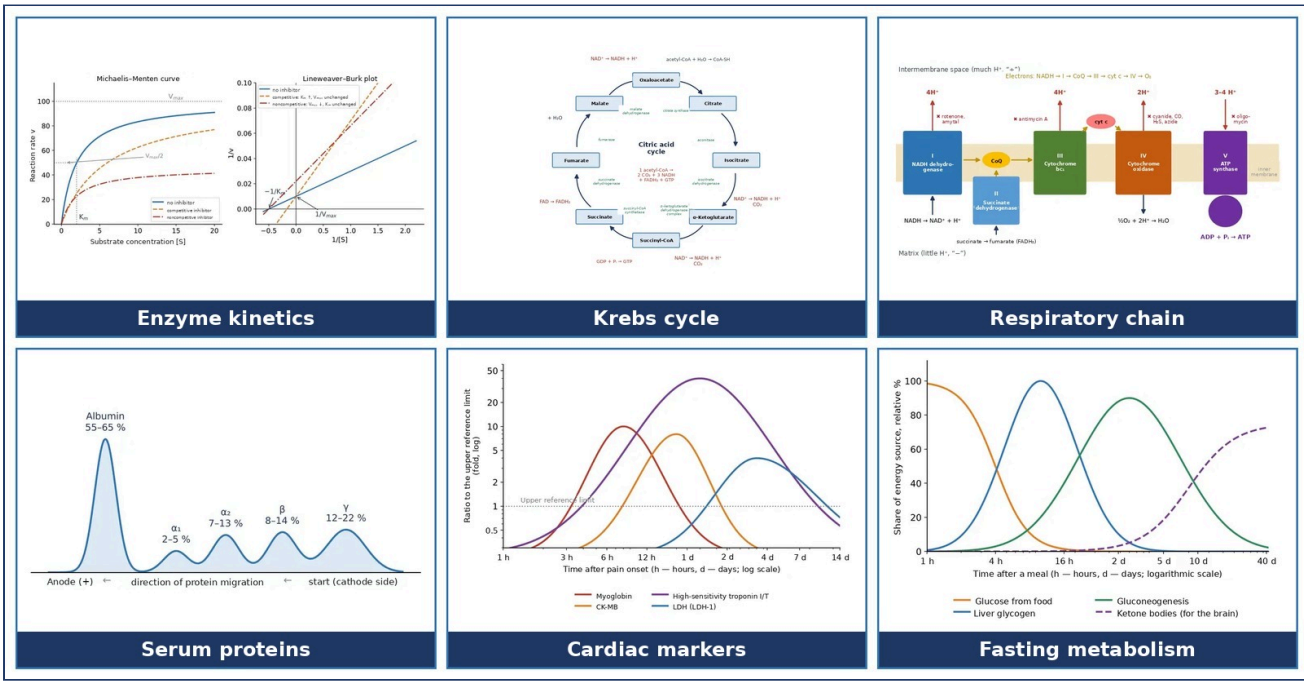
MINISTRY OF HEALTH OF THE REPUBLIC OF UZBEKISTAN
TASHKENT STATE MEDICAL UNIVERSITY
Department of Medical and Biological Chemistry

BIOCHEMISTRY

complete study manual

A study manual for practical classes



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This manual is based on the 30 practical-class topics of the calendar-thematic plan of the subject “Biochemistry” (2nd year, semesters III–IV). It covers an introduction to metabolism, enzymes, biological membranes and biological oxidation, the metabolism of carbohydrates, lipids, proteins, amino acids and nucleotides, molecular biology, oncogenesis, and the biochemistry of blood, connective tissue, muscle, the digestive system, the liver, the endocrine system, the kidneys and the nervous system.

Each chapter consists of the learning objectives, theory (with “Remember”, “Clinical significance”, “Memory aid” and “Caution” boxes), a laboratory (practical) work, algorithms of practical skills, review questions, 15 test questions, 3 clinical cases and a quick review table. The answers to the tests are given at the end of the book.

The manual is based on the textbooks “Biologik kimyo” (2006) and “Biokimyo” by R.A. Sobirova and co-authors, supplemented by “Биохимия” edited by E.S. Severin and by the Harper and Lippincott biochemistry textbooks. The diagrams of metabolic pathways were drawn on the basis of these sources.

The text and structure of the manual were prepared with the help of the artificial intelligence Claude (Anthropic) on the basis of the textbooks listed. The compiler selected the material, defined its structure and checked it. Before applying the clinical information in practice, it is recommended to compare it with the main textbooks.

The manual is intended for 2nd-year students of the general medicine, pediatrics and preventive medicine faculties of higher medical education institutions.

CONTENTS

Preface 10

Introduction to metabolism. Nutritional biochemistry. The coenzyme function of vitamins. The concept of metabolism 11

1. The concept of metabolism 11
2. Stages of catabolism and metabolic pathways 12
3. Nutritional biochemistry 13
4. Vitamins and their coenzyme function 14
5. Laboratory work 16
6. Practical skills 17

Structure and mechanism of action of enzymes. Kinetics of enzyme reactions. Regulation of enzyme activity 22

1. The concept of enzymes. Structure 22
2. Classification and nomenclature of enzymes 23
3. Mechanism of enzyme action 23
4. Properties of enzymes 24
5. Kinetics of enzyme reactions 25
6. Enzyme inhibitors 25
7. Regulation of enzyme activity 26
8. Medical enzymology 26
9. Laboratory work 27
10. Practical skills 28

Biological membranes. Membrane receptors. Signal transduction pathways into the cell 33

1. Structure and chemical composition of biological membranes 33
2. Functions and general properties of membranes 34
3. Transport of substances across the membrane 35
4. Membrane receptors and signal transduction 35
5. Membrane damage 37
6. Laboratory work 38
7. Practical skills 38

Biological oxidation 44

1. Energy metabolism. High-energy compounds 44
2. Biological oxidation. The respiratory chain 45
3. Oxidative phosphorylation 46
4. Respiratory chain inhibitors and uncouplers 47
5. Extramitochondrial oxidation. Reactive oxygen species 47
6. Mitochondrial diseases 48
7. Laboratory work 48
8. Practical skills 49

The common pathways of catabolism 54

1. The concept of the common pathways of catabolism 54
2. Oxidative decarboxylation of pyruvate 54
3. The citric acid cycle (Krebs cycle) 55
4. Energy yield 57

5. Regulation of the Krebs cycle 57
6. Functions of the Krebs cycle. Anaplerotic reactions 57
7. Hypoenergetic states 58
8. Laboratory work 58
9. Practical skills 59
Carbohydrate metabolism and functions 64
1. Dietary carbohydrates and their functions 64
2. Digestion and absorption of carbohydrates 65
3. Entry of glucose into the cell and its phosphorylation 65
4. Glycogen metabolism 66
5. Aerobic oxidation of glucose. Glycolysis 67
6. Glycogenoses and aglycogenoses 69
7. Laboratory work 70
8. Practical skills 70
Gluconeogenesis. The pentose phosphate pathway. Fructose and galactose metabolism. Glucose homeostasis 76
1. Anaerobic glycolysis 76
2. Gluconeogenesis 77
3. The pentose phosphate pathway 79
4. Fructose and galactose metabolism 80
5. Blood glucose homeostasis 80
6. Laboratory work 81
7. Practical skills 82
Structure, functions, digestion and transport of lipids 87
1. Classification and structure of lipids 87
2. Digestion of fats 88
3. Absorption of the products of digestion and their resynthesis 89
4. Blood lipoproteins 90
5. Laboratory work 92
6. Practical skills 92
Intermediary metabolism of fats 98
1. Storage and mobilization of fats 98
2. β-oxidation of fatty acids 99
3. Biosynthesis of fatty acids 101
4. Ketone bodies 101
5. Metabolism of phospholipids and glycolipids 102
6. Laboratory work 103
7. Practical skills 103
Metabolism of complex lipids. Cholesterol metabolism. Regulation of lipid metabolism 109
1. Sterols and steroids. The significance of cholesterol 109
2. Biosynthesis of cholesterol and its regulation 110
3. Transport and removal of cholesterol 111
4. Hypercholesterolemia and atherosclerosis 112
5. Gallstone disease 113

6. Regulation and disorders of fat metabolism 113
7. Laboratory work 114
8. Practical skills 114
Digestion of proteins, absorption of amino acids and their distribution among tissues 119
1. The significance of proteins. Nitrogen balance 119
2. Digestion of proteins in the stomach 120
3. Digestion of proteins in the intestine 121
4. Absorption and distribution of amino acids 122
5. Features of protein digestion in children 122
6. Putrefaction of proteins in the intestine and detoxification of its products 123
7. Laboratory work 124
8. Practical skills 124
The common pathways of amino acid metabolism. Aminotransferases. Decarboxylation of amino acids 130
1. The fate of amino acids 130
2. Transamination 130
3. Deamination 131
4. The fate of the carbon skeleton of amino acids 132
5. Decarboxylation of amino acids. The biogenic amines 133
6. Laboratory work 134
7. Practical skills 135
Detoxification of ammonia. Metabolism of individual amino acids and its hereditary disorders 140
1. Formation and toxicity of ammonia 140
2. Temporary detoxification and transport of ammonia 141
3. Synthesis of urea (the ornithine cycle) 141
4. Hyperammonemia 142
5. Metabolism of one-carbon groups. Serine and glycine 143
6. Metabolism of methionine. Transmethylation 143
7. Metabolism of phenylalanine and tyrosine 144
8. Laboratory work 145
9. Practical skills 146
Metabolism of nucleotides 151
1. Functions of nucleotides. Digestion of nucleoproteins 151
2. Synthesis of purine nucleotides 152
3. Breakdown of purine nucleotides. Uric acid 153
4. Hyperuricemia and gout 153
5. Synthesis and breakdown of pyrimidine nucleotides 154
6. Synthesis of deoxyribonucleotides and thymidylate nucleotides 155
7. Age-related features of nucleotide metabolism 155
8. Laboratory work 156
9. Practical skills 156
The interrelationship of carbohydrate, lipid and amino acid metabolism 161
1. The general scheme of catabolism 161
2. Metabolism of the nitrogen-free residue of amino acids 162

3. Synthesis of glucose from amino acids and glycerol 162
4. Biosynthesis of amino acids from carbohydrates and fats 162
5. Hormonal regulation of metabolism 163
6. Metabolism in the fed and the fasting state 163
7. Disturbance of metabolism in diabetes mellitus 164
8. Metabolic syndrome 165
9. Laboratory work 165
10. Practical skills 166
Molecular biology. DNA structure, replication, transcription 171
1. Structure of nucleic acids 171
2. Organization of the gene and the genome 172
3. DNA replication 173
4. Post-replicative modifications: methylation, telomeres 174
5. Transcription 174
6. Post-transcriptional modifications of RNA (processing) 175
7. Inhibitors of replication and transcription 175
8. Use of recombinant DNA and PCR in medicine 176
9. Laboratory work 176
10. Practical skills 177
Translation. Regulation of gene expression 182
1. The genetic code and its properties 182
2. Components of the protein-synthesizing system 183
3. Stages of translation 183
4. Post-translational modification and protein targeting 184
5. Inhibitors of protein synthesis 185
6. Regulation of gene expression in prokaryotes 185
7. Regulation of gene expression in eukaryotes 186
8. Laboratory work 186
9. Practical skills 187
DNA damage and repair. Apoptosis. Basics of gene therapy 192
1. Causes of DNA damage 192
2. Mutations 193
3. DNA repair 193
4. Genetic variability. The human genome. Protein polymorphism 194
5. Hereditary diseases and DNA technologies 195
6. Apoptosis and necrosis 195
7. Basics of cloning and gene therapy 196
8. Laboratory work 197
9. Practical skills 198
Biochemistry of oncogenesis 203
1. Carcinogenic factors 203
2. Chemical carcinogenesis 204
3. Proto-oncogenes, oncogenes and tumor suppressor genes 204
4. The multistage theory of carcinogenesis 205

5. Biochemical features of tumor cells 206
6. Invasion and metastasis 207
7. Tumor markers 207
8. Biochemical principles of tumor treatment 207
9. Laboratory work 208
10. Practical skills 208
Blood composition, plasma proteins. Biochemistry of the lymphoreticular system 214
1. Functions and chemical composition of blood 214
2. Blood plasma proteins 215
3. Acute phase proteins 216
4. The role of albumin in the distribution of body water. Edema 216
5. Dysproteinemias 216
6. Hematopoiesis and erythrocyte maturation. Heme synthesis 217
7. Features of erythrocyte metabolism 218
8. Hemoglobin and its derivatives 219
9. Buffer systems of blood. Gas transport 219
10. The lymphoreticular system. Metabolism of phagocytes 220
11. Laboratory work 221
12. Practical skills 221
Iron metabolism. Biochemistry of hemostasis 226
1. Distribution and functions of iron in the body 226
2. Absorption, transport and storage of iron 227
3. Regulation of iron metabolism 228
4. Disorders of iron metabolism 228
5. Hemostasis. The vascular–platelet component 229
6. Coagulation hemostasis 229
7. Anticoagulant and fibrinolytic systems 230
8. Disorders of hemostasis and antithrombotic drugs 231
9. Laboratory work 231
10. Practical skills 232
Biochemistry of connective tissue 237
1. The extracellular matrix 237
2. Collagen: composition and structure 237
3. Collagen synthesis 239
4. Breakdown and turnover of collagen 240
5. Disorders of collagen synthesis and maturation 240
6. Elastin 241
7. Glycosaminoglycans and proteoglycans 241
8. Adhesive and antiadhesive proteins 242
9. Laboratory work 243
10. Practical skills 243
Biochemistry of muscle and the cardiovascular system 249
1. Muscle proteins 249
2. Mechanism of muscle contraction and relaxation 250

3. Extractive substances of muscle 251
4. Energy metabolism in muscle. Creatine phosphate 251
5. Biochemical changes in muscular dystrophy and denervation 251
6. Metabolism and secretory function of the heart muscle 252
7. Diagnostic importance of cardiospecific proteins and enzymes 252
8. Secretory function of the endothelium 253
9. Laboratory work 254
10. Practical skills 255
Biochemistry of the gastrointestinal system 260
1. Saliva 260
2. Taste perception. Regulation of appetite and hunger 261
3. Gastric juice 261
4. Pancreatic juice 262
5. Bile 263
6. Intestinal juice and membrane digestion 264
7. Intestinal absorption and its disorders 264
8. Mechanisms of action of drugs in diarrhea and constipation 265
9. Laboratory work 266
10. Practical skills 266
Biochemistry of the liver. Bilirubin metabolism and jaundice 272
1. The role of the liver in metabolism 272
2. Heme catabolism and bilirubin metabolism 273
3. Types of jaundice and their differential diagnosis 274
4. Neonatal jaundice. Hereditary hyperbilirubinemias 275
5. Syndromes of liver damage 276
6. Biochemical mechanisms of hepatic coma 276
7. Laboratory work 277
8. Practical skills 277
Detoxification of toxic substances in the liver 283
1. General principles of biotransformation 283
2. Microsomal oxidation 284
3. Conjugation reactions 284
4. Detoxification of intestinal putrefaction products 285
5. Detoxification of normal metabolites and hormones 285
6. Biotransformation of drugs and the factors that affect it 286
7. P450 polymorphism and personalized treatment 286
8. Detoxification of ethanol in the liver 287
9. Laboratory work 288
10. Practical skills 289
Physiologically active substances and biochemistry of the endocrine system 294
1. Classification of physiologically active substances 294
2. Classification of hormones 295
3. Mechanisms of hormone action 296
4. Hierarchy of regulatory systems. The hypothalamic–pituitary system 296

5. Interrelation of the endocrine and nervous systems. Ontogenesis 297
6. Synthesis and catabolism of steroid hormones 297
7. Mechanism of action of sex hormones 298
8. Chorionic gonadotropin 299
9. Laboratory work 299
10. Practical skills 300
Hormonal regulation of metabolism. Functions of the peripheral glands and their disorders 305
1. Hormonal regulation of carbohydrate, lipid and amino acid metabolism 305
2. Thyroid hormones 306
3. Regulation of calcium–phosphate metabolism 308
4. Regulation of water–salt metabolism 309
5. Locally acting hormones 309
6. Laboratory work 310
7. Practical skills 310
Biochemistry of the kidney. Normal and pathological components of urine 316
1. Mechanism of urine formation 316
2. The role of the kidneys in acid–base balance 318
3. Specific features of metabolism and energy supply in the kidney 318
4. General properties of urine and their age-related features 319
5. Chemical composition of urine 319
6. Pathological components of urine 320
7. Kidney stones and the biochemical basis of their prevention 321
8. Laboratory work 321
9. Practical skills 322
Biochemistry of the central and peripheral nervous system 328
1. Specific features and structural components of the nervous system 328
2. Chemical composition of nervous tissue 328
3. Specific features of energy metabolism in nervous tissue 329
4. Biochemical cooperation between neurons and neuroglia 329
5. Signal transmission across synapses and neurotransmitters 330
6. Biochemistry of sensory systems 331
7. Biochemical basis of mental and nervous system diseases 332
8. Laboratory work 332
9. Practical skills 333
Answers to the test questions 339
References 340

Preface

Dear student! Biochemistry is the science that studies, at the molecular level, the chemical composition of a living organism, the metabolism of substances and energy, the regulation of these processes and the expression of hereditary information. It is the foundation of normal physiology, pathological physiology, pharmacology and all clinical subjects: to understand the essence of a disease, the meaning of laboratory tests and the mechanism of action of drugs, one must first know how substances are transformed in the body.

The manual follows the 30 practical-class topics of the department's calendar-thematic plan: in semester III (topics 1–15) — an introduction to metabolism, enzymes, biological membranes, biological oxidation, the metabolism of carbohydrates, lipids, proteins, amino acids and nucleotides and their integration; in semester IV (topics 16–30) — molecular biology, oncogenesis, blood, iron metabolism and hemostasis, and the biochemistry of connective tissue, muscle, the gastrointestinal tract, the liver, detoxification, the endocrine system, the kidneys and the nervous system. Every chapter is built in the same order:

- **Learning objectives** — what you should know and be able to do by the end of the class;
- **theory** — the main concepts of the topic, diagrams of metabolic pathways, tables and graphs;
- **laboratory work** — qualitative reactions, quantitative methods and analysis of the results performed in class;
- **practical skills** — step-by-step algorithms for interpreting laboratory results;
- **review questions, test questions and clinical cases** — for self-assessment;
- **quick review** — a table for revising at a glance before the exam.

The colored boxes in the text mean the following:

> **REMEMBER**
> The most important definitions, numbers and classifications — they must be memorized.

> **CLINICAL SIGNIFICANCE**
> The significance of the topic in clinical practice, examples related to diseases and laboratory diagnosis.

> **MEMORY AID**
> Techniques and abbreviations that make memorizing easier.

> **CAUTION**
> Common mistakes, dangerous situations and discrepancies between textbooks.

The reference values in the manual may differ slightly between textbooks and laboratories — in such cases, follow the values accepted at the department and in the laboratory that performed the test. Perform laboratory work only under the teacher's supervision and in compliance with the safety rules for working with acids, alkalis and biological materials. It is recommended to read each topic together with the main textbook.

The compiler

TOPIC 1

Introduction to metabolism. Nutritional biochemistry. The coenzyme function of vitamins. The concept of metabolism

Learning objectives

To teach the student the concept of metabolism, its stages, catabolism and anabolism, metabolic pathways and the methods of studying them, the biochemical basis of nutrition, the nonessential and essential components of food, the coenzyme function of vitamins and the biochemistry of milk.

By the end of the class the student should know:

- the concepts of metabolism, catabolism and anabolism;
- the stages of metabolism (digestion, absorption, intermediary metabolism, excretion of end products);
- the three stages of catabolism and the common pathway;
- the functions and daily requirements of the main nutrients;
- the essential amino acids, fatty acids and other factors;
- the water-soluble and fat-soluble vitamins and their coenzyme forms;
- the causes of hypovitaminosis, avitaminosis and hypervitaminosis;
- the methods of studying metabolism.

The student should be able to:

- match vitamins with their coenzymes;
- perform qualitative reactions for vitamins and evaluate the result;
- calculate the energy value of a diet.

1. The concept of metabolism

Metabolism is the sum of the chemical reactions that take place in a living organism and provide its exchange of matter and energy with the external environment. Metabolism consists of two interrelated directions:

- **Catabolism (dissimilation)** — the breakdown of complex substances into simpler ones; mainly oxidation reactions, energy is released (exergonic), part of it is stored in the form of ATP; NADH and NADPH are formed.
- **Anabolism (assimilation)** — the synthesis of complex substances (proteins, nucleic acids, polysaccharides, lipids) from simple ones; energy is consumed (endergonic), and NADPH is used mainly as the reductant. Catabolic and anabolic pathways are often not the same: they proceed through different enzymes, are located in different compartments of the cell (for example, fatty acid oxidation — in the mitochondria, synthesis — in the cytosol) and are regulated separately. **Amphibolic pathways** (for example, the Krebs cycle) serve both catabolism and anabolism.

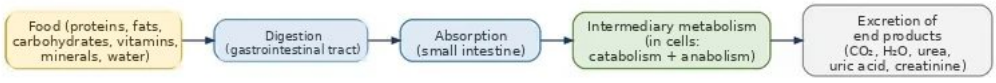


Figure 1.1. *Stages of metabolism (drawn after Sobirova R.A., 2006)*

Stages of metabolism

1. **Digestion** — the breakdown of food polymers into monomers in the gastrointestinal tract by enzymes (proteins → amino acids, polysaccharides → monosaccharides, fats → fatty acids, glycerol, monoacylglycerols).
2. **Absorption** — the passage of monomers through the intestinal wall into the blood and lymph.
3. **Intermediary (intracellular) metabolism** — the conversion of monomers in cells: their breakdown (to obtain energy) and the synthesis of the body's own substances.
4. **Excretion of end products** — CO_2 (lungs), H_2O, urea, uric acid, creatinine, ammonium salts (kidneys), bilirubin derivatives (intestine).

2. Stages of catabolism and metabolic pathways

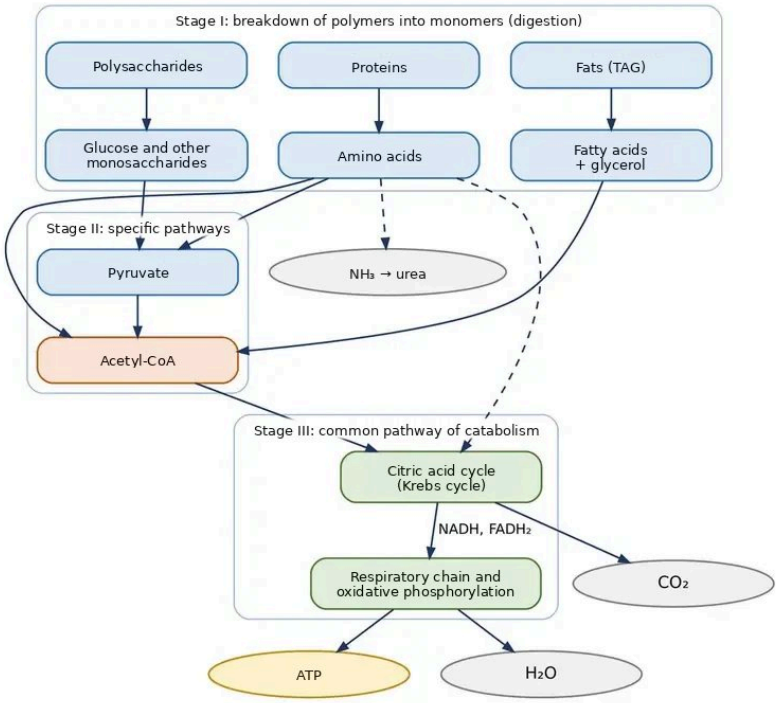


Figure 1.2. *The three stages of catabolism and the common pathway (drawn after Sobirova R.A., 2006; Severin E.S.)*

Table 1.1. *The three stages of catabolism*

Stage	Essence	Site	Energy
I	Hydrolysis of polymers to monomers	Gastrointestinal tract, lysosomes	≈ 1% (dissipated as heat)
II	Conversion of monomers into common intermediates — pyruvate and acetyl-CoA (specific pathways: glycolysis, β-oxidation, amino acid catabolism)	Cytosol, mitochondria	≈ 20–30 %
III	The common pathway of catabolism: oxidative decarboxylation of pyruvate, the Krebs cycle, the respiratory chain	Mitochondria	≈ 70 %

A **metabolic pathway** is a chain of transformations of a substance through successive enzymatic reactions; its starting substance is the **substrate**, the intermediates are **metabolites**, and the end product is the **product**. Metabolic pathways may be linear (glycolysis), cyclic (the Krebs cycle, the

ornithine cycle) or branched. Each pathway usually has one **key (rate-limiting) enzyme**: it catalyzes the slowest reaction, controls an irreversible reaction and determines the overall rate of the pathway.

> **REMEMBER**
> All catabolic pathways converge on **acetyl-CoA** and are completed through the **Krebs cycle + respiratory chain**. Therefore the main part of the body's energy (≈ 90% of ATP) is formed in the mitochondria by oxidative phosphorylation.

Methods of studying metabolism

- **At the level of the whole organism**: balance experiments (nitrogen balance), calorimetry, respiratory quotient; determination of metabolites in blood and urine.
- **At the organ level**: perfusion of an isolated organ, angiostomy (arteriovenous difference).
- **At the tissue and cell level**: tissue slices, homogenates, cell cultures, isolation of organelles by differential centrifugation.
- **Molecular methods**: the **labeled atoms** method (isotopes ^{14}C, ^{3}H, ^{15}N, ^{32}P) — tracing the fate of metabolites; enzyme inhibitors; genetic methods (knockout organisms, mutants); chromatography, mass spectrometry (metabolomics).
- **Clinical methods**: inborn errors of metabolism (the metabolite before the blocked enzyme accumulates) — "experiments of nature."

3. Nutritional biochemistry

Nutrition is a component of metabolism that meets the body's need for energy and plastic material. The main components of food: proteins, fats, carbohydrates (macronutrients), vitamins and minerals (micronutrients), water and dietary fiber.

Table 1.2. *The main nutrients and their functions (adults, moderate physical activity)*

Nutrient	Daily requirement	Energy value	Main functions
Proteins	0.8–1 g/kg (≈ 70–100 g), ≈ 50% of it animal protein	4 kcal/g (17 kJ)	Plastic, catalytic, transport, protective; source of essential amino acids
Fats	80–100 g (25–30% of energy)	9 kcal/g (38 kJ)	Energy, plastic (membranes), source of fat-soluble vitamins and essential fatty acids
Carbohydrates	350–450 g (55–60% of energy)	4 kcal/g (17 kJ)	The main energy source; dietary fiber improves bowel function
Water	2–2.5 l	—	Solvent, medium for reactions, thermoregulation

The daily energy expenditure of adults is 2200–3500 kcal depending on physical activity (in women 1800–2800 kcal).

Essential components of food

Essential (indispensable) substances are not synthesized in the body, or not in sufficient amounts, and must come with food:

- **8 essential amino acids**: valine, leucine, isoleucine, threonine, methionine, lysine, phenylalanine, tryptophan; partly essential in children — **arginine and histidine**.

- **Essential (polyunsaturated) fatty acids**: linoleic (ω-6) and α-linolenic (ω-3); arachidonic acid is formed from linoleic acid.
- **Vitamins** (vitamin D and niacin are partly synthesized).
- **Minerals**: macroelements (Na, K, Ca, Mg, P, Cl) and trace elements (Fe, Zn, Cu, I, Se, F, Co, Mn, Mo).

> **MEMORY AID**
> Essential amino acids: **"Very Lazy Ivan Takes Me Late For Tea"** — Valine, Leucine, Isoleucine, Threonine, Methionine, Lysine, Phenylalanine, Tryptophan.

The biological value of proteins is the degree to which a dietary protein is converted into the body's proteins; it depends on the content of essential amino acids and on digestibility. Egg and milk proteins have the highest biological value; plant proteins are often deficient in lysine (wheat), methionine (legumes) or tryptophan (maize).

Age-related nutrition. Biochemistry of milk

Because of growth, children's requirement for protein (2–4 g/kg) and energy per 1 kg of body weight is 2–3 times higher than in adults. Breast milk is the most appropriate food for the infant.

Table 1.3. *Composition of human milk and cow's milk (per 100 ml, approximately)*

Component	Human milk	Cow's milk
Protein	1.0–1.2 g (mainly whey proteins: α-lactalbumin, lactoferrin, IgA)	3.2–3.5 g (mainly casein)
Fat	3.5–4.5 g (rich in polyunsaturated fatty acids, contains lipase)	3.5 g
Lactose	6.5–7 g	4.5–4.8 g
Minerals	0.2 g	0.7 g (high osmotic load on the kidneys)
Energy	≈ 70 kcal	≈ 65 kcal

Colostrum is secreted in the first days after delivery: it is rich in protein (especially secretory IgA) and provides immune protection.

4. Vitamins and their coenzyme function

Vitamins are low-molecular-weight organic substances that are not synthesized in the body, or not in sufficient amounts, and are needed in small quantities (mg, µg); they are neither an energy source nor plastic material, but most of them are part of **coenzymes**. They are divided into **water-soluble** (B group, C) and **fat-soluble** (A, D, E, K) vitamins.

Table 1.4. *Water-soluble vitamins and their coenzymes*

Vitamin	Coenzyme	Coenzyme function (example reaction)	Deficiency
B_1 (thiamine)	Thiamine diphosphate (TDP, TPP)	Oxidative decarboxylation of keto acids (pyruvate and α-ketoglutarate dehydrogenases), transketolase	Beriberi, Wernicke–Korsakoff syndrome

Vitamin	Coenzyme	Coenzyme function (example reaction)	Deficiency
B_2 (riboflavin)	FMN, FAD	Oxidation-reduction (succinate dehydrogenase, complex I of the respiratory chain)	Cheilitis, glossitis, keratitis
B_3 / PP (niacin)	NAD^+, $NADP^+$	Dehydrogenases (lactate dehydrogenase, glucose-6-phosphate dehydrogenase)	Pellagra: dermatitis, diarrhea, dementia
B_5 (pantothenic acid)	Coenzyme A (CoA-SH)	Transfer of acyl groups (acetyl-CoA)	Rare (“burning feet”)
B_6 (pyridoxine)	Pyridoxal phosphate (PLP)	Transamination, decarboxylation (aminotransferases)	Dermatitis, seizures, anemia
B_7 / H (biotin)	Biotin (carboxybiotin)	Carboxylation (pyruvate carboxylase, acetyl-CoA carboxylase)	Dermatitis (raw egg white — avidin)
B_9 (folic acid)	Tetrahydrofolate (THF)	Transfer of one-carbon groups (purine and thymidylate synthesis)	Megaloblastic anemia, neural tube defects in the fetus
B_{12} (cobalamin)	Methylcobalamin, 5′-deoxyadenosylcobalamin	Methionine synthase, methylmalonyl-CoA mutase	Megaloblastic anemia, subacute combined degeneration of the spinal cord
C (ascorbic acid)	No coenzyme (cofactor)	Hydroxylation of proline and lysine (collagen), antioxidant, reduction of Fe^{3+}	Scurvy

Table 1.5. *Fat-soluble vitamins*

Vitamin	Active form	Main function	Deficiency
A (retinol)	Retinal, retinoic acid	Vision (rhodopsin), epithelial differentiation	“Night blindness,” xerophthalmia
D (calciferol)	1,25-dihydroxycholecalciferol (calcitriol)	Absorption of Ca^{2+} and phosphate in the intestine	Rickets in children, osteomalacia in adults
E (tocopherol)	α-tocopherol	Antioxidant of membrane lipids	Hemolytic anemia, neurological disorders
K (phylloquinone, menaquinone)	Hydroquinone form	γ-carboxylation of clotting factors (II, VII, IX, X)	Bleeding, hemorrhagic disease of the newborn

CAUTION

Hypervitaminosis occurs mainly with the fat-soluble vitamins (A and D), because they accumulate in tissues: vitamin A hypervitaminosis — headache, liver damage, a teratogenic effect in pregnancy; vitamin D hypervitaminosis — hypercalcemia, calcinosis of the kidneys and vessels. An excess of water-soluble vitamins is excreted in the urine (except B_{12}, which accumulates in the liver for years).

Causes of hypovitaminosis and avitaminosis: insufficient intake with food (exogenous); impaired absorption (gastrointestinal diseases, absence of Castle’s intrinsic factor — B_{12}), increased requirement (pregnancy, growth, illness), suppression of the intestinal microflora by antibiotics (K, B group), the action of **antivitamins** (endogenous).

Antivitamins are substances similar in structure to a vitamin that block its coenzyme function: methotrexate (a folate antagonist — antitumor), dicoumarol and warfarin (vitamin K antagonists — anticoagulants), isoniazid (a B_6 antagonist), avidin (binds biotin).

> **CLINICAL SIGNIFICANCE**
> Thiamine deficiency is common in chronic alcoholism: pyruvate dehydrogenase activity falls, pyruvate and lactate accumulate in the blood, and an energy deficit develops in the brain (Wernicke encephalopathy). Such patients must be given thiamine before glucose — otherwise glucose oxidation uses up the thiamine reserve completely and worsens the condition.

5. Laboratory work

Experiment 1: qualitative reaction for vitamin B_1 (thiochrome reaction)

1. Add to a test tube 1 drop of 5% thiamine solution, 5 drops of 10% NaOH and 1–2 drops of 5% potassium ferricyanide ($K_3[Fe(CN)_6]$) solution.
2. Add 5–10 drops of isobutyl (or butyl) alcohol, shake and wait for the layers to separate.
3. Observe the alcohol layer under ultraviolet light. **Result and analysis:** in an alkaline medium thiamine is oxidized to form **thiochrome** — the alcohol layer **fluoresces** blue under ultraviolet light. This reaction is the basis of the quantitative determination of thiamine.

Experiment 2: reduction reaction for vitamin B_2

1. Add to a test tube 10 drops of riboflavin solution (yellow) and 5 drops of concentrated HCl.
2. Drop in a small piece of metallic zinc (Zn). **Result and analysis:** with the release of hydrogen riboflavin is reduced: the yellow color turns first pink and then colorless (leucoflavin). Under the action of atmospheric oxygen the color returns. This shows the property of FMN and FAD to accept and donate hydrogen (oxidation-reduction).

Experiment 3: determination of vitamin C with 2,6-dichlorophenolindophenol (Tillmans reaction)

1. Pour into a test tube 2 ml of ascorbic acid solution (or fruit juice) and 1 ml of 2% HCl.
2. From a burette titrate dropwise with blue 2,6-dichlorophenolindophenol solution until a pink color persists for 30 s. **Result and analysis:** ascorbic acid is a strong reductant — it reduces the blue dye to the colorless leuco form. Once all the ascorbic acid has been oxidized, the excess dye gives a pink color in the acidic medium. The amount of vitamin C is calculated from the volume of dye used.

Experiment 4: calculating the energy value of a diet

1. Determine from the table the amount (g) of protein, fat and carbohydrate in the student's daily food.
2. Calculate the energy: E = protein × 4 + fat × 9 + carbohydrate × 4 (kcal).
3. Find the share (%) of each nutrient in the energy and compare it with the norm (protein 12–15%, fat 25–30%, carbohydrate 55–60%).

6. Practical skills

Skill: evaluating a diet

1. Determine the amount of protein, fat and carbohydrate in the diet.
2. Calculate the total energy value and compare it with expenditure.
3. Check the share of animal protein (≥ 50%) and the sources of essential amino acids.
4. Evaluate the sources of vitamins (especially C, D, B group, folate) and minerals (Fe, Ca, I).
5. Conclusion: the diet is adequate, excessive or deficient; give a recommendation for correction.

Skill: biochemical justification of a vitamin deficiency

1. Identify the clinical signs (for example, anemia, dermatitis, neurological signs).
2. Recall which vitamin is part of which coenzyme.
3. Identify the reaction and enzyme in which this coenzyme takes part and which is impaired.
4. Name the metabolite that accumulates (for example, pyruvate in B_1 deficiency, methylmalonic acid in B_{12} deficiency).
5. Explain the laboratory confirmation and the principle of treatment.

Review questions

1. What is metabolism? What is the difference between catabolism and anabolism?
2. List the four stages of metabolism.
3. Describe the three stages of catabolism.
4. Which processes make up the common pathway of catabolism?
5. What are a metabolic pathway and a key enzyme?
6. What are amphibolic pathways? Give an example.
7. What methods of studying metabolism are there?
8. What are the daily requirements and energy values of proteins, fats and carbohydrates?
9. List the essential amino acids and fatty acids.
10. What does the biological value of proteins depend on?
11. How does human milk differ from cow’s milk?
12. How are vitamins classified?
13. List the coenzymes of the B-group vitamins.
14. What are the causes of hypovitaminosis, avitaminosis and hypervitaminosis?
15. What are antivitamins? Give clinical examples.

Test questions

1. The feature characteristic of catabolism:
    A) Energy is consumed
    B) Complex substances are broken down and energy is released
    C) NADPH is consumed mainly as a reductant
    D) Protein synthesis
    E) Polysaccharide synthesis

2. The common metabolite in which all catabolic pathways converge:
   A) Glucose
   B) Lactate
   C) Acetyl-CoA
   D) Glycerol
   E) Urea

3. The coenzyme of vitamin B_1 (thiamine):
   A) Thiamine diphosphate
   B) FAD
   C) NAD^+
   D) Pyridoxal phosphate
   E) Coenzyme A

4. The coenzyme that takes part in transamination reactions:
   A) Biotin
   B) Tetrahydrofolate
   C) FMN
   D) Pyridoxal phosphate
   E) Thiamine diphosphate

5. The amino acid that is essential for adults:
   A) Alanine
   B) Glutamine
   C) Serine
   D) Glycine
   E) Lysine

6. The energy released by the oxidation of 1 g of fat:
   A) 4 kcal
   B) 7 kcal
   C) 9 kcal
   D) 12 kcal
   E) 2 kcal

7. The vitamin whose deficiency causes pellagra:
   A) B_1
   B) B_2
   C) C
   D) B_{12}
   E) PP (niacin)

8. The vitamins in which hypervitaminosis is observed more often:
   A) B_1 and B_2
   B) C and PP
   C) B_6 and B_9
   D) A and D
   E) Pantothenic acid and biotin

9. The substance that is a vitamin K antagonist (antivitamin):
    A) Methotrexate
    B) Warfarin
    C) Isoniazid
    D) Avidin
    E) Aspirin

10. The vitamin that takes part in carboxylation reactions:
    A) Biotin
    B) Riboflavin
    C) Ascorbic acid
    D) Retinol
    E) Tocopherol

11. The function of the folic acid coenzyme:
    A) Transfer of acyl groups
    B) Transfer of one-carbon groups
    C) Transfer of the amino group
    D) Transfer of electrons
    E) Attachment of a carboxyl group

12. How human milk differs from cow’s milk:
    A) Rich in casein
    B) Rich in minerals
    C) Low in lactose
    D) Rich in whey proteins and lactose, contains secretory IgA
    E) Protein content 3 times higher

13. What the labeled atoms method makes it possible to study:
    A) The pathways of transformation of metabolites in the body
    B) Only energy expenditure
    C) Only nitrogen balance
    D) Only enzyme structure
    E) Only cell structure

14. The vitamin needed for the hydroxylation of proline in collagen synthesis:
    A) A
    B) D
    C) K
    D) E
    E) C

15. An example of an amphibolic pathway:
    A) Urea synthesis
    B) Glycogen synthesis
    C) Krebs cycle
    D) Lipolysis
    E) Proteolysis

Clinical cases

Case 1

A 48-year-old man has abused alcohol for 15 years and is poorly nourished. In the emergency department he was found to have confusion, impaired eye movements and an unsteady gait. Blood pyruvate and lactate levels are increased.

Questions: Which vitamin deficiency is suspected? Why do pyruvate and lactate accumulate? What should be taken into account in treatment?

Answer: Vitamin B_1 (thiamine) deficiency — Wernicke encephalopathy. Thiamine diphosphate is the coenzyme of the pyruvate dehydrogenase complex and α-ketoglutarate dehydrogenase. In its deficiency pyruvate is not converted to acetyl-CoA, accumulates and is reduced to lactate (lactic acidosis); the Krebs cycle slows down and an ATP deficit develops in the brain. In treatment thiamine is given parenterally before glucose.

Case 2

People living in areas where they eat only maize were observed to have dermatitis on sun-exposed areas of skin, diarrhea and memory loss.

Questions: What is the disease? What is its biochemical cause?

Answer: Pellagra — niacin (vitamin PP) deficiency ("3 Ds": dermatitis, diarrhea, dementia). In maize niacin is in a bound (non-absorbable) form and tryptophan is low — so the synthesis of niacin from tryptophan is also insufficient. Niacin is part of NAD^+ and $NADP^+$; when they are deficient many dehydrogenase reactions are impaired, and rapidly renewing tissues (skin, intestinal epithelium) and the nervous system are damaged.

Case 3

A 9-month-old child is fed only cow's milk and semolina porridge and is rarely taken outdoors. Examination shows a flattened occiput, a "rosary" on the ribs and "bracelets" on the wrists. Blood alkaline phosphatase is increased and phosphate is decreased.

Questions: What is the diagnosis? Explain the biochemical mechanism.

Answer: Rickets — vitamin D deficiency. Lack of sunlight reduces the synthesis of cholecalciferol in the skin, and the diet is also poor in vitamin D. The active form calcitriol (hydroxylated in the liver and kidney) provides the absorption of Ca^{2+} and phosphate in the intestine; in its deficiency bone mineralization is impaired and osteoid tissue increases (osteoblast activity — alkaline phosphatase rises). Treatment: vitamin D, sunbathing, proper feeding.

Quick review

Table 1.6. *Key concepts of topic 1*

Concept	Brief definition
Metabolism	The sum of chemical reactions in the body: catabolism + anabolism
Catabolism	Breakdown, oxidation, energy is released, NADH is formed
Anabolism	Synthesis, energy and NADPH are consumed
Stages of catabolism	Hydrolysis → pyruvate and acetyl-CoA → Krebs cycle and respiratory chain

Concept	Brief definition
Key enzyme	The enzyme that determines the rate of a metabolic pathway and catalyzes an irreversible reaction
Energy value	Protein and carbohydrate 4, fat 9 kcal/g
Essential amino acids	Val, Leu, Ile, Thr, Met, Lys, Phe, Trp
B_1	Thiamine diphosphate — oxidative decarboxylation
B_2	FMN, FAD — oxidation-reduction
PP	NAD^+, $NADP^+$ — dehydrogenases
B_6	Pyridoxal phosphate — transamination, decarboxylation
Folate, B_{12}	One-carbon groups; megaloblastic anemia in deficiency
Biotin	Carboxylation
Antivitamins	Methotrexate, warfarin, isoniazid, avidin

TOPIC 2

Structure and mechanism of action of enzymes. Kinetics of enzyme reactions. Regulation of enzyme activity

Learning objectives

To teach the student the chemical nature and structure of enzymes, their active and allosteric sites, cofactors and coenzymes, classification and nomenclature, mechanism of action, properties, the dependence of reaction rate on temperature, pH and the concentrations of enzyme and substrate, inhibitors and types of inhibition, methods of regulating enzyme activity and the basics of medical enzymology; to master the determination of salivary amylase activity.

By the end of the class the student should know:

- the protein nature of enzymes, simple and complex enzymes, the concepts of apoenzyme, cofactor and coenzyme;
- the structure of the active site and the allosteric site;
- the 7 classes of enzymes and the principles of naming;
- the mechanism of enzyme action (lowering of the activation energy, the enzyme–substrate complex);
- the properties of enzymes: specificity, thermolability, pH dependence;
- the Michaelis–Menten equation and the meaning of K_m and V_{max};
- the types of inhibition and drugs that are enzyme inhibitors;
- the methods of regulating enzyme activity;
- the diagnostic significance of isoenzymes and organ-specific enzymes.

The student should be able to:

- determine salivary amylase activity by the Wohlgemuth method;
- determine the type of inhibition from a Lineweaver–Burk plot;
- clinically evaluate changes in serum enzyme activity.

1. The concept of enzymes. Structure

Enzymes are biological catalysts formed in living cells that accelerate chemical reactions. Almost all enzymes are **proteins** (except RNA molecules with catalytic activity — **ribozymes**). The science that studies enzymes is **enzymology**.

- **Simple enzymes** consist only of amino acids (pepsin, trypsin, ribonuclease).
- **Complex enzymes (holoenzyme)** = a protein part (**apoenzyme**) + a nonprotein part (**cofactor**). The cofactor may be a metal ion (Zn^{2+}, Mg^{2+}, Fe^{2+}, Cu^{2+}, Mn^{2+}) or an organic substance — a **coenzyme**. A coenzyme firmly (covalently) bound to the apoenzyme is called a **prosthetic group** (for example, FAD in succinate dehydrogenase). The **active site** is the part of the enzyme molecule that binds the substrate and provides its chemical transformation. It is formed from amino acid residues located in different parts of the polypeptide chain when the tertiary structure forms, and consists of two parts:

- **the binding (contact) part** — recognizes and binds the substrate (determines specificity);
- **the catalytic part** — carries out the chemical transformation (often serine, histidine, cysteine, aspartate and glutamate residues).

The **allosteric site** is a part located outside the active site that binds regulatory molecules (**effectors**). When an effector binds, the conformation of the enzyme and of its active site changes: an **activator** increases activity, an **inhibitor** decreases it. Allosteric enzymes usually consist of several subunits (quaternary structure).

Isoenzymes are forms of an enzyme that catalyze the same reaction but differ in structure, physicochemical properties and tissue distribution. For example, **lactate dehydrogenase (LDH)** is a tetramer built of H and M subunits, with 5 isoenzymes: LDH1 (H4) — heart muscle, LDH5 (M4) — liver and skeletal muscle. **Creatine kinase**: MM — skeletal muscle, MB — heart, BB — brain.

2. Classification and nomenclature of enzymes

According to the classification of the International Union of Biochemistry and Molecular Biology, enzymes are divided into **7 classes** by the type of reaction they catalyze. Each enzyme has a 4-number code (EC 1.1.1.27 — LDH). The working name: substrate name + type of reaction + “-ase” (lactate dehydrogenase).

Table 2.1. *Classes of enzymes*

Class	Type of reaction	Examples
1. Oxidoreductases	Oxidation-reduction	Dehydrogenases, oxidases, catalase, peroxidase
2. Transferases	Transfer of groups from one substance to another	Aminotransferases, kinases, methyltransferases
3. Hydrolases	Hydrolytic cleavage of bonds with the participation of water	Pepsin, amylase, lipase, phosphatases
4. Lyases	Cleavage of bonds without hydrolysis or addition to a double bond	Aldolase, decarboxylases, fumarase
5. Isomerases	Intramolecular rearrangement	Triose phosphate isomerase, mutases, epimerases
6. Ligases (synthetases)	Joining of two molecules using the energy of ATP	Pyruvate carboxylase, DNA ligase, glutamine synthetase
7. Translocases	Transfer of ions and molecules across a membrane	Na^+,K^+-ATPase, ATP synthase

MEMORY AID
Order of the classes: **“Over The Hill Lies Ivan’s Little Town”** — Oxidoreductases, Transferases, Hydrolases, Lyases, Isomerases, Ligases, Translocases.

3. Mechanism of enzyme action

For any chemical reaction to start, molecules must overcome the **activation energy** barrier. Enzymes carry the reaction along another route with a lower activation energy and increase its rate 10^8–10^{12} times. The equilibrium of the reaction and the overall energy change (ΔG) do not change.

Stages of an enzyme reaction:

$E + S \rightleftarrows ES \rightarrow EP \rightarrow E + P$

1. Binding of the substrate to the active site — formation of the **enzyme–substrate complex (ES)**.
2. Strain (deformation) of bonds in the substrate, formation of the transition state.

3. Release of the product; the enzyme remains unchanged. Two models of how the substrate fits the active site: E. Fischer's **"lock and key"** model (rigid fit) and D. Koshland's **"induced fit"** model — when the substrate binds, the conformation of the active site changes and "adapts" to the substrate (the modern view).

Chemical mechanisms of catalysis: acid-base catalysis, covalent catalysis (the substrate temporarily forms a covalent bond with serine or cysteine in the active site), metal ion catalysis, approximation and correct orientation of substrates.

4. Properties of enzymes

Specificity is the selective action of an enzyme on a particular substrate or type of reaction:

- **absolute** — on only one substrate (urease — urea, arginase — arginine);
- **relative (group)** — on a group of substrates with the same bond (pepsin, lipase, phosphatases);
- **stereospecificity** — on only one stereoisomer (L-amino acid oxidase, D-glucose enzymes). **Thermolability.** For every 10 °C rise in temperature the reaction rate doubles, but above 40–50 °C the enzyme protein denatures and its activity falls. The **temperature optimum** of human enzymes is 37–40 °C. At low temperature enzyme activity decreases, but the enzymes do not denature (reversible inactivation) — the cold storage of organs is based on this.

pH dependence. Each enzyme has a **pH optimum**; it depends on the ionization of groups in the active site and of the substrate.

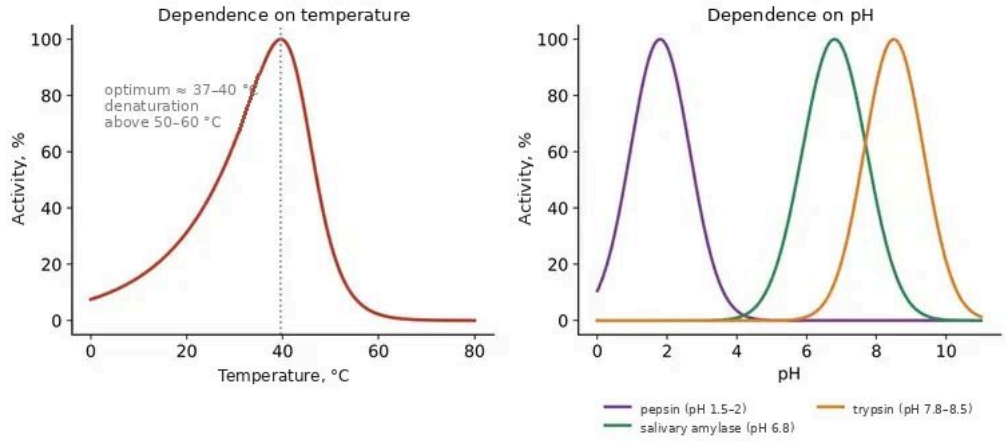


Figure 2.1. *Dependence of enzyme activity on temperature and pH (drawn after Sobirova R.A., 2006; Severin E.S.)*

Table 2.2. *pH optimum of some enzymes*

Enzyme	pH optimum
Pepsin	1,5–2,0
Salivary amylase	6,8–7,0
Trypsin	7,8–8,5
Alkaline phosphatase	9–10
Acid phosphatase (lysosomes)	4,5–5,0

5. Kinetics of enzyme reactions

Dependence on enzyme concentration. When the substrate is in excess, the reaction rate is directly proportional to the amount of enzyme. The determination of enzyme activity is based on this.

Dependence on substrate concentration. When the substrate is low, the rate increases in direct proportion to it; as the substrate increases, all enzyme molecules become occupied and the rate approaches its maximum (V_{max}) — a **hyperbolic curve**. This dependence is expressed by the **Michaelis–Menten equation**:

$v = V_{max} \cdot [S] / (K_m + [S])$

The **Michaelis constant (K_m)** is the substrate concentration (mol/l) at which the reaction rate equals half the maximum rate. K_m shows the **affinity** of the enzyme for the substrate: the smaller the K_m, the higher the affinity. For example, the K_m of hexokinase for glucose is ≈ 0.1 mmol/l (high affinity, it works even when glucose is low), while that of glucokinase is ≈ 10 mmol/l (it works only when glucose is high — in the liver after a meal).

To find K_m and V_{max} precisely, the double-reciprocal plot — the **Lineweaver–Burk plot** (1/v against 1/[S]) — is used: the straight line crosses the y-axis at $1/V_{max}$ and the x-axis at $-1/K_m$.

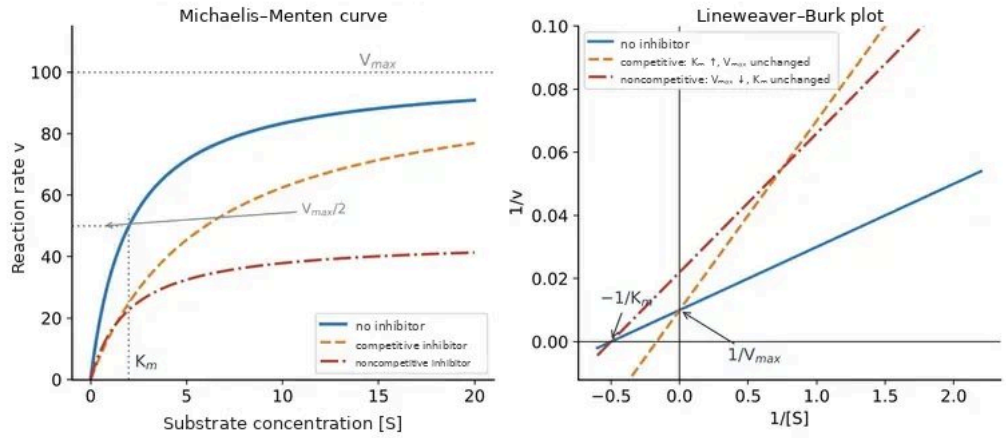


Figure 2.2. *The Michaelis–Menten curve and the Lineweaver–Burk plot; the effect of inhibitors (drawn after Severin E.S.)*

Units of enzyme activity: the international unit (IU, U) — the amount of enzyme that converts 1 μmol of substrate per minute; the **katal** (SI) — the amount of enzyme that converts 1 mol of substrate per second (1 U = 16.67 nkat). **Specific activity** is the activity per 1 mg of protein.

6. Enzyme inhibitors

Inhibitors are substances that reduce enzyme activity.

- **Irreversible inhibition** — the inhibitor binds covalently to a functional group of the active site: organophosphorus compounds (sarin, chlorophos) with the serine of acetylcholinesterase, aspirin acetylates cyclooxygenase, heavy metals (Hg^{2+}, Pb^{2+}) with SH groups, cyanides with the iron of cytochrome oxidase.
- **Reversible inhibition:**
- **Competitive** — the inhibitor is similar in structure to the substrate and competes with the substrate for the active site. Its effect can be abolished by increasing the substrate concentration. **K_m increases, V_{max} does not change.** Examples: malonate — succinate dehydrogenase;

sulfonamides — folate synthesis from para-aminobenzoic acid in bacteria; methotrexate — dihydrofolate reductase; statins — HMG-CoA reductase.

- **Noncompetitive** — the inhibitor binds outside the active site and changes the enzyme's conformation; the effect does not disappear even when the substrate is increased. **V_{max} decreases, K_m does not change.**
- **Allosteric inhibition** — binding of an effector to the allosteric site (see below).

CLINICAL SIGNIFICANCE

Many drugs are enzyme inhibitors: ACE inhibitors (captopril, enalapril — angiotensin-converting enzyme), statins (HMG-CoA reductase), allopurinol (xanthine oxidase — in gout), proton pump inhibitors (H^+,K^+-ATPase), NSAIDs (cyclooxygenase), neostigmine (acetylcholinesterase — in myasthenia). In methanol poisoning ethanol is given: ethanol **competes** with methanol for alcohol dehydrogenase and slows its conversion into toxic formaldehyde.

7. Regulation of enzyme activity

1. **Change in the amount of enzyme** — induction or repression of synthesis (under the action of hormones and substrates; for example, cortisol induces the enzymes of gluconeogenesis, insulin induces glucokinase) and the rate of degradation.
2. **Activation of proenzymes (zymogens)** — partial proteolysis: trypsinogen → trypsin (enteropeptidase), pepsinogen → pepsin (HCl), blood clotting factors, the complement system.
3. **Covalent modification** — **phosphorylation–dephosphorylation**: protein kinases phosphorylate serine, threonine or tyrosine residues at the expense of ATP, and phosphoprotein phosphatases remove the phosphate. For example, glycogen phosphorylase is activated when phosphorylated, whereas glycogen synthase becomes inactive.
4. **Allosteric regulation** — binding of an effector. Often the end product of a metabolic pathway inhibits the first (key) enzyme — **negative feedback** (feedback inhibition).
5. **Protein–protein interaction** — binding or dissociation of regulatory subunits (activation of protein kinase A by cAMP: the regulatory subunits dissociate).
6. **Compartmentalization** — the location of enzymes in different compartments of the cell.

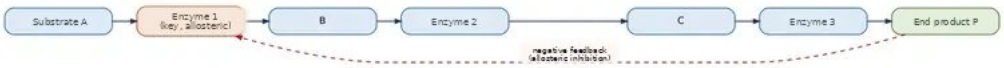


Figure 2.3. *Allosteric regulation of a metabolic pathway by negative feedback*

8. Medical enzymology

Medical enzymology includes three directions:

- **Enzyme diagnostics** — determination of enzyme activity in blood serum. When cells are damaged (cytolysis), intracellular enzymes are released into the blood. **Organ-specific enzymes** indicate the damaged organ.
- **Enzymopathology** — diseases associated with congenital or acquired enzyme deficiency (phenylketonuria — phenylalanine hydroxylase, galactosemia, glucose-6-phosphate dehydrogenase deficiency).
- **Enzyme therapy** — the use of enzymes as drugs: pancreatin (digestive insufficiency), streptokinase and alteplase (dissolving thrombi), asparaginase (in leukemia), hyaluronidase.

Table 2.3. *Enzymes of diagnostic significance*

Enzyme	Mainly in which organ	Increase in blood
ALT (alanine aminotransferase)	Liver	Hepatitis (ALT > AST)
AST (aspartate aminotransferase)	Heart, liver, muscle	Myocardial infarction, liver cirrhosis, alcoholic hepatitis
CK-MB	Heart muscle	Myocardial infarction (from 3–6 hours)
LDH1, LDH2	Heart, erythrocytes	Myocardial infarction, hemolysis
α-amylase, lipase	Pancreas	Acute pancreatitis
Alkaline phosphatase	Liver (bile ducts), bone	Cholestasis, bone tumors, rickets
GGT (γ-glutamyltransferase)	Liver, bile ducts	Cholestasis, alcoholism
Acid phosphatase (prostate)	Prostate gland	Prostate cancer

Immobilized enzymes are enzymes attached to an insoluble carrier; they can be used again and again and are more stable. Glucometer test strips and biosensors (glucose oxidase) work on the basis of such enzymes.

9. Laboratory work

Experiment 1: determination of salivary amylase activity by the Wohlgemuth method

Principle of the method: amylase breaks down starch to dextrins and maltose. Starch gives a blue color with iodine, dextrins — violet-red, maltose — no color. Saliva is serially diluted and the smallest amount that completely breaks down the starch is found.

1. Rinse the mouth with water, collect 1 ml of saliva and dilute it with 9 ml of distilled water (1 : 10).
2. Pour 1 ml of saline into each of 10 test tubes. Add 1 ml of diluted saliva to tube 1, mix and transfer 1 ml of it to tube 2; serially dilute in this way up to tube 10 (1 ml is discarded from the last one). Degree of dilution: 1 : 20, 1 : 40 ... 1 : 10 240.
3. Add 2 ml of 0.1% starch solution to each tube, mix and place in a 38 °C thermostat for 30 minutes.
4. Cool the tubes (to stop the reaction) and add 1 drop of 0.1% iodine solution to each.
5. Identify the most diluted tube in which the starch has been completely broken down (yellow color, no blue). **Calculation:** amylase activity (A) is the number of ml of 0.1% starch solution broken down by 1 ml of undiluted saliva at 38 °C in 30 minutes. For example, if the yellow color is in the tube diluted 1 : 160: A = 2 ml × 160 = 320.

Evaluation: normal salivary amylase activity is **160–320 units** (Wohlgemuth). Urinary amylase (diastase) is normally 16–64 units; it rises sharply in acute pancreatitis.

Experiment 2: demonstrating the specificity of enzymes

1. Pour 1 ml of diluted saliva into each of two test tubes.
2. Add 1 ml of 1% starch to tube 1 and 1 ml of 1% sucrose solution to tube 2; keep at 38 °C for 15 minutes.

3. Perform the Trommer (or Fehling) reaction in both tubes: add 1 ml of 10% NaOH and 3–5 drops of 1% $CuSO_4$ and heat. **Result and analysis:** in tube 1 a red precipitate (Cu_2O) — starch has been broken down to maltose, a reducing sugar. In tube 2 the reaction is negative — amylase does not break down sucrose (sucrose is acted on by sucrase). This shows the relative (group) specificity of the enzyme.

Experiment 3: effect of temperature, activators and inhibitors on amylase activity

1. Keep a saliva + starch mixture in three tubes at 0 °C (ice), 38 °C and 100 °C (previously boiled saliva) for 10 minutes; test with iodine.
2. To three other tubes add, respectively, 1% NaCl (activator), 1% $CuSO_4$ (inhibitor) and water, and keep with saliva + starch at 38 °C; test with iodine from time to time. **Result and analysis:** at 38 °C starch is broken down quickly; on ice — very slowly (reversible inactivation); with boiled saliva — not at all (denaturation). Cl^- ions activate amylase (breakdown speeds up), Cu^{2+} ions inhibit it.

10. Practical skills

Skill: determining the type of inhibition from a Lineweaver–Burk plot

1. Draw the straight lines of 1/v against 1/[S] without inhibitor and with inhibitor.
2. If the lines cross on the y-axis ($1/V_{max}$) — V_{max} is unchanged and K_m has increased: **competitive inhibition**.
3. If the lines cross on the x-axis ($-1/K_m$) — K_m is unchanged and V_{max} has decreased: **noncompetitive inhibition**.
4. Conclude whether increasing the substrate concentration can abolish the inhibition.

Skill: evaluating the result of enzyme diagnostics

1. Determine which enzymes are increased.
2. Recall which organ they are characteristic of (organ specificity, isoenzymes).
3. Calculate the De Ritis ratio: AST/ALT (normal 1.33 ± 0.42): < 1 — liver damage (hepatitis), > 2 — myocardial infarction or alcoholic hepatitis.
4. Take into account the dynamics of the enzyme rise (CK-MB rises in 3–6 hours, peaks in 1–2 days).
5. Compare the conclusion with the clinical data.

Review questions

1. What are enzymes? What is their chemical nature?
2. What are simple and complex enzymes, apoenzyme, cofactor and coenzyme?
3. How is the active site structured?
4. What are the allosteric site and effectors?
5. What are isoenzymes? Give examples.
6. How are enzymes classified?
7. Explain the mechanism of enzyme action.
8. List the types of enzyme specificity.
9. How does the reaction rate depend on temperature and pH?

10. Write the Michaelis–Menten equation. What does K_m mean?
11. What is the difference between competitive and noncompetitive inhibition?
12. Which drugs are enzyme inhibitors?
13. By what methods is enzyme activity regulated?
14. Explain the directions of medical enzymology.
15. How is amylase activity determined by the Wohlgemuth method?

Test questions

1. What the apoenzyme and coenzyme together are called:
   A) Proenzyme
   B) Isoenzyme
   C) Prosthetic group
   D) Ribozyme
   E) Holoenzyme

2. What the Michaelis constant (K_m) means:
   A) The maximum reaction rate
   B) The substrate concentration at which the rate equals half of V_{max}
   C) The enzyme concentration
   D) The inhibitor concentration
   E) The equilibrium constant of the reaction

3. How the kinetic parameters change in competitive inhibition:
   A) K_m does not change, V_{max} decreases
   B) K_m and V_{max} decrease
   C) Neither changes
   D) V_{max} increases
   E) K_m increases, V_{max} does not change

4. The class to which aminotransferases belong:
   A) Oxidoreductases
   B) Hydrolases
   C) Transferases
   D) Lyases
   E) Ligases

5. The pH optimum of pepsin:
   A) 1,5–2,0
   B) 4,5–5,0
   C) 6,8–7,0
   D) 7,8–8,5
   E) 9–10

6. The enzyme inhibited by statins:
   A) HMG-CoA reductase
   B) Cyclooxygenase
   C) Xanthine oxidase
   D) Acetylcholinesterase
   E) Dihydrofolate reductase

7. How proenzymes (zymogens) are activated:
    A) By phosphorylation
    B) By partial proteolysis
    C) By binding of an allosteric activator
    D) By attachment of a coenzyme
    E) By denaturation

8. The type of specificity shown by urease acting only on urea:
    A) Relative
    B) Absolute
    C) Stereospecificity
    D) Group
    E) Reaction specificity

9. The covalent modification that activates glycogen phosphorylase:
    A) Acetylation
    B) Methylation
    C) Partial proteolysis
    D) Phosphorylation
    E) Glycosylation

10. The early and specific enzyme marker of myocardial infarction:
    A) Alkaline phosphatase
    B) α-amylase
    C) GGT
    D) CK-MB
    E) Lipase

11. The type of inhibitor methotrexate is:
    A) Irreversible, covalent
    B) Noncompetitive
    C) A competitive inhibitor of dihydrofolate reductase
    D) An allosteric activator
    E) An irreversible inhibitor of cyclooxygenase

12. The normal activity of salivary amylase by the Wohlgemuth method:
    A) 1–5 units
    B) 16–64 units
    C) 160–320 units
    D) 1000–2000 units
    E) More than 5000 units

13. How organophosphorus compounds inhibit acetylcholinesterase:
    A) Irreversibly, by covalently binding the serine of the active site
    B) Competitively, reversibly
    C) By binding to the allosteric site
    D) By removing the coenzyme
    E) By denaturing the enzyme

14. What inhibition of the first enzyme by the end product of a metabolic pathway is called:
    A) Induction
    B) Repression
    C) Compartmentalization
    D) Partial proteolysis
    E) Negative feedback (feedback inhibition)

15. The number of isoenzymes of lactate dehydrogenase:
    A) 2
    B) 3
    C) 4
    D) 5
    E) 7

Clinical cases

Case 1

After working in a newly renovated warehouse, a patient was brought to hospital with profuse salivation, sweating, constricted pupils, bronchospasm and muscle twitching. Serum cholinesterase activity is sharply reduced. An organophosphorus pesticide had been used in the warehouse.

Questions: Explain the biochemical mechanism of the condition. On what principle is treatment based?

Answer: Organophosphorus compounds phosphorylate the serine residue in the active site of acetylcholinesterase and inhibit the enzyme irreversibly. Acetylcholine is not broken down and accumulates in the synapses — muscarinic (salivation, sweating, miosis, bronchospasm, bradycardia) and nicotinic (fasciculations, paralysis) signs develop. Treatment: atropine (blocks muscarinic receptors) and enzyme reactivators — oximes (pralidoxime): they remove the phosphate group from the serine and reactivate the enzyme (effective in the first hours, before the phosphorylated enzyme “ages”).

Case 2

A patient drank methyl alcohol by mistake. He was prescribed intravenous ethyl alcohol solution and fomepizole.

Questions: On what biochemical principle is this treatment based?

Answer: Methanol is converted by alcohol dehydrogenase into formaldehyde and then into formic acid — they damage the optic nerve and cause metabolic acidosis. Ethanol is a substrate of the same enzyme and competes with methanol (it has a higher affinity for the enzyme); fomepizole is a competitive inhibitor of alcohol dehydrogenase. As a result the conversion of methanol into toxic products slows down, and it is excreted unchanged through the kidneys and lungs (hemodialysis is also used).

Case 3

A 35-year-old patient suddenly developed severe girdle-like pain in the upper abdomen. Urinary diastase is 1024 units, and serum lipase and α-amylase are sharply increased.

Questions: Which organ is damaged? Why has enzyme activity increased?

Answer: Acute pancreatitis. When pancreatic cells are damaged (premature activation of proteolytic enzymes inside the gland — conversion of trypsinogen to trypsin, autolysis), amylase and lipase are released into the blood. Because amylase is a small molecule, it passes through the kidneys into the urine — urinary diastase rises (normal 16–64 units). Lipase is more specific for pancreatitis and persists longer in the blood.

Quick review

Table 2.4. *Key concepts of topic 2*

Concept	Brief definition
Enzyme	A biological catalyst of protein nature; lowers the activation energy
Holoenzyme	Apoenzyme + cofactor (coenzyme or metal ion)
Active site	Binding + catalytic part
Allosteric site	The place where effectors (activator, inhibitor) bind
Isoenzymes	Forms that catalyze one reaction but differ in structure (LDH1–5, CK)
7 classes	Oxidoreductase, transferase, hydrolase, lyase, isomerase, ligase, translocase
K_m	[S] at v = $V_{max}/2$; low K_m — high affinity
Competitive inhibition	K_m ↑, V_{max} unchanged (statins, methotrexate, malonate)
Noncompetitive inhibition	V_{max} ↓, K_m unchanged
Irreversible inhibition	Organophosphorus compounds, aspirin, cyanide, heavy metals
Regulation	Amount, partial proteolysis, phosphorylation, allostery, feedback
Optimum	Temperature 37–40 °C; pH specific to the enzyme (pepsin 1.5–2, trypsin 7.8–8.5)
Amylase (Wohlgemuth)	In saliva 160–320, in urine 16–64 units

TOPIC 3

Biological membranes. Membrane receptors. Signal transduction pathways into the cell

Learning objectives

To teach the student the chemical composition, structure, functions and general properties of biological membranes, the routes of transport of substances across the membrane, the types of membrane receptors, signal transduction into the cell and the systems of second messengers, and the significance of membrane damage in pathology.

By the end of the class the student should know:

- the lipid and protein composition of membranes and the fluid mosaic model;
- the functions of membranes;
- the general properties of membranes: transverse asymmetry, fluidity, selective permeability;
- the types of passive and active transport, endocytosis and exocytosis;
- the types of membrane receptors;
- the systems of G proteins, adenylate cyclase and phospholipase C, and the second messengers;
- tyrosine kinase receptors and receptors linked to ion channels;
- the causes and consequences of membrane damage.

The student should be able to:

- identify the type of transport according to the substance;
- explain the mechanism of hormone action by the signal transduction system;
- determine the osmotic resistance of erythrocytes.

1. Structure and chemical composition of biological membranes

Biological membranes are lipid-protein structures 6–10 nm thick that surround the cell and its organelles. The cell has a plasma membrane and internal membranes (the membranes of the nucleus, endoplasmic reticulum, Golgi apparatus, mitochondria and lysosomes).

Membranes contain **lipids** (≈ 40%), **proteins** (≈ 50–60%) and **carbohydrates** (2–10%, as part of glycolipids and glycoproteins). In the myelin sheath lipids make up 75–80%, in the inner mitochondrial membrane proteins ≈ 75%.

Membrane lipids:

- **Phospholipids** — glycerophospholipids (phosphatidylcholine, phosphatidylethanolamine, phosphatidylserine, phosphatidylinositol) and sphingomyelin. They are **amphiphilic**: a hydrophilic “head” and two hydrophobic “tails” (fatty acid residues). In water they form a **lipid bilayer**.
- **Glycolipids** — cerebrosides, gangliosides (mainly in the outer layer; receptor and antigen function).
- **Cholesterol** — regulates the rigidity and fluidity of the membrane (a “buffer”): at high temperature it reduces fluidity, at low temperature it prevents solidification. **Membrane proteins:**
- **Integral (transmembrane)** proteins — span the bilayer (channels, carriers, pumps, receptors).

- **Peripheral** proteins — bound to the membrane surface by ionic and hydrogen bonds.
- **Anchored** proteins — attached through a lipid "anchor" (a fatty acid, prenyl, glycosylphosphatidylinositol). **Fluid mosaic model** (S. Singer and G. Nicolson, 1972): the membrane is a fluid lipid bilayer in which proteins "float" and can move laterally.

2. Functions and general properties of membranes

Functions:

1. **Barrier** — separates the internal environment of the cell and organelles, compartmentalization.
2. **Transport** — selective passage of substances.
3. **Receptor (signaling)** — reception of hormones, mediators and other signals.
4. **Enzymatic** — membrane-bound enzymes (respiratory chain, ATP synthase, adenylate cyclase).
5. **Energy conversion** — formation of an electrochemical gradient in the mitochondria and ATP synthesis.
6. **Intercellular contact** — gap junctions, adhesion molecules, antigenicity (blood groups, histocompatibility antigens).
7. **Electrogenic** — generation of the membrane potential and conduction of excitation. **General properties:**

- **Transverse asymmetry** — the composition of the outer and inner leaflets differs: the outer leaflet is rich in phosphatidylcholine, sphingomyelin and glycolipids, the inner in phosphatidylserine and phosphatidylethanolamine. The movement of lipids from one leaflet to the other ("flip-flop") is very slow and is carried out by special enzymes (flippases). Carbohydrate parts are only on the outer surface (glycocalyx).
- **Fluidity** — lateral movement of lipids and proteins; fluidity increases when unsaturated fatty acids increase and when the temperature rises, while cholesterol and saturated fatty acids reduce it.
- **Selective permeability** — small hydrophobic molecules (O_2, CO_2, steroids) pass freely; ions and polar molecules pass only through special proteins.

> **CLINICAL SIGNIFICANCE**
> In apoptosis phosphatidylserine moves to the outer leaflet of the membrane — this is an "eat me" signal for macrophages. When platelets are activated, phosphatidylserine also moves outward and forms a surface for the clotting factor complexes.

3. Transport of substances across the membrane

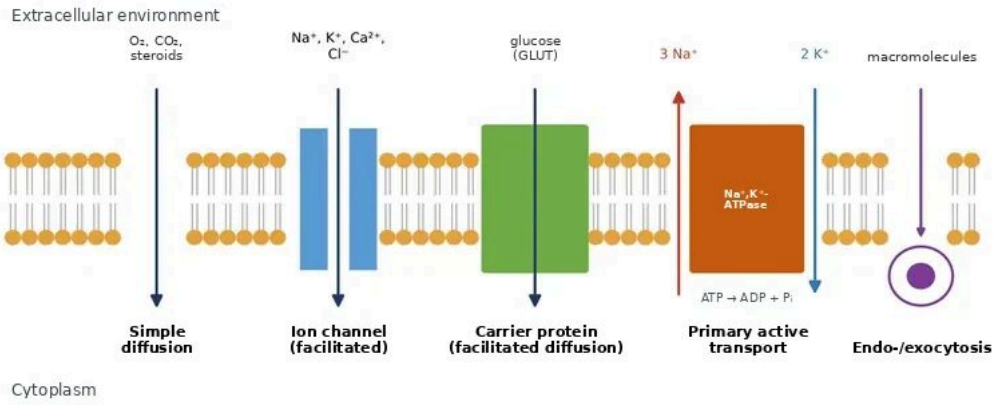


Figure 3.1. *Types of transport of substances across the membrane (drawn after Severin E.S.)*

Table 3.1. *Types of membrane transport*

Type	Energy	Relative to the gradient	Examples
Simple diffusion	Not consumed	Down the concentration gradient	O_2, CO_2, urea, ethanol, steroid hormones
Facilitated diffusion: channels	Not consumed	Relative to the gradient	Na^+, K^+, Ca^{2+}, Cl^- ion channels; aquaporins (water)
Facilitated diffusion: carriers	Not consumed	Relative to the gradient	Glucose transporters GLUT1–GLUT5
Primary active transport	ATP hydrolysis	Against the gradient	Na^+,K^+-ATPase (3 Na^+ out, 2 K^+ in), Ca^{2+}-ATPase, H^+,K^+-ATPase (stomach)
Secondary active transport	Energy of the Na^+ gradient	Against the gradient	Symport: Na^+-glucose (SGLT1 in the intestine, SGLT2 in the kidney), Na^+-amino acids; antiport: Na^+/Ca^{2+}, Na^+/H^+
Endocytosis and exocytosis	Consumed	—	Pinocytosis, phagocytosis, receptor-mediated endocytosis (LDL, transferrin), secretion

Carriers (uniport, symport, antiport) have enzyme-like properties: specificity, saturation (V_{max}), competitive inhibition. For example, GLUT4 is insulin-sensitive (muscle, adipose tissue): insulin moves it from intracellular vesicles to the membrane.

> **REMEMBER**
> **Na^+,K^+-ATPase** uses 30–40% of the cell's ATP (up to 70% in the brain). It maintains the membrane potential, cell volume and the Na^+ gradient for secondary active transport. **Cardiac glycosides** (digoxin) inhibit this enzyme: Na^+ increases inside the cell, Na^+/Ca^{2+} exchange weakens, Ca^{2+} accumulates and myocardial contraction becomes stronger.

4. Membrane receptors and signal transduction

Signaling molecules (**ligands**, first messengers) are hormones, neurotransmitters, cytokines and growth factors. Hydrophilic ligands (peptide hormones, catecholamines) do not cross the membrane

and act through **membrane receptors**; hydrophobic ligands (steroid and thyroid hormones, calcitriol) cross the membrane, bind to **intracellular (nuclear) receptors** and change gene transcription.

Table 3.2. *The main types of membrane receptors*

Receptor type	Mechanism of signal transduction	Examples
Ion channel–linked (ionotropic)	The channel opens when the ligand binds — a fast response (milliseconds)	N-cholinergic receptor, GABAA, NMDA glutamate receptor
G protein–coupled (crossing the membrane 7 times)	The G protein is activated and regulates an effector enzyme; second messengers	β- and α-adrenergic receptors, glucagon, vasopressin, M-cholinergic receptors
With tyrosine kinase activity	After ligand binding — dimerization and autophosphorylation; signaling proteins (Ras, MAP kinases, PI3 kinase)	Insulin receptor, epidermal growth factor (EGF) receptor
Linked to tyrosine kinases (JAK-STAT)	Receptor-associated JAK kinases phosphorylate STAT proteins	Growth hormone, prolactin, erythropoietin, interferons
With guanylate cyclase activity	cGMP is formed	Natriuretic peptide receptor

The G protein system

G proteins are GTP-binding heterotrimeric proteins (α, β, γ subunits). At rest the α subunit is bound to GDP. When the receptor is activated, GDP is exchanged for GTP, α-GTP dissociates and acts on the effector enzyme; the α subunit's own GTPase activity hydrolyzes GTP to GDP and terminates the signal.

- **G_s** — activates adenylate cyclase (cAMP ↑);
- **G_i** — inhibits adenylate cyclase (cAMP ↓);
- **G_q** — activates phospholipase C. **The adenylate cyclase system:** adenylate cyclase forms **cAMP** from ATP. cAMP binds to the regulatory subunits of **protein kinase A** and releases the catalytic subunits; these phosphorylate target proteins (phosphorylase kinase, hormone-sensitive lipase and others). cAMP is broken down to AMP by **phosphodiesterase** (caffeine and theophylline inhibit phosphodiesterase and prolong the action of cAMP).

The inositol phosphate system: phospholipase C cleaves membrane phosphatidylinositol-4,5-bisphosphate (PIP_2) into two second messengers: **inositol-1,4,5-trisphosphate (IP_3)** — releases Ca^{2+} from the endoplasmic

reticulum; **diacylglycerol (DAG)** — together with Ca^{2+} activates **protein kinase C**. Many effects of Ca^{2+} are carried out through the protein **calmodulin**.

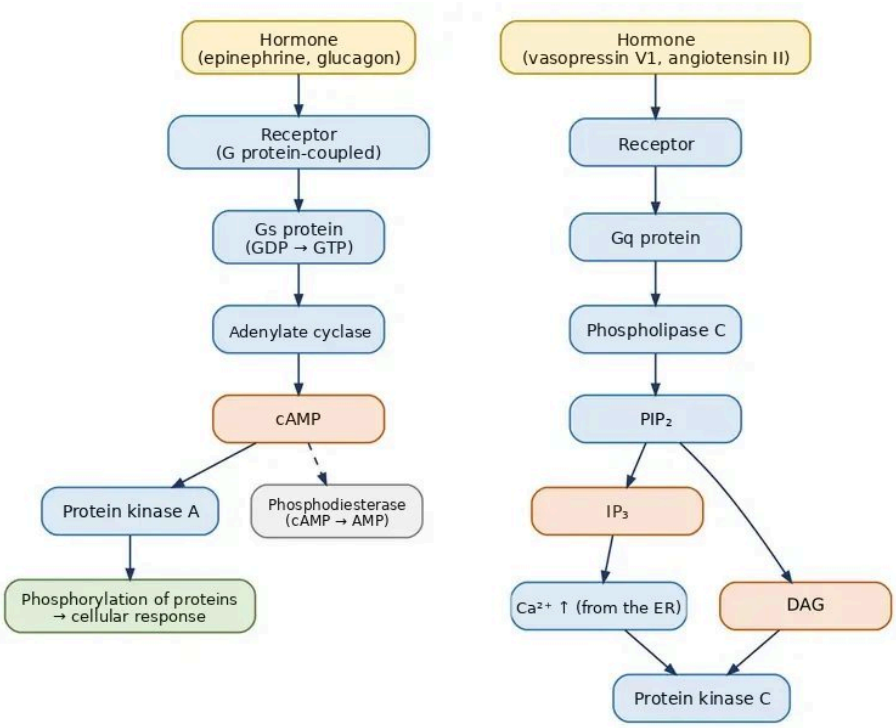


Figure 3.2. *The adenylate cyclase and inositol phosphate signaling systems (drawn after Severin E.S.)*

Table 3.3. *Second messengers*

Second messenger	Enzyme that forms it	Target of action
cAMP	Adenylate cyclase	Protein kinase A, ion channels
cGMP	Guanylate cyclase (NO, natriuretic peptide)	Protein kinase G — smooth muscle relaxation
IP_3	Phospholipase C	Ca^{2+} channels of the endoplasmic reticulum
DAG	Phospholipase C	Protein kinase C
Ca^{2+}	Enters through channels or is released from stores	Calmodulin, protein kinase C, muscle contraction
NO	NO synthase (from arginine)	Soluble guanylate cyclase

CLINICAL SIGNIFICANCE

Cholera toxin ADP-ribosylates the α subunit of the G_s protein and abolishes its GTPase activity — the G protein stays permanently active, cAMP rises sharply in the intestinal epithelium, and Cl^- and water move into the intestinal lumen — severe diarrhea. Pertussis toxin, in contrast, blocks the G_i protein. Nitroglycerin releases NO and dilates vessels through cGMP; sildenafil inhibits phosphodiesterase-5, which breaks down cGMP.

5. Membrane damage

Membrane damage is a key link in the pathogenesis of many diseases:

- **Lipid peroxidation (LPO)** — free radicals (O_2
- −, $^{\bullet}OH$) oxidize polyunsaturated fatty acids; membrane permeability increases, enzymes and receptors fail. Antioxidant defense: vitamins E and C, glutathione, superoxide dismutase, catalase, glutathione peroxidase.

- **Activation of phospholipases** (ischemia, excess Ca^{2+}) — phospholipids are broken down.
- **Detergent action** — excess bile acids and fatty acids.
- **Mechanical and osmotic damage**, the membrane attack complex of the complement system, bacterial toxins.
- **Congenital defects** — a defect of membrane skeleton proteins (spectrin) — hereditary spherocytosis; ion channel diseases (channelopathies) — cystic fibrosis (the CFTR Cl^- channel).

> **CAUTION**
> The consequence of cell damage is the release of intracellular enzymes into the blood (cytolysis). Therefore an increase in blood ALT, AST, CK and LDH activity is a biochemical sign of membrane damage.

6. Laboratory work

Experiment 1: determination of the osmotic resistance of erythrocytes

1. Pour 1 ml each of NaCl solutions from 0.7% to 0.2% (in steps of 0.02–0.05%) into test tubes.
2. Add 1 drop of blood to each tube, mix and leave for 1 hour (or centrifuge).
3. The tube with the highest concentration in which the solution has turned pink and there is a sediment — **minimal resistance** (onset of hemolysis).
4. The tube in which there is no sediment at all and the solution is clear red ("lacquered blood") — **maximal resistance** (complete hemolysis). **Evaluation:** normally minimal resistance is 0.46–0.48% and maximal 0.32–0.34% NaCl. In hereditary spherocytosis resistance decreases (hemolysis begins at a higher concentration), in thalassemia and iron deficiency anemia it increases (thin, flat erythrocytes).

Analysis: in a hypotonic medium water enters the erythrocyte by osmosis, the cell swells, its membrane ruptures and hemoglobin is released. Resistance depends on the membrane composition, the state of the cytoskeleton and the cell shape (surface/volume ratio).

Experiment 2: effect of a detergent and temperature on membrane permeability (demonstration)

1. Pour 2 ml of an erythrocyte suspension in 0.9% NaCl into three test tubes.
2. Add 3 drops of 1% bile salt (or saponin) to tube 1 and 3 drops of ethanol to tube 2; heat tube 3 at 60 °C for 10 minutes.
3. Compare with the control tube. **Result and analysis:** under the action of a detergent and heating, membrane lipids and proteins are disrupted — hemolysis (the solution is clear red). This shows the consequence of loss of membrane integrity.

7. Practical skills

Skill: identifying the type of transport of a substance

1. Determine the nature of the substance: a small hydrophobic molecule, an ion, a polar molecule or a macromolecule.
2. Determine the direction of the gradient: down the gradient — passive, against the gradient — active.

3. In active transport determine the energy source: ATP (primary) or another ion gradient (secondary).
4. Check whether there is saturation and specificity — the participation of a carrier.
5. Conclude with an example (for example, glucose in the intestine — SGLT1 secondary active, then facilitated diffusion into the blood through GLUT2).

Skill: explaining the mechanism of hormone action

1. Determine the chemical nature of the hormone (hydrophilic or hydrophobic).
2. Determine the location of the receptor (membrane or intracellular).
3. If it is a membrane receptor — identify the type of G protein and the second messenger.
4. Indicate the protein kinase and the target protein that is phosphorylated.
5. Explain how the signal is terminated (GTPase, phosphodiesterase, phosphatases).

Review questions

1. What is the chemical composition of biological membranes?
2. What are the types and properties of membrane lipids?
3. What is the difference between integral and peripheral proteins?
4. What does the fluid mosaic model explain?
5. List the main functions of membranes.
6. What is transverse asymmetry?
7. What factors affect membrane fluidity?
8. Describe the types of passive transport.
9. What is the difference between primary and secondary active transport?
10. What is the significance of Na^+,K^+-ATPase?
11. List the main types of membrane receptors.
12. How do G proteins work?
13. Explain the adenylate cyclase system.
14. How does the inositol phosphate system work?
15. What are the causes and consequences of membrane damage?

Test questions

1. The property of phospholipids that causes them to form a lipid bilayer in the membrane:
   A) Good solubility in water
   B) Positive charge
   C) Hydrophobicity alone
   D) Formation of covalent bonds
   E) Amphiphilicity
2. The “buffer” lipid that regulates membrane fluidity:
   A) Phosphatidylcholine
   B) Sphingomyelin
   C) Cholesterol
   D) Ganglioside
   E) Triacylglycerol

3. The leaflet of the membrane in which phosphatidylserine is mainly located:
   A) In the outer leaflet
   B) In the inner (cytoplasmic) leaflet
   C) Equally in both leaflets
   D) In the glycocalyx
   E) Only in the mitochondria

4. The number of ions transported by Na^+,K^+-ATPase per ATP:
   A) 3 Na^+ out, 2 K^+ in
   B) 2 Na^+ out, 3 K^+ in
   C) 1 Na^+ and 1 K^+
   D) 3 Na^+ in, 2 K^+ out
   E) Only Na^+

5. The type of transport of glucose absorbed together with Na^+ in the intestine:
   A) Simple diffusion
   B) Facilitated diffusion
   C) Primary active transport
   D) Secondary active transport (symport)
   E) Pinocytosis

6. Where the receptors of steroid hormones are located:
   A) In the plasma membrane
   B) In the outer mitochondrial membrane
   C) In the Golgi apparatus
   D) In lysosomes
   E) Inside the cell (in the cytoplasm and nucleus)

7. The second messenger formed by adenylate cyclase:
   A) cAMP
   B) cGMP
   C) IP_3
   D) DAG
   E) NO

8. The messengers formed from PIP_2 by phospholipase C:
   A) IP_3 and DAG
   B) cAMP and cGMP
   C) Arachidonic acid and prostaglandins
   D) Ca^{2+} and calmodulin
   E) NO and cGMP

9. The type to which the insulin receptor belongs:
   A) Ion channel–linked
   B) G protein–coupled
   C) Guanylate cyclase
   D) Intracellular receptor
   E) With tyrosine kinase activity

10. How cholera toxin acts:
    A) Blocks the G_i protein
    B) Keeps the G_s protein permanently active
    C) Activates phosphodiesterase
    D) Inhibits Na^+,K^+-ATPase
    E) Breaks down adenylate cyclase

11. The enzyme inhibited by cardiac glycosides:
    A) Ca^{2+}-ATPase
    B) Na^+,K^+-ATPase
    C) H^+,K^+-ATPase
    D) Adenylate cyclase
    E) Phospholipase C

12. How caffeine increases cAMP:
    A) Activates adenylate cyclase
    B) Activates the G_s protein
    C) Inhibits phosphodiesterase
    D) Activates protein kinase A
    E) Increases the number of receptors

13. The normal lower limit (minimal) of erythrocyte osmotic resistance:
    A) 0.9% NaCl
    B) 0.7% NaCl
    C) 0.2% NaCl
    D) 0.46–0.48% NaCl
    E) 0.1% NaCl

14. The insulin-sensitive glucose transporter:
    A) GLUT1
    B) GLUT2
    C) GLUT4
    D) GLUT5
    E) SGLT2

15. The fat-soluble vitamin that protects against lipid peroxidation:
    A) Vitamin A
    B) Vitamin D
    C) Vitamin K
    D) Vitamin E
    E) Vitamin B_{12}

Clinical cases

Case 1

A patient who came from an endemic area has watery “rice-water” diarrhea of up to 15–20 liters a day, severe dehydration and a fall in blood pressure. Vibrio cholerae was detected in the stool.

Questions: Explain the biochemical mechanism of the diarrhea. Why is glucose added to the rehydration solution?

Answer: Cholera toxin ADP-ribosylates the α subunit of the G_s protein in enterocytes; its GTPase activity is lost and adenylate cyclase is permanently activated. cAMP rises and, through protein kinase A, phosphorylates the CFTR Cl^- channel — Cl^-, followed by Na^+ and water, moves into the intestinal lumen. The glucose in the oral rehydration solution provides, through SGLT1 (Na^+-glucose symport), the absorption of Na^+ and, with it, of water by osmosis; this carrier is not impaired by cAMP.

Case 2

A patient with chronic heart failure takes digoxin. After losing potassium because of diarrhea, he developed nausea, yellowish spots in his vision and heart rhythm disturbances.

Questions: What is the mechanism of action of digoxin? Why does hypokalemia increase the toxicity?

Answer: Digoxin inhibits Na^+,K^+-ATPase in cardiomyocytes — Na^+ increases inside the cell, the Na^+/Ca^{2+} exchanger removes less Ca^{2+}, Ca^{2+} accumulates in the cell and contraction becomes stronger (a positive inotropic effect). Digoxin competes with K^+ for the same site on the ATPase: when K^+ falls, digoxin binds to the enzyme more and its effect (toxicity) increases — arrhythmias develop.

Case 3

An 8-year-old child has recurrent jaundice, anemia and an enlarged spleen. The blood smear shows small spherical erythrocytes, and osmotic resistance is reduced: hemolysis begins at 0.60% NaCl. His father has the same disease.

Questions: What is the diagnosis? What is its biochemical basis?

Answer: Hereditary spherocytosis (Minkowski–Chauffard disease). Because of a defect of membrane skeleton proteins (spectrin, ankyrin), the lipid bilayer is poorly bound to the cytoskeleton, pieces of membrane are lost, the erythrocyte becomes spherical and its surface/volume ratio decreases. Such cells tolerate swelling poorly (reduced osmotic resistance) and are trapped and destroyed in the spleen — hemolytic anemia, raised bilirubin (jaundice), splenomegaly.

Quick review

Table 3.4. *Key concepts of topic 3*

Concept	Brief definition
Membrane composition	Lipids ≈ 40%, proteins 50–60%, carbohydrates 2–10%
Fluid mosaic model	A fluid lipid bilayer + proteins moving in it (Singer–Nicolson)
Asymmetry	Outside phosphatidylcholine, sphingomyelin, glycolipids; inside phosphatidylserine
Cholesterol	The “buffer” of membrane fluidity
Passive transport	Simple and facilitated diffusion (channels, GLUT)
Primary active	Na^+,K^+-ATPase: 3 Na^+ out, 2 K^+ in
Secondary active	At the expense of the Na^+ gradient: SGLT1, SGLT2
G_s / G_i / G_q	Adenylate cyclase ↑ / ↓ / phospholipase C ↑
Second messengers	cAMP, cGMP, IP_3, DAG, Ca^{2+}, NO
Tyrosine kinase receptors	Insulin, EGF

Concept	Brief definition
Intracellular receptors	Steroid and thyroid hormones, calcitriol
LPO	Free radicals oxidize membrane lipids; protection — vitamins E and C, glutathione
Osmotic resistance	Normal: 0.46–0.48% (onset) — 0.32–0.34% (complete)

TOPIC 4

Biological oxidation

Learning objectives

To give the student knowledge of the concepts of energy metabolism and biological oxidation, high-energy compounds and the significance of ATP, the structure and components of the respiratory (electron transport) chain, the mechanism of oxidative phosphorylation (the chemiosmotic theory), substrate-level phosphorylation, respiratory control and the P/O ratio, respiratory chain inhibitors and uncouplers, and mitochondrial diseases.

By the end of the class the student should know:

- exergonic and endergonic reactions and their coupling;
- high-energy compounds and the ATP–ADP cycle;
- the types of biological oxidation (dehydrogenation, electron donation, oxygen addition);
- the complexes and carriers of the respiratory chain and their arrangement according to redox potentials;
- the chemiosmotic theory of oxidative phosphorylation and how ATP synthase works;
- the reactions of substrate-level phosphorylation;
- the P/O ratio and respiratory control;
- respiratory chain inhibitors and uncouplers;
- extramitochondrial oxidation and reactive oxygen species.

The student should be able to:

- determine succinate dehydrogenase activity by the Thunberg method;
- determine catalase and peroxidase activity;
- give a biochemical explanation of respiratory chain disorders in poisonings.

1. Energy metabolism. High-energy compounds

Reactions in the living cell are of two types: **exergonic** (free energy is released, $\Delta G < 0$, proceeds spontaneously) and **endergonic** (energy is consumed, $\Delta G > 0$). Endergonic reactions proceed **coupled** with exergonic reactions; the universal mediator of this coupling is **ATP**.

High-energy compounds are substances that release a large amount of free energy on hydrolysis ($\Delta G^{\circ\prime} < -30$ kJ/mol): ATP and other nucleoside triphosphates (GTP, UTP, CTP), creatine phosphate, phosphoenolpyruvate, 1,3-bisphosphoglycerate, acetyl-CoA, succinyl-CoA.

Table 4.1. *Standard free energy of hydrolysis of some phosphate compounds*

Compound	$\Delta G^{\circ\prime}$, kJ/mol
Phosphoenolpyruvate	−61,9
1,3-bisphosphoglycerate	−49,4
Creatine phosphate	−43,1
ATP → ADP + P_i	−30,5
Glucose-6-phosphate	−13,8
Glycerol-3-phosphate	−9,2

The intermediate value of ATP allows it to accept energy from compounds with a higher potential and pass it on to those with a lower potential. In the human body an amount of ATP close to body weight (≈ 40–60 kg) is formed and broken down per day, yet at any moment the body contains only ≈ 50–100 g of ATP — the **ATP–ADP cycle** turns over very rapidly.

Routes of ATP formation:

1. **Oxidative phosphorylation** — in the mitochondria at the expense of the respiratory chain energy (≈ 90%).
2. **Substrate-level phosphorylation** — direct transfer of a phosphate group from a high-energy substrate to ADP: in glycolysis (1,3-bisphosphoglycerate → 3-phosphoglycerate, phosphoenolpyruvate → pyruvate) and in the Krebs cycle (succinyl-CoA → succinate, GTP is formed).

2. Biological oxidation. The respiratory chain

Biological oxidation is the sum of the oxidation-reduction reactions that take place in living cells. Oxidation is the removal of electrons (and often of hydrogen atoms), reduction is their addition. Hydrogen atoms removed from substrates ($2H^+ + 2e^-$) pass to **NAD$^+$** or **FAD**, are then delivered to oxygen through the **respiratory (electron transport) chain**, and water is formed. This process is called **tissue respiration**.

The respiratory chain is located in the **inner mitochondrial membrane**. Electrons move along the carriers from the side with a low (negative) **redox potential** to the side with a high (positive) one: $NAD^+/NADH$ (−0.32 V) → ... → $½O_2/H_2O$ (+0.82 V). The total potential difference is ≈ 1.14 V and the free energy released is ≈ 220 kJ/mol (from NADH).

Table 4.2. *Complexes of the respiratory chain*

Complex	Name	Composition (prosthetic groups)	Function	Protons
I	NADH dehydrogenase (NADH:CoQ oxidoreductase)	FMN, iron-sulfur centers	Passes electrons from NADH to CoQ	4 H^+
II	Succinate dehydrogenase	FAD, Fe-S centers	Passes electrons from succinate to CoQ (a Krebs cycle enzyme)	0
—	Coenzyme Q (ubiquinone)	A lipid-soluble mobile carrier	Carries electrons from complexes I and II to III	—
III	CoQ:cytochrome c oxidoreductase (cytochrome bc1)	Cytochromes b, c1, Fe-S	Passes electrons to cytochrome c	4 H^+
—	Cytochrome c	Heme (Fe)	A mobile carrier on the outer surface of the membrane	—
IV	Cytochrome oxidase	Cytochromes a, a3, CuA, CuB	Passes electrons to O_2, water is formed	2 H^+
V	ATP synthase (FoF1)	F_o — proton channel, F_1 — catalytic part	ATP synthesis at the expense of the energy of the H^+ flow	≈ 3–4 H^+ / ATP

Cytochromes are heme-containing proteins; their iron passes between the Fe^{3+} ⇄ Fe^{2+} states and carries one electron at a time.

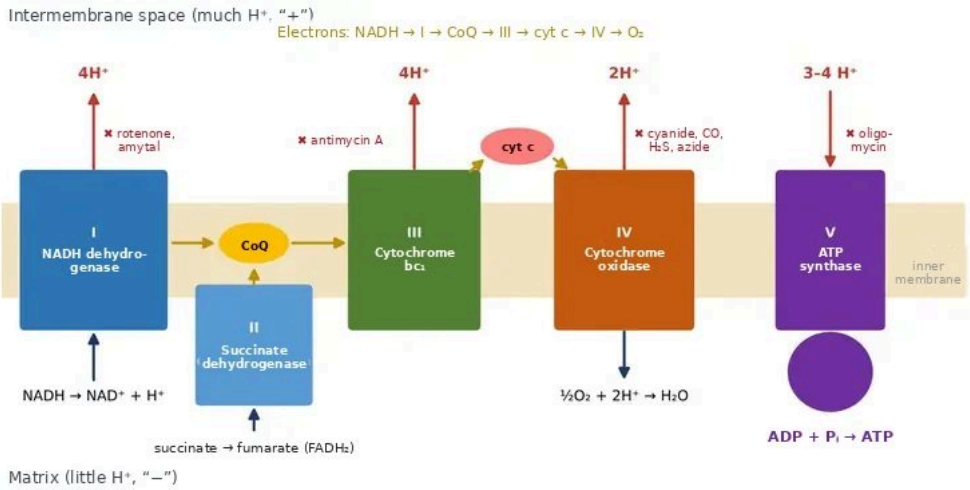


Figure 4.1. *The respiratory chain in the inner mitochondrial membrane, the proton gradient and ATP synthase; sites of action of inhibitors (drawn after Severin E.S.)*

3. Oxidative phosphorylation

Oxidative phosphorylation is ATP synthesis coupled to electron transport in the respiratory chain. Its mechanism is explained by **P. Mitchell's chemiosmotic theory** (1961, Nobel Prize 1978):

1. Complexes I, III and IV pump protons from the matrix into the **intermembrane space** at the expense of the energy of electron transport.
2. The inner membrane is impermeable to protons — an **electrochemical proton gradient** ($\Delta\mu H+$) is formed: a chemical part ($\Delta pH \approx 0.5–1$) and an electrical part ($\Delta\psi \approx 150–180$ mV).
3. Protons return to the matrix only through the **F_o channel of ATP synthase**; this flow drives the rotation of F_1 (rotational catalysis, P. Boyer) and the synthesis of ATP from ADP + P_i. The **P/O ratio** is the number of ATP molecules formed per 1 atom of oxygen (2 electrons). According to current data, oxidation of NADH yields **≈ 2.5 ATP** and oxidation of $FADH_2$ (succinate) **≈ 1.5 ATP** (older textbooks give whole numbers — 3 and 2).

Respiratory control is the dependence of the rate of respiration (O_2 consumption) on the ADP concentration. When ATP is used and ADP increases, ATP synthase speeds up, the proton gradient decreases and the respiratory chain speeds up. When ADP is low, the gradient is high and the electron flow slows down. Thanks to this, ATP formation is matched to the cell's needs.

ATP is exported to the cytosol by the **ADP/ATP translocase** (antiport); phosphate enters by symport with H^+.

Transfer of cytosolic NADH into the mitochondria

The inner membrane is impermeable to NADH; the hydrogen of NADH formed in the cytosol in glycolysis enters through **shuttle mechanisms**:

- **The malate-aspartate shuttle** (liver, heart, kidney) — hydrogen passes to NAD^+ in the mitochondria (≈ 2.5 ATP).
- **The glycerol phosphate shuttle** (skeletal muscle, brain) — hydrogen passes to FAD (≈ 1.5 ATP).

4. Respiratory chain inhibitors and uncouplers

Table 4.3. *Substances that affect oxidative phosphorylation*

Substance	Site of action	Result
Rotenone, amytal (barbiturates), piericidin	Complex I	NADH is not oxidized; succinate oxidation is preserved
Malonate	Complex II (competitive)	Succinate is not oxidized
Antimycin A	Complex III	Electrons do not pass to cytochrome c
Cyanides, CO, H_2S, azides	Complex IV (cytochrome oxidase)	Electrons do not pass to O_2 — tissue hypoxia
Oligomycin	ATP synthase (F_o)	ATP synthesis and (because of respiratory control) respiration stop
2,4-dinitrophenol, high-dose thyroxine, thermogenin (UCP-1), salicylates (in large doses)	Uncouplers — carry protons bypassing ATP synthase	Respiration speeds up, ATP synthesis decreases, energy is dissipated as heat

Uncouplers are substances that uncouple oxidation from phosphorylation. In **brown adipose tissue** the protein thermogenin (UCP-1) is a physiological uncoupler: it returns protons to the matrix and the energy is released as heat — **non-shivering thermogenesis** in newborns and hibernating animals.

CLINICAL SIGNIFICANCE

Cyanide poisoning: cytochrome oxidase is blocked, the tissues cannot use oxygen (venous blood is as red as arterial blood, O_2 is not used), anaerobic glycolysis increases and lactic acidosis develops. Treatment: sodium nitrite (converts hemoglobin to methemoglobin, which binds cyanide — cyanmethemoglobin), sodium thiosulfate (converts cyanide into the less toxic thiocyanate with the participation of rhodanese), hydroxocobalamin. **In CO poisoning** hemoglobin is converted to carboxyhemoglobin and cytochrome oxidase is also inhibited; treatment — oxygen (hyperbaric oxygenation).

CAUTION

Uncouplers used as a weight-loss agent (2,4-dinitrophenol) have caused fatal hyperthermia — because energy is converted not into ATP but into heat, and body temperature rises uncontrollably. In hyperthyroidism oxidation and phosphorylation are also partly uncoupled: patients are heat-intolerant, sweat, and their basal metabolic rate is increased.

5. Extramitochondrial oxidation. Reactive oxygen species

≈ 90% of oxygen is reduced to water in the respiratory chain; the rest is used by other routes:

- **Oxidases** — pass hydrogen directly to O_2, forming H_2O_2 (amino acid oxidases, xanthine oxidase).
- **Oxygenases** — attach oxygen atoms to the substrate: **monooxygenases** (hydroxylases, cytochrome P450 — in the endoplasmic reticulum, metabolism of xenobiotics and steroids) and **dioxygenases**. **Reactive oxygen species (ROS)** are incompletely reduced active forms of oxygen: the superoxide anion radical (O_2^-), hydrogen peroxide (H_2O_2), the hydroxyl radical ($^{\bullet}OH$). They are formed by the "leakage" of electrons from the respiratory chain, in phagocytes (NADPH oxidase — the "respiratory burst") and under ionizing radiation. Normally ROS take part in killing bacteria and in signaling, but in excess they damage lipids, proteins and DNA (oxidative stress).

Antioxidant defense:

- enzymes: **superoxide dismutase** ($2O_2^- + 2H^+ \rightarrow H_2O_2 + O_2$), **catalase** ($2H_2O_2 \rightarrow 2H_2O + O_2$), **glutathione peroxidase** (contains selenium; converts H_2O_2 to water at the expense of reduced glutathione), glutathione reductase (uses NADPH);
- nonenzymatic antioxidants: vitamins E and C, β-carotene, glutathione, uric acid, bilirubin.

6. Mitochondrial diseases

The mitochondria have their own DNA (mtDNA); it encodes 13 proteins of the respiratory chain and is inherited only **through the mother**. Mutations of mtDNA or of nuclear genes encoding respiratory chain proteins cause **mitochondrial diseases**. They mainly affect tissues with high energy needs — the brain, skeletal and heart muscle, the optic nerve: Leber hereditary optic neuropathy, MELAS syndrome (myopathy, encephalopathy, lactic acidosis, stroke-like episodes), Kearns–Sayre syndrome. Their common biochemical sign is increased blood lactate and pyruvate.

7. Laboratory work

Experiment 1: determination of succinate dehydrogenase activity by the Thunberg method

Principle of the method: succinate dehydrogenase (SDH) removes hydrogen from succinate and passes it to an artificial acceptor — methylene blue; methylene blue is reduced and becomes colorless (the leuco form).

1. Mince muscle tissue and wash it with water (to remove endogenous substrates).
2. Put equal amounts of tissue into two test tubes. Add 2 ml of 2% sodium succinate to tube 1 and 2 ml of water to tube 2 (control).
3. Add 2 drops of 0.01% methylene blue to each and cover with a layer of petroleum jelly oil to keep air out.
4. Place the tubes in a 37 °C water bath and observe the time of decolorization. **Result and analysis:** in the tube with succinate methylene blue decolorizes faster — SDH oxidizes succinate to fumarate. In the control decolorization is very slow. If **malonate** is added to a 3rd tube together with succinate, decolorization slows down — competitive inhibition.

Experiment 2: determination of catalase activity

1. Pour 1 ml of 3% hydrogen peroxide solution into a test tube.
2. Add 1 drop of blood (or liver homogenate).
3. Repeat in another tube with boiled blood. **Result and analysis:** when fresh blood is added, vigorous foaming (O_2 bubbles) occurs — catalase breaks down H_2O_2 into water and oxygen. With boiled blood there is no reaction — the enzyme is denatured. Catalase activity is determined quantitatively by the Bach–Zubkova method: the undecomposed H_2O_2 is titrated with potassium permanganate.

Experiment 3: determination of peroxidase activity (benzidine or guaiac test)

1. Pour 1 ml of diluted blood (or potato juice) and 5 drops of benzidine (or guaiac resin) solution into a test tube.
2. Add 3–5 drops of 3% H_2O_2.

Result and analysis: a blue (or blue-green) color appears — peroxidase (in blood, the pseudoperoxidase activity of hemoglobin) oxidizes the substrate at the expense of the oxygen of

H_2O_2. This reaction underlies the detection of occult blood in stool and urine.

8. Practical skills

Skill: calculating the ATP formed in the oxidation of a substrate

1. Determine the number of NADH and $FADH_2$ formed in the oxidation of the substrate.
2. Multiply the number of NADH by 2.5 and of $FADH_2$ by 1.5.
3. Add the ATP (GTP) formed at the substrate level.
4. Subtract the ATP used (for example, for activation).
5. Take into account through which shuttle the cytosolic NADH passes.

Skill: analyzing respiratory chain disorders in poisoning

1. Determine which complex or ATP synthase the poison acts on.
2. Distinguish whether it is an inhibitor (electron flow stops, O_2 consumption decreases) or an uncoupler (O_2 consumption increases, ATP decreases, heat increases).
3. State the expected biochemical changes: rise in lactate, increase of O_2 in venous blood, hyperthermia.
4. Explain the antidote or the principle of treatment.

Review questions

1. How are exergonic and endergonic reactions coupled?
2. Give examples of high-energy compounds.
3. What routes of ATP formation are there?
4. What is biological oxidation? What is tissue respiration?
5. List the complexes and mobile carriers of the respiratory chain.
6. In what order do electrons move in the respiratory chain, and why?
7. What is the essence of the chemiosmotic theory?
8. How does ATP synthase work?
9. What is the P/O ratio? What is it for NADH and $FADH_2$?
10. What is respiratory control?
11. How does cytosolic NADH enter the mitochondria?
12. Name the respiratory chain inhibitors and their sites of action.
13. How do uncouplers act? What is the significance of thermogenin?
14. What are reactive oxygen species and the antioxidant defense system?
15. What is the biochemical basis of mitochondrial diseases?

Test questions

1. The part of the mitochondrion in which the respiratory chain is located:
   A) In the outer membrane
   B) In the inner membrane
   C) In the matrix
   D) In the intermembrane space
   E) In ribosomes

2. The compound that is not high-energy:
   A) ATP
   B) Creatine phosphate
   C) Phosphoenolpyruvate
   D) Glucose-6-phosphate
   E) Acetyl-CoA

3. The part of the respiratory chain blocked by cyanides:
   A) Cytochrome oxidase (complex IV)
   B) NADH dehydrogenase (complex I)
   C) Succinate dehydrogenase
   D) ATP synthase
   E) Cytochrome bc1

4. The theory that explains the mechanism of oxidative phosphorylation:
   A) Mitchell’s chemiosmotic theory
   B) Fischer’s “lock and key” theory
   C) Koshland’s induced fit theory
   D) The Watson–Crick model
   E) The Singer–Nicolson model

5. The P/O ratio for the oxidation of NADH (current data):
   A) 1
   B) 1,5
   C) 2
   D) 4
   E) 2,5

6. The respiratory chain complex inhibited by rotenone:
   A) IV
   B) III
   C) II
   D) I
   E) V

7. The physiological uncoupling protein of brown adipose tissue:
   A) Cytochrome c
   B) Myoglobin
   C) Thermogenin (UCP-1)
   D) Ubiquinone
   E) Calmodulin

8. An example of substrate-level phosphorylation:
   A) Oxidation of NADH in the respiratory chain
   B) Oxidation of succinate to fumarate
   C) Conversion of glucose to glucose-6-phosphate
   D) Conversion of phosphoenolpyruvate to pyruvate
   E) Phosphorylation of creatine

9. What is observed under the action of uncouplers:
   A) Respiration stops
   B) ATP synthesis increases
   C) The proton gradient increases
   D) Oxygen consumption decreases
   E) Respiration speeds up, ATP synthesis decreases, a lot of heat is released

10. The lipid-soluble mobile electron carrier:
   A) Cytochrome c
   B) NAD^+
   C) Coenzyme Q (ubiquinone)
   D) FMN
   E) Cytochrome a3

11. How oligomycin acts:
   A) Blocks complex I
   B) Blocks complex III
   C) Carries protons across the membrane
   D) Binds cytochrome c
   E) Blocks the proton channel (F_o) of ATP synthase

12. The enzyme that breaks down hydrogen peroxide into water and oxygen:
   A) Superoxide dismutase
   B) Peroxidase
   C) Catalase
   D) Cytochrome oxidase
   E) Xanthine oxidase

13. The trace element that is part of glutathione peroxidase:
   A) Copper
   B) Selenium
   C) Zinc
   D) Manganese
   E) Cobalt

14. How mitochondrial DNA is inherited:
   A) Only through the mother
   B) Only through the father
   C) Autosomal dominant
   D) X-linked
   E) Autosomal recessive

15. What is meant by respiratory control:
   A) The sensitivity of the respiratory center to CO_2
   B) The dependence of the rate of respiration on the ADP concentration
   C) The binding of O_2 to hemoglobin
   D) The action of uncouplers
   E) The Pasteur effect of glycolysis

Clinical cases

Case 1

A person rescued from a fire was brought in unconscious. The skin is pink-red, blood lactate is sharply increased, and the oxygen content of venous blood is close to that of arterial blood. Cyanide and CO were released from the burning synthetic materials.

Questions: Why is there a lot of oxygen in the venous blood? What is the cause of the lactic acidosis? Which antidotes are used?

Answer: Cyanide and CO block cytochrome oxidase (complex IV) — electrons do not pass to oxygen and the tissues cannot use it (tissue, or histotoxic, hypoxia), so the venous blood remains rich in oxygen. Once oxidative phosphorylation stops, ATP is formed only by anaerobic glycolysis — pyruvate is reduced to lactate and lactic acidosis develops. CO also forms carboxyhemoglobin with hemoglobin (red skin). Treatment is given by the set emergency protocol: oxygen (including hyperbaric oxygenation) and the antidotes used against cyanide (hydroxocobalamin, nitrite and thiosulfate preparations).

Case 2

A gym member bought a “fat burner” online — 2,4-dinitrophenol — and took it. A few days later his body temperature rose to 41 °C, with profuse sweating, tachycardia and shortness of breath.

Questions: How does this substance act? Why did hyperthermia develop?

Answer: 2,4-dinitrophenol is an uncoupler of oxidative phosphorylation: it carries protons into the matrix bypassing ATP synthase, and the proton gradient is lost. The respiratory chain works continuously (O_2 consumption rises sharply), but little ATP is formed and the energy is released as heat. The body cannot dissipate this heat — uncontrollable hyperthermia, a life-threatening condition. The substance is not a medicine and its use is dangerous.

Case 3

A 14-year-old adolescent has exercise intolerance, muscle weakness and recurrent stroke-like attacks. Blood lactate is increased at rest. His mother also has a mild myopathy; his father is healthy.

Questions: Which group of diseases is suspected? Why is lactate increased? What does the pattern of inheritance indicate?

Answer: A mitochondrial disease (MELAS syndrome is suspected). Because of a defect of respiratory chain proteins, NADH oxidation and ATP formation are impaired; the NADH/NAD^+ ratio rises and pyruvate is reduced to lactate — lactic acidosis. The tissues with the greatest energy needs (brain, muscles) are affected. Transmission of the disease through the mother points to a mutation of mitochondrial DNA, because mitochondria pass to the zygote only from the egg cell.

Quick review

Table 4.4. *Key concepts of topic 4*

Concept	Brief definition
High-energy compounds	ATP, creatine phosphate, phosphoenolpyruvate, 1,3-bisphosphoglycerate, acetyl-CoA, succinyl-CoA
ATP formation	Oxidative phosphorylation (≈ 90%) and substrate-level phosphorylation
Respiratory chain	I → CoQ → III → cyt c → IV → O_2; II — from succinate to CoQ

Concept	Brief definition
Protons	I — 4, III — 4, IV — 2 H^+ into the intermembrane space
Chemiosmotic theory	The energy of the proton gradient is converted into ATP through ATP synthase (Mitchell)
P/O	NADH — 2.5; $FADH_2$ — 1.5
Respiratory control	High ADP — respiration speeds up
Inhibitors	I — rotenone, amytal; III — antimycin A; IV — cyanide, CO, H_2S; V — oligomycin
Uncouplers	2,4-dinitrophenol, thermogenin, excess thyroxine — heat ↑, ATP ↓
Shuttle mechanisms	Malate-aspartate (2.5 ATP), glycerol phosphate (1.5 ATP)
ROS	$O_2^{\bullet -}$, H_2O_2, $^{\bullet}OH$
Antioxidant enzymes	Superoxide dismutase, catalase, glutathione peroxidase (Se)
Mitochondrial diseases	mtDNA — through the mother; lactic acidosis, myopathy, encephalopathy

TOPIC 5

The common pathways of catabolism

Learning objectives

To teach the student the common pathways of catabolism — the oxidative decarboxylation of pyruvate and the citric acid (Krebs) cycle, their enzymes, coenzymes, energy yield, regulation, anabolic (amphibolic) functions and link with the respiratory chain, and the biochemical basis of hypoenergetic states.

By the end of the class the student should know:

- the significance of the common pathways of catabolism;
- the structure, coenzymes and reaction of the pyruvate dehydrogenase complex;
- the reactions, enzymes and products of the Krebs cycle;
- the regulation of the Krebs cycle and the pyruvate dehydrogenase complex;
- the energy yield of the cycle and its link with the respiratory chain;
- the anabolic functions of the cycle and the anaplerotic reactions;
- the causes and consequences of hypoenergetic states.

The student should be able to:

- calculate the amount of ATP formed in the oxidation of glucose and pyruvate;
- explain the disturbance of the common pathways in vitamin deficiencies;
- evaluate the result of determining citric acid and pyruvate in urine.

1. The concept of the common pathways of catabolism

The specific catabolic pathways of carbohydrates, fats and amino acids (glycolysis, β-oxidation, amino acid breakdown) lead to the common metabolites — **pyruvate** and **acetyl-CoA**. Their further oxidation proceeds through reactions common to all substances:

1. **Oxidative decarboxylation of pyruvate** — pyruvate → acetyl-CoA;
2. **The citric acid cycle (Krebs cycle, tricarboxylic acid cycle)** — oxidation of the acetyl residue to CO_2;
3. **The respiratory chain and oxidative phosphorylation** — transfer of hydrogen atoms to oxygen and ATP synthesis. These processes take place in the **mitochondrial matrix** (the Krebs cycle enzymes, except succinate dehydrogenase, which is in the inner membrane) and provide the main part of the body’s energy.

2. Oxidative decarboxylation of pyruvate

Pyruvate formed in the cytosol in glycolysis enters the mitochondria through a special carrier and is converted to acetyl-CoA by the **pyruvate dehydrogenase complex (PDC)**:

$$\text{Pyruvate} + NAD^+ + \text{CoA-SH} \rightarrow \text{Acetyl-CoA} + NADH + H^+ + CO_2$$

The reaction is **irreversible**; therefore glucose cannot be synthesized from acetyl-CoA (the reason why glucose is not formed from fatty acids).

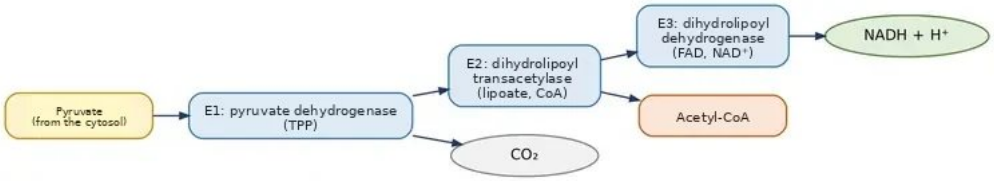


Figure 5.1. *The pyruvate dehydrogenase complex: enzymes and coenzymes (drawn after Severin E.S.)*

Table 5.1. *Composition of the pyruvate dehydrogenase complex*

Enzyme	Coenzyme	Vitamin	Function
E_1 — pyruvate dehydrogenase (pyruvate decarboxylase)	Thiamine diphosphate (TDP)	B_1	Decarboxylates pyruvate (CO_2 is released)
E_2 — dihydrolipoyl transacetylase	Lipoic acid, CoA-SH	Lipoate, pantothenic acid (B_5)	Transfers the acetyl group to CoA
E_3 — dihydrolipoyl dehydrogenase	FAD, NAD^+	B_2, PP	Reoxidizes lipoic acid, NADH is formed

MEMORY AID

The 5 coenzymes of the PDC: **“Tender Loving Care For Nancy”** — TDP (B_1), Lipoate, CoA (B_5), FAD (B_2), NAD (PP). The same coenzymes also work in the **α-ketoglutarate dehydrogenase complex**.

Regulation. PDC activity is:

- **inhibited** by the products — acetyl-CoA and NADH (high ATP/ADP and NADH/NAD^+), and also by **phosphorylation** (PDH kinase phosphorylates E_1 and makes it inactive; acetyl-CoA, NADH and ATP activate the kinase);
- **activated** by pyruvate, ADP, NAD^+, CoA, Ca^{2+} (during muscle contraction) and **dephosphorylation** (PDH phosphatase; insulin activates it).

CLINICAL SIGNIFICANCE

In thiamine deficiency (alcoholism, beriberi) the PDC and α-ketoglutarate dehydrogenase do not work — pyruvate and lactate increase in the blood, and an energy deficit develops in the brain and heart (Wernicke encephalopathy, “wet” beriberi — heart failure). **Arsenite** binds the SH groups of lipoic acid and inhibits the PDC. Congenital PDC deficiency — severe lactic acidosis and neurological disorders in children; a ketogenic diet is used.

3. The citric acid cycle (Krebs cycle)

The cycle was discovered in 1937 by **H. Krebs** (Nobel Prize 1953). In the cycle the two-carbon acetyl residue of acetyl-CoA combines with oxaloacetate and is oxidized in 8 successive reactions to two molecules of CO_2, while oxaloacetate is regenerated.

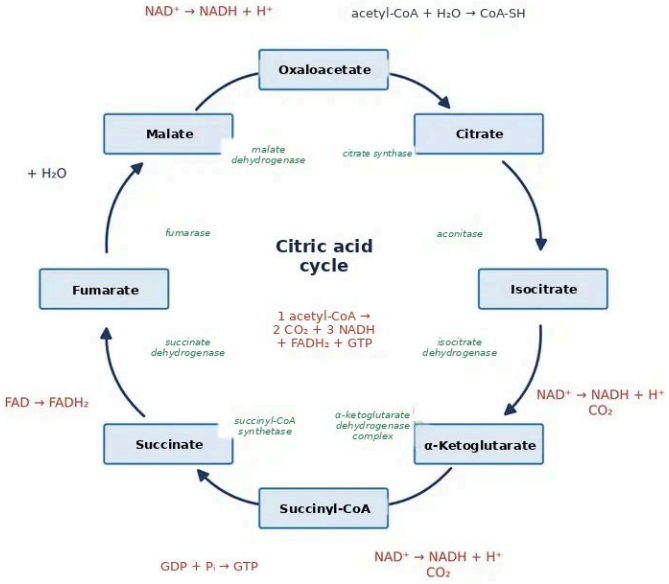


Figure 5.2. *The citric acid cycle: metabolites, enzymes and the products formed (drawn after Sobirova R.A., 2006; Severin E.S.)*

Table 5.2. *Reactions of the Krebs cycle*

№	Reaction	Enzyme	Product and feature
1	Acetyl-CoA + oxaloacetate + H_2O → citrate + CoA-SH	Citrate synthase	Irreversible, key reaction
2	Citrate ⇄ isocitrate (via cis-aconitate)	Aconitase (Fe-S)	Isomerization
3	Isocitrate + NAD^+ → α-ketoglutarate + CO_2 + NADH	Isocitrate dehydrogenase	1st CO_2, 1st NADH; irreversible, the main regulated reaction
4	α-Ketoglutarate + NAD^+ + CoA-SH → succinyl-CoA + CO_2 + NADH	α-Ketoglutarate dehydrogenase complex	2nd CO_2, 2nd NADH; irreversible (coenzymes as in the PDC)
5	Succinyl-CoA + GDP + P_i → succinate + GTP + CoA-SH	Succinyl-CoA synthetase (succinyl thiokinase)	Substrate-level phosphorylation
6	Succinate + FAD → fumarate + $FADH_2$	Succinate dehydrogenase (complex II)	$FADH_2$; malonate is a competitive inhibitor
7	Fumarate + H_2O ⇄ L-malate	Fumarase (fumarate hydratase)	Hydration
8	L-malate + NAD^+ ⇄ oxaloacetate + NADH + H^+	Malate dehydrogenase	3rd NADH; oxaloacetate is regenerated

Overall equation of one turn:

Acetyl-CoA + $3NAD^+$ + FAD + GDP + P_i + $2H_2O$ → $2CO_2$ + 3NADH + $3H^+$ + $FADH_2$ + GTP + CoA-SH

MEMORY AID

Order of the metabolites: **“Our City Is Kept Spotless, Shiny, Fresh, Modern”** — Oxaloacetate, Citrate, Isocitrate, α-Ketoglutarate, Succinyl-CoA, Succinate, Fumarate, Malate.

4. Energy yield

Table 5.3. *ATP formed in the oxidation of one acetyl-CoA*

Source	Number	ATP (current calculation)
NADH (isocitrate, α-ketoglutarate, malate)	3	3 × 2,5 = 7,5
$FADH_2$ (succinate)	1	1 × 1,5 = 1,5
GTP (substrate level)	1	1
Total		10 ATP (12 by the old calculation)

Starting from pyruvate (adding the NADH of the PDC): 10 + 2.5 = **12.5 ATP** (15 by the old calculation). The aerobic oxidation of glucose yields a total of **30–32 ATP** (36–38 by the old calculation) (topic 6).

> **REMEMBER**
> Oxygen does not take part in the Krebs cycle itself, but the cycle works only under **aerobic conditions**: NAD^+ and FAD are reoxidized only in the respiratory chain. When oxygen is lacking, NADH accumulates and the cycle stops.

5. Regulation of the Krebs cycle

The rate of the cycle is matched to the cell's energy needs. The main regulated enzymes are those that catalyze the irreversible reactions:

Table 5.4. *Regulation of the Krebs cycle enzymes*

Enzyme	Activated by	Inhibited by
Citrate synthase	Oxaloacetate (availability of substrate)	ATP, NADH, succinyl-CoA, citrate
Isocitrate dehydrogenase	ADP, Ca^{2+}	ATP, NADH
α-Ketoglutarate dehydrogenase	Ca^{2+}	Succinyl-CoA, NADH, ATP

General principle: **high ATP and NADH** (enough energy) — the cycle slows down; **high ADP, NAD^+ and Ca^{2+}** (energy expenditure, muscle contraction) — the cycle speeds up. In addition, the rate of the cycle depends on the availability of oxaloacetate.

6. Functions of the Krebs cycle. Anaplerotic reactions

Catabolic (energy) function — oxidizing the acetyl residue to CO_2 and supplying NADH and $FADH_2$ to the respiratory chain.

Anabolic functions — the cycle metabolites are raw material for synthesis:

- oxaloacetate → phosphoenolpyruvate → **glucose** (gluconeogenesis);
- oxaloacetate → **aspartate**, α-ketoglutarate → **glutamate** (transamination);
- citrate moves to the cytosol and gives acetyl-CoA → **synthesis of fatty acids and cholesterol**;
- succinyl-CoA + glycine → **heme synthesis**. For this reason the cycle is considered an **amphibolic** pathway.

Anaplerotic ("filling-up") reactions replenish the metabolites withdrawn for synthesis:

- **Pyruvate carboxylase** (biotin, ATP): pyruvate + CO_2 → oxaloacetate — the most important; acetyl-CoA activates it allosterically (when acetyl-CoA increases, its "partner" oxaloacetate

increases too);

- formation of α-ketoglutarate from glutamate (glutamate dehydrogenase, transamination);
- formation of succinyl-CoA from odd-chain fatty acids, valine, isoleucine, methionine and threonine (propionyl-CoA → methylmalonyl-CoA → succinyl-CoA; requires B_{12}).

CLINICAL SIGNIFICANCE

In starvation and diabetes mellitus oxaloacetate is used up in gluconeogenesis; the acetyl-CoA formed in the liver from the β-oxidation of fatty acids cannot fully enter the Krebs cycle and is directed to the synthesis of **ketone bodies** (topic 9).

7. Hypoenergetic states

A **hypoenergetic state** is a fall in ATP formation in the cell below its needs. Causes:

- **Alimentary** — starvation, lack of nutrients and vitamins (B_1, B_2, PP, B_5);
- **Hypoxic** — lack of oxygen (respiratory and circulatory failure, anemia, CO poisoning);
- **Histotoxic** — inhibition of respiratory chain enzymes (cyanides), the action of uncouplers;
- **Mitochondrial diseases** — congenital defects of respiratory chain and Krebs cycle enzymes. Consequences: ion pumps fail (cell swelling), synthetic processes weaken, anaerobic glycolysis increases and lactic acidosis develops; the most sensitive organs are the brain and the heart.

8. Laboratory work

Experiment 1: determination of citric acid in urine (Deniges reaction)

1. Pour 2 ml of urine, 0.5 ml of 10% H_2SO_4 and 0.5 ml of 5% $KMnO_4$ solution into a test tube and heat to boiling.
2. After decolorization add 1 ml of mercuric sulfate reagent (Deniges reagent). **Result and analysis:** citric acid is oxidized to acetone dicarboxylic acid, which gives a white precipitate with mercuric sulfate. Normally 0.2–1.2 g of citrate is excreted in the urine per day; it binds calcium and prevents the formation of kidney stones (hypocitraturia is a risk factor for stone formation).

Experiment 2: determination of pyruvate in urine (sodium nitroprusside test)

1. Add to a test tube 2 ml of urine, 0.5 ml of 10% NaOH and 3–5 drops of freshly prepared 5% sodium nitroprusside solution.
2. Observe the color that appears, then add 1 ml of concentrated acetic acid. **Result and analysis:** if pyruvate is present, a deep orange-red color forms in the alkaline medium and turns violet when acetic acid is added. An increase of pyruvate in the urine (normal 10–25 mg per day) is characteristic of B_1 hypovitaminosis, diabetes mellitus, hypoxia and PDC deficiency.

Experiment 3: inhibition of succinate dehydrogenase in muscle tissue by malonate

1. Put equal amounts of washed minced muscle into three test tubes.
2. Add 2 ml of succinate to tube 1, 2 ml of succinate + 1 ml of malonate to tube 2, and 2 ml of water to tube 3 (control); add 2 drops of methylene blue and a layer of petroleum jelly oil to all of them.
3. Compare the time of decolorization at 37 °C.

Result and analysis: in the tube with malonate decolorization slows down — malonate is similar in structure to succinate and competes with it for the active site. Increasing the amount of substrate (succinate) reduces the inhibition.

9. Practical skills

Skill: calculating the ATP formed in the oxidation of acetyl-CoA and pyruvate

1. Take into account that 1 NADH (2.5 ATP) is formed in the PDC reaction.
2. Add the 3 NADH (7.5 ATP), 1 $FADH_2$ (1.5 ATP) and 1 GTP (1 ATP) formed in the Krebs cycle.
3. Find that the total is 12.5 ATP from pyruvate and 10 ATP from acetyl-CoA.
4. If the calculation starts from another substrate (for example, lactate), add the extra NADH.

Skill: identifying the disturbance of the common pathways in vitamin deficiency

1. Determine which coenzyme the vitamin is part of (B_1 — TDP, B_2 — FAD, PP — NAD^+, B_5 — CoA).
2. Indicate in which enzymes this coenzyme works (PDC, α-ketoglutarate dehydrogenase, succinate dehydrogenase, malate dehydrogenase).
3. Name the metabolites that accumulate (pyruvate, lactate, α-ketoglutarate).
4. Explain the clinical consequences (energy deficit, lactic acidosis, neurological disorders).

Review questions

1. Which processes make up the common pathways of catabolism?
2. Where and how is pyruvate converted into acetyl-CoA?
3. List the enzymes and coenzymes of the pyruvate dehydrogenase complex.
4. How is the pyruvate dehydrogenase complex regulated?
5. Why is glucose not synthesized from acetyl-CoA?
6. Write the reactions of the Krebs cycle in order.
7. In which reactions of the cycle is CO_2 released?
8. In which reactions of the cycle are NADH, $FADH_2$ and GTP formed?
9. How much ATP is formed in the oxidation of one acetyl-CoA?
10. Why does the Krebs cycle work only under aerobic conditions?
11. Explain the key enzymes of the Krebs cycle and their regulation.
12. List the anabolic functions of the Krebs cycle.
13. What are anaplerotic reactions?
14. What are the causes of hypoenergetic states?
15. What biochemical changes are observed in thiamine deficiency?

Test questions

1. The part of the cell in which the oxidative decarboxylation of pyruvate takes place:
   A) In the cytosol
   B) In the mitochondrial matrix
   C) In the endoplasmic reticulum
   D) In lysosomes
   E) In the nucleus

2. The coenzyme of enzyme E_1 of the pyruvate dehydrogenase complex:
   A) Thiamine diphosphate
   B) FAD
   C) Lipoic acid
   D) Biotin
   E) Pyridoxal phosphate

3. The enzyme that catalyzes the first reaction of the Krebs cycle:
   A) Aconitase
   B) Malate dehydrogenase
   C) Citrate synthase
   D) Fumarase
   E) Isocitrate dehydrogenase

4. The reaction of the Krebs cycle in which substrate-level phosphorylation takes place:
   A) Succinyl-CoA → succinate
   B) Isocitrate → α-ketoglutarate
   C) Fumarate → malate
   D) Malate → oxaloacetate
   E) Citrate → isocitrate

5. The number of NADH formed in one turn of the Krebs cycle:
   A) 1
   B) 2
   C) 3
   D) 4
   E) 5

6. The FAD-dependent enzyme of the Krebs cycle:
   A) Malate dehydrogenase
   B) Isocitrate dehydrogenase
   C) Aconitase
   D) Citrate synthase
   E) Succinate dehydrogenase

7. The ATP formed in the oxidation of one acetyl-CoA in the Krebs cycle (current calculation):
   A) 2,5
   B) 10
   C) 12,5
   D) 30
   E) 38

8. The substances that inhibit the pyruvate dehydrogenase complex:
   A) ADP and NAD^+
   B) Pyruvate and CoA
   C) Ca^{2+} and insulin
   D) Acetyl-CoA and NADH
   E) AMP and P_i

9. The enzyme that catalyzes the most important anaplerotic reaction:
   A) Citrate synthase
   B) Pyruvate kinase
   C) Lactate dehydrogenase
   D) Pyruvate carboxylase
   E) Phosphoenolpyruvate carboxykinase

10. The metabolite of the Krebs cycle used for heme synthesis:
    A) Succinyl-CoA
    B) Citrate
    C) Fumarate
    D) Malate
    E) Isocitrate

11. The enzyme for which malonate is a competitive inhibitor:
    A) Citrate synthase
    B) Aconitase
    C) Fumarase
    D) Malate dehydrogenase
    E) Succinate dehydrogenase

12. The substance that activates isocitrate dehydrogenase:
    A) ATP
    B) ADP
    C) NADH
    D) Succinyl-CoA
    E) Citrate

13. Why the Krebs cycle is called an amphibolic pathway:
    A) Because it takes part only in breakdown
    B) Because it takes part only in synthesis
    C) Because it serves both catabolism and anabolism
    D) Because it takes place in the cytosol
    E) Because it works without oxygen

14. The vitamin that is not needed by the α-ketoglutarate dehydrogenase complex:
    A) B_1
    B) B_2
    C) PP
    D) Pantothenic acid
    E) Biotin

15. The metabolite that increases in the blood in thiamine deficiency:
    A) Acetyl-CoA
    B) Oxaloacetate
    C) Citrate
    D) Pyruvate
    E) Succinate

Clinical cases

Case 1

A 6-month-old child has had low muscle tone, developmental delay and seizures since birth. Blood lactate and pyruvate are sharply increased, the lactate/pyruvate ratio is normal, and alanine is also increased. The condition worsens after carbohydrate food.

Questions: Which enzyme deficiency is suspected? Why do carbohydrates worsen the condition? What diet is recommended?

Answer: Congenital deficiency of the pyruvate dehydrogenase complex. Pyruvate is not converted to acetyl-CoA and accumulates; it is converted to lactate (LDH) and to alanine (transamination). Carbohydrates form pyruvate through glycolysis and increase the acidosis. Because the brain obtains energy mainly from glucose oxidation, the neurological disorders are severe. Treatment: a ketogenic diet (low in carbohydrates, high in fats) — fatty acids and ketone bodies give acetyl-CoA bypassing the PDC; in some patients high-dose thiamine is effective.

Case 2

After rats were given an inhibitor of one of the Krebs cycle enzymes, succinate accumulated in the tissues while the amounts of fumarate and malate decreased.

Questions: Which enzyme is inhibited? Name a possible inhibitor. How does this affect energy metabolism?

Answer: Succinate dehydrogenase (complex II) is inhibited; malonate is its classic competitive inhibitor. Succinate is not converted to fumarate and the cycle stops: the following metabolites and oxaloacetate decrease, acetyl-CoA cannot enter the cycle; less $FADH_2$ and NADH are formed and ATP synthesis falls. Increasing the succinate concentration partly overcomes the inhibition.

Case 3

In a patient after prolonged starvation, blood glucose is 3.0 mmol/l, ketone bodies are sharply increased and acetone is detected in the urine.

Questions: How does the activity of the Krebs cycle in the liver change in starvation? Why do ketone bodies increase?

Answer: In starvation, under the action of glucagon and cortisol, gluconeogenesis in the liver increases and oxaloacetate is used for it (it leaves for the cytosol as malate). At the same time the β-oxidation of fatty acids coming from adipose tissue increases and a lot of acetyl-CoA is formed. Because oxaloacetate is lacking, acetyl-CoA cannot form citrate and is not fully oxidized in the Krebs cycle — in the liver mitochondria it is converted into acetoacetate and β-hydroxybutyrate (ketone bodies); these serve as an energy source for other tissues (brain, muscle).

Quick review

Table 5.5. *Key concepts of topic 5*

Concept	Brief definition
Common pathways	PDC + Krebs cycle + respiratory chain (in the mitochondria)
PDC	Pyruvate → acetyl-CoA + CO_2 + NADH; irreversible
Coenzymes of the PDC	TDP (B_1), lipoate, CoA (B_5), FAD (B_2), NAD^+ (PP)
Regulation of the PDC	Acetyl-CoA, NADH, phosphorylation ↓; ADP, pyruvate, Ca^{2+}, insulin (dephosphorylation) ↑
Krebs cycle	8 reactions: OAA → citrate → isocitrate → α-KG → succinyl-CoA → succinate → fumarate → malate → OAA
Products	2 CO_2, 3 NADH, 1 $FADH_2$, 1 GTP
ATP	10 from acetyl-CoA, 12.5 from pyruvate
Key enzymes	Citrate synthase, isocitrate dehydrogenase, α-ketoglutarate dehydrogenase
Regulation	ATP, NADH ↓; ADP, Ca^{2+} ↑
Anabolic function	Metabolites for the synthesis of glucose, amino acids, fatty acids and heme
Anaplerotic reaction	Pyruvate carboxylase (biotin): pyruvate → oxaloacetate
Hypoenergetic state	Starvation, hypoxia, vitamin deficiency, poisons, mitochondrial diseases

TOPIC 6

Carbohydrate metabolism and functions

Learning objectives

To give the student knowledge of dietary carbohydrates and their significance, the digestion and absorption of carbohydrates (including age-related features), the fate of the absorbed monosaccharides in the tissues, the synthesis and breakdown of glycogen in the liver and muscles and their regulation, aerobic glycolysis, the transfer of the hydrogen of cytosolic NADH into the mitochondria, and the glycogenoses and aglycogenoses.

By the end of the class the student should know:

- the main dietary carbohydrates and their functions;
- the digestion of carbohydrates, its enzymes and the mechanisms of absorption;
- the causes of lactase deficiency (milk intolerance);
- the entry of glucose into the cell and its phosphorylation (hexokinase, glucokinase);
- the reactions of glycogenesis and glycogenolysis and their hormonal regulation;
- the stages, key enzymes and energy yield of aerobic glycolysis;
- the shuttle mechanisms;
- the biochemical basis of the glycogenoses and aglycogenoses.

The student should be able to:

- calculate the ATP formed in the aerobic oxidation of glucose;
- perform qualitative reactions for carbohydrates;
- identify the type of glycogenosis according to the enzyme.

1. Dietary carbohydrates and their functions

Carbohydrates provide 55–60% of the energy of food; the daily requirement of adults is 350–450 g. The main dietary carbohydrates are: **polysaccharides** — starch (amylose and amylopectin), glycogen (little in meat), cellulose and other dietary fiber; **disaccharides** — sucrose (glucose + fructose, α1→β2), lactose (galactose + glucose, β1→4), maltose (2 glucose, α1→4); **monosaccharides** — glucose, fructose.

Functions of carbohydrates in the body: energy (1 g → 4 kcal), storage (glycogen), plastic and structural (glycoproteins, proteoglycans, glycolipids, the ribose and deoxyribose of nucleotides), protective (mucous substances), receptor and recognition (glycocalyx, blood group antigens), and a carbon skeleton for the synthesis of other substances (amino acids, fats).

Dietary fiber (cellulose, pectin, hemicellulose) is not digested in humans, but it increases intestinal peristalsis, binds and removes cholesterol and bile acids, and serves as food for the beneficial microflora.

2. Digestion and absorption of carbohydrates

Table 6.1. *Enzymes that digest carbohydrates*

Site	Enzyme	Substrate → product
Oral cavity	Salivary α-amylase (pH 6.8; activated by Cl^-)	Starch → dextrins, maltose (partly)
Stomach	— (amylase is inactivated in the acidic medium)	Amylase works for a short time inside the food bolus
Lumen of the small intestine	Pancreatic α-amylase	Starch, dextrins → maltose, isomaltose, maltotriose
Surface of the enterocytes (membrane digestion)	Sucrase-isomaltase complex	Sucrose → glucose + fructose; isomaltose → 2 glucose
	Maltase (glucoamylase)	Maltose → 2 glucose
	Lactase (β-galactosidase)	Lactose → galactose + glucose

α-Amylase cleaves only α1→4 glycosidic bonds; the α1→6 bonds at the branch points are broken down by isomaltase. Humans have no cellulase, which cleaves the β-glycosidic bond.

Absorption takes place in the enterocytes of the small intestine (mainly the duodenum and jejunum):

- **Glucose and galactose** — symport with Na^+ through **SGLT1** (secondary active transport, against the gradient);
- **Fructose** — facilitated diffusion through **GLUT5**;
- from the enterocyte into the blood — through **GLUT2** in the basolateral membrane. The absorbed monosaccharides reach the liver first through the **portal vein**.

> **CLINICAL SIGNIFICANCE**
> **Lactase deficiency** (milk intolerance) — in a large part of the world's population lactase activity falls after childhood for genetic reasons (primary); in intestinal diseases (enteritis, celiac disease) it is secondary. Undigested lactose passes into the large intestine, where bacteria ferment it and form gases (H_2, CO_2, CH_4) and organic acids — bloating, pain, osmotic diarrhea. Diagnosis: the hydrogen breath test, a lactose load. Treatment: a lactose-free diet, fermented milk products (in which lactose is partly broken down by bacteria), lactase preparations. **In newborns** congenital lactase deficiency is rare but runs severely.

3. Entry of glucose into the cell and its phosphorylation

Glucose enters the cell through **GLUT** transporters: GLUT1 (erythrocytes, the blood-brain barrier), GLUT2 (liver, pancreatic β-cells, enterocytes, kidney — high K_m), GLUT3 (neurons — low K_m), **GLUT4** (skeletal muscle, heart, adipose tissue — **insulin-dependent**).

In the cell glucose is phosphorylated at once and can no longer leave the cell:

Glucose + ATP → Glucose-6-phosphate + ADP

Table 6.2. *Hexokinase and glucokinase*

Feature	Hexokinase (I–III)	Glucokinase (hexokinase IV)
Site	Almost all tissues	Liver, pancreatic β-cells
K_m for glucose	≈ 0.1 mmol/l (high affinity)	≈ 10 mmol/l (low affinity)

Feature	Hexokinase (I–III)	Glucokinase (hexokinase IV)
Inhibition by glucose-6-phosphate	Inhibited	Not inhibited
Induction	—	Insulin (carbohydrate food)
Significance	Supplies the tissue even when glucose is low	Retains excess glucose in the liver after a meal

Glucose-6-phosphate is the "crossroads" of carbohydrate metabolism: it can enter glycolysis, glycogen synthesis or the pentose phosphate pathway, or (in the liver) be converted into free glucose by glucose-6-phosphatase.

4. Glycogen metabolism

Glycogen is a branched polymer of glucose (α1→4 bonds, with an α1→6 branch every 8–12 residues). It is stored mainly in the **liver** (≈ 100–120 g, 5–6% of its mass; to maintain the blood glucose level) and in **skeletal muscle** (≈ 300–400 g, 1–2% of its mass; only for the muscle's own energy).

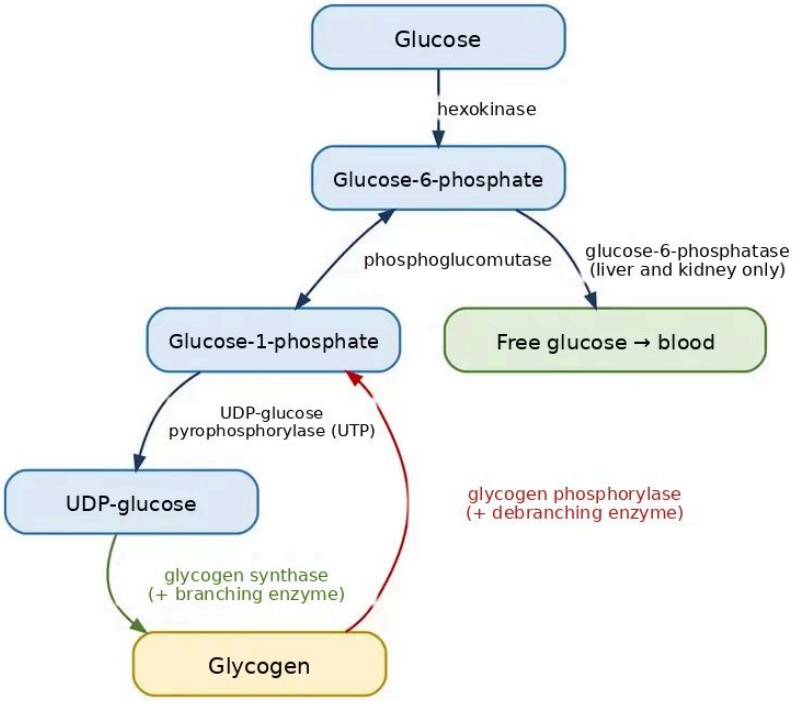


Figure 6.1. *Synthesis and breakdown of glycogen (drawn after Sobirova R.A., 2006; Severin E.S.)*

Glycogen synthesis (glycogenesis)

1. Glucose → glucose-6-phosphate (hexokinase/glucokinase, ATP).
2. Glucose-6-phosphate ⇄ glucose-1-phosphate (phosphoglucomutase).
3. Glucose-1-phosphate + UTP → **UDP-glucose** + PP_i (UDP-glucose pyrophosphorylase); the pyrophosphate is hydrolyzed — the reaction is irreversible.
4. The glucose residue of UDP-glucose is attached to the end of the glycogen chain by an α1→4 bond — **glycogen synthase** (the key enzyme). The "primer" protein for a new chain is **glycogenin**.

5. The **branching enzyme** (amylo-1,4→1,6-transglycosylase) transfers a piece of the chain and forms an α1→6 branch. Adding one glucose to glycogen costs 2 high-energy bonds (ATP and UTP).

Breakdown of glycogen (glycogenolysis)

1. **Glycogen phosphorylase** (the key enzyme; coenzyme — pyridoxal phosphate) cleaves α1→4 bonds from the end of the chain with phosphate (phosphorolysis): glycogen(n) + P_i → glycogen(n−1) + glucose-1-phosphate.
2. The **debranching enzyme** (transferase + amylo-1,6-glucosidase) removes the branch point, and free glucose is released.
3. Glucose-1-phosphate → glucose-6-phosphate (phosphoglucomutase).
4. **In the liver**: glucose-6-phosphatase → free glucose → into the blood. **In muscle** there is no glucose-6-phosphatase — glucose-6-phosphate enters glycolysis (muscle glycogen does not serve to maintain blood glucose).

Regulation of glycogen metabolism

Glycogen synthase and glycogen phosphorylase are regulated in **opposite** directions — through **phosphorylation–dephosphorylation**:

- **When phosphorylated**: phosphorylase is **active**, glycogen synthase is **inactive** → glycogen is broken down.
- **When dephosphorylated**: phosphorylase is inactive, glycogen synthase is **active** → glycogen is synthesized.

Table 6.3. *Hormonal regulation of glycogen metabolism*

Hormone	Tissue	Mechanism	Result
Glucagon (starvation)	Liver	G_s → adenylate cyclase → cAMP → protein kinase A → phosphorylase kinase → phosphorylase is phosphorylated; glycogen synthase is phosphorylated	Glycogenolysis ↑, glucose into the blood
Adrenaline (stress, physical work)	Liver, muscle	β-receptors → cAMP (in the liver also α1 → IP_3, Ca^{2+})	Glycogenolysis ↑
Insulin (after a meal)	Liver, muscle	Activates phosphodiesterase and phosphoprotein phosphatase; induces glucokinase	Glycogenesis ↑

Allosteric regulation: in muscle AMP and Ca^{2+} (during contraction) activate phosphorylase; in the liver glucose inhibits phosphorylase; glucose-6-phosphate activates glycogen synthase.

> **REMEMBER**
> **The two glycogen "stores" serve different purposes**: liver glycogen maintains the level of **blood glucose** (between meals, ≈ 12–24 hours); muscle glycogen is an energy source only for the muscle's **own contraction**.

5. Aerobic oxidation of glucose. Glycolysis

Glycolysis is the breakdown of glucose in the cytosol to **pyruvate** (10 reactions). Under aerobic conditions pyruvate is converted in the mitochondria into acetyl-CoA and oxidized in the Krebs cycle

to CO_2 and H_2O — **aerobic oxidation**. Under anaerobic conditions pyruvate is reduced to lactate (topic 7).

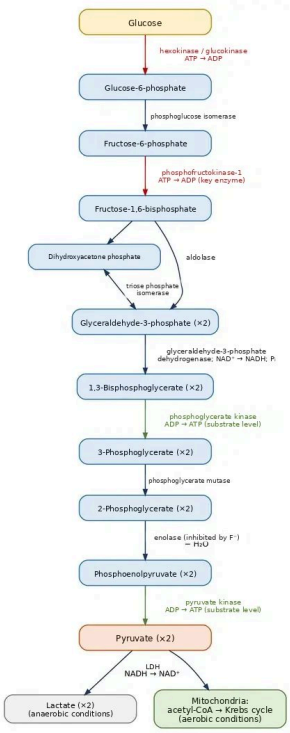


Figure 6.2. *Reactions of glycolysis; the key enzymes and the reactions in which ATP is formed (drawn after Sobirova R.A., 2006; Severin E.S.)*

The two stages of glycolysis:

1. **The preparatory stage** (energy is consumed): glucose is phosphorylated at the expense of 2 ATP and split into two triose phosphates.
2. **The stage of oxidation and energy production**: glyceraldehyde-3-phosphate is oxidized (NADH is formed) and 4 ATP are formed in two substrate-level phosphorylation reactions (phosphoglycerate kinase, pyruvate kinase). **The overall equation of glycolysis (aerobic):**

$$\text{Glucose} + 2NAD^+ + 2ADP + 2Pi \rightarrow 2\ \text{Pyruvate} + 2NADH + 2H^+ + 2ATP + 2H_2O$$

Key (irreversible) enzymes: hexokinase (glucokinase), **phosphofructokinase-1** (the main one), pyruvate kinase.

Table 6.4. *Regulation of phosphofructokinase-1*

Activators	Inhibitors
AMP, ADP (little energy)	ATP (plenty of energy)
Fructose-2,6-bisphosphate (increases under the action of insulin; the strongest activator)	Citrate (enough products of the Krebs cycle)
—	H^+ (acidosis)

Pyruvate kinase is activated by fructose-1,6-bisphosphate (a “feed-forward signal” from the preceding step) and inhibited by ATP and alanine; in the liver it is phosphorylated and inactivated under the action of glucagon.

Transfer of the hydrogen of cytosolic NADH into the mitochondria

Under aerobic conditions the NADH formed in glycolysis must be oxidized in the respiratory chain, but the inner mitochondrial membrane is impermeable to NADH. The hydrogen passes through **shuttle mechanisms**:

- **The malate-aspartate shuttle** (liver, kidney, heart): in the cytosol oxaloacetate is reduced to malate at the expense of NADH, malate enters the mitochondria, is oxidized back to oxaloacetate and NADH is formed in the matrix (≈ 2.5 ATP). Oxaloacetate returns in the form of aspartate.
- **The glycerol phosphate shuttle** (skeletal muscle, brain): dihydroxyacetone phosphate is reduced to glycerol-3-phosphate at the expense of NADH; the FAD-dependent glycerol-3-phosphate dehydrogenase of the mitochondrial membrane reoxidizes it and the electrons pass to CoQ (≈ 1.5 ATP).

Table 6.5. *Energy yield of the aerobic oxidation of glucose*

Stage	Products	ATP (current calculation)
Glycolysis (net)	2 ATP + 2 NADH	2 + 2 × (1.5 or 2.5) = 5–7
2 pyruvate → 2 acetyl-CoA	2 NADH	5
2 acetyl-CoA → Krebs cycle	6 NADH, 2 $FADH_2$, 2 GTP	15 + 3 + 2 = 20
Total		30–32 ATP (36–38 by the old calculation)

The Pasteur effect is the fall in glucose consumption under aerobic conditions: when oxygen is present, more ATP and citrate are formed and they inhibit phosphofructokinase.

6. Glycogenoses and aglycogenoses

Glycogenoses are congenital deficiencies of the enzymes of glycogen breakdown; glycogen (sometimes of abnormal structure) accumulates in excess in the tissues.

Table 6.6. *The main glycogenoses*

Type	Name	Deficient enzyme	Main features
I	Von Gierke disease	Glucose-6-phosphatase (liver, kidney)	Severe fasting hypoglycemia, hepatomegaly, lactic acidosis, hyperlipidemia, hyperuricemia
II	Pompe disease	Lysosomal α-1,4-glucosidase (acid maltase)	Cardiomegaly, muscle weakness; heart failure in infants
III	Cori (Forbes) disease	Debranching enzyme	Glycogen with short branches accumulates; hepatomegaly, milder hypoglycemia
V	McArdle disease	Muscle glycogen phosphorylase	Muscle pain and cramps on exercise, myoglobinuria; lactate does not rise
VI	Hers disease	Liver glycogen phosphorylase	Hepatomegaly, mild hypoglycemia

Aglycogenosis (glycogenosis 0) is a deficiency of liver **glycogen synthase**: almost no glycogen accumulates, there is severe hypoglycemia between meals (morning seizures), and hyperglycemia with a rise in lactate after a meal. Frequent feeding and starch at night are recommended.

CLINICAL SIGNIFICANCE

In von Gierke disease glucose-6-phosphate is not converted into glucose — the liver cannot release glucose into the blood by glycogenolysis or gluconeogenesis. The accumulated glucose-6-phosphate goes into glycolysis and lactate and pyruvate increase (lactic acidosis); through the pentose phosphate pathway more purines are formed and uric acid rises (gout); the synthesis of fats from acetyl-CoA increases (hyperlipidemia, fatty degeneration of the liver). Treatment — frequent feeding, raw corn starch at night (it is broken down slowly).

7. Laboratory work

Experiment 1: iodine reaction for starch and glycogen

1. Pour 1 ml of 1% starch and 1 ml of 1% glycogen solution into two test tubes.
2. Add 1 drop of Lugol's solution to each; then heat and cool. **Result and analysis:** starch gives a blue color and glycogen a red-brown one (iodine enters the spiral of the polysaccharide and forms a complex; because the branches of glycogen are short, the color is different). On heating the spiral is destroyed and the color disappears; on cooling it returns.

Experiment 2: the Trommer reaction for reducing carbohydrates

1. Pour 1 ml each of 1% glucose, sucrose and lactose solutions into test tubes.
2. Add to each 1 ml of 10% NaOH and 3–5 drops of 1% $CuSO_4$ (the blue Cu(OH)2 precipitate dissolves).
3. Heat the upper part to boiling. **Result and analysis:** with glucose and lactose a yellow (CuOH) and then a red-brick (Cu_2O) precipitate forms — they are reducing sugars (they have a free hemiacetal hydroxyl). Sucrose gives no reaction (both hemiacetal hydroxyls are used in the glycosidic bond), but after hydrolysis with acid it gives a positive reaction.

Experiment 3: the Seliwanoff reaction for fructose

1. Add to a test tube 1 ml of 1% fructose solution and 1 ml of Seliwanoff reagent (a solution of resorcinol in concentrated HCl); heat to boiling.
2. Compare with glucose. **Result and analysis:** fructose (a ketose) quickly gives a red color — in acid it is converted into hydroxymethylfurfural, which forms a colored compound with resorcinol. With glucose (an aldose) the color develops very slowly and is weak. Sucrose is also positive (it contains fructose). This reaction is used to detect fructose in urine (fructosuria).

8. Practical skills

Skill: calculating the ATP formed in the oxidation of glucose

1. Write down that glycolysis gives a net 2 ATP and 2 NADH.
2. Determine through which shuttle the cytosolic NADH passes (malate-aspartate — 2.5, glycerol phosphate — 1.5 ATP).
3. Add 2 NADH (5 ATP) for the conversion of 2 pyruvate into acetyl-CoA.
4. Add 20 ATP for the oxidation of 2 acetyl-CoA in the Krebs cycle.
5. Show that the total is 30–32 ATP; compare it with anaerobic conditions (2 ATP).

Skill: identifying the type of glycogenosis

1. Determine which organ is affected: the liver (hepatomegaly, hypoglycemia) or the muscle (exercise intolerance, myoglobinuria).
2. Evaluate the degree of hypoglycemia and the rise in lactate, uric acid and lipids.
3. Analyze the result of the glucagon test: if glucose does not rise — glycogen is not being broken down.
4. Determine the deficient enzyme and the type of glycogenosis.
5. Explain the principle of the diet.

Review questions

1. List the main dietary carbohydrates. What are their functions in the body?
2. Where and by which enzymes is starch digested?
3. Which enzymes take part in membrane digestion?
4. How are glucose, galactose and fructose absorbed?
5. What are the causes and signs of lactase deficiency?
6. What are the types and locations of the GLUT transporters?
7. What is the difference between hexokinase and glucokinase?
8. Write the reactions of glycogen synthesis.
9. How is glycogen broken down? What is the difference between liver and muscle glycogen?
10. How is glycogen metabolism regulated?
11. List the reactions and key enzymes of glycolysis.
12. How is phosphofructokinase-1 regulated?
13. Explain the shuttle mechanisms.
14. How much ATP is formed in the aerobic oxidation of glucose?
15. What is the biochemical basis of the glycogenoses and aglycogenoses?

Test questions

1. The enzyme that cleaves the α1→6 glycosidic bonds of starch:
   A) α-Amylase
   B) Maltase
   C) Lactase
   D) Sucrase
   E) Isomaltase

2. The transporter by which glucose is absorbed in the intestine together with Na^+:
   A) GLUT1
   B) GLUT2
   C) GLUT4
   D) SGLT1
   E) GLUT5

3. The tissues in which the insulin-dependent glucose transporter is found:
   A) In the liver and erythrocytes
   B) In skeletal muscle and adipose tissue
   C) In neurons
   D) In enterocytes
   E) In the renal tubules

4. How glucokinase differs from hexokinase:
   A) It is found in all tissues
   B) Its K_m is very low
   C) It is inhibited by glucose-6-phosphate
   D) High K_m, not inhibited by glucose-6-phosphate, induced by insulin
   E) It is found only in muscle

5. The active form of glucose in glycogen synthesis:
   A) Glucose-6-phosphate
   B) Glucose-1-phosphate
   C) Fructose-6-phosphate
   D) UDP-glucose
   E) GDP-glucose

6. The reason why free glucose is not released into the blood when muscle glycogen is broken down:
   A) There is no glycogen phosphorylase
   B) There is no phosphoglucomutase
   C) There is no glucose-6-phosphatase
   D) There is no GLUT4
   E) There is little glycogen

7. How glucagon affects glycogen metabolism in the liver:
   A) Through cAMP it activates phosphorylase and increases glycogenolysis
   B) It activates glycogen synthase
   C) Activates phosphodiesterase
   D) It induces glucokinase
   E) It has no effect

8. The main key enzyme of glycolysis:
   A) Aldolase
   B) Enolase
   C) Phosphoglucoisomerase
   D) Triose phosphate isomerase
   E) Phosphofructokinase-1

9. The strongest allosteric activator of phosphofructokinase-1:
   A) ATP
   B) Citrate
   C) Fructose-2,6-bisphosphate
   D) H^+
   E) Glucose-6-phosphate

10. The enzyme that catalyzes substrate-level phosphorylation in glycolysis:
    A) Pyruvate kinase
    B) Hexokinase
    C) Phosphofructokinase-1
    D) Aldolase
    E) Enolase

11. The ATP formed in the aerobic oxidation of glucose (current calculation):
    A) 2
    B) 30–32
    C) 12
    D) 15
    E) 50

12. The enzyme that is deficient in von Gierke disease:
    A) Glucose-6-phosphatase
    B) Lysosomal α-1,4-glucosidase
    C) Muscle glycogen phosphorylase
    D) Debranching enzyme
    E) Glycogen synthase

13. The significance of the malate-aspartate shuttle:
    A) It exports acetyl-CoA to the cytosol
    B) It transfers the hydrogen of cytosolic NADH into the mitochondria
    C) It exports ATP to the cytosol
    D) It brings fatty acids into the mitochondria
    E) It brings pyruvate into the mitochondria

14. The mechanism of diarrhea in lactase deficiency:
    A) Secretory (a rise in cAMP)
    B) Inflammatory
    C) Weakening of intestinal motility
    D) The action of bile acids
    E) The osmotic and fermentative effect of undigested lactose

15. The feature characteristic of McArdle disease (glycogenosis type V):
    A) Severe fasting hypoglycemia
    B) Cardiomegaly
    C) Muscle pain and myoglobinuria on exercise
    D) Hyperuricemia
    E) Hepatomegaly

Clinical cases

Case 1

A 1-year-old child has an enlarged abdomen (the liver extends 5 cm below the costal margin), a “doll’s face” and growth retardation. In the morning, fasting glucose is 1.8 mmol/l, and lactate, triglycerides and uric acid are increased. After glucagon was given, glucose did not rise and lactate rose further.

Questions: What is the diagnosis? Explain all the biochemical changes.

Answer: Von Gierke disease (glycogenosis type I) — glucose-6-phosphatase deficiency. Glycogen accumulates in the liver (hepatomegaly); because the last reaction of glycogenolysis and gluconeogenesis is blocked, glucose is not released into the blood — severe fasting hypoglycemia; glucagon increases glycogenolysis, but glucose-6-phosphate goes into glycolysis and lactate rises. The excess of glucose-6-phosphate increases purine synthesis through the pentose phosphate pathway (uric acid ↑), and through glycolysis more acetyl-CoA and glycerol are formed and fat synthesis increases (hyperlipidemia). Low insulin and high glucagon also raise lipids.

Case 2

A 22-year-old man feels severe pain and cramps in his muscles during the first minutes of a gym session; after a rest he can continue exercising (the “second wind” phenomenon). After heavy exercise his urine was brown. In the ischemic exercise test blood lactate did not rise.

Questions: Which enzyme is deficient? Why does lactate not rise? Explain the “second wind” phenomenon.

Answer: McArdle disease (glycogenosis type V) — deficiency of muscle glycogen phosphorylase. Muscle glycogen is not broken down, and at the start of exercise no glucose-6-phosphate is formed for glycolysis — an ATP deficit, muscle cramps, damage to muscle cells and release of myoglobin into the urine (brown urine). Because there is no glycolysis, lactate does not rise. The “second wind” — after a few minutes blood flow increases and more glucose and fatty acids reach the muscle from the blood, so the energy supply improves.

Case 3

A 25-year-old woman complains of bloating, rumbling and loose stools 1–2 hours after drinking milk. She tolerates kefir and cheese well. In the hydrogen breath test, hydrogen in the exhaled air rose sharply after a lactose load.

Questions: What is the diagnosis? Explain the mechanism of the signs and why she tolerates fermented milk products.

Answer: Lactase deficiency (in adults, primary: a genetically determined fall in lactase activity). Lactose is not broken down or absorbed in the small intestine; in the large intestine it is osmotically active and retains water (osmotic diarrhea), and bacteria ferment it to H_2, CO_2 and short-chain acids (flatulence, pain; a rise in H_2 in the breath). In fermented milk products most of the lactose has been broken down by lactic acid bacteria, and cheese contains very little lactose.

Quick review

Table 6.7. *Key concepts of topic 6*

Concept	Brief definition
Digestive enzymes	Amylase (saliva, pancreas); sucrase-isomaltase, maltase, lactase (enterocytes)
Absorption	Glucose, galactose — SGLT1; fructose — GLUT5; into the blood — GLUT2
GLUT4	Muscle, adipose tissue; insulin-dependent
Hexokinase / glucokinase	Low / high K_m; inhibited / not inhibited by G-6-P
Glycogenesis	G-1-P + UTP → UDP-glucose → glycogen synthase (key) + branching enzyme
Glycogenolysis	Glycogen phosphorylase (key, PLP) + debranching enzyme

Concept	Brief definition
Liver and muscle	The liver has glucose-6-phosphatase — glucose into the blood; the muscle does not
Regulation	Glucagon, adrenaline — phosphorylation → breakdown; insulin — dephosphorylation → synthesis
Key enzymes of glycolysis	Hexokinase, phosphofructokinase-1, pyruvate kinase
PFK-1	Activators — AMP, fructose-2,6-bisphosphate; inhibitors — ATP, citrate
Result of glycolysis	2 pyruvate, 2 NADH, net 2 ATP
Aerobic oxidation	30–32 ATP
Glycogenoses	I — von Gierke (G-6-phosphatase), II — Pompe, III — Cori, V — McArdle, VI — Hers

TOPIC 7

Gluconeogenesis. The pentose phosphate pathway. Fructose and galactose metabolism. Glucose homeostasis

Learning objectives

To teach the student anaerobic glycolysis and its significance, gluconeogenesis, the Cori and glucose-alanine cycles, the reciprocal regulation of glycolysis and gluconeogenesis, the pentose phosphate pathway of glucose oxidation, the metabolism of fructose and galactose and their hereditary disorders, and the regulation and age-related features of blood glucose homeostasis; to master the determination of blood glucose with a glucometer and by the enzymatic method.

By the end of the class the student should know:

- the reactions of anaerobic glycolysis and alcoholic fermentation and their energy yield;
- the substrates of gluconeogenesis, the “bypass” reactions and the energy cost;
- the Cori and glucose-alanine cycles;
- the reciprocal (opposite) regulation of glycolysis and gluconeogenesis;
- the stages and significance of the pentose phosphate pathway;
- the metabolism of fructose and galactose, hereditary fructose intolerance and galactosemia;
- the normal blood glucose level and its hormonal regulation.

The student should be able to:

- determine blood glucose with a glucometer and by the glucose oxidase method;
- evaluate a glucose tolerance test;
- give a biochemical explanation of the causes of hyper- and hypoglycemia.

1. Anaerobic glycolysis

When oxygen is lacking (intense muscle work, hypoxia) or in cells without mitochondria (**erythrocytes**), pyruvate is reduced to lactate by **lactate dehydrogenase**:

Pyruvate + NADH + H^+ → Lactate + NAD^+

This reaction regenerates NAD^+ for glyceraldehyde-3-phosphate dehydrogenase and allows glycolysis to continue without oxygen.

The overall equation of anaerobic glycolysis: Glucose + 2ADP + 2Pi → 2 Lactate + 2ATP + $2H_2O$

The energy yield is **2 ATP** (≈ 15 times less than in aerobic oxidation), but it is fast and needs no oxygen. Anaerobic glycolysis is characteristic mainly of erythrocytes, the retina and lens of the eye, the renal medulla, white muscle fibers and tumor cells.

Alcoholic fermentation (in yeast): pyruvate → acetaldehyde + CO_2 (pyruvate decarboxylase, TDP) → ethanol (alcohol dehydrogenase, NADH). Fermentation is used in baking bread and in making wine and beer.

CLINICAL SIGNIFICANCE

Lactic acidosis — blood lactate > 5 mmol/l and pH < 7.35 (normal lactate 0.5–2.2 mmol/l). Causes: tissue hypoxia (shock, heart failure, sepsis, severe anemia — type A), liver failure (lactate is not used in gluconeogenesis), metformin, thiamine deficiency, mitochondrial diseases (type B). Tumor cells break glucose down to lactate even in the presence of oxygen

— the **Warburg effect**; this underlies the detection of tumors in PET scanning (^{18}F-fluorodeoxyglucose).

2. Gluconeogenesis

Gluconeogenesis is the synthesis of glucose from non-carbohydrate substances. It takes place mainly in the **liver** (≈ 90%) and in the **renal cortex** (up to ≈ 40% in prolonged starvation). Its significance: maintaining blood glucose during starvation, when the glycogen store is exhausted (after 12–24 hours) — the brain (≈ 120 g per day), the erythrocytes and the renal medulla need glucose.

Its substrates: lactate (working muscle, erythrocytes), **glucogenic amino acids** (alanine, glutamine and others — from the breakdown of muscle proteins), **glycerol** (from lipolysis in adipose tissue), propionate (from odd-chain fatty acids). **Fatty acids and acetyl-CoA do not give glucose** (the pyruvate dehydrogenase reaction is irreversible).

Gluconeogenesis uses the reversible reactions of glycolysis, but the three irreversible reactions of glycolysis are bypassed by **four special "bypass" reactions**:

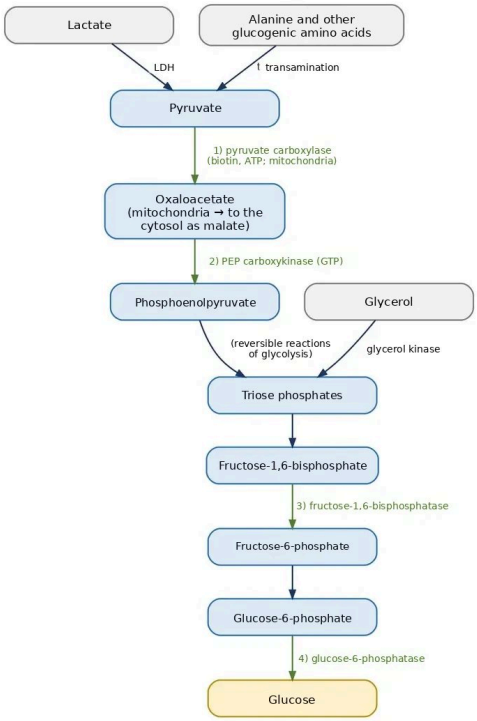


Figure 7.1. *Gluconeogenesis: substrates and the reactions that bypass the irreversible steps of glycolysis (drawn after Severin E.S.)*

Table 7.1. *Key enzymes of gluconeogenesis*

Glycolytic enzyme	The "bypass" of gluconeogenesis	Site and feature
Pyruvate kinase	1) Pyruvate carboxylase: pyruvate + CO_2 + ATP → oxaloacetate (biotin)	Mitochondria; allosterically activated by acetyl-CoA
	2) Phosphoenolpyruvate carboxykinase (PEPCK): oxaloacetate + GTP → phosphoenolpyruvate + CO_2	Cytosol (and mitochondria); induced by cortisol and glucagon

Glycolytic enzyme	The “bypass” of gluconeogenesis	Site and feature
Phosphofructokinase-1	3) Fructose-1,6-bisphosphatase	Cytosol; inhibited by AMP and fructose-2,6-bisphosphate
Hexokinase/glucokinase	4) Glucose-6-phosphatase	Endoplasmic reticulum; only in the liver and kidney

Oxaloacetate does not cross the mitochondrial membrane — it leaves for the cytosol in the form of malate (or aspartate).

Energy cost: the synthesis of 1 glucose from 2 pyruvate consumes **6 high-energy bonds** (4 ATP + 2 GTP) and 2 NADH.

Regulation of glycolysis and gluconeogenesis

Glycolysis and gluconeogenesis are regulated **reciprocally**: when one is activated the other is inhibited (otherwise a “futile cycle” would waste ATP).

Table 7.2. *Hormonal and allosteric regulation of glycolysis and gluconeogenesis*

Factor	Glycolysis	Gluconeogenesis
Insulin (fed state)	↑ induction of glucokinase, PFK-1 and pyruvate kinase; fructose-2,6-bisphosphate ↑	↓ synthesis of the key enzymes
Glucagon (starvation)	↓ fructose-2,6-bisphosphate falls; pyruvate kinase is phosphorylated and inactive	↑ induction of PEPCK
Cortisol	—	↑ induction of PEPCK, glucose-6-phosphatase and aminotransferases; breakdown of proteins (substrate)
Fructose-2,6-bisphosphate	↑ PFK-1	↓ fructose-1,6-bisphosphatase
AMP	↑ PFK-1	↓ fructose-1,6-bisphosphatase
Acetyl-CoA	↓ PDC	↑ pyruvate carboxylase

REMEMBER

Fructose-2,6-bisphosphate is the “key” to glycolysis and gluconeogenesis in the liver: insulin increases its formation and thereby glycolysis, while glucagon (through cAMP) decreases it and increases gluconeogenesis.

The Cori cycle and the glucose-alanine cycle

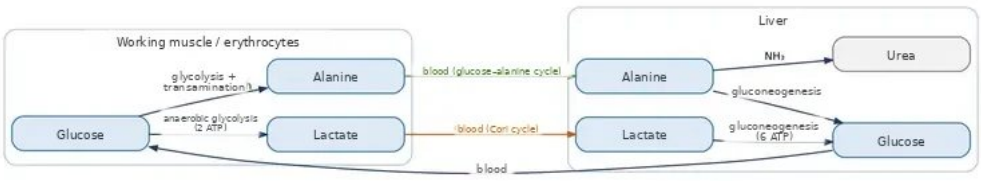


Figure 7.2. *The Cori cycle and the glucose-alanine cycle (drawn after Severin E.S.)*

- **The Cori cycle** — lactate formed in working muscle and erythrocytes reaches the liver through the blood, is converted into glucose and returns to the muscle. The muscle’s “oxygen debt” is repaid at the expense of the liver’s energy.

- **The glucose-alanine cycle** — in muscle, pyruvate takes the amino group from amino acids and is converted into alanine; in the liver alanine is converted back into pyruvate (gluconeogenesis), while the ammonia goes into urea synthesis. This cycle carries nitrogen from the muscle in a harmless form.

3. The pentose phosphate pathway

The pentose phosphate pathway (PPP) is an alternative route of glucose-6-phosphate oxidation; it takes place in the cytosol. It does not form ATP; its main products are **NADPH** and **ribose-5-phosphate**.

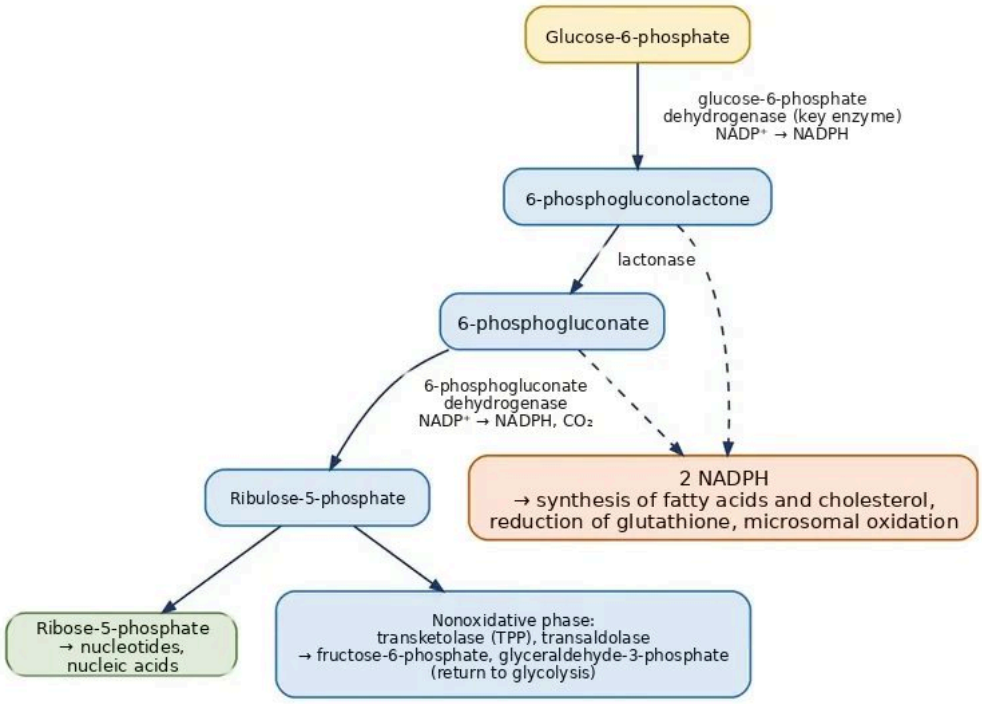


Figure 7.3. *The oxidative and non-oxidative stages of the pentose phosphate pathway (drawn after Severin E.S.)*

1. **The oxidative stage** (irreversible): glucose-6-phosphate → 6-phosphogluconolactone → 6-phosphogluconate → ribulose-5-phosphate + CO_2. **2 NADPH** are formed in the two dehydrogenase reactions. The key enzyme is **glucose-6-phosphate dehydrogenase (G6PD)**; it is activated by $NADP^+$ and inhibited by NADPH; insulin induces it.
2. **The non-oxidative stage** (reversible): with the participation of **transketolase** (coenzyme — TDP, B_1) and **transaldolase**, pentoses are converted into fructose-6-phosphate and glyceraldehyde-3-phosphate (returning to glycolysis) or, conversely, ribose-5-phosphate is formed from glycolytic intermediates. **The significance of NADPH:** synthesis of fatty acids, cholesterol and steroid hormones (the liver, adipose tissue, mammary gland and adrenal cortex are the organs with the most active PPP); keeping glutathione in the reduced state (protection of erythrocytes from oxidative stress); microsomal oxidation (cytochrome P450); NADPH oxidase in phagocytes (the bactericidal “respiratory burst”).

CLINICAL SIGNIFICANCE

Glucose-6-phosphate dehydrogenase deficiency is the most common enzymopathy (X-linked; frequent in areas where malaria is widespread). Little NADPH is formed in the erythrocytes and glutathione is not reduced; when oxidizing drugs (primaquine, sulfonamides, nitrofurans), infections or **fava beans** are involved, hemoglobin is oxidized and precipitates (Heinz bodies), and an **acute hemolytic crisis** develops.

4. Fructose and galactose metabolism

Fructose (in fruit, honey and sucrose) is metabolized mainly in the liver:

Fructose → (**fructokinase**, ATP) → fructose-1-phosphate → (**aldolase B**) → dihydroxyacetone phosphate + glyceraldehyde → glycolysis or gluconeogenesis

Fructose "bypasses" phosphofructokinase-1, so it is broken down rapidly in the liver and its excess is directed to fatty acid synthesis (high fructose intake is a risk factor for fatty degeneration of the liver and hyperlipidemia).

Galactose (part of lactose):

Galactose → (**galactokinase**) → galactose-1-phosphate → (+ UDP-glucose, **galactose-1-phosphate uridyltransferase, GALT**) → UDP-galactose + glucose-1-phosphate; UDP-galactose → (**epimerase**) → UDP-glucose

Table 7.3. *Hereditary disorders of fructose and galactose metabolism*

Disease	Deficient enzyme	Features
Essential fructosuria	Fructokinase	Harmless; fructose in the urine (the Seliwanoff test is positive)
Hereditary fructose intolerance (fructosemia)	Aldolase B	Vomiting after fructose (juice, fruit, sucrose), severe hypoglycemia, liver and kidney damage; fructose-1-phosphate accumulates and P_i and ATP fall
Classic galactosemia	GALT	When the newborn starts milk feeding — vomiting, jaundice, hepatomegaly, cataract, mental retardation, sepsis (E. coli)
Galactokinase deficiency	Galactokinase	Mainly cataract (galactose → galactitol accumulates in the lens)

CLINICAL SIGNIFICANCE

Galactosemia is part of the newborn screening program. As soon as the diagnosis is made, the child is switched to **lactose-free and galactose-free** feeding (soy formulas) — this prevents fatal complications. The cause of the cataract: galactose is reduced by aldose reductase to **galactitol**, which does not leave the lens and, by its osmotic effect, draws in water and damages the fibers.

5. Blood glucose homeostasis

The blood glucose level is strictly regulated, because the brain and erythrocytes obtain their energy mainly from glucose.

Table 7.4. *Normal blood glucose values (adults, fasting)*

Material	Normal value
Capillary blood	3.3–5.5 mmol/l
Venous blood plasma	3.9–6.1 mmol/l (< 5.6 in many recommendations)

Material	Normal value
Newborns	2.8–4.4 mmol/l

Sources of blood glucose: dietary carbohydrates (absorption — 2–3 hours), liver glycogenolysis (between meals, ≈ 12–24 hours), gluconeogenesis (in prolonged starvation). **Its consumption:** oxidation in the tissues, synthesis of glycogen and fats; above 10 mmol/l glucose passes into the urine (the **renal threshold** is 8.9–10 mmol/l).

Table 7.5. *Hormones that regulate blood glucose*

Hormone	Effect on glucose	Mechanism
Insulin	Lowers it (the only hypoglycemic hormone)	Moves GLUT4 to the membrane; increases glycogenesis, glycolysis and lipogenesis; inhibits gluconeogenesis and glycogenolysis
Glucagon	Raises it	Glycogenolysis and gluconeogenesis in the liver
Adrenaline	Raises it	Glycogenolysis in the liver and muscle; reduces insulin secretion
Cortisol	Raises it	Induction of gluconeogenesis enzymes; breakdown of proteins in muscle; reduced uptake of glucose in the periphery
Growth hormone, thyroid hormones	Raises it	Counter-insular action, absorption in the intestine (T_3, T_4)

Age-related features: in the first days of life newborns have physiological hypoglycemia (small glycogen store, immature gluconeogenic enzymes); the risk is higher in preterm infants and in infants of mothers with diabetes. In old age glucose tolerance decreases.

Glycoproteins and proteoglycans are complex proteins that contain carbohydrate: glycoproteins (immunoglobulins, hormones, membrane proteins) have short branched oligosaccharides; proteoglycans have many long glycosaminoglycan chains (connective tissue — topic 22). Their carbohydrate part is formed from glucose through UDP-sugar nucleotides.

6. Laboratory work

Experiment 1: determination of blood glucose by the glucose oxidase method

Principle of the method: glucose oxidase oxidizes glucose to gluconolactone and H_2O_2; peroxidase, at the expense of H_2O_2, oxidizes a chromogen (4-aminoantipyrine + phenol) and forms a pink compound (the Trinder reaction). The intensity of the color is proportional to the glucose concentration.

1. Pour 2 ml of working reagent (glucose oxidase, peroxidase, chromogen) into each of three test tubes.
2. Add 20 µl of serum (plasma) to tube 1, 20 µl of standard glucose solution (5.55 mmol/l) to tube 2 and 20 µl of water to tube 3 (control).
3. Mix and keep at 37 °C for 15 minutes.
4. Measure the optical density against the control at 500–510 nm on a photometer (or a biochemical analyzer). **Calculation:** C (mmol/l) = (Esample / Estandard) × 5.55

Evaluation: in venous plasma the fasting norm is 3.9–6.1 mmol/l; 6.1–6.9 — impaired fasting glycemia; ≥ 7.0 (on two occasions) — diabetes mellitus; < 3.3 — hypoglycemia.

Experiment 2: determination of glucose in capillary blood with a glucometer

1. Wash and dry the hand, wipe the pad of the finger with alcohol and wait until the alcohol evaporates.
2. Insert the test strip into the glucometer (check that the code matches).
3. Puncture the side of the finger with a lancet and wipe away the first drop.
4. Touch the second drop to the application area of the test strip.
5. After 5–10 s read and record the result. **Analysis:** the glucose oxidase (or glucose dehydrogenase) immobilized on the test strip oxidizes glucose; the electrons pass to the electrode and an electric current is generated (an electrochemical biosensor). The glucometer is intended for self-monitoring; the diagnosis is made in the laboratory in venous plasma. Causes of error: alcohol left on the finger, the first drop, squeezing out the blood (tissue fluid), an expired test strip, a sharp change in hematocrit.

Experiment 3: evaluating a glucose tolerance test (from a case)

The subject gives blood on an empty stomach (after 8–14 hours of fasting), then within 5 minutes drinks **75 g of glucose** dissolved in 250–300 ml of water (in children 1.75 g/kg, up to 75 g); blood is taken again after 2 hours.

Table 7.6. *Evaluation of the glucose tolerance test (venous plasma, WHO)*

Condition	Fasting	After 2 hours
Normal value	< 6,1	< 7,8
Impaired fasting glycemia	6,1–6,9	< 7,8
Impaired glucose tolerance	< 7,0	7,8–11,0
Diabetes mellitus	≥ 7,0	≥ 11,1

7. Practical skills

Skill: measuring glucose with a glucometer

1. Check that the glucometer works and that the test strips are in date.
2. Have the patient wash the hand, wipe the finger with alcohol and dry it.
3. Puncture with a lancet and wipe away the first drop.
4. Touch the second drop to the test strip and record the result.
5. Evaluate the result and, if necessary, recommend confirmation in venous blood in the laboratory.

Skill: giving a biochemical explanation of the cause of hypoglycemia

1. Determine when the hypoglycemia occurs: after a meal (reactive) or in the fasting state.
2. Assess the state of the liver (glycogen store, gluconeogenesis), the hormones (excess insulin, cortisol or glucagon deficiency) and medicines.
3. Exclude congenital diseases (glycogenoses, fructosemia, galactosemia).
4. Take into account the effect of alcohol (NADH increases and inhibits gluconeogenesis).
5. Emergency care: drinking glucose or intravenous 40% glucose, glucagon.

Review questions

1. Under what conditions does anaerobic glycolysis take place and what is its significance?
2. How much ATP is formed in anaerobic glycolysis?
3. How does alcoholic fermentation proceed?
4. Where does gluconeogenesis take place and from which substrates is glucose formed?
5. Why is glucose not formed from fatty acids?
6. Write the “bypass” reactions of gluconeogenesis.
7. How much energy is used for gluconeogenesis?
8. How are glycolysis and gluconeogenesis regulated?
9. Explain the Cori cycle and the glucose-alanine cycle.
10. What are the stages and products of the pentose phosphate pathway?
11. In which processes is NADPH used?
12. Why does hemolysis develop in glucose-6-phosphate dehydrogenase deficiency?
13. How are fructose and galactose metabolized? Name their hereditary disorders.
14. What are the normal level and the sources of blood glucose?
15. By which hormones is blood glucose regulated?

Test questions

1. How is energy formed in erythrocytes?
   A) By oxidative phosphorylation
   B) By anaerobic glycolysis
   C) By the β-oxidation of fatty acids
   D) By the Krebs cycle
   E) By the oxidation of ketone bodies

2. The reaction in which NAD^+ is regenerated for glyceraldehyde-3-phosphate dehydrogenase in anaerobic glycolysis:
   A) The pyruvate kinase reaction
   B) The phosphoglycerate kinase reaction
   C) The malate dehydrogenase reaction
   D) The lactate dehydrogenase reaction
   E) The aldolase reaction

3. The substance that cannot be a substrate of gluconeogenesis:
   A) Lactate
   B) Alanine
   C) Glycerol
   D) Propionate
   E) Palmitic acid

4. The coenzyme of pyruvate carboxylase, the first “bypass” enzyme of gluconeogenesis:
   A) Thiamine diphosphate
   B) Pyridoxal phosphate
   C) Tetrahydrofolate
   D) Biotin
   E) Lipoic acid

5. How cortisol increases gluconeogenesis:
   A) By inducing PEP carboxykinase and the other key enzymes and supplying substrate (amino acids)
   B) By inducing glucokinase
   C) By increasing fructose-2,6-bisphosphate
   D) By activating glycogen synthase
   E) By moving GLUT4 to the membrane
6. The number of high-energy bonds used for the synthesis of 1 glucose from 2 molecules of pyruvate:
   A) 2
   B) 3
   C) 4
   D) 8
   E) 6
7. The essence of the Cori cycle:
   A) Transport of lactate from the liver to the muscle
   B) Transport of nitrogen from the muscle in the form of alanine
   C) Conversion of muscle lactate into glucose in the liver and its return to the muscle
   D) Formation of glucose from fatty acids
   E) Breakdown of glycogen in muscle
8. The key enzyme of the pentose phosphate pathway:
   A) Transketolase
   B) Transaldolase
   C) 6-phosphogluconate dehydrogenase
   D) Lactonase
   E) Glucose-6-phosphate dehydrogenase
9. The main products of the pentose phosphate pathway:
   A) ATP and NADH
   B) Lactate and ATP
   C) NADPH and ribose-5-phosphate
   D) Acetyl-CoA and CO_2
   E) Glucose and glycerol
10. The vitamin from which the coenzyme of transketolase is derived:
    A) B_2
    B) B_1
    C) B_6
    D) PP
    E) B_{12}
11. The enzyme that is deficient in hereditary fructose intolerance:
    A) Fructokinase
    B) Hexokinase
    C) Galactokinase
    D) Aldolase B
    E) GALT

12. The cause of cataract in classic galactosemia:
    A) Accumulation of glucose
    B) Accumulation of galactitol
    C) Accumulation of fructose
    D) Lack of sorbitol
    E) Accumulation of lactose

13. The normal fasting glucose level in capillary blood:
    A) 1.5–2.5 mmol/l
    B) 6.5–8.0 mmol/l
    C) 3.3–5.5 mmol/l
    D) 8.9–10 mmol/l
    E) Above 11.1 mmol/l

14. The only hormone that lowers blood glucose:
    A) Insulin
    B) Glucagon
    C) Adrenaline
    D) Cortisol
    E) Somatotropin

15. The value in venous plasma 2 hours after a glucose tolerance test that indicates diabetes mellitus:
    A) ≥ 11.1 mmol/l
    B) Less than 7.8 mmol/l
    C) 5.5 mmol/l
    D) Less than 3.3 mmol/l
    E) 6.1 mmol/l

Clinical cases

Case 1

Two days after taking a sulfonamide drug, a 30-year-old man developed weakness, jaundice and dark urine. Blood hemoglobin is 85 g/l, indirect bilirubin is increased, and Heinz bodies were found in the erythrocytes. He has had the same condition before after eating a bean dish.

Questions: Which enzyme deficiency is suspected? Explain the mechanism of the hemolysis.

Answer: Glucose-6-phosphate dehydrogenase deficiency. In erythrocytes NADPH is formed only in the pentose phosphate pathway; through glutathione reductase it keeps glutathione in the reduced state. An oxidizing drug (sulfonamide) or the substances of beans generate H_2O_2 and free radicals; because glutathione is not regenerated, hemoglobin and membrane proteins are oxidized, hemoglobin precipitates and forms Heinz bodies, and the erythrocytes are destroyed — acute hemolytic anemia, a rise in indirect bilirubin (jaundice) and hemoglobinuria.

Case 2

When a 6-month-old child was started on fruit juice and sweetened porridge as complementary food, pallor, sweating, vomiting, seizures and liver enlargement were observed after meals. While breastfed, the child had no complaints.

Questions: Which disease is suspected? Explain the mechanism of the hypoglycemia.

Answer: Hereditary fructose intolerance — aldolase B deficiency. Fructose (juice, sucrose) is converted in the liver by fructokinase into fructose-1-phosphate, but it is not broken down and accumulates. Phosphate and ATP are consumed in this process (hypophosphatemia, ATP deficit — liver damage). Fructose-1-phosphate inhibits glycogen phosphorylase and aldolase — glycogenolysis and gluconeogenesis stop and severe hypoglycemia develops. Breast milk contains no fructose. Treatment — complete exclusion of fructose, sucrose and sorbitol from the diet.

Case 3

A 45-year-old man drank a lot of alcohol for 2 days without eating. He was brought in unconscious: blood glucose 2.1 mmol/l, lactate increased.

Questions: Explain the mechanism of alcoholic hypoglycemia.

Answer: In starvation the liver glycogen is exhausted within 24 hours and blood glucose is maintained by gluconeogenesis. In the liver ethanol is oxidized by alcohol dehydrogenase and aldehyde dehydrogenase and generates a lot of NADH; the NADH/NAD^+ ratio rises sharply. As a result pyruvate is reduced to lactate and oxaloacetate to malate — the main substrates for gluconeogenesis (pyruvate, oxaloacetate) are lacking and gluconeogenesis stops. Hypoglycemia and lactic acidosis develop. Help: intravenous glucose (thiamine first).

Quick review

Table 7.7. *Key concepts of topic 7*

Concept	Brief definition
Anaerobic glycolysis	Glucose → 2 lactate + 2 ATP; LDH regenerates NAD^+
Gluconeogenesis	Glucose from lactate, glucogenic amino acids and glycerol; liver, kidney
Bypass enzymes	Pyruvate carboxylase, PEPCK, fructose-1,6-bisphosphatase, glucose-6-phosphatase
Energy cost	2 pyruvate → glucose: 4 ATP + 2 GTP
Reciprocal regulation	Fructose-2,6-bisphosphate: glycolysis ↑, gluconeogenesis ↓
Cori cycle	Muscle lactate → glucose in the liver → back to the muscle
Glucose-alanine cycle	Muscle nitrogen to the liver in the form of alanine
PPP	G-6-P → 2 NADPH + ribose-5-phosphate; key enzyme — G6PD
Transketolase	TDP (B_1) — the non-oxidative stage
G6PD deficiency	Oxidizing drugs, beans — hemolytic crisis
Fructosemia	Aldolase B; hypoglycemia after fructose
Galactosemia	GALT; jaundice, cataract, mental retardation
Blood glucose	Capillary 3.3–5.5; venous plasma 3.9–6.1 mmol/l
GTT	After 2 hours: < 7.8 normal; 7.8–11.0 impaired tolerance; ≥ 11.1 diabetes

TOPIC 8

Structure, functions, digestion and transport of lipids

Learning objectives

To teach the student the classification, structure and significance of the lipids of food and of the human body, the digestion and absorption of fats (including the features of the child's body), the significance of bile acids, the resynthesis of fats in the intestinal wall, the formation and transport of chylomicrons and blood lipoproteins, the significance of lipoprotein lipase, and the disorders of fat digestion.

By the end of the class the student should know:

- the classification of lipids and the structure of their main representatives;
- the functions of lipids in the body;
- the essential fatty acids;
- the stages and enzymes of fat digestion and the role of bile acids;
- the absorption of the products of fat digestion and their resynthesis in the intestinal wall;
- the types, composition and functions of lipoproteins;
- the significance of lipoprotein lipase;
- the causes of steatorrhea.

The student should be able to:

- demonstrate the emulsifying action of bile acids;
- determine lipase activity;
- evaluate the indices of the blood lipid profile.

1. Classification and structure of lipids

Lipids are a group of substances insoluble in water but soluble in organic solvents (ether, chloroform, benzene) that contain fatty acids or their derivatives.

Table 8.1. *Classification of lipids*

Group	Representatives	Structure
Fatty acids	Palmitic (C16:0), stearic (C18:0), oleic (C18:1), linoleic (C18:2, ω-6), α-linolenic (C18:3, ω-3), arachidonic (C20:4, ω-6)	Long-chain carboxylic acids; saturated and unsaturated
Triacylglycerols (neutral fats, TAG)	Tristearin, triolein	Glycerol + 3 fatty acids (an ester)
Glycerophospholipids	Phosphatidylcholine (lecithin), phosphatidylethanolamine, phosphatidylserine, phosphatidylinositol, cardiolipin	Glycerol + 2 fatty acids + phosphate + a nitrogenous base or inositol
Sphingolipids	Sphingomyelin, cerebrosides, gangliosides	Sphingosine + fatty acid (ceramide) + phosphocholine or carbohydrates
Sterols and steroids	Cholesterol, bile acids, steroid hormones, vitamin D	The cyclopentanoperhydrophenanthrene ring

Group	Representatives	Structure
Eicosanoids	Prostaglandins, thromboxanes, leukotrienes	Derivatives of arachidonic acid

Functions in the body:

- **energy** — the oxidation of 1 g of fat gives 9 kcal; TAG is the main energy store (10–15 kg in adipose tissue — enough for 40 days of starvation);
- **structural** — phospholipids, glycolipids and cholesterol form the basis of membranes;
- **regulatory** — steroid hormones, eicosanoids, second messengers (DAG, IP_3);
- **protective and heat-insulating** — the subcutaneous fat layer, the fat around internal organs;
- a source of fat-soluble vitamins (A, D, E, K) and necessary for their absorption;
- **bile acids** — digestion and absorption of fats; **surfactant** (dipalmitoylphosphatidylcholine) — keeps the alveoli from collapsing. **Essential fatty acids** are linoleic (ω-6) and α-linolenic (ω-3): humans have no enzymes that introduce a double bond further than carbon 9 from the carboxyl group. Sources: vegetable oils (sunflower, corn, linseed), fish oil (eicosapentaenoic and docosahexaenoic acids — ω-3). Their deficiency causes dermatitis, growth retardation and impaired wound healing.

2. Digestion of fats

The daily fat requirement of adults is 80–100 g; ≈ 90% of it is TAG.

In the oral cavity and the stomach fats are hardly digested in adults: gastric lipase has low activity (its pH optimum is 4–5, while gastric juice is strongly acidic) and the fat is not emulsified. **In newborns**, however, the lingual and gastric lipases play an important role: the gastric medium is less acidic, the fat of milk is already emulsified, and breast milk also contains its own lipase.

In the small intestine the main digestion of fats takes place:

1. **Emulsification** — the breaking up of large fat droplets into small ones; this is helped by the salts of **bile acids** (surface-active substances), peristalsis and the CO_2 bubbles formed by bicarbonate. Emulsification increases many times over the surface on which lipase acts.
2. **Hydrolysis** — by the enzymes of the pancreas: – **pancreatic lipase** (its activating protein is **colipase**) cleaves the ester bonds of TAG at

positions 1 and 3: TAG → 2-monoacylglycerol + 2 fatty acids; – **phospholipase A_2** (activated by trypsin) removes the fatty acid at position 2 from a

phospholipid → lysophospholipid; – **cholesterol esterase** breaks down cholesterol esters into cholesterol and a fatty acid.

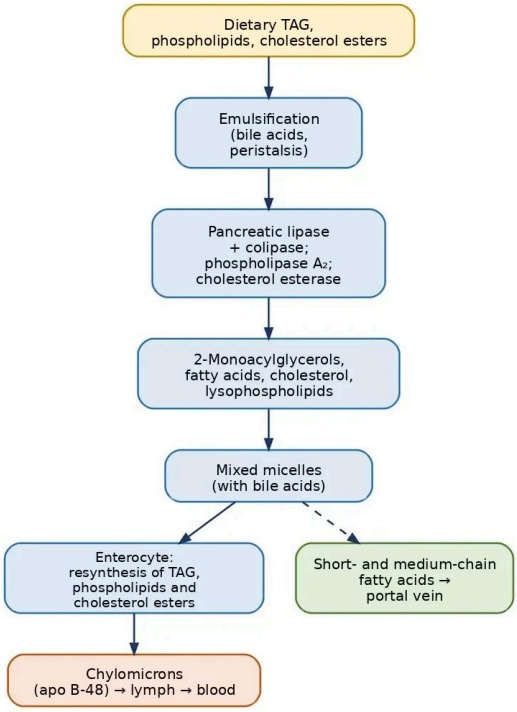


Figure 8.1. *Digestion and absorption of fats and their resynthesis in the intestinal wall (drawn after Severin E.S.)*

Bile acids

Bile acids are synthesized in the liver from cholesterol (the key enzyme is 7α-hydroxylase). The **primary** bile acids are cholic and chenodeoxycholic acid; they are **conjugated** with glycine or taurine (glycocholic and taurocholic acids) and enter the intestine with the bile as sodium and potassium salts. In the intestine, under the action of bacteria, the **secondary** bile acids — deoxycholic and lithocholic acid — are formed.

The functions of bile acids: they emulsify fats; activate lipase; form **mixed micelles** with the products of digestion and provide for their absorption; are necessary for the absorption of fat-soluble vitamins; keep cholesterol dissolved in the bile.

The enterohepatic circulation: ≈ 95% of the bile acids are reabsorbed in the terminal ileum, return to the liver through the portal vein and are secreted into the bile again (6–10 times a day). Only ≈ 0.5 g/day is lost in the feces — this is the main route by which cholesterol is removed from the body.

3. Absorption of the products of digestion and their resynthesis

Monoacylglycerols, long-chain fatty acids, cholesterol and lysophospholipids form **mixed micelles** with bile acids and are delivered to the surface of the enterocytes; the lipid components cross the membrane by diffusion (and through special carriers), while the bile acids remain in the intestinal lumen. **Short- and medium-chain** fatty acids (C4–C12) are absorbed without micelles and pass directly into the blood of the portal vein (they are abundant in milk fat; in malabsorption, food with medium-chain TAG is prescribed).

Resynthesis in the enterocytes (in the endoplasmic reticulum):

- the fatty acids are activated: fatty acid + CoA + ATP → acyl-CoA (acyl-CoA synthetase);
- **the monoacylglycerol pathway**: 2-monoacylglycerol + 2 acyl-CoA → TAG;
- phospholipids are formed from lysophospholipids, and cholesterol esters from cholesterol (the ACAT enzyme). **The significance of resynthesis:** dietary fats are converted into fats

characteristic of the human body (the fatty acid composition changes); the products of digestion are “carried away” from the intestinal cell, which allows absorption to continue; the TAG becomes part of the transport form — the chylomicrons.

The resynthesized lipids are assembled together with the protein **apo B-48** into **chylomicrons**, which leave by exocytosis into the **lymph** (the intestinal lymph capillaries — the “lacteals”) and enter the blood through the thoracic duct (this is why the serum is cloudy after a fatty meal — **alimentary lipemia**).

CLINICAL SIGNIFICANCE

Steatorrhea is an increase of fat in the feces (> 7 g per day). Causes: **lack of bile** (obstructive jaundice, obstruction of the bile ducts, liver diseases) — emulsification and micelle formation are impaired; **lipase deficiency** (chronic pancreatitis, cystic fibrosis, a pancreatic tumor); **impaired absorption** (celiac disease, Crohn disease, resection of the ileum). The feces are shiny, sticky, large in amount and have a sharp odor. Consequences: deficiency of fat-soluble vitamins (night blindness, osteomalacia, bleeding), a shortage of essential fatty acids, weight loss.

4. Blood lipoproteins

Lipids are insoluble in water, so they are transported in the blood in **lipoproteins** — lipid-protein complexes. The core of a lipoprotein particle contains hydrophobic lipids (TAG, cholesterol esters), and its surface amphiphilic substances — phospholipids, free cholesterol and **apoproteins**.

Functions of the apoproteins: forming the structure of the particle (apo B-48, B-100), activating enzymes (apo C-II — lipoprotein lipase, apo A-I — LCAT), binding to receptors (apo B-100 and apo E — LDL receptors).

Table 8.2. *Blood lipoproteins*

Lipoprotein	Where it is formed	Main lipid	Main apoproteins	Function
Chylomicrons (CM)	Intestine	Dietary TAG (≈ 85%)	B-48, C-II, E	Transport of dietary fats from the intestine to the tissues
Very low density lipoproteins (VLDL)	Liver	Endogenous TAG (≈ 55%)	B-100, C-II, E	Transport of TAG synthesized in the liver to the tissues
Intermediate density lipoproteins (IDL)	In the blood from VLDL	TAG and cholesterol	B-100, E	An intermediate form between VLDL and LDL
Low density lipoproteins (LDL)	In the blood from IDL	Cholesterol (≈ 45–50%)	B-100	Transport of cholesterol from the liver to the tissues (“bad” cholesterol)
High density lipoproteins (HDL)	Liver, intestine	Phospholipids, cholesterol; protein ≈ 50%	A-I, A-II	Return of cholesterol from the tissues to the liver (“good” cholesterol)

As the density increases, the particle becomes smaller and the share of protein larger (CM are the largest and lightest, HDL the smallest and densest).

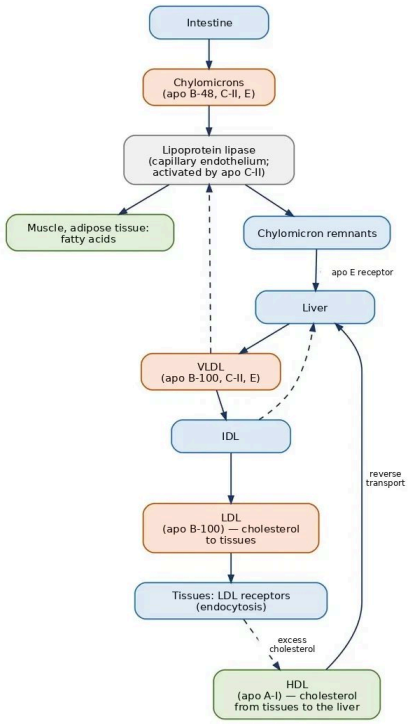


Figure 8.2. *Metabolism of lipoproteins: the exogenous and endogenous pathways and reverse cholesterol transport (drawn after Severin E.S.)*

Lipoprotein lipase

Lipoprotein lipase (LPL) is an enzyme located on the surface of the capillary endothelium (bound to heparan sulfate); it hydrolyzes the TAG of chylomicrons and VLDL. The fatty acids formed enter the tissues: in **muscle** they are oxidized (energy), in **adipose tissue** they are converted back into TAG and stored. LPL is activated by **apo C-II**; its synthesis in adipose tissue is increased by **insulin** (after a meal fat goes into store), and in muscle by starvation and physical activity. Heparin releases LPL from the endothelium into the blood (the **"clearing factor"** — lipemic serum becomes clear).

The **remnant chylomicrons**, emptied of TAG, are taken up by the liver (through apo E); VLDL are converted into IDL and LDL. LDL enter the tissues by endocytosis through **LDL receptors** (topic 10).

CLINICAL SIGNIFICANCE

Congenital deficiency of LPL or apo C-II (type I hyperlipoproteinemia) — chylomicrons rise sharply in the blood: the serum is "milky", triglycerides > 10–20 mmol/l, with attacks of acute pancreatitis, xanthomas on the skin and enlargement of the liver and spleen. Treatment — a diet with sharply restricted fat (medium-chain TAG are allowed, because they do not form chylomicrons).

Table 8.3. *Normal values of the blood lipid profile (adults)*

Index	Desirable value
Total cholesterol	< 5.2 mmol/l
Triglycerides	< 1.7 mmol/l
LDL cholesterol	< 3.0 mmol/l (< 1.8 or < 1.4 in high-risk patients)
HDL cholesterol	> 1.0 in men, > 1.2 mmol/l in women
Atherogenic index: (total CH − HDL-CH) /	< 3,0

Index Desirable value

HDL-CH

In children total cholesterol is lower (2.5–4.5 mmol/l); in breastfed newborns lipids rise quickly.

5. Laboratory work

Experiment 1: the emulsifying action of bile acids

1. Pour 2 ml of water into each of three test tubes.
2. Add 2 drops of vegetable oil to tube 1, 2 drops of oil + 5 drops of bile (or 1% sodium cholate) to tube 2, and 2 drops of oil + 5 drops of 1% sodium carbonate to tube 3.
3. Shake all of them vigorously and leave for 5–10 minutes. **Result and analysis:** in tube 1 the oil quickly rises to the top (the emulsion is unstable). In the tube with bile a stable, milky emulsion forms — bile acids lower the surface tension of the droplets. In the tube with sodium carbonate the emulsion is also more stable — soap is formed with the free fatty acids of the oil.

Experiment 2: determination of pancreatic lipase activity and the effect of bile

1. Pour 10 ml of milk (or a fat emulsion) and 2 drops of phenolphthalein into each of two flasks; neutralize with 0.1 N NaOH to a pale pink color.
2. Add 1 ml of pancreatin solution to flask 1 and 1 ml of pancreatin + 1 ml of bile to flask 2; keep at 37 °C.
3. Every 10 minutes take 1 ml from the flasks and titrate with 0.05 N NaOH to a pink color (for 30–40 minutes). **Result and analysis:** with time the amount of NaOH used for titration increases — lipase breaks down TAG and forms fatty acids. In the flask with bile this happens faster: the bile acids emulsify the fat, increase the surface on which lipase acts and activate the lipase.

Experiment 3: assessing the lipemia of serum and interpreting the results of the lipid profile (from a case)

1. Compare serum taken on an empty stomach and 3–4 hours after a fatty meal (demonstration or figure).
2. Leave the lipemic serum in the refrigerator (+4 °C) overnight: if a “cream” layer forms on top — chylomicrons; if the serum itself remains cloudy — there is a lot of VLDL.
3. Compare the lipid profile results given by the teacher with the normal values and calculate the atherogenic index.

6. Practical skills

Skill: identifying the cause of steatorrhea

1. Confirm the increase of fat in the feces (coprogram: neutral fat or fatty acids).
2. If there is a lot of neutral fat (TAG) — lipase deficiency (pancreatitis): fecal elastase-1 is reduced.
3. If there are many fatty acids and soaps — lack of bile or impaired absorption.
4. If there is jaundice, acholic feces and raised conjugated bilirubin in the blood — obstruction of the bile ducts.
5. Check for signs of fat-soluble vitamin deficiency and recommend a diet.

Skill: evaluating the lipid profile

1. Compare the values of total cholesterol, triglycerides, LDL and HDL cholesterol with the normal values.
2. Calculate the atherogenic index.
3. Determine which lipoproteins are increased (CM, VLDL or LDL).
4. Suggest the cause: diet, obesity, diabetes mellitus, hypothyroidism, hereditary factors.
5. Assess the risk of atherosclerosis.

Review questions

1. How are lipids classified?
2. List the functions of lipids in the body.
3. Which are the essential fatty acids and what are their sources?
4. How does the digestion of fats in the stomach differ between adults and newborns?
5. What is emulsification and which factors help it?
6. Which enzymes of the pancreas break down lipids?
7. How are bile acids formed? Which are the primary and the secondary bile acids?
8. List the functions of bile acids.
9. What is the enterohepatic circulation?
10. What are mixed micelles?
11. How does the resynthesis of fats proceed in the intestinal wall and what is its significance?
12. How are lipoproteins built? What are the functions of the apoproteins?
13. Describe the types and functions of the blood lipoproteins.
14. What are the significance and the regulation of lipoprotein lipase?
15. What are the causes and consequences of steatorrhea?

Test questions

1. The fatty acid that is essential:
   A) Palmitic
   B) Stearic
   C) Linoleic
   D) Oleic
   E) Palmitoleic

2. The activating protein of pancreatic lipase:
   A) Colipase
   B) Apo C-II
   C) Apo A-I
   D) Enteropeptidase
   E) Trypsin

3. The products into which pancreatic lipase breaks down TAG:
   A) Glycerol and 3 fatty acids
   B) 1-monoacylglycerol and 2 fatty acids
   C) Diacylglycerol and 1 fatty acid
   D) Phosphatidic acid
   E) 2-monoacylglycerol and 2 fatty acids

4. The primary bile acids:
   A) Cholic and chenodeoxycholic acids
   B) Deoxycholic and lithocholic acids
   C) Glycocholic and deoxycholic acids
   D) Lithocholic and cholic acids
   E) Taurine and glycine

5. Where bile acids are reabsorbed in their enterohepatic circulation:
   A) In the terminal ileum
   B) In the stomach
   C) In the duodenum
   D) In the large intestine
   E) In the esophagus

6. The form in which TAG resynthesized in the intestinal wall enters the blood:
   A) Free
   B) In chylomicrons through the lymph
   C) Bound to albumin
   D) In LDL through the portal vein
   E) In HDL

7. The main structural apoprotein of chylomicrons:
   A) Apo A-I
   B) Apo B-100
   C) Apo C-II
   D) Apo B-48
   E) Apo E

8. The apoprotein that activates lipoprotein lipase:
   A) Apo A-I
   B) Apo B-100
   C) Apo C-II
   D) Apo E
   E) Apo B-48

9. The lipoproteins that return cholesterol from the tissues to the liver:
   A) Chylomicrons
   B) VLDL
   C) IDL
   D) LDL
   E) HDL

10. The lipoproteins that transport endogenous TAG synthesized in the liver to the tissues:
    A) Chylomicrons
    B) VLDL
    C) HDL
    D) LDL
    E) Albumin

11. How short- and medium-chain fatty acids are absorbed:
    A) In chylomicrons into the lymph
    B) Only with micelles
    C) Only by pinocytosis
    D) In LDL
    E) Without micelles, directly into the portal vein

12. The main cause of steatorrhea in obstructive jaundice:
    A) Lack of lipase
    B) Lack of lipoprotein lipase
    C) A change in the intestinal microflora
    D) Bile acids do not reach the intestine
    E) Lack of colipase

13. The desirable value of serum triglycerides:
    A) < 5.2 mmol/l
    B) < 3.0 mmol/l
    C) < 1.7 mmol/l
    D) > 1.2 mmol/l
    E) < 10 mmol/l

14. What the clearing of lipemic serum after heparin is due to:
    A) Precipitation of the lipids
    B) A halt in cholesterol synthesis
    C) The action of bile acids
    D) Release of lipoprotein lipase into the blood
    E) An increase of albumin

15. The type of lipids that forms the basis of membranes:
    A) Triacylglycerols
    B) Phospholipids
    C) Free fatty acids
    D) Cholesterol esters
    E) Waxes

Clinical cases

Case 1

A 50-year-old patient has yellow skin and sclerae, itching, colorless (acholic), shiny and bulky stools and dark urine. Ultrasound revealed a stone in the common bile duct. The clotting time is prolonged.

Questions: Why has the fat in the feces increased? Explain the clotting disorder.

Answer: Because the stone has blocked the bile duct, bile acids do not reach the intestine: fats are not emulsified, the surface on which lipase acts is small, and mixed micelles are not formed — the

products of digestion are not absorbed and leave with the feces (steatorrhea). There are no bilirubin derivatives (stercobilin) in the feces — they are acholic. Fat-soluble vitamins are also not absorbed; because of vitamin K deficiency the γ-carboxylation of clotting factors II, VII, IX and X in the liver is impaired — the clotting time is prolonged. Vitamin K is given parenterally.

Case 2

An 8-year-old child has recurrent attacks of severe abdominal pain (acute pancreatitis), yellow nodules on the skin and buttocks (eruptive xanthomas) and enlargement of the liver and spleen. The fasting serum is milky and, after being left in the refrigerator, formed a creamy layer on top; triglycerides are 25 mmol/l.

Questions: What is the disease? Explain its biochemical mechanism and the principle of treatment.

Answer: Type I hyperlipoproteinemia — deficiency of lipoprotein lipase (or of its activator apo C-II). The TAG of the chylomicrons is not hydrolyzed and the chylomicrons stay in the blood for a long time (even in the fasting state) — the serum is milky and in the refrigerator a “cream” layer floats to the top. Excess triglycerides are broken down by pancreatic lipase in the capillaries of the pancreas and form toxic fatty acids — pancreatitis. Treatment: sharp restriction of long-chain fats in the diet; medium-chain TAG are allowed, because they are absorbed into the portal vein without chylomicrons.

Case 3

A 5-year-old child with cystic fibrosis is underweight and thin even though the appetite is good. The stools are bulky, fatty and strong-smelling; there is a lot of neutral fat in the feces and elastase-1 is sharply reduced. He sees poorly in the evening.

Questions: What is the cause of the steatorrhea? Explain the “night blindness.” What treatment is needed?

Answer: In cystic fibrosis a defect of the CFTR Cl^- channel makes the pancreatic secretion thick and it blocks the ducts; the gland tissue undergoes fibrosis — lipase and the other enzymes do not reach the intestine (exocrine insufficiency). TAG is not broken down — there is a lot of neutral fat in the feces. Fat-soluble vitamins are not absorbed: vitamin A deficiency impairs rhodopsin synthesis and vision in the dark deteriorates. Treatment: pancreatic enzyme preparations (pancreatin) with meals, fat-soluble vitamins (A, D, E, K), a high-calorie diet.

Quick review

Table 8.4. *Key concepts of topic 8*

Concept	Brief definition
Classification of lipids	Fatty acids, TAG, phospholipids, sphingolipids, steroids, eicosanoids
Essential FA	Linoleic (ω-6), α-linolenic (ω-3)
Emulsification	Bile acid salts, peristalsis, CO_2
Pancreatic lipase	+ colipase; TAG → 2-MAG + 2 FA
Phospholipase A_2	Phospholipid → lysophospholipid + FA
Bile acids	Primary: cholic, chenodeoxycholic; conjugation — glycine, taurine
Enterohepatic circulation	≈ 95% is reabsorbed in the ileum

Concept	Brief definition
Resynthesis	In the enterocyte: 2-MAG + 2 acyl-CoA → TAG → chylomicrons (apo B-48)
Lipoproteins	CM, VLDL — transport TAG; LDL — cholesterol to the tissues; HDL — to the liver
LPL	On the capillary endothelium; activated by apo C-II, induced by insulin (in adipose tissue)
Normal values	Cholesterol < 5.2; TG < 1.7; LDL-CH < 3.0; HDL-CH > 1.0–1.2 mmol/l
Steatorrhea	Deficiency of bile or lipase, or impaired absorption

TOPIC 9

Intermediary metabolism of fats

Learning objectives

To give the student knowledge of the storage and mobilization of fats in adipose tissue (lipolysis and lipogenesis) and their regulation, the formation of fats from carbohydrates, the transport and β-oxidation of fatty acids, the oxidation of glycerol, the biosynthesis of fatty acids and its regulation, the synthesis and breakdown of ketone bodies, and the metabolism of the main phospho- and glycolipids.

By the end of the class the student should know:

- the synthesis of TAG and lipolysis in adipose tissue and their hormonal regulation;
- the transport of fatty acids in the blood and their entry into the cell;
- the carnitine shuttle system and the four reactions of β-oxidation;
- the energy yield of the oxidation of palmitic acid;
- the oxidation pathway of glycerol;
- the stages, enzymes and regulation of fatty acid synthesis;
- the synthesis and use of ketone bodies and the causes of ketosis;
- the main pathways of synthesis and breakdown of phospholipids and glycolipids.

The student should be able to:

- determine ketone bodies in urine and evaluate the result;
- calculate the ATP formed in the oxidation of a fatty acid;
- explain the changes in fat metabolism in starvation and diabetes mellitus.

1. Storage and mobilization of fats

Adipose tissue is the body’s main energy store (15–25% of body weight in healthy adults). In adipocytes TAG is formed from two sources: from the fatty acids released from chylomicrons and VLDL by lipoprotein lipase, and from glucose (glucose → glycerol-3-phosphate, excess glucose → acetyl-CoA → fatty acids).

TAG synthesis (lipogenesis) — in the liver and adipose tissue:

glycerol-3-phosphate + 2 acyl-CoA → phosphatidic acid → (phosphatase) → diacylglycerol + acyl-CoA → **TAG**

In adipose tissue glycerol-3-phosphate is formed only from **glucose** (reduction of dihydroxyacetone phosphate), because adipocytes have almost no glycerol kinase. For this reason the accumulation of TAG in adipose tissue depends on the availability of glucose and insulin. The liver does have glycerol kinase.

Lipolysis (mobilization of fats) — the hydrolysis of TAG to glycerol and fatty acids in starvation, physical work and stress:

1. **Adipose triglyceride lipase (ATGL)**: TAG → diacylglycerol + FA;
2. **Hormone-sensitive lipase (HSL)**: DAG → monoacylglycerol + FA (the key, regulated enzyme);
3. **Monoacylglycerol lipase**: MAG → glycerol + FA.

Table 9.1. *Hormonal regulation of lipolysis*

Hormone	Effect	Mechanism
Adrenaline, noradrenaline (β-receptors), glucagon	Lipolysis ↑	cAMP → protein kinase A → HSL and perilipin are phosphorylated
Cortisol, growth hormone, thyroid hormones	Lipolysis ↑ (a permissive effect)	Synthesis of lipase and β-receptors ↑
Insulin	Lipolysis ↓, lipogenesis ↑	Phosphodiesterase is activated (cAMP ↓), HSL is dephosphorylated; LPL, glucose entry and lipogenic enzymes ↑

The **fatty acids** released by lipolysis are bound in the blood to **albumin** (free, "non-esterified" fatty acids) and transported to muscle, heart, liver and kidney. **Glycerol** goes to the liver, where glycerol kinase converts it into glycerol-3-phosphate and then into dihydroxyacetone phosphate, which enters glycolysis (energy) or gluconeogenesis (glucose).

> **REMEMBER**
>
> **The brain and erythrocytes do not use fatty acids as an energy source**: fatty acids bound to albumin hardly cross the blood-brain barrier, and erythrocytes have no mitochondria. In prolonged starvation the brain obtains 2/3 of its energy from **ketone bodies**.

2. β-oxidation of fatty acids

β-oxidation is the oxidation of fatty acids in the mitochondrial matrix with the successive removal of two-carbon fragments (acetyl-CoA) (F. Knoop, 1904). It takes place mainly in the heart, skeletal muscle, liver and kidney.

1. Activation of the fatty acid (in the cytosol, on the outer mitochondrial membrane):

Fatty acid + CoA-SH + ATP → Acyl-CoA + AMP + PP_i (acyl-CoA synthetase; 2 high-energy bonds are used)

2. Transport of the acyl residue into the mitochondria — the carnitine shuttle system. The inner membrane is impermeable to acyl-CoA:

- **carnitine palmitoyltransferase I (CPT-I)** (outer membrane) transfers the acyl group to carnitine → acylcarnitine; this is the **key (regulated) step of β-oxidation**;
- **carnitine-acylcarnitine translocase** carries the acylcarnitine into the matrix;
- **CPT-II** in the matrix transfers the acyl group back to CoA. CPT-I is inhibited by **malonyl-CoA**, the first product of fatty acid synthesis — so synthesis and breakdown do not occur at the same time.

3. The β-oxidation cycle — four reactions:

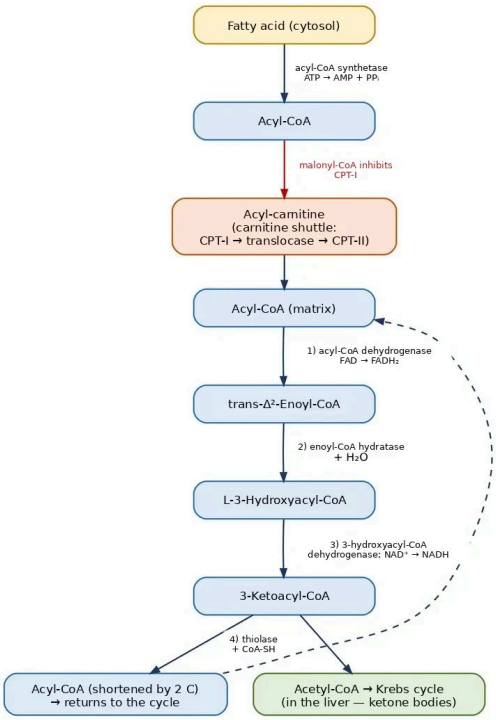


Figure 9.1. *Activation of fatty acids, the carnitine shuttle system and the reactions of β-oxidation (drawn after Severin E.S.)*

1. **Dehydrogenation** — acyl-CoA dehydrogenase (FAD): acyl-CoA → trans-Δ2-enoyl-CoA + **$FADH_2$**;
2. **Hydration** — enoyl-CoA hydratase: → L-3-hydroxyacyl-CoA;
3. **Dehydrogenation** — 3-hydroxyacyl-CoA dehydrogenase (NAD^+): → 3-ketoacyl-CoA + **NADH**;
4. **Thiolytic cleavage** — thiolase (+ CoA-SH): → **acetyl-CoA** + an acyl-CoA shortened by two carbons. The shortened acyl-CoA starts the cycle again. For a Cn fatty acid with an even number of carbons there are (n/2 − 1) cycles and n/2 acetyl-CoA are formed.

Table 9.2. *Energy yield of the complete oxidation of palmitic acid (C16)*

Stage	Products	ATP (current calculation)
7 cycles of β-oxidation	7 $FADH_2$	7 × 1,5 = 10,5
	7 NADH	7 × 2,5 = 17,5
8 acetyl-CoA → Krebs cycle		8 × 10 = 80
Activation		−2
Total		106 ATP (129–131 by the old calculation)

The high energy value of fats is due to the fact that fatty acids are more reduced than carbohydrates.

The oxidation of **unsaturated fatty acids** needs additional enzymes (isomerase, reductase); each double bond means one $FADH_2$ less. From **odd-chain fatty acids** propionyl-CoA is formed at the end → methylmalonyl-CoA → (B_{12}) succinyl-CoA → it can enter gluconeogenesis. **Very long chain** fatty acids (C22 and longer) are first shortened in the **peroxisomes** (here H_2O_2 is formed instead of $FADH_2$).

CLINICAL SIGNIFICANCE

Carnitine or CPT deficiency and **medium-chain acyl-CoA dehydrogenase (MCAD) deficiency**: during starvation or illness fatty acids are not oxidized — **hypoketotic hypoglycemia** (ketone bodies are not formed and there is not enough ATP for gluconeogenesis), muscle weakness, cardiomyopathy and liver damage; one of the causes of sudden infant death syndrome. Treatment: avoiding long fasts, frequent carbohydrate feeding, carnitine preparations.

3. Biosynthesis of fatty acids

Fatty acids are synthesized in the **cytosol**, mainly in the **liver**, **adipose tissue** and the **lactating mammary gland**; the source is **acetyl-CoA** formed from excess carbohydrates. Synthesis increases under the action of **insulin**, after a carbohydrate-rich meal.

1. Export of acetyl-CoA from the mitochondria to the cytosol (the citrate shuttle). Acetyl-CoA does not cross the mitochondrial membrane: it forms citrate with oxaloacetate, citrate leaves for the cytosol and is split again into acetyl-CoA and oxaloacetate by **citrate lyase** (ATP). Oxaloacetate is converted into malate and then, by **malic enzyme**, into pyruvate, and **NADPH** is formed.

2. Carboxylation of acetyl-CoA (the key reaction):

Acetyl-CoA + CO_2 + ATP → **Malonyl-CoA** + ADP + P_i — **acetyl-CoA carboxylase** (coenzyme — **biotin**)

3. Fatty acid synthase — a multienzyme complex (a dimer with 7 active sites in each subunit and an acyl-carrier part — the **pantetheine group** (B_5)). In each cycle 2 carbons are added from malonyl-CoA (CO_2 is released); **2 NADPH** are used for the reduction. In 7 cycles **palmitic acid** (C16:0) is formed.

The overall equation of palmitate synthesis:

8 Acetyl-CoA + 7 ATP + 14 NADPH + 14 H^+ → Palmitate + 8 CoA-SH + 7 ADP + 7 P_i + 14 $NADP^+$ + 6 H_2O

Sources of NADPH: the pentose phosphate pathway (the main one) and malic enzyme.

Elongation and desaturation: palmitate is elongated in the endoplasmic reticulum to stearate and longer; desaturases introduce a double bond at position Δ9 (stearate → oleate). Linoleic and linolenic acids are not synthesized.

Table 9.3. *Regulation of acetyl-CoA carboxylase*

Activated by	Inhibited by
Citrate (allosteric; promotes polymerization)	Palmitoyl-CoA (long-chain acyl-CoA)
Insulin (dephosphorylation; induction of enzyme synthesis)	Glucagon, adrenaline (cAMP → phosphorylation); AMP kinase (low energy)
A carbohydrate-rich, low-fat diet	Starvation, a fatty diet

4. Ketone bodies

Ketone bodies are acetoacetate, β-hydroxybutyrate and acetone. They are synthesized from acetyl-CoA only in the **liver mitochondria** (ketogenesis), are carried by the blood to other tissues and are oxidized there as an energy source.

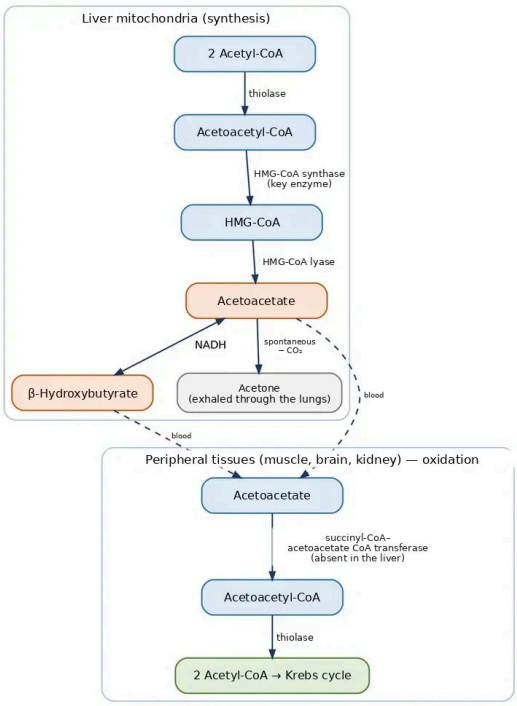


Figure 9.2. *Synthesis of ketone bodies in the liver and their use in peripheral tissues (drawn after Severin E.S.)*

Synthesis: 2 acetyl-CoA → acetoacetyl-CoA (thiolase) → **HMG-CoA** (HMG-CoA synthase — the key enzyme) → acetoacetate + acetyl-CoA (HMG-CoA lyase). Acetoacetate is reduced to β-hydroxybutyrate at the expense of NADH or spontaneously decarboxylated to acetone.

Use: the heart, skeletal muscle, kidney and, in prolonged starvation, the **brain**. β-Hydroxybutyrate is oxidized to acetoacetate; acetoacetate is converted by **succinyl-CoA-acetoacetate-CoA transferase** into acetoacetyl-CoA and then into 2 acetyl-CoA. **The liver has no such transferase** — it does not use ketone bodies, it only “exports” them.

Ketogenesis increases when a lot of acetyl-CoA is formed and oxaloacetate is lacking: starvation, a carbohydrate-free (ketogenic) diet, prolonged physical work, **diabetes mellitus** (insulin deficiency — lipolysis and β-oxidation increase sharply and oxaloacetate is used for gluconeogenesis).

Normally blood ketone bodies are 0.1–0.5 mmol/l and they are not detected in urine by routine methods. **Ketonemia** and **ketonuria** are seen in starvation and diabetes; because ketone bodies are strong acids, when a lot of them accumulate **ketoacidosis** develops (the pH falls); acetone leaves with the breath and gives a “rotten apple” odor from the mouth.

> **CLINICAL SIGNIFICANCE**
> **Diabetic ketoacidosis** is a life-threatening complication of type 1 diabetes mellitus: hyperglycemia (> 13–15 mmol/l), ketone bodies > 3 mmol/l, pH < 7.3, reduced bicarbonate; Kussmaul breathing, dehydration, vomiting, abdominal pain, coma. Treatment — insulin (it stops lipolysis and ketogenesis), fluids and potassium.

5. Metabolism of phospholipids and glycolipids

Synthesis of glycerophospholipids (in the endoplasmic reticulum) — on the basis of phosphatidic acid or diacylglycerol:

- **phosphatidylcholine** and **phosphatidylethanolamine** — through the CDP-choline (CDP-ethanolamine) pathway; phosphatidylcholine is also formed from phosphatidylethanolamine by

three methylations at the expense of S-adenosylmethionine;

- **phosphatidylinositol** — CDP-diacylglycerol + inositol;
- **lipotropic factors** (choline, methionine, vitamin B_{12}, folate, serine) are needed for the synthesis of phosphatidylcholine and for the export of TAG from the liver in VLDL; when they are lacking, **fatty degeneration of the liver** develops. **Sphingolipids** are synthesized on the basis of **sphingosine**, formed from serine and palmitoyl-CoA: sphingosine + fatty acid → ceramide; ceramide + phosphocholine → **sphingomyelin**; ceramide + carbohydrates → **cerebrosides, gangliosides**.

Breakdown: phospholipids by phospholipases (A_1, A_2, C, D); sphingolipids are broken down in the lysosomes by specific hydrolases. Congenital deficiencies of these enzymes are the **sphingolipidoses** (lysosomal storage diseases, topic 10).

6. Laboratory work

Experiment 1: determination of ketone bodies in urine (Legal test)

1. Pour into a test tube 1 ml of urine, 0.5 ml of 10% NaOH and 0.5 ml of freshly prepared 10% sodium nitroprusside solution.
2. Observe the red color that appears.
3. Add 1 ml of concentrated acetic acid. **Result and analysis:** if acetone and acetoacetate are present, a red color forms in the alkaline medium; when acetic acid is added it turns **cherry-red (violet)** — a positive result. Creatinine also gives a red color in alkali, but after acetic acid it turns yellow (which excludes a false positive). β-Hydroxybutyrate is not detected by this test.

Experiment 2: determination of acetoacetate in urine (Gerhardt test)

1. Pour 2 ml of urine into a test tube and add 10% iron (III) chloride solution dropwise (the phosphates precipitate; filter off the precipitate).
2. Add a few more drops of $FeCl_3$ to the filtrate. **Result and analysis:** if acetoacetate is present, a **red-brown (wine) color** appears. On heating, acetoacetate is converted into acetone and the color disappears (whereas the color given by salicylates persists on heating).

Experiment 3: detection of ketonuria with test strips (from a case)

1. Dip the test strip into the urine for 1–2 s and shake off the excess on the edge of the container.
2. After 60 s compare the color with the scale (−, +, ++, +++). **Evaluation:** positive in starvation, vomiting (especially in children — the acetonemic state), a carbohydrate-free diet and decompensated diabetes mellitus. In diabetes ketonuria is seen together with glucosuria.

7. Practical skills

Skill: calculating the ATP formed in the oxidation of a fatty acid

1. From the number of carbons find the number of acetyl-CoA (n/2) and the number of β-oxidation cycles (n/2 − 1).
2. Calculate that each cycle gives 1 $FADH_2$ (1.5 ATP) and 1 NADH (2.5 ATP).
3. Add 10 ATP for each acetyl-CoA in the Krebs cycle.
4. Subtract 2 ATP for the activation.
5. If the fatty acid is unsaturated, subtract 1.5 ATP for each double bond.

Skill: identifying the cause of ketosis

1. Assess the glycemia: hyperglycemia + ketosis — diabetes mellitus; normo- or hypoglycemia + ketosis — starvation, vomiting, a ketogenic diet.
2. Determine the acid-base state (pH, bicarbonate) — is there ketoacidosis?
3. Take the history into account (alcohol, not eating, pregnancy, acetonemic vomiting in children).
4. Explain the biochemical mechanism (the insulin/glucagon ratio, lack of oxaloacetate).
5. State the principle of treatment (insulin, glucose, fluids).

Review questions

1. How is TAG synthesized in adipose tissue? Where does the glycerol-3-phosphate come from?
2. By which enzymes does lipolysis proceed?
3. By which hormones is lipolysis regulated?
4. How are fatty acids transported in the blood?
5. How is glycerol oxidized?
6. How is a fatty acid activated?
7. Explain the carnitine shuttle system. How is it regulated?
8. Write the reactions of β-oxidation.
9. How much ATP is formed in the oxidation of palmitic acid?
10. How are odd-chain fatty acids oxidized?
11. Where and from which substrates does the synthesis of fatty acids proceed?
12. How is acetyl-CoA carboxylase regulated?
13. What are the significance and the sources of NADPH in fatty acid synthesis?
14. How are ketone bodies synthesized and used?
15. Why does ketosis develop in diabetes mellitus and in starvation?

Test questions

1. The key (regulated) enzyme of lipolysis in adipose tissue:
   A) Hormone-sensitive lipase
   B) Lipoprotein lipase
   C) Pancreatic lipase
   D) Phospholipase A_2
   E) Acyl-CoA synthetase

2. The hormone that inhibits lipolysis:
   A) Adrenaline
   B) Glucagon
   C) Insulin
   D) Cortisol
   E) Growth hormone

3. The substance to which fatty acids are bound for transport in the blood:
   A) Hemoglobin
   B) Albumin
   C) Transferrin
   D) Ceruloplasmin
   E) Fibrinogen

4. The regulated (key) step of β-oxidation:
   A) The carnitine palmitoyltransferase I reaction
   B) The enoyl-CoA hydratase reaction
   C) The thiolase reaction
   D) Krebs cycle
   E) The acetyl-CoA carboxylase reaction

5. The substance that inhibits carnitine palmitoyltransferase I:
   A) Acetyl-CoA
   B) Citrate
   C) Carnitine
   D) Malonyl-CoA
   E) Palmitate

6. The coenzyme that is reduced in the first reaction of β-oxidation:
   A) FAD
   B) NAD^+
   C) $NADP^+$
   D) Biotin
   E) Thiamine diphosphate

7. The ATP formed in the complete oxidation of palmitic acid (current calculation):
   A) 12
   B) 38
   C) 106
   D) 80
   E) 30

8. The key enzyme of fatty acid synthesis:
   A) Fatty acid synthase
   B) Citrate lyase
   C) Malic enzyme
   D) Thiolase
   E) Acetyl-CoA carboxylase

9. The reducing equivalent used in fatty acid synthesis:
   A) NADH
   B) $FADH_2$
   C) $FMNH_2$
   D) NADPH
   E) Glutathione

10. The form in which acetyl-CoA leaves the mitochondria for the cytosol:
    A) Acylcarnitine
    B) Malate
    C) Citrate
    D) Pyruvate
    E) Acetoacetate

11. The organ in which ketone bodies are synthesized:
    A) In the brain
    B) In the liver
    C) In skeletal muscle
    D) In the kidney
    E) In adipose tissue

12. Why the liver cannot use ketone bodies:
    A) It has no thiolase
    B) It has few mitochondria
    C) Ketone bodies do not enter the liver
    D) The liver lacks NADH
    E) The liver has no succinyl-CoA-acetoacetate-CoA transferase

13. The key enzyme of ketone body synthesis:
    A) Thiolase
    B) HMG-CoA reductase
    C) HMG-CoA lyase
    D) HMG-CoA synthase
    E) β-Hydroxybutyrate dehydrogenase

14. What the Legal test detects in urine:
    A) Glucose
    B) Acetone and acetoacetate
    C) β-Hydroxybutyrate
    D) Protein
    E) Bilirubin

15. The condition that develops when lipotropic factors are lacking:
    A) Atherosclerosis
    B) Ketoacidosis
    C) Gallstone disease
    D) Steatorrhea
    E) Fatty degeneration of the liver

Clinical cases

Case 1

A 17-year-old adolescent has complained for 3 days of thirst, passing a lot of urine, vomiting and abdominal pain. The breathing is deep and noisy and there is an odor of acetone from the mouth. Blood glucose is 22 mmol/l, pH 7.15, bicarbonate 8 mmol/l; glucose and ketone bodies (+++) were found in the urine.

Questions: What is the diagnosis? Explain the mechanism of the rise in ketone bodies.

Answer: Ketoacidotic decompensation of type 1 diabetes mellitus. When insulin is lacking, glucagon and the counter-insular hormones predominate: hormone-sensitive lipase in adipose tissue is activated and lipolysis increases sharply, so a lot of fatty acids reach the liver; because malonyl-CoA is low, CPT-I is not inhibited, β-oxidation increases and a lot of acetyl-CoA is formed. Since oxaloacetate is used for gluconeogenesis, acetyl-CoA cannot enter the Krebs cycle and is converted into ketone bodies. They are acids — metabolic acidosis (compensated by Kussmaul breathing), and acetone leaves with the breath. Hyperglycemia causes osmotic diuresis and dehydration. Treatment: insulin, fluids, potassium.

Case 2

A 2-year-old child with influenza, after not eating for 18 hours, developed lethargy, vomiting and seizures. Blood glucose is 1.9 mmol/l, ketone bodies were not detected, and ammonia and liver enzymes are increased. Medium-chain acylcarnitines are increased in the blood.

Questions: Which enzyme deficiency is suspected? Why does the hypoglycemia occur without ketosis?

Answer: Medium-chain acyl-CoA dehydrogenase (MCAD) deficiency. In starvation energy is obtained mainly from the oxidation of fatty acids; because the enzyme is lacking, β-oxidation stops — acetyl-CoA and ketone bodies are not formed (hypoketosis), and there is not enough ATP or acetyl-CoA (an activator of pyruvate carboxylase) for gluconeogenesis — hypoglycemia. Medium-chain acyl-CoAs bind to carnitine and accumulate in the blood (they are detected by tandem mass spectrometry). Treatment: avoiding long fasts, giving glucose during illness.

Case 3

A 30-year-old woman on a diet with almost all carbohydrates excluded (a “keto diet”) has an odor of acetone from the mouth and ketone bodies (++) in the urine. Blood glucose is 4.2 mmol/l and the pH is normal.

Questions: Is this condition dangerous? Explain its mechanism and how it differs from diabetic ketoacidosis.

Answer: Physiological (dietary) ketosis. When carbohydrates are low, insulin is low and glucagon relatively high: lipolysis and β-oxidation increase, the liver forms a lot of ketone bodies and they serve as an energy source for muscle and brain. But insulin is not absent altogether — it keeps lipolysis in check, so ketone bodies stay at a moderate level (0.5–3 mmol/l), the buffer systems neutralize them, the pH is normal and glucose is normal. In diabetic ketoacidosis, by contrast, there is almost no insulin: ketones exceed 3 mmol/l and acidosis and hyperglycemia develop.

Quick review

Table 9.4. *Key concepts of topic 9*

Concept	Brief definition
Lipogenesis	Glycerol-3-phosphate + 3 acyl-CoA → TAG; in adipose tissue glycerol-3-phosphate comes only from glucose
Lipolysis	ATGL, hormone-sensitive lipase (key), MAG lipase
Regulation	Adrenaline, glucagon — lipolysis ↑ (cAMP); insulin — lipolysis ↓
Transport of FA	With albumin; the brain and erythrocytes do not use them

Concept	Brief definition
Carnitine shuttle system	CPT-I (key; inhibited by malonyl-CoA), translocase, CPT-II
β-oxidation	Dehydrogenation (FAD) → hydration → dehydrogenation (NAD^+) → thiolase
Palmitate	7 cycles, 8 acetyl-CoA, 106 ATP
FA synthesis	Cytosol; acetyl-CoA carboxylase (biotin) — key; FA synthase; NADPH
Citrate shuttle	Acetyl-CoA leaves for the cytosol in the form of citrate
Ketone bodies	Acetoacetate, β-hydroxybutyrate, acetone; synthesis in the liver (HMG-CoA synthase)
Use	Muscle, heart, kidney and, in starvation, the brain; the liver has no transferase
Ketosis	Starvation, diabetes, a carbohydrate-free diet; ketoacidosis > 3 mmol/l
Lipotropic factors	Choline, methionine, B_{12}, folate — prevent fatty degeneration of the liver

TOPIC 10

Metabolism of complex lipids. Cholesterol metabolism. Regulation of lipid metabolism

Learning objectives

To teach the student sterols and steroids, the significance and biological functions of cholesterol in the body, the biosynthesis of cholesterol and its regulation, its transport in the blood, the significance of LDL receptors and lecithin-cholesterol acyltransferase, the routes of removal of cholesterol from the body, the biochemical basis of hypercholesterolemia, atherosclerosis and gallstone disease, and the dyslipoproteinemias and sphingolipidoses; to master the determination of serum cholesterol by the enzymatic method.

By the end of the class the student should know:

- the structure, sources and functions of cholesterol;
- the stages of cholesterol biosynthesis and the regulation of HMG-CoA reductase;
- the transport of cholesterol in the blood, LDL receptors and reverse transport;
- the routes of removal of cholesterol from the body;
- the biochemical basis and risk factors of atherosclerosis;
- the biochemical basis of gallstone disease;
- the types of dyslipoproteinemia and the sphingolipidoses;
- the age-related normal values of the lipid indices.

The student should be able to:

- determine total cholesterol in serum by the enzymatic method;
- calculate the atherogenic index and evaluate the lipid profile.

1. Sterols and steroids. The significance of cholesterol

Steroids are substances based on the cyclopentanoperhydrophenanthrene ring. **Sterols** are steroid alcohols with a hydroxyl group at position 3; the main sterol of animals is **cholesterol** ($C_{27}H_{45}OH$). The body contains ≈ 140 g of cholesterol; ≈ 90% of it is in the cells (in membranes) and the rest in the blood.

The biological functions of cholesterol:

- **structural** — a component of membranes, regulating their fluidity and permeability; abundant in the myelin sheath;
- a basis for the synthesis of **bile acids** (liver);
- synthesis of the **steroid hormones**: cortisol, aldosterone (adrenal cortex), testosterone, estrogens, progesterone (gonads, placenta);
- synthesis of **vitamin D_3** — in the skin from 7-dehydrocholesterol under ultraviolet light. **Sources:** food (0.3–0.5 g per day: egg yolk, meat, liver, butter) and endogenous synthesis (≈ 0.8–1 g per day; ≈ 50% in the liver, the rest in the intestine, skin and adrenal cortex). Plant foods contain no cholesterol (phytosterols reduce the absorption of cholesterol in the intestine).

Age-related features: in childhood a lot of cholesterol is needed for myelination and growth; breast milk is rich in cholesterol. Blood total cholesterol rises with age: in children 2.5–4.5 mmol/l, in adults

the desirable value is < 5.2 mmol/l; in women, after the menopause, LDL rises as estrogens fall.

2. Biosynthesis of cholesterol and its regulation

Cholesterol is synthesized in all nucleated cells, mainly in the **liver**, in the cytosol and endoplasmic reticulum, from **acetyl-CoA**. One molecule of cholesterol costs 18 acetyl-CoA, 36 ATP and 16 NADPH.

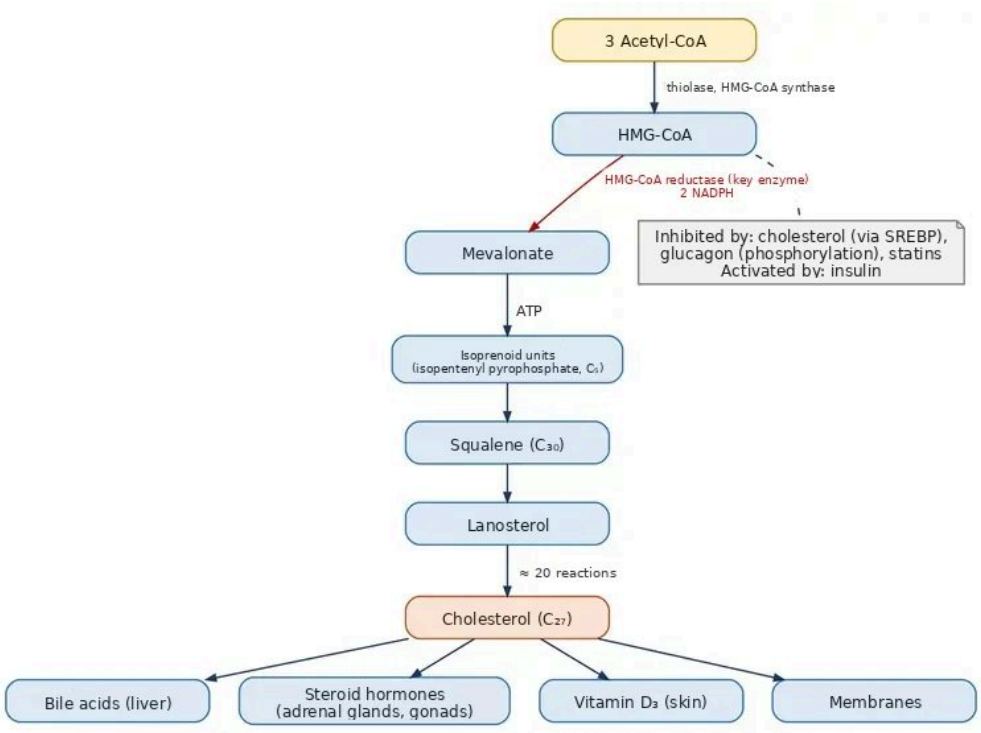


Figure 10.1. *The main stages of cholesterol biosynthesis and its regulation (drawn after Severin E.S.)*

1. 3 acetyl-CoA → **HMG-CoA** (β-hydroxy-β-methylglutaryl-CoA) — in the cytosol (unlike ketone body synthesis, which takes place in the mitochondria).
2. HMG-CoA + 2 NADPH → **mevalonate** — **HMG-CoA reductase** (the key enzyme, in the membrane of the endoplasmic reticulum).
3. Mevalonate → activated isoprenoid units (C5, at the expense of ATP) → geranyl and farnesyl pyrophosphate → **squalene** (C30).
4. Squalene is cyclized to **lanosterol** and then, in ≈ 20 reactions, converted into **cholesterol**.

Table 10.1. *Regulation of HMG-CoA reductase*

Mechanism	Effect
Intracellular cholesterol (through the transcription factor SREBP)	When it is high — the synthesis of the enzyme and of LDL receptors falls and the enzyme is degraded faster
Insulin	Activates it by dephosphorylation; increases its synthesis
Glucagon, AMP kinase (low energy)	Inactivate it by phosphorylation
Thyroid hormones	Increase the synthesis of the enzyme (but the removal of cholesterol and the number of LDL receptors increase more — hence hypercholesterolemia in hypothyroidism)

Mechanism	Effect
Statins (atorvastatin, rosuvastatin)	Competitive inhibitors; cholesterol in the cell falls and LDL receptors increase — LDL in the blood decreases

3. Transport and removal of cholesterol

Cholesterol is transported in the blood in lipoproteins (topic 8); ≈ 2/3 of total cholesterol is in the form of **cholesterol esters**.

Transport to the tissues. The VLDL synthesized in the liver are converted in the blood by LPL into IDL and then into **LDL** — ≈ 70% of blood cholesterol is in LDL. LDL enter cells by endocytosis through **LDL receptors** (which recognize apo B-100 and apo E) (M. Brown and J. Goldstein, Nobel Prize 1985).

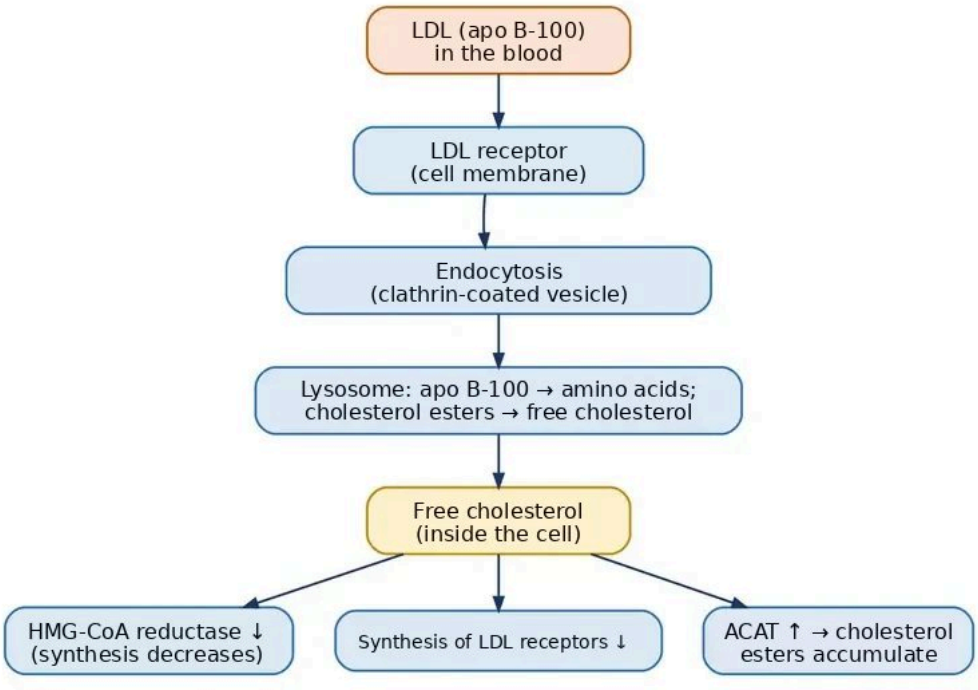


Figure 10.2. *Entry of cholesterol into the cell through the LDL receptor and regulation of the intracellular cholesterol level (drawn after Severin E.S.)*

When free cholesterol increases inside the cell: the synthesis of HMG-CoA reductase and of LDL receptors falls, and **ACAT** (acyl-CoA:cholesterol acyltransferase) is activated, so the excess cholesterol is stored as esters.

Reverse transport. HDL (apo A-I) are formed in the liver and intestine as “flat discs”; they take free cholesterol from peripheral cells through the ABCA1 transporter. **Lecithin-cholesterol acyltransferase (LCAT)** (activated by apo A-I) transfers a fatty acid from phosphatidylcholine to cholesterol and forms cholesterol esters; these move into the core of the HDL and the particle becomes spherical. HDL deliver cholesterol to the **liver** (SR-B1 receptors) or pass the cholesterol esters to other lipoproteins. For this reason HDL are **antiatherogenic** lipoproteins.

Removal. Cholesterol is not broken down to CO_2 and H_2O in the body. It:

- is converted in the liver into **bile acids** (≈ 0.5 g per day; the key enzyme is 7α-hydroxylase) and leaves with the bile;
- passes unchanged with the bile into the intestine; part of it is reabsorbed, and the rest is converted by bacteria into **coprostanol** and excreted with the feces;

- a small amount leaves with sebum, and the metabolites of steroid hormones with the urine.

4. Hypercholesterolemia and atherosclerosis

Atherosclerosis is a chronic disease in which lipids (mainly cholesterol and its esters) accumulate in the intima of large and medium arteries and fibrous plaques form. Its complications — ischemic heart disease, myocardial infarction, stroke — are among the main causes of death in the world.

The biochemical stages of atherogenesis:

1. Damage to the endothelium (hypercholesterolemia, arterial hypertension, smoking, hyperglycemia, homocysteine).
2. LDL enter the intima and are **oxidized** or glycated there (modified LDL).
3. Monocytes move into the intima and become macrophages; they take up modified LDL without limit through **scavenger receptors** (these receptors are not regulated by cholesterol) — **foam cells** are formed.
4. Under the action of inflammatory cytokines smooth muscle cells proliferate and collagen accumulates — a plaque with a fibrous cap forms; rupture of the plaque causes thrombosis. **Risk factors:** raised LDL cholesterol, low HDL cholesterol, high lipoprotein(a), arterial hypertension, smoking, diabetes mellitus, obesity, physical inactivity, heredity, age and male sex.

The atherogenic index (A.N. Klimov): AI = (total CH − HDL-CH) / HDL-CH; normal < 3.0; > 4 — high risk.

> **CLINICAL SIGNIFICANCE**
> **Familial hypercholesterolemia** (type IIa hyperlipoproteinemia) — a mutation of the LDL receptor gene (autosomal dominant). In heterozygotes (≈ 1:250) total cholesterol is 8–12 mmol/l and ischemic heart disease appears at 30–50 years; in homozygotes (≈ 1:1 000 000) cholesterol is 15–25 mmol/l, with tendon xanthomas and myocardial infarction in childhood. LDL do not enter the cells, and the cell, sensing cholesterol as “low”, does not reduce its synthesis. Treatment: statins, ezetimibe (reduces absorption in the intestine), PCSK9 inhibitors and, in severe cases, LDL apheresis.

Table 10.2. *The Fredrickson classification of the dyslipoproteinemias (main types)*

Type	Increased lipoproteins	In the blood	Cause (primary)	Atherogenicity
I	Chylomicrons	TG ↑↑↑	Deficiency of LPL or apo C-II	Low; risk of pancreatitis
IIa	LDL	CH ↑↑	Defect of the LDL receptor (familial hypercholesterolemia)	Very high
IIb	LDL + VLDL	CH ↑↑, TG ↑	Mixed (combined)	High
III	Remnant CM, IDL	CH ↑, TG ↑	Defect of apo E (E2/E2)	High
IV	VLDL	TG ↑↑	Familial hypertriglyceridemia; often secondary (diabetes, obesity, alcohol)	Moderate
V	CM + VLDL	TG ↑↑↑	Mixed	Moderate; risk of pancreatitis

Secondary dyslipoproteinemias: diabetes mellitus, hypothyroidism, nephrotic syndrome, cholestasis, obesity, alcoholism, certain drugs.

5. Gallstone disease

In bile, cholesterol is kept dissolved by forming micelles with bile acids and phosphatidylcholine. When their ratio is disturbed (too much cholesterol or too few bile acids and phospholipids), the bile becomes **supersaturated with cholesterol ("lithogenic")**; cholesterol crystals precipitate in the gallbladder and **cholesterol stones** form (≈ 80% of all stones). Risk factors: female sex, pregnancy, estrogens, obesity, rapid weight loss, a high-calorie diet, reduced motility of the gallbladder, diseases of the ileum (bile acids are not reabsorbed).

The biochemical basis of treatment: ursodeoxycholic (and chenodeoxycholic) acid preparations — they increase the bile acid pool, improve the solubility of cholesterol in the bile and reduce cholesterol synthesis in the liver (they dissolve small cholesterol stones); surgery (cholecystectomy). **Pigment stones** (calcium bilirubinate) form in hemolytic anemias and infection of the bile ducts.

6. Regulation and disorders of fat metabolism

In the fed state (insulin ↑): the synthesis of fatty acids and TAG from glucose increases (in the liver) and VLDL are formed; in adipose tissue LPL is activated and fat is stored; lipolysis is inhibited.

In starvation (glucagon, adrenaline, cortisol ↑): lipolysis in adipose tissue increases, fatty acids are oxidized in muscle and liver, and ketone bodies are formed in the liver.

Obesity is an increase of adipose tissue resulting from a prolonged excess of energy intake over expenditure (body mass index ≥ 30 kg/m^2). Adipose tissue is an endocrine organ: it produces **leptin** (a satiety signal acting on the hypothalamus), adiponectin and cytokines. In obesity, leptin resistance, insulin resistance, dyslipidemia and arterial hypertension develop — the **metabolic syndrome** (topic 15). **Obesity in children** is usually related to overeating and physical inactivity; rarely it is caused by a defect of the leptin gene or of its receptor.

Sphingolipidoses are congenital deficiencies of the enzymes that break down sphingolipids in the lysosomes; the lipids accumulate mainly in nervous tissue, the liver and the spleen.

Table 10.3. *The main sphingolipidoses*

Disease	Deficient enzyme	Lipid that accumulates	Main features
Tay–Sachs disease	Hexosaminidase A	GM2 ganglioside	Developmental regression, blindness, a "cherry-red spot" on the retina
Gaucher disease	β-Glucocerebrosidase	Glucocerebroside	Hepatosplenomegaly, bone damage, cytopenias
Niemann–Pick disease	Sphingomyelinase	Sphingomyelin	Hepatosplenomegaly, neurological disorders, a "cherry-red spot"
Fabry disease	α-Galactosidase A	Globotriaosylceramide	Painful neuropathy, kidney and heart damage (X-linked)
Krabbe disease	Galactocerebrosidase	Galactocerebroside	Demyelination

In some sphingolipidoses (Gaucher, Fabry) **enzyme replacement therapy** is used.

7. Laboratory work

Experiment 1: determination of total cholesterol in serum by the enzymatic method (on a biochemical analyzer)

Principle of the method (CHOD-PAP):

1. **Cholesterol esterase**: cholesterol esters + H_2O → cholesterol + fatty acids;
2. **Cholesterol oxidase**: cholesterol + O_2 → cholest-4-en-3-one + H_2O_2;
3. **Peroxidase**: $2H_2O_2$ + 4-aminoantipyrine + phenol → pink quinoneimine + $4H_2O$. The intensity of the color is proportional to the amount of total cholesterol (500 nm).

1. Pour 1 ml of working reagent into each of three test tubes.
2. Add 10 µl of serum to tube 1, 10 µl of standard (5.17 mmol/l) to tube 2 and 10 µl of water to tube 3.
3. Mix and keep at 37 °C for 10 minutes.
4. Measure the optical density against the control at 500 nm (automatically on the analyzer). **Calculation:** C (mmol/l) = (Esample / Estandard) × 5.17

Evaluation: < 5.2 mmol/l — desirable; 5.2–6.2 — borderline high; ≥ 6.2 — high. The sample is taken after 12–14 hours of fasting.

Experiment 2: qualitative reaction for cholesterol (Liebermann–Burchard reaction)

1. Pour 1 ml of a solution of cholesterol in chloroform into a dry test tube.
2. Add 10 drops of acetic anhydride and 1–2 drops of concentrated sulfuric acid (take care!).
3. Observe the change of color. **Result and analysis:** first a pink, then a blue and finally a **green** color appears — cholesterol is dehydrated by sulfuric acid and forms colored compounds. This reaction used to be the basis of the quantitative determination of cholesterol (the Ilk method).

Experiment 3: interpreting the results of the lipid profile (from a case)

Using the results given by the teacher: compare total cholesterol, triglycerides, LDL and HDL cholesterol with the normal values; calculate the atherogenic index; determine the type of dyslipoproteinemia by the Fredrickson classification. If LDL-CH was not measured directly, calculate it with the Friedewald formula (when TG < 4.5 mmol/l): LDL-CH = total CH − HDL-CH − TG / 2.2.

8. Practical skills

Skill: assessing the risk of atherosclerosis from the lipid profile

1. Make sure the blood is taken after 12–14 hours of fasting.
2. Compare total CH, TG, HDL-CH and LDL-CH with the normal values.
3. Calculate the atherogenic index.
4. Assess them together with the other risk factors (age, sex, smoking, hypertension, diabetes, heredity).
5. Determine the type of dyslipidemia and recommend excluding secondary causes (hypothyroidism, diabetes, nephrotic syndrome).

Skill: explaining the mechanism of action of statins

1. Show that HMG-CoA reductase is the key enzyme of cholesterol synthesis.

2. State that statins are its competitive inhibitors.
3. Explain that when cholesterol falls in the cell, SREBP is activated and the synthesis of LDL receptors increases.
4. Show that as a result the uptake of LDL from the blood increases and LDL-CH falls.
5. Name the adverse effects (myopathy — a rise in CK, a rise in transaminases) and the monitoring tests.

Review questions

1. What are the structure and the biological functions of cholesterol?
2. What are the exogenous and endogenous sources of cholesterol?
3. List the main stages of cholesterol biosynthesis.
4. How is HMG-CoA reductase regulated?
5. How do statins act?
6. How is cholesterol transported in the blood?
7. What is the significance of LDL receptors?
8. Which reaction does LCAT catalyze and what is its significance?
9. What is reverse cholesterol transport?
10. How is cholesterol removed from the body?
11. Explain the biochemical mechanism of atherosclerosis.
12. How is the atherogenic index calculated?
13. What is the biochemical basis of familial hypercholesterolemia?
14. How does gallstone disease arise and how is it treated?
15. What are the sphingolipidoses? Give examples.

Test questions

1. The key enzyme of cholesterol synthesis:
   A) HMG-CoA synthase
   B) HMG-CoA lyase
   C) Squalene synthase
   D) Acetyl-CoA carboxylase
   E) HMG-CoA reductase

2. The substances for whose synthesis cholesterol is not a basis:
   A) Bile acids
   B) Vitamin D_3
   C) Prostaglandins
   D) Cortisol
   E) Testosterone

3. The mechanism of action of statins:
   A) They reduce the absorption of cholesterol in the intestine
   B) They bind bile acids
   C) They competitively inhibit HMG-CoA reductase
   D) They activate LPL
   E) They inhibit LCAT

4. The lipoproteins that carry the main part of blood cholesterol:
    A) Chylomicrons
    B) LDL
    C) VLDL
    D) HDL
    E) IDL

5. The apoprotein recognized by LDL receptors:
    A) Apo A-I
    B) Apo C-II
    C) Apo B-48
    D) Apo B-100
    E) Apo A-II

6. The apoprotein that activates lecithin-cholesterol acyltransferase (LCAT):
    A) Apo A-I
    B) Apo B-100
    C) Apo C-II
    D) Apo E
    E) Apo B-48

7. The main route of removal of cholesterol from the body:
    A) Conversion into bile acids and removal with the bile
    B) Oxidation to CO_2 and H_2O
    C) Excretion unchanged in the urine
    D) Removal with sweat
    E) Removal through the lungs

8. How macrophages turn into foam cells:
    A) By taking up chylomicrons
    B) By taking up modified LDL without limit through scavenger receptors
    C) By taking up HDL
    D) Because of an increase in LDL receptors
    E) By taking up glucose

9. The antiatherogenic lipoproteins:
    A) LDL
    B) HDL
    C) VLDL
    D) IDL
    E) Lipoprotein(a)

10. The cause of familial hypercholesterolemia:
    A) LPL deficiency
    B) Apo C-II deficiency
    C) LCAT deficiency
    D) A mutation of the LDL receptor gene
    E) Apo A-I deficiency

11. The normal value of the atherogenic index:
    A) > 5
    B) 4–6
    C) More than 1.7
    D) Less than 5.2
    E) < 3,0

12. The drug used to dissolve cholesterol stones:
    A) Ursodeoxycholic acid
    B) Statins
    C) Pancreatin
    D) Fibrates
    E) Heparin

13. The enzyme that is deficient in Gaucher disease:
    A) Hexosaminidase A
    B) Sphingomyelinase
    C) α-Galactosidase A
    D) β-Glucocerebrosidase
    E) Galactocerebrosidase

14. The enzyme that forms H_2O_2 in the enzymatic determination of cholesterol:
    A) Cholesterol esterase
    B) Peroxidase
    C) Cholesterol oxidase
    D) Lipase
    E) Catalase

15. The main cause of hypercholesterolemia in hypothyroidism:
    A) Increased cholesterol synthesis
    B) Increased absorption in the intestine
    C) Increased LCAT activity
    D) Increased synthesis of bile acids
    E) A fall in the number of LDL receptors and reduced removal of cholesterol

Clinical cases

Case 1

A 12-year-old child was found to have hard nodules over the Achilles tendons (tendon xanthomas) and yellow patches on the eyelids (xanthelasmas). Total cholesterol is 16 mmol/l and triglycerides are normal. His father died of myocardial infarction at 38, and his mother’s cholesterol is 9 mmol/l.

Questions: What is the diagnosis? Explain its biochemical mechanism. Why may the effect of statins be limited?

Answer: Homozygous familial hypercholesterolemia (type IIa) — a mutation in both alleles of the LDL receptor gene (both parents are heterozygotes). LDL do not enter cells and accumulate in the blood; the cells sense a “shortage” of cholesterol and do not reduce HMG-CoA reductase and synthesis. The excess LDL is taken up by macrophages through scavenger receptors — xanthomas and early atherosclerosis. Statins act mainly by increasing the number of LDL receptors; if the

receptors do not work at all, the effect is small — LDL apheresis, PCSK9 inhibitors and other methods are needed.

Case 2

A 45-year-old obese woman (BMI 34 kg/m^2) complains of attacks of severe pain under the right costal margin after fatty food. Ultrasound revealed small radiolucent stones in the gallbladder.

Questions: What is the likely chemical composition of the stones? Explain the mechanism of their formation.

Answer: Cholesterol stones (radiolucent). Obesity, female sex and estrogens increase cholesterol synthesis in the liver and its secretion into the bile, while the synthesis of bile acids falls relatively; the bile becomes supersaturated with cholesterol (lithogenic). Reduced motility of the gallbladder helps crystals to form. Small cholesterol stones can be dissolved with ursodeoxycholic acid; with frequent attacks — cholecystectomy.

Case 3

A 55-year-old man smokes and his blood pressure is 160/95 mm Hg. Tests: total cholesterol 6.8 mmol/l, HDL-CH 0.9 mmol/l, triglycerides 2.2 mmol/l.

Questions: Calculate the atherogenic index and LDL-CH. Assess the risk.

Answer: AI = (6.8 − 0.9) / 0.9 ≈ 6.6 (normal < 3.0). Friedewald formula: LDL-CH = 6.8 − 0.9 − 2.2 / 2.2 = 6.8 − 0.9 − 1.0 = 4.9 mmol/l (desirable < 3.0). HDL is low, LDL and TG are high — atherogenic dyslipidemia. Together with smoking and arterial hypertension the risk of cardiovascular disease is very high. Recommendations: stopping smoking, diet, physical activity, statins, control of blood pressure.

Quick review

Table 10.4. *Key concepts of topic 10*

Concept	Brief definition
Functions of cholesterol	Membranes, bile acids, steroid hormones, vitamin D_3
Synthesis	Acetyl-CoA → HMG-CoA → mevalonate → squalene → lanosterol → cholesterol
Key enzyme	HMG-CoA reductase; cholesterol, glucagon, statins ↓; insulin ↑
LDL	≈ 70% of blood cholesterol; into the cell through the LDL receptor (apo B-100)
HDL	Reverse transport; LCAT (apo A-I) forms cholesterol esters
Removal	Bile acids and unchanged cholesterol with the bile; coprostanol
Atherosclerosis	Oxidized LDL → macrophages (scavenger receptors) → foam cells
AI	(total CH − HDL-CH) / HDL-CH; normal < 3.0
Friedewald	LDL-CH = CH − HDL-CH − TG/2.2
Type IIa	Mutation of the LDL receptor; xanthomas, early atherosclerosis
Gallstones	Cholesterol high, bile acids low; ursodeoxycholic acid
Sphingolipidoses	Tay–Sachs, Gaucher, Niemann–Pick, Fabry, Krabbe — lysosomal enzymes
Enzymatic method	Cholesterol esterase → cholesterol oxidase (H_2O_2) → peroxidase

TOPIC 11

Digestion of proteins, absorption of amino acids and their distribution among tissues

Learning objectives

To teach the student the significance of proteins in the body, nitrogen balance and the biological value of proteins, the dynamic state of body proteins, the digestion of proteins in the gastrointestinal tract and its regulation, normal and pathological gastric juice, the absorption of amino acids and their distribution to the tissues, the features of protein digestion in children, and the putrefaction of proteins in the intestine and the detoxification of its products; to master the determination of the acidity of gastric juice.

By the end of the class the student should know:

- the functions of proteins and the daily requirement;
- the types of nitrogen balance and their causes;
- the biological value of proteins and protein deficiency;
- the composition of gastric juice, the formation and functions of HCl;
- the proteolytic enzymes, their activation and specificity;
- the mechanisms of amino acid absorption;
- the features of protein digestion in children and food allergy;
- the products of putrefaction and their detoxification in the liver.

The student should be able to:

- determine the total acidity and free and bound HCl of gastric juice by titration;
- determine indican in urine and evaluate the result;
- calculate the nitrogen balance.

1. The significance of proteins. Nitrogen balance

Proteins carry out many functions in the body: **plastic** (the structure of cells and tissues), **catalytic** (enzymes), **transport** (hemoglobin, albumin, transferrin), **protective** (immunoglobulins, clotting factors), **regulatory** (peptide hormones), **contractile** (actin, myosin), **energy** (1 g → 4 kcal; in starvation). Proteins are the source of **nitrogen** and of the **essential amino acids** for the body.

Nitrogen balance is the difference between the nitrogen taken in with food and the nitrogen excreted from the body (mainly as urea). 1 g of nitrogen corresponds to ≈ 6.25 g of protein.

Table 11.1. *Types of nitrogen balance*

Type	Essence	When it is seen
Nitrogen equilibrium	Intake = output	In healthy adults on an adequate diet
Positive nitrogen balance	Intake > output (nitrogen is retained)	Growth in children, pregnancy, recovery after illness, the action of anabolic hormones
Negative nitrogen balance	Intake < output	Starvation, protein deficiency, severe illness, burns, injuries, sepsis, tumors, diabetes mellitus, excess cortisol, old age

The daily requirement for protein in adults is 0.8–1.0 g/kg (≈ 70–100 g), at least half of it animal protein; in children 1.5–4 g/kg (most in newborns), and more in pregnant and breastfeeding women, in athletes and during recovery. The **protein minimum** (the smallest amount needed to maintain nitrogen equilibrium) is ≈ 30–50 g.

The biological value of proteins depends on their content of essential amino acids and on their digestibility: egg and milk proteins are high; plant proteins are lower (the limiting amino acids: wheat — lysine, legumes — methionine, maize — tryptophan and lysine). Eating different plant proteins together (for example, grain + legumes) increases the biological value.

> **CLINICAL SIGNIFICANCE**
> **Protein-energy malnutrition** is one of the main causes of child mortality in developing countries. **Kwashiorkor** — energy in the diet is relatively adequate but protein is sharply reduced: edema because of hypoalbuminemia, fatty degeneration of the liver, changes of the skin and hair, reduced immunity. **Marasmus** — both protein and energy are lacking: severe wasting, loss of muscle and adipose tissue, no edema.

The dynamic state of body proteins. The body's proteins are constantly broken down and resynthesized: ≈ 300–400 g of protein is renewed per day (more than the amount that comes with food) — ≈ 75% of the amino acids released by breakdown are reused. The half-life of proteins differs: pancreatic enzymes and insulin — minutes to hours; liver enzymes — hours to days; hemoglobin — 120 days (the life of the erythrocyte); collagen — months to years. Intracellular proteins are broken down in the **lysosomes** (cathepsins) and in the **proteasomes** (the ubiquitin-dependent pathway).

The species and tissue specificity of proteins: the protein composition of every species and tissue is genetically determined; therefore, if foreign proteins enter the body undigested (for example, parenterally), they cause an immune response. Dietary proteins are broken down to amino acids and lose their "species specificity".

2. Digestion of proteins in the stomach

Proteins are not digested in the oral cavity. Gastric juice (1.5–2.5 l per day) contains **hydrochloric acid**, **pepsinogens**, mucus (mucin) and Castle's intrinsic factor.

The formation of HCl — in the **parietal cells** of the fundic glands: by carbonic anhydrase $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$

; H^+ is pumped into the gastric lumen in exchange for K^+ by **H^+,K^+-ATPase** (the "proton pump"), and Cl^- leaves through channels; HCO_3^-

passes into the blood in exchange for Cl^- (the "alkaline tide" after a meal).

The functions of HCl: it denatures and swells proteins (opening the peptide bonds to enzymes); converts pepsinogen into pepsin and creates the optimal pH for pepsin (1.5–2); has a bactericidal action; regulates gastric emptying and the release of secretin in the duodenum; improves the absorption of iron (Fe^{3+}

$\rightarrow Fe^{2+}$).

Pepsin is synthesized in the chief cells as inactive **pepsinogen**; under the action of HCl and by **autocatalysis** a fragment of 42 amino acid residues is removed from the N-terminus and active pepsin is formed. Pepsin is an **endopeptidase**: it cleaves the peptide bonds formed by aromatic amino acids (phenylalanine, tyrosine, tryptophan) and leucine; proteins are broken down into large

polypeptides. **Gastricsin** (pH 3–3.5) is important in the newborn's stomach. Newborns also have **chymosin (rennin)**, the enzyme that curdles milk casein.

Regulation of gastric juice secretion: nervous (the vagus — acetylcholine), humoral (**gastrin** — released by the G cells of the antrum under the action of protein products and distension of the stomach; **histamine** — through H_2 receptors), and inhibitors — somatostatin, secretin, a low pH.

Table 11.2. *Normal acidity of gastric juice (fasting, in titration units)*

Index	Normal (t.u.)
Total acidity	40–60
Free HCl	20–40
Bound HCl	10–20
pH	1,5–2,0

Pathological gastric juice: a hyperacid state (gastritis, peptic ulcer, Zollinger–Ellison syndrome — gastrinoma); **hyposecretion and achylia** (atrophic gastritis, gastric cancer, pernicious anemia — no HCl and no intrinsic factor). Pathological components: **lactic acid** (fermentation by bacteria in the absence of HCl — suspicion of gastric cancer), **blood**, bile.

CLINICAL SIGNIFICANCE

The biochemical basis of treating acid-related diseases: antacids (aluminium and magnesium hydroxides) neutralize HCl; **H_2 blockers** (famotidine) block the action of histamine; **proton pump inhibitors** (omeprazole) irreversibly inhibit H^+,K^+-ATPase — the most powerful antisecretory agent. The main etiological factor in peptic ulcer is Helicobacter pylori (with the help of urease it breaks urea down to ammonia and survives in the acidic medium; the urea breath test is based on this).

3. Digestion of proteins in the intestine

The pancreatic proteases are synthesized as inactive **proenzymes** (this protects the gland tissue from self-digestion; the gland also contains a **trypsin inhibitor** protein). In the duodenum **enteropeptidase** (enterokinase) removes a hexapeptide from trypsinogen and forms **trypsin**; trypsin then activates the remaining trypsinogen by autocatalysis, as well as the other proenzymes.

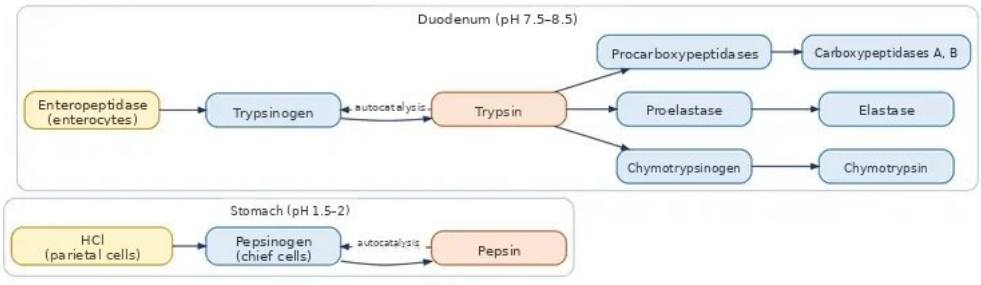


Figure 11.1. *Activation of the proteolytic enzymes (drawn after Severin E.S.)*

Table 11.3. *Specificity of the proteolytic enzymes*

Enzyme	Type	Peptide bonds cleaved
Pepsin	Endopeptidase	On the amino side of aromatic amino acids and leucine
Trypsin	Endopeptidase	On the carboxyl side of lysine and arginine (basic amino acids)

Enzyme	Type	Peptide bonds cleaved
Chymotrypsin	Endopeptidase	On the carboxyl side of aromatic amino acids (Phe, Tyr, Trp)
Elastase	Endopeptidase	Bonds of small neutral amino acids (glycine, alanine, serine)
Carboxypeptidase A and B	Exopeptidase (Zn^{2+})	Removes amino acids from the C-terminus (B — lysine and arginine)
Aminopeptidases (enterocytes)	Exopeptidase	Removes amino acids from the N-terminus
Di- and tripeptidases (enterocytes)	Exopeptidase	Break di- and tripeptides down into amino acids

Endopeptidases break the chain into fragments from inside, while exopeptidases release free amino acids from its ends. The last stage of digestion takes place on the surface of the enterocytes (membrane digestion) and inside them.

CLINICAL SIGNIFICANCE

Acute pancreatitis — premature activation of the pancreatic enzymes inside the gland (diseases of the bile ducts, alcohol, trauma): trypsin activates the other proenzymes, phospholipase A_2 and elastase, and the gland tissue digests itself (autolysis, necrosis) and the blood vessels are damaged. Amylase and lipase rise in the blood. Principles of treatment: fasting (to reduce secretion), protease inhibitors (aprotinin — historically), pain relief, infusion therapy.

4. Absorption and distribution of amino acids

Free amino acids and di- and tripeptides are absorbed in the small intestine:

- free L-amino acids — by **symport with Na^+** (secondary active transport) — with separate carriers for neutral, basic and acidic amino acids and imino acids;
- di- and tripeptides — by **symport with H^+** (PepT1), and they are broken down inside the enterocyte;
- **the γ-glutamyl cycle (Meister cycle)** — entry of amino acids into the cell with the participation of glutathione (renal tubules, intestine, brain). The absorbed amino acids reach the liver through the **portal vein**; the liver retains part of them (synthesis of plasma proteins, gluconeogenesis, urea) and passes the rest into the general circulation. **The amino acid pool** is the free amino acids in the blood and cells (≈ 35 g). Its sources: dietary proteins, breakdown of body proteins, synthesis of nonessential amino acids. Its uses: protein synthesis, synthesis of nitrogenous compounds (heme, creatine, nucleotides, biogenic amines), production of energy and glucose.

Distribution among tissues: in starvation muscle proteins are a source of amino acids (they are released mainly as alanine and glutamine); the liver is the center of amino acid metabolism; the kidney uses glutamine and excretes ammonium ions.

Congenital absorption disorders: Hartnup disease (a defect of the carrier of neutral amino acids, especially tryptophan — pellagra-like signs), **cystinuria** (a defect of the carrier of cystine, ornithine, lysine and arginine — cystine stones in the kidney).

5. Features of protein digestion in children

- The acidity of gastric juice is low (pH 3–5) and pepsin activity is low; **gastricsin** and **chymosin** play an important role.

- In the first months of life the permeability of the intestinal mucosa is high — some proteins (**IgA** and IgG of the colostrum) are absorbed undigested by **pinocytosis** (passive immunity).
- For this reason foreign dietary proteins (cow's milk proteins, egg, wheat gluten) may also be absorbed partly undigested and cause **food allergy**. This is why complementary food should be introduced at the right time and gradually.
- **Celiac disease** — an immune reaction to the gluten (gliadin) of wheat, barley and rye: atrophy of the villi of the small intestine, malabsorption; treatment — a gluten-free diet.

6. Putrefaction of proteins in the intestine and detoxification of its products

Amino acids that have not been absorbed undergo **putrefaction** in the large intestine under the action of bacteria (decarboxylation, deamination, breakdown of side chains) and toxic substances are formed.

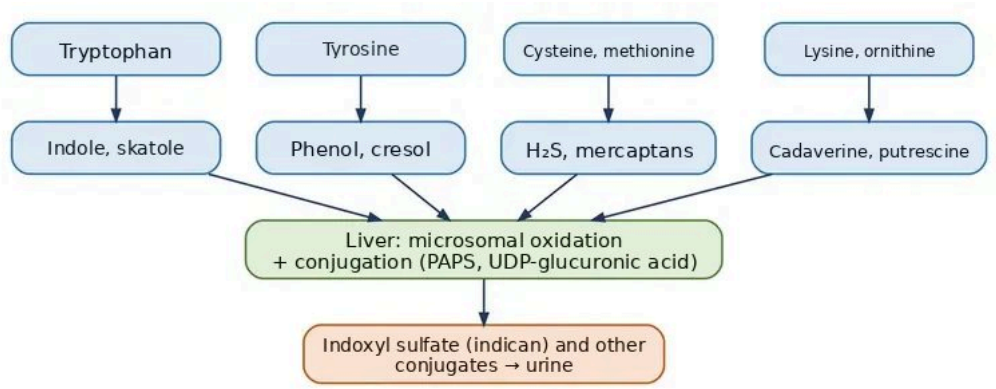


Figure 11.2. *The products of putrefaction and their detoxification in the liver (drawn after Severin E.S.)*

Table 11.4. *The main products of putrefaction*

Amino acid	Product of putrefaction
Tryptophan	Indole, skatole (the odor of feces)
Tyrosine	Phenol, cresol
Cysteine, methionine	H_2S, methyl mercaptan
Lysine	Cadaverine
Ornithine, arginine	Putrescine
All amino acids	Ammonia

The products of putrefaction reach the **liver** through the portal vein and are detoxified there (topic 26): first by **microsomal oxidation** (for example, indole → indoxyl, skatole → skatoxyl), then by **conjugation** — with active sulfate (PAPS) or with **UDP-glucuronic acid**. The soluble conjugates formed are excreted in the urine. For example, the potassium salt of indoxyl sulfate is **indican**.

The amount of indican in the urine (normally 10–20 mg per day) shows the intensity of putrefaction: it rises in intestinal obstruction, constipation, lack of HCl in the stomach (achylia), bacterial overgrowth in the large intestine and a diet rich in protein. In liver failure the products of putrefaction and ammonia are not detoxified — **hepatic encephalopathy** (topic 25).

7. Laboratory work

Experiment 1: determination of the acidity of gastric juice by titration (Töpfer method)

Principle of the method: the acids of gastric juice are titrated with 0.1 N NaOH in the presence of two indicators. **Dimethylaminoazobenzene** (which changes from red to yellow at pH 2.9–4.0) shows free HCl; **phenolphthalein** (pink at pH 8.2–10) shows the total acidity.

1. Measure 5 ml of filtered gastric juice into a flask and add 1–2 drops of dimethylaminoazobenzene and 1–2 drops of phenolphthalein (a red color means free HCl is present).
2. Titrate from a burette with 0.1 N NaOH until the red color turns **orange** — mark 1 (free HCl).
3. Continue the titration until a **yellow** color appears — mark 2.
4. Titrate until a **pink** color appears — mark 3 (total acidity). **Calculation:** the result is expressed in titration units (t.u.) — the number of ml of 0.1 N NaOH needed to neutralize 100 ml of gastric juice: X = NaOH used (ml) × 20 (for 5 ml of juice).

- Free HCl = NaOH used up to mark 1 × 20;
- Total acidity = NaOH used up to mark 3 × 20;
- Total HCl (free + bound) = the mean of marks 2 and 3 × 20; bound HCl = total HCl − free HCl. **Evaluation:** total acidity 40–60, free HCl 20–40, bound HCl 10–20 t.u. If there is no free HCl — **achlorhydria**; if there is neither HCl nor pepsin — **achylia**.

Experiment 2: determination of lactic acid in gastric juice (Uffelmann reaction)

1. Add to a test tube 1 ml of 1% phenol solution and 1 drop of $FeCl_3$ (a violet color — iron phenolate).
2. Add gastric juice dropwise. **Result and analysis:** if lactic acid is present, the violet color turns **yellow-green** (iron lactate). In a healthy person there is no lactic acid in the gastric juice; it appears as a result of bacterial fermentation in the absence of HCl (achylia, gastric cancer).

Experiment 3: determination of indican in urine (Obermayer reaction)

1. Pour 2 ml of urine and 2 ml of Obermayer reagent (a solution of $FeCl_3$ in concentrated HCl) into a test tube and mix.
2. Add 1 ml of chloroform, invert several times and wait for the layers to separate. **Result and analysis:** indican is hydrolyzed to indoxyl, which is oxidized to **indigo blue** and colors the chloroform layer blue. The intensity of the color corresponds to the amount of indican; a deep blue color shows that putrefaction in the intestine has increased.

8. Practical skills

Skill: calculating the nitrogen balance

1. Determine the amount of protein taken in with food (g) and convert it into nitrogen: N = protein / 6.25.
2. Determine the urea nitrogen in the daily urine and add 2–4 g (losses with feces, skin and other nitrogenous substances).
3. Compare the nitrogen taken in and excreted.
4. Determine the type of balance (equilibrium, positive, negative).

5. In a negative balance analyze the cause (starvation, a catabolic state) and reassess the protein requirement.

Skill: evaluating a gastric juice analysis

1. Compare the total acidity and the free and bound HCl with the normal values.
2. Assess the pathological components (lactic acid, blood, bile).
3. Determine the state: hyperacid, hyposecretion, achlorhydria, achylia.
4. Name the possible diseases (peptic ulcer, atrophic gastritis, gastric cancer, pernicious anemia).
5. Give the biochemical basis of the principle of treatment (antisecretory agents or replacement therapy).

Review questions

1. List the functions of proteins in the body.
2. What is nitrogen balance? What are its types and causes?
3. What does the daily requirement and the biological value of protein depend on?
4. What is the difference between kwashiorkor and marasmus?
5. What is the dynamic state of body proteins?
6. How is HCl formed and what functions does it carry out?
7. How is pepsinogen activated? What is the specificity of pepsin?
8. How is the secretion of gastric juice regulated?
9. How are the pancreatic proteases activated?
10. Explain the specificity of the proteolytic enzymes.
11. How are amino acids absorbed?
12. What are the features of protein digestion in children?
13. What products does putrefaction in the intestine form?
14. How are the products of putrefaction detoxified in the liver?
15. What is the normal acidity of gastric juice and how is it determined?

Test questions

1. When is a positive nitrogen balance seen:
   A) In starvation
   B) In severe burns
   C) During the growth of children
   D) In diabetes mellitus
   E) In old age

2. The factor that converts pepsinogen into pepsin:
   A) Enteropeptidase
   B) Trypsin
   C) Gastrin
   D) Histamine
   E) HCl and autocatalysis

3. The enzyme that activates trypsinogen:
   A) Enteropeptidase
   B) Pepsin
   C) Chymotrypsin
   D) Carboxypeptidase
   E) Elastase

4. The peptide bonds of which amino acids trypsin cleaves:
   A) Aromatic amino acids
   B) Glycine and alanine
   C) Proline
   D) Aspartate and glutamate
   E) Lysine and arginine

5. The protein that pumps H^+ into the gastric lumen in the parietal cells:
   A) H^+,K^+-ATPase
   B) Na^+,K^+-ATPase
   C) Carbonic anhydrase
   D) Ca^{2+}-ATPase
   E) The Na^+/H^+ exchanger

6. The exopeptidase:
   A) Pepsin
   B) Trypsin
   C) Chymotrypsin
   D) Carboxypeptidase
   E) Elastase

7. The normal value of free HCl in gastric juice (in titration units):
   A) 0–5
   B) 5–10
   C) 40–60
   D) 60–100
   E) 20–40

8. The substance formed from tryptophan during putrefaction in the intestine:
   A) Cadaverine
   B) Phenol
   C) Indole
   D) Putrescine
   E) Cresol

9. The process whose intensity is shown by indican in the urine:
   A) Glycolysis
   B) Putrefaction in the intestine
   C) Urea synthesis
   D) Lipolysis
   E) Gluconeogenesis

10. The feature characteristic of kwashiorkor:
    A) Loss of adipose tissue only
    B) Absence of edema
    C) Edema because of hypoalbuminemia
    D) Hyperglycemia
    E) Hyperproteinemia

11. The mechanism by which free amino acids enter the enterocytes:
    A) Simple diffusion
    B) Symport with Na^+ (secondary active transport)
    C) Endocytosis
    D) Primary active transport
    E) In chylomicrons

12. What the appearance of lactic acid in gastric juice indicates:
    A) A hyperacid state
    B) Bacterial fermentation in the absence of HCl (achylia, gastric cancer)
    C) A normal state
    D) Pancreatitis
    E) A gastrinoma

13. The mechanism of action of proton pump inhibitors (omeprazole):
    A) They irreversibly inhibit H^+,K^+-ATPase
    B) They block histamine H_2 receptors
    C) They neutralize HCl
    D) They increase gastrin secretion
    E) They inhibit pepsin

14. The biochemical basis of food allergy in children:
    A) High pepsin activity
    B) Lack of trypsin
    C) Excess HCl
    D) Absorption of undigested proteins because of the high permeability of the intestine
    E) Impaired absorption of amino acids

15. The substance with which the products of putrefaction are conjugated in the liver:
    A) With glycine
    B) With acetyl-CoA
    C) With taurine
    D) With UDP-glucuronic acid and active sulfate (PAPS)
    E) With glutathione

Clinical cases

Case 1

A 60-year-old patient has a reduced appetite, aversion to meat, weight loss and a feeling of heaviness in the epigastrium. Gastric juice analysis: total acidity 10 t.u., no free HCl, the Uffelmann reaction positive. Blood hemoglobin is 95 g/l.

Questions: Evaluate the gastric juice analysis. Why has lactic acid appeared? Which disease is suspected?

Answer: Achlorhydria (no free HCl, sharply reduced total acidity) and lactic acid in the gastric juice. Without HCl the stomach loses its bactericidal property; when food is retained in the stomach, bacteria ferment carbohydrates and form lactic acid. Aversion to meat, weight loss, anemia and lactic acid raise the suspicion of gastric cancer — endoscopy and biopsy are needed. The anemia is related to chronic blood loss and impaired iron absorption (no HCl).

Case 2

A 40-year-old man complains of pain in the upper abdomen on an empty stomach and at night; the pain decreases after food or after taking soda. Gastric juice: total acidity 90 t.u., free HCl 70 t.u. The urea breath test is positive.

Questions: Evaluate the condition. Explain the biochemical basis of treatment.

Answer: A hyperacid state — duodenal ulcer (associated with Helicobacter pylori). With the help of urease H. pylori breaks urea down into NH_3 and CO_2 (the labeled CO_2 is detected in the breath test), surrounds itself with an alkaline "cloud", damages the mucosa and increases gastrin secretion. Treatment: a proton pump inhibitor (it blocks H^+,K^+-ATPase and sharply reduces HCl secretion) + antibiotics (eradication); antacids are symptomatic. Soda ($NaHCO_3$) neutralizes HCl and relieves the pain, but it forms CO_2 and causes acid rebound — it is not recommended.

Case 3

In a patient with chronic constipation the Obermayer reaction in the urine is strongly positive (a deep blue color). He eats a lot of protein food.

Questions: What does this result show? Write the pathway of indican formation.

Answer: Indican in the urine is increased — putrefaction of proteins in the large intestine has increased (because of constipation the intestinal contents stay for a long time, and there is a lot of protein). The pathway of formation: tryptophan → (bacteria) → indole → portal vein → microsomal oxidation in the liver → indoxyl → conjugation with active sulfate (PAPS) → indoxyl sulfate → its potassium salt (indican) → urine. Recommendation: more dietary fiber and fluid, treatment of the constipation.

Quick review

Table 11.5. *Key concepts of topic 11*

Concept	Brief definition
Nitrogen balance	Positive — growth, pregnancy; negative — starvation, illness, catabolism
Protein requirement	Adults 0.8–1 g/kg; children 1.5–4 g/kg
Biological value	Content of essential amino acids and digestibility
Dynamic state	300–400 g of protein is renewed per day; lysosomes and proteasomes
HCl	Parietal cells; H^+,K^+-ATPase; denaturation, activation of pepsinogen, bactericidal action
Pepsin	Bonds of aromatic amino acids; pH 1.5–2
Activation	Enteropeptidase → trypsin → the other proenzymes
Trypsin / chymotrypsin	Lysine, arginine / aromatic amino acids
Exopeptidases	Carboxypeptidase, aminopeptidase, dipeptidases

Concept	Brief definition
Absorption	Na^+ symport; di- and tripeptides — H^+ symport; the γ-glutamyl cycle
Children	Gastricsin, chymosin; high permeability — absorption of IgA, food allergy
Putrefaction	Indole, skatole, phenol, cresol, H_2S, cadaverine, putrescine
Detoxification	Microsomal oxidation + PAPS or UDP-glucuronic acid; indican
Gastric juice	Total 40–60, free HCl 20–40, bound 10–20 t.u.

TOPIC 12

The common pathways of amino acid metabolism. Aminotransferases. Decarboxylation of amino acids

Learning objectives

To teach the student the fate of the absorbed amino acids, the common pathways of amino acid catabolism — transamination (the aminotransferases and the coenzyme function of vitamin B_6), direct and indirect oxidative deamination, the decarboxylation of amino acids, the formation, functions and detoxification of the biogenic amines, and the significance of histamine in allergy and inflammation; to master the determination of aminotransferase activity in serum.

By the end of the class the student should know:

- the pathways of transformation of amino acids in the cell;
- the transamination reaction, the aminotransferases and the role of pyridoxal phosphate;
- the clinical and diagnostic significance of ALT and AST and the De Ritis ratio;
- direct and indirect (transdeamination) deamination;
- the significance and regulation of glutamate dehydrogenase;
- the fate of the carbon skeleton of amino acids (glucogenic and ketogenic amino acids);
- the decarboxylation reaction and the biogenic amines;
- the detoxification of the biogenic amines (MAO, COMT) and the significance of inhibitors of these enzymes.

The student should be able to:

- determine ALT and AST activity in serum on a biochemical analyzer;
- evaluate aminotransferase activity clinically;
- explain the mechanism of drugs that affect the metabolism of the biogenic amines.

1. The fate of amino acids

The amino acid pool of the cell is used in the following ways:

1. **Protein synthesis** — the main route.
2. **Synthesis of nitrogenous compounds**: purine and pyrimidine bases, heme, creatine, carnitine, biogenic amines, hormones (thyroxine, adrenaline), neurotransmitters, glutathione, nicotinamide.
3. **Catabolism** — removal of the amino group (transamination, deamination) and oxidation of the carbon skeleton or its conversion into glucose and ketone bodies. The catabolism of amino acids increases especially when the diet is rich in protein, in starvation and in diabetes mellitus. The **ammonia** formed is detoxified (topic 13).

2. Transamination

Transamination is the transfer of the α-amino group of an amino acid to an α-keto acid; a new amino acid and a new α-keto acid are formed. The reaction is **reversible** and no ammonia is released. It was discovered by A.E. Braunstein and M.G. Kritzman (1937).

L-amino acid + α-ketoglutarate ⇄ α-keto acid + L-glutamate

Aminotransferases (transaminases) belong to the class of transferases; they are found in almost all tissues, in the cytosol and the mitochondria. Their coenzyme is **pyridoxal phosphate (PLP)**, a derivative of vitamin B_6. PLP accepts the amino group from the amino acid and is converted into **pyridoxamine phosphate**, then passes it to the keto acid (a “ping-pong” mechanism).

Most aminotransferases use **α-ketoglutarate** as the acceptor of the amino group — as a result the amino groups are collected in **glutamate**.

Table 12.1. *The main aminotransferases*

Enzyme	Reaction	Main location
Alanine aminotransferase (ALT)	Alanine + α-ketoglutarate ⇄ pyruvate + glutamate	Liver (cytosol) — a lot; less in the kidney, muscle and heart
Aspartate aminotransferase (AST)	Aspartate + α-ketoglutarate ⇄ oxaloacetate + glutamate	Heart, liver (cytosol and mitochondria), skeletal muscle, kidney

The significance of transamination: synthesis of nonessential amino acids (from keto acids); collection of the amino groups of amino acids in glutamate (for subsequent deamination); the link between amino acid catabolism and gluconeogenesis (alanine → pyruvate, aspartate → oxaloacetate); the malate-aspartate shuttle; transport of nitrogen between tissues.

The clinical significance of the aminotransferases

Normally the activity of ALT and AST in the blood is low. When cells are damaged (cytolysis) they are released into the blood:

Table 12.2. *Aminotransferases in serum (37 °C, kinetic method, adults)*

Enzyme	Men	Women
ALT	≤ 41 IU/l	≤ 33 IU/l
AST	≤ 37 IU/l	≤ 31 IU/l

- **In hepatitis** (viral, toxic) **ALT rises sharply** (10–100 times), ALT > AST; the rise of ALT begins 1–2 weeks before the jaundice.
- **In myocardial infarction AST** rises after 6–12 hours, peaks at 24–48 hours and returns to normal in 4–7 days; ALT changes little.
- **In alcoholic hepatitis and cirrhosis** AST > ALT (mitochondrial AST is released; in alcoholism a lack of PLP lowers ALT more). **The De Ritis ratio** = AST / ALT; normally 1.33 ± 0.42. **< 1** — liver damage (viral hepatitis); **> 2** — myocardial infarction, alcoholic liver damage.

3. Deamination

Deamination is the removal of the amino group from an amino acid in the form of **ammonia**. Its types: oxidative, reductive, hydrolytic and intramolecular. In humans the main type is **oxidative deamination**.

Direct oxidative deamination

Only **glutamate** is actively deaminated — by **glutamate dehydrogenase** (GDH; in the mitochondria, especially active in the liver):

L-Glutamate + NAD^+ ($NADP^+$) + H_2O ⇄ α-Ketoglutarate + NH_3 + NADH + H^+

The reaction is reversible: in the reverse direction it binds ammonia and synthesizes glutamate. GDH is an **allosteric** enzyme: **ADP and GDP** (low energy) activate it — the catabolism of amino acids increases; **ATP, GTP and NADH** inhibit it.

The direct deamination of other amino acids is of little importance: **L-amino acid oxidase** (FMN; very low activity), D-amino acid oxidase (FAD; for D-amino acids coming from bacteria), serine and threonine — through dehydratases (PLP), histidine — through histidase.

Indirect deamination (transdeamination)

Most amino acids are deaminated in **two steps**:

1. **Transamination**: the amino acid passes its amino group to α-ketoglutarate → glutamate;
2. **Oxidative deamination of glutamate** (GDH): glutamate → α-ketoglutarate + NH_3.

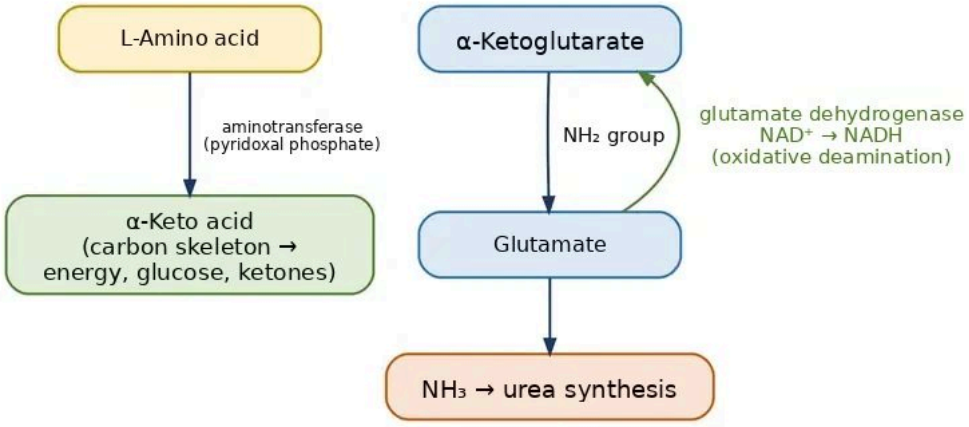


Figure 12.1. *Indirect deamination (transdeamination): transamination and the oxidative deamination of glutamate (drawn after Severin E.S.)*

REMEMBER

Transdeamination is the **main route** of deamination of amino acids. α-Ketoglutarate and glutamate act as "carriers" in this process and deliver the amino groups to GDH. In skeletal muscle there is little GDH, and deamination there proceeds through the **purine nucleotide cycle** (AMP deaminase).

Reductive amination (the reverse of transdeamination) — the synthesis of nonessential amino acids: α-ketoglutarate + NH_3 (GDH) → glutamate → other amino acids through transamination.

4. The fate of the carbon skeleton of amino acids

After the amino group has been removed, the carbon skeletons of the 20 amino acids are converted into 7 common products: pyruvate, acetyl-CoA, acetoacetyl-CoA, α-ketoglutarate, succinyl-CoA, fumarate and oxaloacetate. They are oxidized in the Krebs cycle (energy) or:

- **glucogenic amino acids** (those converted into pyruvate or Krebs cycle metabolites) — give glucose through gluconeogenesis: alanine, serine, glycine, cysteine, aspartate, asparagine, glutamate, glutamine, arginine, histidine, proline, methionine, valine, threonine;
- **ketogenic amino acids** (those that give acetyl-CoA or acetoacetyl-CoA) — only **leucine** and **lysine**;
- **mixed** (gluco-ketogenic): isoleucine, phenylalanine, tyrosine, tryptophan.

5. Decarboxylation of amino acids. The biogenic amines

Decarboxylation is the removal of the carboxyl group from an amino acid as CO_2; the product is a **biogenic amine**. The reaction is catalyzed by **decarboxylases** (lyases); their coenzyme is **pyridoxal phosphate** (B_6). The reaction is irreversible.

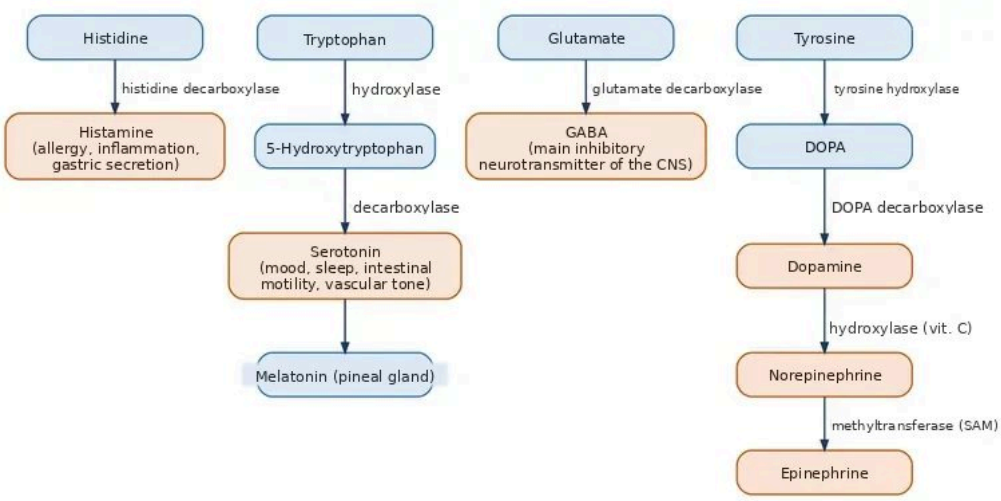


Figure 12.2. *Formation of the main biogenic amines (drawn after Severin E.S.)*

Table 12.3. *The main biogenic amines*

Amino acid	Biogenic amine	Function
Histidine	Histamine	A mediator of allergy and inflammation (vasodilation, increased permeability, bronchospasm, itching — H_1); secretion of gastric juice (H_2); a neurotransmitter in the CNS
Tryptophan (→ 5-hydroxytryptophan)	Serotonin	A neurotransmitter (mood, sleep, appetite); intestinal motility; constricts vessels; platelet aggregation; melatonin is formed from it in the pineal gland
Glutamate	GABA (γ-aminobutyric acid)	The main inhibitory mediator of the CNS
Tyrosine (→ DOPA)	Dopamine, noradrenaline, adrenaline (catecholamines)	Neurotransmitters; hormones of the adrenal medulla
Serine	Ethanolamine	Synthesis of phospholipids
Cysteine (through cysteine sulfinic acid)	Taurine	Conjugation of bile acids; an antioxidant
Ornithine	Putrescine → spermidine, spermine (polyamines)	Cell growth and division, stabilization of nucleic acids

CLINICAL SIGNIFICANCE

Histamine is released from mast cells and basophils in allergic reactions: urticaria, Quincke edema, bronchospasm, and a sharp fall of blood pressure in anaphylactic shock. **Antihistamines**: H_1 blockers (diphenhydramine, loratadine, cetirizine) — in allergy; H_2 blockers (famotidine) — reduce the secretion of gastric juice. In vitamin B_6 deficiency the activity of glutamate decarboxylase falls and GABA decreases — **seizures** in newborns (B_6 is given together with isoniazid treatment).

Detoxification of the biogenic amines

So that the action of the biogenic amines is short, they are broken down quickly:

- **Monoamine oxidases (MAO)** — on the outer mitochondrial membrane, containing FAD; oxidative deamination: $R\text{-}CH_2\text{-}NH_2 + O_2 + H_2O \rightarrow R\text{-}CHO + NH_3 + H_2O_2$. MAO-A breaks down serotonin and noradrenaline, MAO-B dopamine. The aldehydes formed are oxidized to acids (serotonin → 5-hydroxyindoleacetic acid; catecholamines → vanillylmandelic acid) and excreted in the urine.
- **Diamine oxidase (histaminase)** — oxidizes histamine, putrescine and cadaverine.
- **Methylation — catechol-O-methyltransferase (COMT)** methylates catecholamines at the expense of S-adenosylmethionine; histamine is also methylated.

CLINICAL SIGNIFICANCE

MAO inhibitors (moclobemide — MAO-A, selegiline — MAO-B) and **selective serotonin reuptake inhibitors** (fluoxetine) are antidepressants: serotonin and noradrenaline increase in the synapse. In Parkinson disease (a lack of dopamine in the substantia nigra) **L-DOPA** (levodopa) is used — it crosses the blood-brain barrier, which dopamine cannot, and is converted into dopamine in the brain; it is given together with a peripheral decarboxylase inhibitor (carbidopa). If a patient taking MAO inhibitors eats food rich in tyramine (hard cheese, wine), a hypertensive crisis (the “cheese reaction”) may develop. **Pheochromocytoma** (a catecholamine-producing tumor) is diagnosed by a rise in vanillylmandelic acid and metanephrines in the urine; a **carcinoid tumor** by a rise in 5-hydroxyindoleacetic acid in the urine.

6. Laboratory work

Experiment 1: determination of ALT and AST activity in serum on a biochemical analyzer (kinetic method)

Principle of the method (IFCC):

- **ALT**: L-alanine + α-ketoglutarate → pyruvate + L-glutamate; pyruvate + NADH + H^+ → lactate + NAD^+ (the indicator enzyme is **lactate dehydrogenase**);
- **AST**: L-aspartate + α-ketoglutarate → oxaloacetate + L-glutamate; oxaloacetate + NADH + H^+ → malate + NAD^+ (the indicator enzyme is **malate dehydrogenase**). NADH absorbs light at 340 nm while NAD^+ does not; the **rate of fall** of the optical density is proportional to the enzyme activity. Pyridoxal phosphate is added to the reagent (to avoid a falsely low result in B_6 deficiency).

1. Pour 1 ml of working reagent (substrates, NADH, LDH or MDH, PLP) into a cuvette and warm it at 37 °C for 5 minutes.
2. Add 100 µl of serum and mix.
3. After 1 minute measure the optical density at 340 nm and record it every minute for 3 minutes (automatically on the analyzer).
4. Calculate the mean ΔE/min. **Calculation:** activity (IU/l) = ΔE/min × F (F is the factor for the reagent and cuvette, derived from the molar absorption coefficient of NADH; for example, 1746).

Evaluation: compare with the normal values in the table and calculate the De Ritis ratio. Hemolyzed serum falsely raises AST (erythrocytes contain a lot of AST).

Experiment 2: determination of the aminotransferases by the Reitman–Frankel method (colorimetric) (a historical method)

Principle of the method: the pyruvate formed in transamination (or the pyruvate formed spontaneously from oxaloacetate) forms a hydrazone with **2,4-dinitrophenylhydrazine**; in an alkaline medium it gives a red-brown color. The intensity of the color is proportional to the amount of pyruvate and therefore to the enzyme activity. The result is expressed in μmol pyruvate / (hour · ml); normal values: ALT 0.1–0.68, AST 0.1–0.45 mmol/(hour · l).

Experiment 3: the ninhydrin reaction for amino acids

1. Add to a test tube 1 ml of 1% glycine solution (or a protein hydrolysate) and 5 drops of 0.1% ninhydrin solution.
2. Boil for 1–2 minutes. **Result and analysis:** α-amino acids give a **blue-violet** color with ninhydrin (Ruhemann's purple): the amino acid undergoes oxidative deamination and decarboxylation, and the ammonia released forms a colored compound with ninhydrin. Proline and hydroxyproline give a yellow color. This reaction underlies the detection of amino acids in chromatography.

7. Practical skills

Skill: clinical evaluation of an aminotransferase result

1. Compare ALT and AST with the normal values for the sex; determine how many times they are increased.
2. Calculate the De Ritis ratio.
3. If it is < 1 — think of cytolysis of hepatocytes (hepatitis); if > 2 — damage to the myocardium or muscle, alcoholic hepatitis.
4. Compare with the other indices: bilirubin, alkaline phosphatase, GGT (liver); CK-MB, troponins (heart).
5. Check that there is no hemolysis in the sample.

Skill: explaining the mechanism of a drug that affects the metabolism of a biogenic amine

1. Determine which step the drug acts on: synthesis (decarboxylase), the receptor, reuptake or breakdown (MAO, COMT).
2. State how the amount of the biogenic amine changes in the synapse or in the tissue.
3. Explain the clinical effect (antidepressant, antihistamine, antiparkinsonian).
4. Indicate the adverse effects and the interaction with food (tyramine).

Review questions

1. In what ways are the amino acids of the cell used?
2. What is transamination? Write the equation of the reaction.
3. What is the role of pyridoxal phosphate in transamination?
4. Which reactions do ALT and AST catalyze and where are they located?
5. What is the clinical and diagnostic significance of the aminotransferases?
6. What is the De Ritis ratio and how should it be evaluated?
7. How does oxidative deamination proceed?

8. How is glutamate dehydrogenase regulated?
9. What is transdeamination and what is its significance?
10. Name the glucogenic and ketogenic amino acids.
11. How does the decarboxylation reaction proceed and what is its coenzyme?
12. List the main biogenic amines and their functions.
13. What is the significance of histamine in allergy and inflammation?
14. How are the biogenic amines detoxified?
15. Explain the mechanism of action of MAO inhibitors and L-DOPA.

Test questions

1. The coenzyme of the aminotransferases:
   A) Pyridoxal phosphate
   B) Thiamine diphosphate
   C) Biotin
   D) Tetrahydrofolate
   E) FAD

2. The main acceptor of the amino group in transamination:
   A) Pyruvate
   B) α-Ketoglutarate
   C) Oxaloacetate
   D) Glutamine
   E) Fumarate

3. The reaction catalyzed by ALT:
   A) Aspartate + α-ketoglutarate ⇄ oxaloacetate + glutamate
   B) Glutamate → α-ketoglutarate + NH_3
   C) Alanine + α-ketoglutarate ⇄ pyruvate + glutamate
   D) Histidine → histamine + CO_2
   E) Glutamate + NH_3 → glutamine

4. The enzyme that rises more in viral hepatitis:
   A) AST
   B) CK-MB
   C) ALT
   D) α-Amylase
   E) Lipase

5. The normal value of the De Ritis ratio:
   A) 1,33 ± 0,42
   B) Less than 0.5
   C) 3–5
   D) More than 10
   E) 0

6. The only amino acid that undergoes active oxidative deamination:
   A) Alanine
   B) Aspartate
   C) Glutamate
   D) Glycine
   E) Leucine

7. The substance that allosterically activates glutamate dehydrogenase:
   A) ATP
   B) ADP
   C) NADH
   D) GTP
   E) Glutamine

8. The main route of deamination of amino acids:
   A) Hydrolytic deamination
   B) Reductive deamination
   C) Through L-amino acid oxidase
   D) Transdeamination (indirect deamination)
   E) Intramolecular deamination

9. The amino acids that are purely ketogenic:
   A) Alanine and serine
   B) Glutamate and aspartate
   C) Phenylalanine and tyrosine
   D) Valine and isoleucine
   E) Leucine and lysine

10. The substance formed by the decarboxylation of histidine:
    A) Histamine
    B) Serotonin
    C) GABA
    D) Dopamine
    E) Taurine

11. The amino acid from which the main inhibitory mediator of the CNS is formed:
    A) Tryptophan
    B) Tyrosine
    C) Histidine
    D) Glutamate
    E) Serine

12. The enzyme that detoxifies biogenic amines by oxidative deamination:
    A) Glutamate dehydrogenase
    B) Monoamine oxidase
    C) Decarboxylase
    D) Aminotransferase
    E) Xanthine oxidase

13. The amino acid from which serotonin is formed:
    A) Tyrosine
    B) Histidine
    C) Glutamate
    D) Phenylalanine
    E) Tryptophan

14. Why L-DOPA is given instead of dopamine in the treatment of Parkinson disease:
    A) Dopamine is not absorbed in the intestine
    B) L-DOPA acts more strongly
    C) Dopamine is not broken down in the liver
    D) L-DOPA inhibits MAO
    E) Dopamine does not cross the blood-brain barrier, while L-DOPA does and is converted into dopamine in the brain

15. The enzyme that is falsely increased in hemolyzed serum:
    A) ALT
    B) α-Amylase
    C) Lipase
    D) AST
    E) GGT

Clinical cases

Case 1

A 25-year-old patient has weakness, loss of appetite, nausea and heaviness under the right costal margin; jaundice appeared 3 days later. Tests: ALT 1200 IU/l, AST 680 IU/l, bilirubin 85 μmol/l (mainly conjugated and free), HBsAg positive.

Questions: Calculate the De Ritis ratio and interpret the result. Why has ALT risen more?

Answer: De Ritis = 680 / 1200 ≈ 0.57 (< 1) — cytolysis of hepatocytes (acute viral hepatitis B). ALT is located mainly in the cytosol of hepatocytes and is most abundant in the liver; when the membrane is damaged it is the first to be released into the blood. A large part of AST is in the mitochondria — it is released in quantity only in deep (necrotic) damage. The rise of ALT begins before the jaundice and is a sensitive sign of the activity of hepatitis.

Case 2

A 58-year-old man says he had severe pain behind the sternum 8 hours ago. Tests: AST 180 IU/l, ALT 45 IU/l, CK-MB and troponin I sharply increased.

Questions: Calculate the De Ritis ratio. Which organ is damaged?

Answer: De Ritis = 180 / 45 = 4 (> 2) — damage to the myocardium. The heart muscle contains a lot of AST and little ALT; in myocardial infarction the cardiomyocytes undergo necrosis and AST, CK-MB, LDH1 and troponins are released into the blood. AST rises after 6–12 hours, peaks at 24–48 hours and normalizes in 4–7 days; today the troponins are considered the most specific marker.

Case 3

A patient taking a MAO inhibitor for depression ate hard cheese and drank red wine at a party, after which he developed a severe headache, sweating and palpitations; his blood pressure is 210/120 mm

Hg.

Questions: Explain the biochemical mechanism of this condition.

Answer: The “cheese reaction” — a hypertensive crisis. Cheese and wine contain a lot of tyramine (the decarboxylation product of tyrosine). Normally it is broken down by MAO in the intestinal wall and the liver. When MAO is inhibited, tyramine passes into the blood and displaces noradrenaline from the sympathetic nerve endings (an indirect sympathomimetic); because noradrenaline is also not broken down by MAO, its effect is stronger — the vessels constrict and blood pressure rises sharply. People taking MAO inhibitors are forbidden foods rich in tyramine.

Quick review

Table 12.4. *Key concepts of topic 12*

Concept	Brief definition
Transamination	Amino acid + α-KG ⇄ α-keto acid + glutamate; reversible; no NH_3 is released
Coenzyme	Pyridoxal phosphate (B_6) — in transamination and decarboxylation
ALT	Alanine ⇄ pyruvate; liver cytosol; ↑↑ in hepatitis
AST	Aspartate ⇄ oxaloacetate; heart, liver; ↑ in myocardial infarction
De Ritis	AST/ALT: normal 1.33; < 1 — hepatitis; > 2 — infarction, alcohol
Oxidative deamination	Glutamate dehydrogenase: glutamate → α-KG + NH_3 + NADH
Regulation of GDH	ADP, GDP ↑; ATP, GTP, NADH ↓
Transdeamination	Transamination + GDH — the main route
Ketogenic amino acids	Leucine, lysine
Decarboxylation	Amino acid → biogenic amine + CO_2; PLP
Biogenic amines	Histamine (histidine), serotonin (tryptophan), GABA (glutamate), catecholamines (tyrosine)
Detoxification	MAO (FAD), diamine oxidase, COMT (methylation)
Drugs	Antihistamines, MAO inhibitors, L-DOPA + carbidopa

TOPIC 13

Detoxification of ammonia. Metabolism of individual amino acids and its hereditary disorders

Learning objectives

To give the student knowledge of the sources and toxic action of ammonia, the routes of its temporary and final detoxification, the synthesis of urea and its link with transamination and the Krebs cycle, the types and consequences of hyperammonemia, the metabolism of serine, glycine, methionine, phenylalanine and tyrosine, one-carbon groups and transmethylation, and the congenital diseases of amino acid metabolism; to master the determination of urea in serum.

By the end of the class the student should know:

- the sources of ammonia and the mechanisms of its toxic action;
- the temporary binding of ammonia (glutamine, alanine) and the formation of ammonium salts in the kidney;
- the reactions, enzymes, energy cost and regulation of the ornithine cycle;
- the normal values and the changes of urea in blood and urine;
- congenital and acquired hyperammonemias;
- the significance of tetrahydrofolate and one-carbon groups;
- the metabolism of methionine, S-adenosylmethionine and transmethylation, and homocysteine;
- the metabolism of phenylalanine and tyrosine and its congenital diseases.

The student should be able to:

- determine urea in serum and evaluate the result;
- determine phenylpyruvate in urine (the Fölling test);
- give a biochemical explanation of the cause of hyperammonemia.

1. Formation and toxicity of ammonia

Sources of ammonia:

- deamination of amino acids (mainly the oxidative deamination of glutamate);
- hydrolysis of amides (glutamine, asparagine);
- oxidation of biogenic amines by MAO;
- deamination of purine and pyrimidine bases (adenine, cytosine);
- putrefaction of proteins and breakdown of urea by bacteria in the intestine (it reaches the liver through the portal vein). About 15–20 g of ammonia is formed per day in the human body, but its level in the blood is very low: **11–35 µmol/l** (normally < 50 µmol/l).

Its toxic action (especially on the brain):

- in the brain ammonia forms glutamate with α-ketoglutarate (reductive amination) — **α-ketoglutarate falls**, the Krebs cycle slows down and ATP formation decreases;
- glutamate and glutamine increase — **osmotic swelling** of the astrocytes (glutamine) and a disturbed balance of neurotransmitters (glutamate ↓ or ↑, GABA ↑);
- NH_4

+ disrupts K^+ channels and Na^+,K^+-ATPase, and the membrane potential changes;

- the blood pH shifts to the alkaline side. Clinical signs: nausea, vomiting, lethargy, tremor, seizures, coma.

2. Temporary detoxification and transport of ammonia

The ammonia formed in the tissues is bound at once and transported to the liver and kidney in nontoxic forms:

- **Synthesis of glutamine** (the main route; brain, muscle and other tissues): glutamate + NH_3 + ATP → **glutamine** + ADP + P_i — **glutamine synthetase**. Glutamine is the main transport form of ammonia (the most abundant amino acid in the blood).
- **Formation of alanine** (muscles): pyruvate is transaminated into alanine and goes to the liver — the **glucose-alanine cycle** (topic 7).
- **Synthesis of asparagine** (less important). **In the kidney** glutamine is broken down by **glutaminase** into glutamate and NH_3; NH_3 passes into the tubular lumen, binds H^+ and is excreted in the urine as **ammonium salts** (NH_4

+) (≈ 0.5–1 g per day). This process **increases sharply in acidosis** — an important mechanism by which the kidney maintains the acid-base balance (topic 29).

3. Synthesis of urea (the ornithine cycle)

The **final detoxification** of ammonia is the synthesis of urea in the **liver**. The cycle was discovered by H. Krebs and K. Henseleit (1932). Urea is a slightly toxic, highly water-soluble, neutral substance; ≈ 80–90% of the body's nitrogen is excreted in this form.

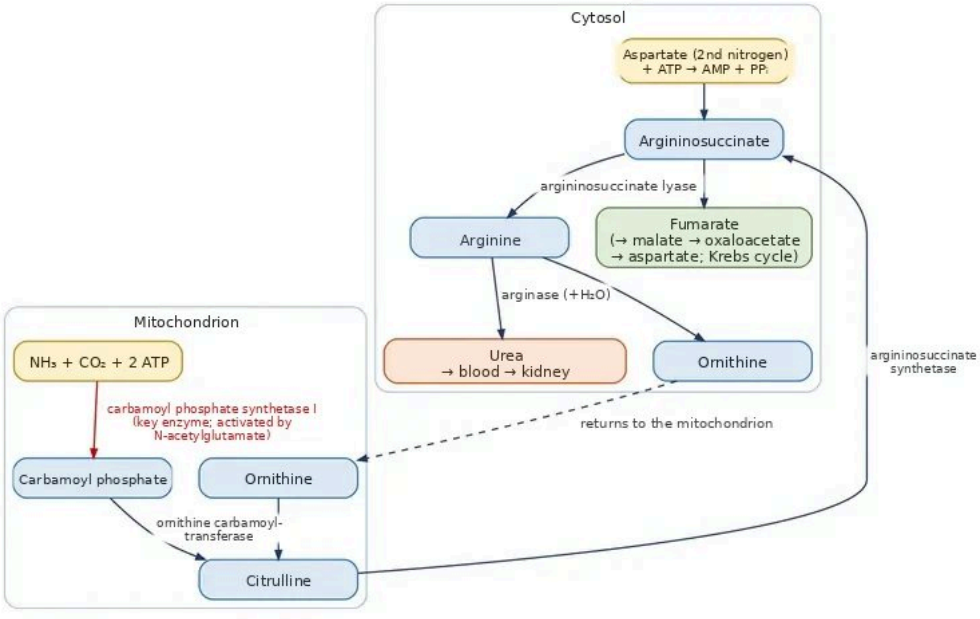


Figure 13.1. *The ornithine (urea) cycle (drawn after Sobirova R.A., 2006; Severin E.S.)*

Table 13.1. *Reactions of the ornithine cycle*

№	Reaction	Enzyme	Site
1	$NH_3 + CO_2 + 2ATP$ → carbamoyl phosphate + $2ADP + P_i$	Carbamoyl phosphate synthetase I (the key enzyme)	Mitochondria

№	Reaction	Enzyme	Site
2	Carbamoyl phosphate + ornithine → citrulline + P_i	Ornithine carbamoyltransferase	Mitochondria
3	Citrulline + aspartate + ATP → argininosuccinate + AMP + PP_i	Argininosuccinate synthetase	Cytosol
4	Argininosuccinate → arginine + fumarate	Argininosuccinate lyase	Cytosol
5	Arginine + H_2O → urea + ornithine	Arginase (Mn^{2+}; almost only in the liver)	Cytosol

The sources of the two nitrogens of the urea molecule: the first is **ammonia** (from the deamination of glutamate and the glutaminase reaction), the second is **aspartate** (from the transamination of oxaloacetate). In this way the nitrogen of all amino acids is delivered to the cycle through transamination.

Energy cost: 3 ATP for one urea (**4 high-energy bonds**, because one ATP is split into AMP + PP_i).

The link with the Krebs cycle (the "Krebs bicycle"): the **fumarate** formed in the ornithine cycle is converted in the Krebs cycle into malate and oxaloacetate; oxaloacetate is transaminated back into **aspartate** and returns to the ornithine cycle. The NADH formed in the malate dehydrogenase reaction recovers part of the energy spent on urea synthesis.

Regulation:

- **N-acetylglutamate** (acetyl-CoA + glutamate; its synthesis is activated by arginine) is the obligatory allosteric activator of carbamoyl phosphate synthetase I;
- with a protein-rich diet and in starvation (when amino acid catabolism increases) the synthesis of the cycle enzymes is induced (glucagon, cortisol). **Normal values of urea:** in serum **2.5–8.3 mmol/l**; in the urine **20–35 g** (333–583 mmol) per day.

Table 13.2. *Changes in the blood urea level*

Increase	Decrease
Renal failure (acute and chronic) — the most important cause	Liver failure (cirrhosis, acute liver dystrophy)
Dehydration, shock, heart failure (prerenal)	A protein-free diet, the late stage of starvation
Obstruction of the urinary tract (postrenal)	Pregnancy (hydremia, anabolism)
A protein-rich diet, strong catabolism (burns, fever, tumors), gastrointestinal bleeding	Congenital deficiency of the ornithine cycle enzymes

4. Hyperammonemia

Hyperammonemia is an increase of the ammonia level in the blood.

Congenital (primary) hyperammonemias are congenital deficiencies of the ornithine cycle enzymes (≈ 1:25 000–35 000 newborns):

Table 13.3. *Congenital deficiency of the ornithine cycle enzymes*

Disease	Deficient enzyme	Changes in blood/urine
Hyperammonemia type I	Carbamoyl phosphate synthetase I	NH_3 ↑↑; orotic acid normal

Disease	Deficient enzyme	Changes in blood/urine
Hyperammonemia type II (the most common, X-linked)	Ornithine carbamoyltransferase	NH_3 ↑↑, glutamine ↑; orotic acid ↑ in the urine (carbamoyl phosphate goes into pyrimidine synthesis)
Citrullinemia	Argininosuccinate synthetase	Citrulline ↑↑
Argininosuccinic aciduria	Argininosuccinate lyase	Argininosuccinate ↑↑
Argininemia	Arginase	Arginine ↑↑; spastic paraparesis, NH_3 rises less

In newborns, in the first days of life (when protein intake begins), refusal to feed, vomiting, lethargy, seizures and coma are seen. Treatment: restriction of protein, **sodium benzoate** and **phenylbutyrate** (they conjugate with glycine and glutamine and remove nitrogen bypassing urea), supplements of arginine or citrulline, hemodialysis, liver transplantation.

Acquired (secondary) hyperammonemia — mainly in **liver cirrhosis and failure**: urea synthesis falls and the ammonia of the portal vein bypasses the liver through portocaval anastomoses and enters the general circulation — **hepatic encephalopathy** (topic 25). In children, taking aspirin during viral infections may cause **Reye syndrome** (damage to the mitochondria, hyperammonemia, fatty degeneration of the liver, encephalopathy); for this reason aspirin is not recommended for children with fever.

5. Metabolism of one-carbon groups. Serine and glycine

Tetrahydrofolate (THF) is the active form of folic acid (B_9); it is formed by dihydrofolate reductase (NADPH). THF carries **one-carbon groups** (methyl $-CH_3$, methylene $=CH_2$, methenyl $=CH-$, formyl $-CHO$, formimino). The main sources of one-carbon groups are **serine** (serine + THF → glycine + N5,N10-methylene-THF; serine hydroxymethyltransferase, PLP) and **glycine** (the glycine cleavage system).

One-carbon groups are used for: the synthesis of purine nucleotides (atoms C2 and C8), the synthesis of **thymidylate** (dUMP → dTMP), the regeneration of methionine (homocysteine → methionine), and the interconversion of serine and glycine.

Syntheses in which **glycine** takes part: proteins (every third amino acid of collagen), heme (succinyl-CoA + glycine), creatine, purines, glutathione, conjugation of bile acids (glycocholic acid), detoxification of benzoic acid (hippuric acid).

> **CLINICAL SIGNIFICANCE**
> **Sulfonamides** competitively inhibit the synthesis of folate from para-aminobenzoic acid in bacteria (humans obtain folate from food) — a bacteriostatic effect. **Methotrexate** inhibits dihydrofolate reductase — a lack of THF stops the synthesis of thymidylate and purines; rapidly dividing tumor cells die. **Folate deficiency** — megaloblastic anemia; in pregnancy — neural tube defects in the fetus (which is why folic acid is prescribed when a pregnancy is planned).

6. Metabolism of methionine. Transmethylation

Methionine is an essential amino acid and the main **donor of methyl groups** in the body. With ATP it forms **S-adenosylmethionine (SAM)**. SAM gives its methyl group to various acceptors — **transmethylation** (by methyltransferases):

- noradrenaline → **adrenaline**;

- guanidinoacetate → **creatine**;
- phosphatidylethanolamine → **phosphatidylcholine** (choline);
- **methylation** of nucleotides and DNA (epigenetic regulation of gene expression);
- synthesis of carnitine and melatonin; detoxification of foreign substances and hormones (catecholamines — COMT). After giving up its methyl group, SAM is converted into **S-adenosylhomocysteine** and then into **homocysteine**. The fate of homocysteine:
- **remethylation** — it returns to methionine: the methyl donor is N5-methyl-THF and the coenzyme is **methylcobalamin (B_{12})** (methionine synthase);
- **transsulfuration** — with serine it forms cystathionine (cystathionine β-synthase, **B_6**) and then **cysteine**; cysteine → taurine, glutathione, sulfates (PAPS).

CLINICAL SIGNIFICANCE

Hyperhomocysteinemia (normal blood level 5–15 μmol/l) is seen in deficiency of B_{12}, folate or B_6, in renal failure or in congenital enzyme defects (cystathionine β-synthase — **homocystinuria**: dislocation of the lens, skeletal changes, mental retardation, early thrombosis). Homocysteine damages the endothelium and is an independent risk factor for **atherosclerosis and thrombosis**. In B_{12} deficiency N5-methyl-THF is not used and accumulates (the "folate trap") — the other forms of THF fall and megaloblastic anemia develops.

7. Metabolism of phenylalanine and tyrosine

Phenylalanine (essential) is converted into **tyrosine** mainly in the liver by **phenylalanine hydroxylase** (coenzyme — **tetrahydrobiopterin, BH_4**). From tyrosine are formed: the **catecholamines** (dopamine, noradrenaline, adrenaline), **melanin** (tyrosinase in melanocytes) and the **thyroid hormones** (T_3, T_4); the final catabolism of tyrosine leads through homogentisic acid to **fumarate** and **acetoacetate**.

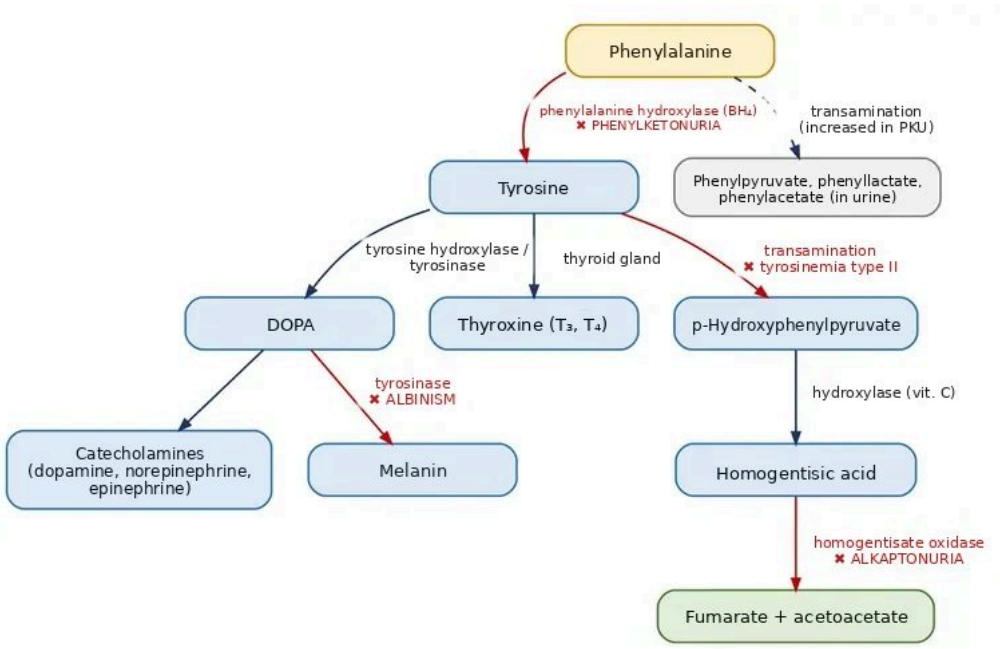


Figure 13.2. *Metabolism of phenylalanine and tyrosine; the reactions blocked in the congenital diseases (drawn after Severin E.S.)*

Table 13.4. *Congenital diseases of phenylalanine and tyrosine metabolism*

Disease	Deficient enzyme	Features
Phenylketonuria (PKU)	Phenylalanine hydroxylase (classic); rarely — synthesis of BH_4	Phenylalanine ↑↑ in the blood, phenylpyruvate, phenyllactate and phenylacetate in the urine (a “mousy” odor); if untreated — severe mental retardation, seizures, pale skin and hair (little melanin)
Alkaptonuria	Homogentisate oxidase	The urine darkens in air (homogentisic acid is oxidized); ochronosis — darkening of cartilage and connective tissue, arthropathy
Albinism (oculocutaneous)	Tyrosinase (in melanocytes)	No melanin in the skin, hair and eyes; photophobia, reduced visual acuity, risk of skin cancer
Tyrosinemia type I	Fumarylacetoacetate hydrolase	Severe liver and kidney damage (Fanconi syndrome); treatment — nitisinone, diet
Tyrosinemia type II	Tyrosine aminotransferase	Damage to the eyes and skin, mental retardation

CLINICAL SIGNIFICANCE

Screening for phenylketonuria is carried out in all newborns on days 3–5 of life (the Guthrie test or tandem mass spectrometry). Once the diagnosis is made, the child is put on a **diet low in phenylalanine** (special formulas; tyrosine becomes essential for such patients). If the diet is started early, mental development is almost normal. Mothers with PKU must follow the diet during pregnancy as well — otherwise a high phenylalanine level causes microcephaly and heart defects in the fetus. One source of phenylalanine is **aspartame** (an artificial sweetener).

8. Laboratory work

Experiment 1: determination of urea in serum by the urease method

Principle of the method: urease breaks urea down into ammonia and CO_2; the ammonia forms **blue indophenol** with phenol and hypochlorite (catalyzed by sodium nitroprusside) (the Berthelot reaction). The intensity of the color is proportional to the amount of urea (546–600 nm). Analyzers use a kinetic method: the ammonia is measured by the consumption of NADH in the glutamate dehydrogenase reaction (340 nm).

1. Pour 1 ml of urease-containing buffer solution into each of three test tubes.
2. Add 10 µl of serum to tube 1, 10 µl of standard (for example, 8.33 mmol/l) to tube 2 and 10 µl of water to tube 3; keep at 37 °C for 10 minutes.
3. Add the phenol-nitroprusside and alkaline hypochlorite reagents and keep at 37 °C for another 10 minutes.
4. Measure the optical density against the control on a photometer. **Calculation:** C (mmol/l) = (Esample / Estandard) × Cstandard

Evaluation: the normal value is 2.5–8.3 mmol/l. It is evaluated together with creatinine (kidney function).

Experiment 2: determination of phenylpyruvate in urine (Fölling test)

1. Pour 1 ml of fresh urine into a test tube and add 2–3 drops of 10% iron (III) chloride solution.
2. Observe the change of color within 30–60 s. **Result and analysis:** if phenylpyruvate is present, a **blue-green** color appears (it fades within a few minutes) — suspicion of phenylketonuria. False

positives: some drugs (salicylates — violet), bilirubin, homogentisic acid (a transient blue color). The result is confirmed by measuring phenylalanine in the blood.

Experiment 3: determination of homogentisic acid in urine (demonstration)

1. Pour 2 ml of urine from a patient with alkaptonuria (or a solution of homogentisic acid) into each of two test tubes.
2. Add 1 ml of 10% NaOH to one of them and shake it in air. **Result and analysis:** in an alkaline medium homogentisic acid is oxidized quickly and gives a **black-brown** color; in the tube without alkali the urine darkens slowly over several hours in air.

9. Practical skills

Skill: identifying the cause of a change in blood urea

1. Compare the urea level with the normal value.
2. If it is increased — assess creatinine: if both are increased — renal failure; if the urea/creatinine ratio is very high — a prerenal cause (dehydration, gastrointestinal bleeding, catabolism).
3. If it is decreased — check liver function (albumin, prothrombin time, ammonia).
4. Take the diet into account (the amount of protein).
5. Draw a conclusion and recommend further tests.

Skill: identifying the cause of hyperammonemia

1. Consider the patient’s age: in newborns — congenital enzyme defects; in adults — liver diseases.
2. Analyze the results of testing the blood amino acids (citrulline, arginine, glutamine) and the urinary orotic acid.
3. Determine the enzyme that is blocked (orotic acid ↑ — ornithine carbamoyltransferase; citrulline ↑ — argininosuccinate synthetase).
4. Explain the principles of treatment (protein restriction, benzoate, phenylbutyrate, arginine, dialysis).

Review questions

1. From which sources is ammonia formed in the body?
2. By which mechanisms is the toxic action of ammonia brought about?
3. In which forms is ammonia transported from the tissues to the liver?
4. How are ammonium salts formed in the kidney and what is their significance?
5. Write the reactions of the ornithine cycle and name its enzymes.
6. What are the sources of the nitrogen atoms of urea?
7. How much energy is used for the synthesis of urea?
8. How is the ornithine cycle linked with the Krebs cycle?
9. How is the ornithine cycle regulated?
10. In which conditions does the blood urea level change?
11. Describe the congenital hyperammonemias.
12. What is the significance of tetrahydrofolate and what are the sources of one-carbon groups?
13. What is transmethylation? In which syntheses does S-adenosylmethionine take part?

14. How is homocysteine metabolized and what is the significance of hyperhomocysteinemia?
15. Describe the congenital diseases of phenylalanine and tyrosine metabolism.

Test questions

1. The main transport form of ammonia in the tissues:
   A) Glutamine
   B) Urea
   C) Ammonium chloride
   D) Creatine
   E) Uric acid

2. The key enzyme of urea synthesis:
   A) Arginase
   B) Ornithine carbamoyltransferase
   C) Argininosuccinate lyase
   D) Carbamoyl phosphate synthetase I
   E) Glutaminase

3. The obligatory allosteric activator of carbamoyl phosphate synthetase I:
   A) Arginine
   B) Ornithine
   C) Fumarate
   D) ATP
   E) N-acetylglutamate

4. The source of the second nitrogen atom of the urea molecule:
   A) Glutamine
   B) Ornithine
   C) Aspartate
   D) Arginine
   E) Citrulline

5. Where the synthesis of urea takes place:
   A) In the liver (in the mitochondria and the cytosol)
   B) In the kidney
   C) In the brain
   D) In skeletal muscle
   E) In the intestinal wall

6. The normal level of urea in serum:
   A) 2.5–8.3 mmol/l
   B) 0.1–0.5 mmol/l
   C) 11–35 μmol/l
   D) 20–35 mmol/l
   E) 150–450 μmol/l

7. The most important cause of a rise in blood urea:
   A) Liver cirrhosis
   B) A protein-free diet
   C) Renal failure
   D) Pregnancy
   E) Deficiency of the ornithine cycle enzymes

8. The substance that increases in the urine in ornithine carbamoyltransferase deficiency:
   A) Urea
   B) Orotic acid
   C) Citrulline
   D) Phenylpyruvate
   E) Homogentisic acid

9. The active coenzyme of folic acid:
   A) S-adenosylmethionine
   B) Tetrahydrofolate
   C) Methylcobalamin
   D) Pyridoxal phosphate
   E) Tetrahydrobiopterin

10. The main donor of methyl groups in transmethylation:
    A) Methylcobalamin
    B) Serine
    C) Choline
    D) S-adenosylmethionine
    E) Glycine

11. The vitamin needed for the remethylation of homocysteine to methionine:
    A) B_1
    B) B_2
    C) B_{12}
    D) C
    E) K

12. The enzyme that is deficient in phenylketonuria:
    A) Tyrosinase
    B) Homogentisate oxidase
    C) Tyrosine aminotransferase
    D) Fumarylacetoacetate hydrolase
    E) Phenylalanine hydroxylase

13. The characteristic sign of alkaptonuria:
    A) Absence of melanin in the skin and hair
    B) Darkening of the urine in air and ochronosis
    C) A “mousy” odor
    D) Hypoglycemia
    E) Hyperammonemia

14. The enzyme that is deficient in albinism:
    A) Phenylalanine hydroxylase
    B) Homogentisate oxidase
    C) Tyrosine hydroxylase
    D) DOPA decarboxylase
    E) Tyrosinase

15. The color that appears in the Fölling test when phenylpyruvate is present in the urine:
    A) Red
    B) Yellow
    C) Black
    D) Blue-green
    E) Violet

Clinical cases

Case 1

A 3-day-old boy refused to breastfeed and developed vomiting, lethargy and seizures. Blood ammonia is 450 µmol/l (normal < 50), glutamine is increased, citrulline is low, and orotic acid is sharply increased in the urine.

Questions: Which enzyme is deficient? Why is orotic acid increased? State the principles of treatment.

Answer: Ornithine carbamoyltransferase deficiency (hyperammonemia type II; X-linked, which is why it is severe in a boy). Carbamoyl phosphate is not converted into citrulline (citrulline is low), accumulates in the mitochondria, leaves for the cytosol and enters pyrimidine synthesis, increasing orotic acid. Ammonia is not detoxified — in the brain α-ketoglutarate falls, glutamine accumulates (swelling) and encephalopathy develops. Treatment: temporarily stopping protein, glucose (to reduce catabolism), sodium benzoate and phenylbutyrate (which remove nitrogen as hippuric acid and phenylacetylglutamine), arginine and, in severe cases, hemodialysis; later a protein-restricted diet and liver transplantation.

Case 2

Newborn screening found a blood phenylalanine level of 1200 µmol/l (normal < 120). Tyrosine is reduced.

Questions: What is the diagnosis? What are the consequences if it is not treated? What is the essence of the diet?

Answer: Classic phenylketonuria — phenylalanine hydroxylase deficiency. Phenylalanine is not converted into tyrosine, accumulates and is transaminated into phenylpyruvate, phenyllactate and phenylacetate (a “mousy” odor in the urine). A high phenylalanine level blocks the transport of other large neutral amino acids (tyrosine, tryptophan) into the brain, myelination is impaired and the synthesis of neurotransmitters (dopamine, serotonin) falls — if untreated, severe mental retardation and seizures; melanin is low — the skin and hair are pale. Treatment: from the first weeks of life a diet low in phenylalanine (special formulas) with tyrosine supplements; blood phenylalanine is monitored regularly.

Case 3

A 60-year-old vegetarian woman has weakness, numbness in the legs and memory loss. The blood shows megaloblastic anemia, homocysteine 45 μmol/l (normal 5–15) and increased methylmalonic acid.

Questions: Which vitamin is deficient? Why are homocysteine and methylmalonic acid increased?

Answer: Vitamin B_{12} (cobalamin) deficiency (low in the diet of vegetarians — it is found only in animal products). Methylcobalamin is the coenzyme of methionine synthase: when it is lacking, homocysteine is not converted into methionine (hyperhomocysteinemia — a risk of thrombosis and atherosclerosis), N5-methyl-THF accumulates (the "folate trap") and the other forms of folate fall — the synthesis of thymidylate and DNA is impaired (megaloblastic anemia). 5′-deoxyadenosylcobalamin is the coenzyme of methylmalonyl-CoA mutase: when it is lacking, methylmalonic acid accumulates; abnormal fatty acids are incorporated into myelin and damage the nervous system (subacute combined degeneration). Treatment — parenteral B_{12}.

Quick review

Table 13.5. *Key concepts of topic 13*

Concept	Brief definition
Sources of ammonia	Deamination, hydrolysis of amides, MAO, deamination of bases, the intestine
Toxicity	In the brain α-KG ↓ (Krebs cycle ↓), glutamine ↑ (swelling)
Temporary binding	Glutamine synthetase (glutamine), alanine
Kidney	Glutaminase → excretion of NH_4^+; ↑ in acidosis
Ornithine cycle	CPS-I (key) → OCT → ASS → ASL → arginase; in the liver
Nitrogens	1 — NH_3, 2 — aspartate
Energy	3 ATP (4 high-energy bonds)
Activator	N-acetylglutamate
Urea	In blood 2.5–8.3 mmol/l; in urine 20–35 g/day
Hyperammonemia type II	OCT deficiency; orotic acid ↑
THF	One-carbon groups; purines, thymidylate; sulfonamides, methotrexate
SAM	Methyl donor: adrenaline, creatine, choline, DNA methylation
Homocysteine	B_{12}, folate, B_6; its rise — a risk of thrombosis and atherosclerosis
PKU	Phenylalanine hydroxylase (BH_4); screening, diet
Alkaptonuria / albinism	Homogentisate oxidase / tyrosinase

TOPIC 14

Metabolism of nucleotides

Learning objectives

To give the student knowledge of the digestion of nucleoproteins, the functions of nucleotides, the synthesis of purine nucleotides (the de novo and salvage pathways), its regulation and their breakdown, the formation of uric acid, hyperuricemia and gout, the synthesis and breakdown of pyrimidine nucleotides, the biosynthesis of deoxyribonucleotides and thymidylate nucleotides, the disorders of nucleotide metabolism and the drugs that act on these pathways; to master the determination of uric acid in blood and urine.

By the end of the class the student should know:

- the digestion of nucleoproteins and the functions of nucleotides;
- the sources of the atoms of the purine ring and the synthesis of IMP;
- the regulation of purine nucleotide synthesis and the salvage pathway;
- the breakdown of purines to uric acid;
- the types of hyperuricemia, gout and Lesch–Nyhan syndrome;
- the synthesis of pyrimidine nucleotides and orotic aciduria;
- the ribonucleotide reductase and thymidylate synthase reactions;
- the mechanism of action of the antimetabolites (antitumor and antiviral drugs).

The student should be able to:

- determine uric acid in serum and evaluate the result;
- perform the murexide test for uric acid;
- explain the biochemical basis of the treatment of hyperuricemia.

1. Functions of nucleotides. Digestion of nucleoproteins

Nucleotides are compounds consisting of a nitrogenous base (purine: adenine, guanine; pyrimidine: cytosine, uracil, thymine), a pentose (ribose or deoxyribose) and 1–3 phosphate residues.

Functions:

- monomers of the nucleic acids (DNA, RNA);
- **energy** — ATP is the universal energy carrier; GTP (protein synthesis, gluconeogenesis), UTP (glycogen synthesis), CTP (phospholipid synthesis);
- components of **coenzymes** — NAD^+, $NADP^+$, FAD, CoA;
- **second messengers** — cAMP, cGMP;
- **activated intermediates** — UDP-glucose, CDP-choline, S-adenosylmethionine, PAPS;
- **allosteric effectors** — they regulate the activity of many enzymes. **Digestion of nucleoproteins:** in the stomach nucleoproteins are split into the protein part (histones) and the nucleic acids; in the intestine the pancreatic **nucleases** (DNase, RNase) break the nucleic acids down into oligonucleotides, **phosphodiesterases** into mononucleotides, **nucleotidases** into nucleosides, and **nucleosidases (phosphorylases)** into the nitrogenous base and the pentose. The purine and pyrimidine bases that come with food are almost not used for the synthesis of nucleic acids in the body — they are broken down. Therefore nucleotides are **not an essential factor**: they are synthesized from amino acids and other simple substances.

2. Synthesis of purine nucleotides

De novo synthesis

Purine nucleotides are synthesized mainly in the **liver**, in the cytosol, on the basis of **phosphoribosyl pyrophosphate (PRPP)** — the purine ring is "assembled" step by step on ribose-5-phosphate.

Sources of the atoms of the purine ring: glycine (C4, C5, N7), the amide nitrogen of glutamine (N3, N9), aspartate (N1), CO_2 (C6), N10-formyl-THF (C2, C8).

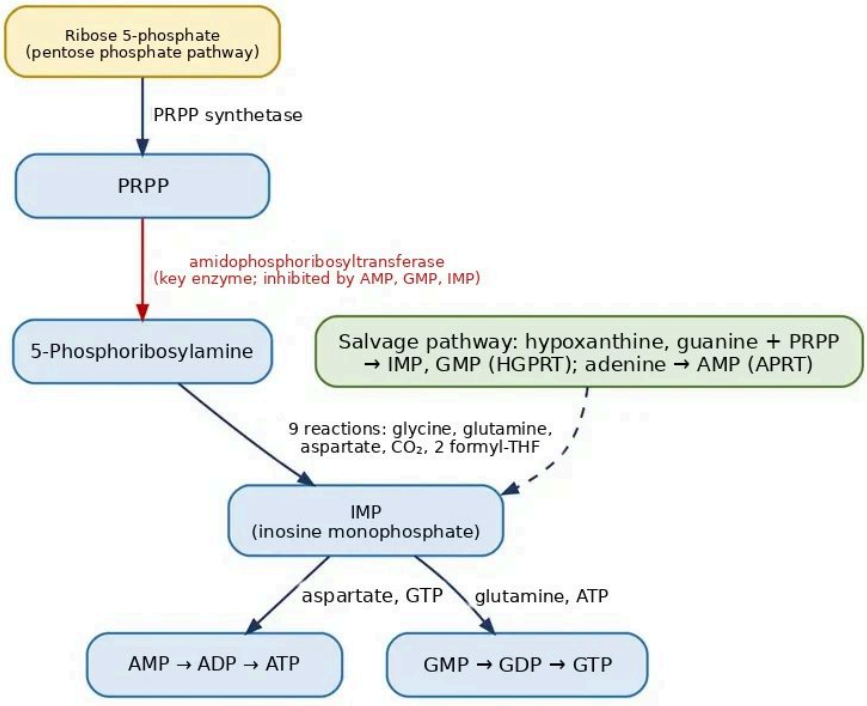


Figure 14.1. *De novo synthesis of purine nucleotides and the salvage pathway (drawn after Severin E.S.)*

1. Ribose-5-phosphate + ATP → **PRPP** (PRPP synthetase).
2. PRPP + glutamine → 5-phosphoribosylamine — **amidophosphoribosyltransferase** (the **key enzyme**).
3. In 9 reactions **inosine monophosphate (IMP)** is formed (6 ATP are used).
4. From IMP: **AMP** (the amino group of aspartate, the energy of GTP) and **GMP** (the amide group of glutamine, the energy of ATP). **Regulation:** PRPP synthetase and amidophosphoribosyltransferase are inhibited by the end products — **AMP, GMP and IMP** (negative feedback), while PRPP activates them. In addition there is **reciprocal regulation**: GTP is needed for the synthesis of AMP and ATP for the synthesis of GMP — this keeps the balance of adenine and guanine nucleotides.

The salvage (reutilization) pathway

The free purine bases formed by the breakdown of nucleotides combine with PRPP and are converted back into nucleotides — energy is saved:

- **hypoxanthine-guanine phosphoribosyltransferase (HGPRT)**: hypoxanthine + PRPP → IMP; guanine + PRPP → GMP;
- **adenine phosphoribosyltransferase (APRT)**: adenine + PRPP → AMP. The salvage pathway is especially important in the **brain** and erythrocytes (where de novo synthesis is low).

CLINICAL SIGNIFICANCE

Lesch–Nyhan syndrome — complete congenital deficiency of **HGPRT** (X-linked recessive, in boys). The salvage pathway does not work: PRPP accumulates and sharply increases de novo synthesis, so purines

are formed in excess and broken down — **severe hyperuricemia**, uric acid stones, gouty arthritis; neurological disorders (choreoathetosis, mental retardation) and **self-injury** (biting the lips and fingers). Allopurinol reduces the hyperuricemia but does not correct the neurological signs.

3. Breakdown of purine nucleotides. Uric acid

Purine nucleotides are converted by nucleotidases, adenosine deaminase, purine nucleoside phosphorylase and guanase into **hypoxanthine** and **xanthine**, and then by **xanthine oxidase** (which contains molybdenum, FAD and Fe; active in the liver and intestine) into **uric acid**.

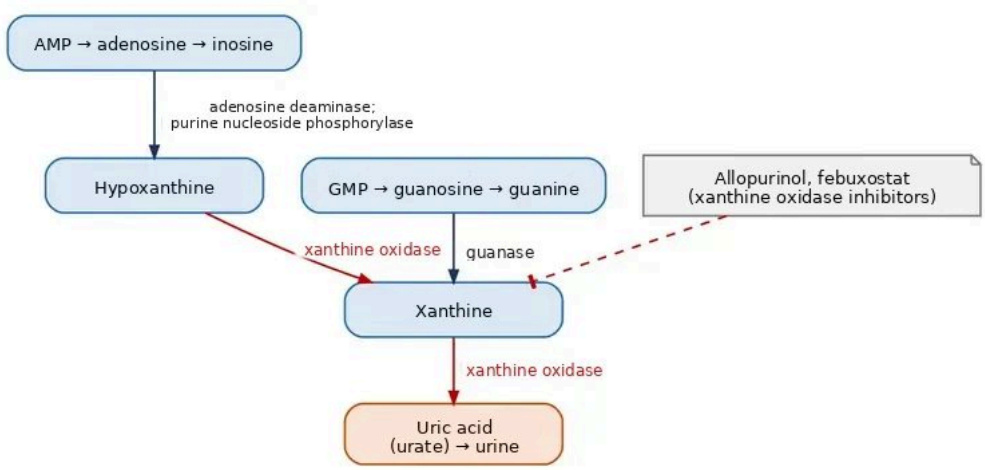


Figure 14.2. *Breakdown of the purine bases to uric acid and the xanthine oxidase inhibitors (drawn after Severin E.S.)*

In humans and other primates uric acid is the **end product** of purine metabolism (in most mammals it is converted by **uricase** into the readily soluble allantoin; humans do not have this enzyme). Uric acid is a weak acid and at physiological pH is in the form of **urate** (sodium urate); it is poorly soluble in water. About 2/3 of it is excreted by the kidneys and 1/3 through the intestine. Uric acid is also a strong **antioxidant**.

Table 14.1. *Normal values of uric acid*

Material	Normal value
Serum: men	200–420 µmol/l
Serum: women	140–360 µmol/l
Children	120–320 µmol/l
Daily urine	1.5–4.5 mmol (≈ 0.25–0.75 g)

Adenosine deaminase deficiency — dATP accumulates in lymphocytes and inhibits ribonucleotide reductase, DNA synthesis stops — **severe combined immunodeficiency** (no T and B lymphocytes); the first gene therapy was used in exactly this disease (topic 18).

4. Hyperuricemia and gout

Hyperuricemia is a rise of uric acid in the blood (> 420 µmol/l in men, > 360 in women).

Primary hyperuricemia (in ≈ 90% of cases — a hereditary reduction of urate excretion by the kidney; less often — increased synthesis: raised PRPP synthetase activity, partial HGPRT deficiency).

Secondary hyperuricemia:

- increased formation — extensive breakdown of cells: leukemias, lymphomas, chemotherapy of tumors (**tumor lysis syndrome**), hemolytic anemia, psoriasis; food rich in purines (meat, liver, kidney, fish, beer), excess fructose, type I glycogenosis;
- reduced excretion — renal failure, thiazide diuretics, low-dose aspirin, lactic acidosis and ketoacidosis (lactate and ketone bodies compete with urate for secretion), alcohol, lead poisoning. **Gout** is a disease in which urate crystals (monosodium urate) are deposited in joints and tissues. **Acute gouty arthritis** — most often the 1st metatarsophalangeal joint of the big toe; the crystals are taken up by neutrophils and inflammatory mediators are released (severe pain, swelling, redness). In chronic gout **tophi** (deposits of urate under the skin), **urate stones** in the kidney and urate nephropathy develop.

Table 14.2. *The biochemical basis of the treatment of hyperuricemia and gout*

Group	Drug	Mechanism
Xanthine oxidase inhibitors	Allopurinol (an analogue of hypoxanthine; alloxanthine — its active metabolite), febuxostat	Reduce the formation of uric acid; hypoxanthine and xanthine (which are more soluble) are excreted; the salvage pathway uses up PRPP and so also reduces de novo synthesis
Uricosuric agents	Probenecid, benzbromarone	Reduce the reabsorption of urate in the renal tubules
Uricase preparations	Rasburicase, pegloticase	Break uric acid down into allantoin (in tumor lysis syndrome)
In an acute attack	Colchicine, NSAIDs, glucocorticoids	Reduce inflammation (migration of neutrophils, tubulin)

Diet: restriction of purine-rich food, alcohol (especially beer) and fructose-containing drinks, drinking plenty of fluid, alkalinization of the urine (urate becomes more soluble).

5. Synthesis and breakdown of pyrimidine nucleotides

The pyrimidine ring is first assembled free and then attached to PRPP:

1. Glutamine + CO_2 + 2ATP → carbamoyl phosphate — **carbamoyl phosphate synthetase II** (in the cytosol; the **key enzyme**; inhibited by UTP, activated by PRPP and ATP). (Unlike carbamoyl phosphate synthetase I of the ornithine cycle: that one is in the mitochondria and uses NH_3.)
2. Carbamoyl phosphate + aspartate → carbamoyl aspartate (aspartate carbamoyltransferase) → dihydroorotate → **orotic acid** (dihydroorotate dehydrogenase — in the mitochondria).
3. Orotate + PRPP → orotidine monophosphate → **UMP** (a bifunctional enzyme — **UMP synthase**: orotate phosphoribosyltransferase and OMP decarboxylase).
4. UMP → UDP → **UTP** → **CTP** (CTP synthetase, the amide group of glutamine). In humans the first three enzymes are combined in one multifunctional protein (CAD).

Breakdown: cytosine and uracil → β-alanine, NH_3, CO_2; thymine → β-aminoisobutyric acid. The products are readily soluble in water and do not cause diseases like hyperuricemia.

CLINICAL SIGNIFICANCE

Hereditary orotic aciduria — UMP synthase deficiency: a lot of **orotic acid** in the urine (crystals), megaloblastic anemia (not enough pyrimidine nucleotides and DNA synthesis; it is not cured by folate or B_{12}),

growth retardation. Treatment — **uridine**: it is converted into UMP and makes up the shortage of nucleotides, and through UTP it inhibits carbamoyl phosphate synthetase II and reduces the formation of orotic acid. Orotic aciduria is also seen in ornithine carbamoyltransferase deficiency (topic 13), but there is hyperammonemia there and no anemia.

6. Synthesis of deoxyribonucleotides and thymidylate nucleotides

Ribonucleotide reductase reduces ribonucleoside **diphosphates** (ADP, GDP, CDP, UDP) to deoxyribonucleoside diphosphates. The reductant is **thioredoxin** (which is regenerated at the expense of thioredoxin reductase and NADPH). The enzyme is regulated allosterically in a complex way: **dATP** is an inhibitor for all substrates and ATP an activator; the various dNTPs change the substrate specificity and keep the balance of deoxyribonucleotides. The activity of the enzyme is highest in the S phase of the cell cycle.

Thymidylate nucleotides: dUMP + N5,N10-methylene-THF → **dTMP** + dihydrofolate — **thymidylate synthase**. Dihydrofolate is converted back into THF by **dihydrofolate reductase** (NADPH). These reactions are vital for rapidly dividing cells (tumors, bone marrow, intestinal epithelium).

Table 14.3. *Drugs that block the synthesis of nucleotides (antimetabolites)*

Drug	Target enzyme or process	Use
Methotrexate	Dihydrofolate reductase	Tumors, rheumatoid arthritis, psoriasis
Trimethoprim	Bacterial dihydrofolate reductase	Antibacterial (with sulfamethoxazole)
5-Fluorouracil	Thymidylate synthase (irreversibly)	Tumors of the colon and the breast
Hydroxyurea	Ribonucleotide reductase	Myeloproliferative diseases, sickle cell anemia
6-Mercaptopurine, 6-thioguanine	Purine synthesis (PRPP amidotransferase, the conversions of IMP)	Leukemias
Mycophenolic acid	IMP dehydrogenase (synthesis of GMP)	Immunosuppressant (in transplantation)
Leflunomide	Dihydroorotate dehydrogenase	Rheumatoid arthritis
Acyclovir, zidovudine	Viral DNA polymerase / reverse transcriptase (chain terminators)	Herpes, HIV

7. Age-related features of nucleotide metabolism

In the first days of life the blood uric acid of newborns is high (a lot of cells, especially erythrocytes, are broken down) and urate crystals may be deposited in the renal tubules — a **uric acid infarct** (a physiological state; the urine is brick-red and it passes within 1–2 weeks). During growth the synthesis of nucleotides is intense. Hyperuricemia is more common in old age and, in women, after the menopause (estrogens increase the excretion of urate).

8. Laboratory work

Experiment 1: determination of uric acid in serum by the uricase method

Principle of the method: uricase oxidizes uric acid to allantoin, CO_2 and **H_2O_2**; peroxidase, at the expense of H_2O_2, oxidizes a chromogen and forms a colored compound (the Trinder reaction, 520 nm). The intensity of the color is proportional to the amount of uric acid.

1. Pour 1 ml of working reagent (uricase, peroxidase, chromogen) into each of three test tubes.
2. Add 25 µl of serum to tube 1, 25 µl of standard (for example, 357 µmol/l) to tube 2 and 25 µl of water to tube 3.
3. Keep at 37 °C for 10 minutes and measure against the control at 520 nm. **Calculation:** C (µmol/l) = (Esample / Estandard) × Cstandard

Evaluation: 200–420 µmol/l in men, 140–360 in women. Ascorbic acid (a strong reductant) can use up the H_2O_2 and falsely lower the result.

Experiment 2: the murexide test for uric acid

1. Put a small amount of uric acid (or scrapings from a urate stone) into a porcelain dish and add 2–3 drops of concentrated nitric acid.
2. Evaporate carefully on a water bath (a red-yellow residue).
3. After cooling, add 1 drop of 10% ammonia solution. **Result and analysis: violet-red** color appears (murexide — the ammonium salt of purpuric acid); with NaOH it is blue-violet. This reaction is characteristic of purines and uric acid; it is used to determine the chemical composition of kidney stones.

Experiment 3: microscopic examination of uric acid crystals in urine (demonstration)

1. Place a drop of the urine sediment (after centrifugation) on a slide.
2. Examine under the microscope. **Result and analysis:** in acidic urine, uric acid crystals are yellow-brown and rhomboid, barrel-shaped or in rosettes; amorphous urates form a brick-red sediment. Large amounts indicate a urate diathesis or a tendency to urate stones.

9. Practical skills

Skill: identifying the cause of hyperuricemia

1. Compare blood uric acid with the normal value for the sex.
2. Assess uric acid excretion in daily urine: high — increased synthesis (tumors, enzyme defect); low — impaired excretion (kidneys, drugs).
3. Assess creatinine and kidney function.
4. Take drugs (diuretics, aspirin), alcohol, diet and chemotherapy into account.
5. Choose the principle of treatment: a xanthine oxidase inhibitor or a uricosuric agent.

Skill: explaining the mechanism of action of an antimetabolite

1. Determine which natural metabolite the structure of the drug resembles.
2. Write down which enzyme is inhibited and which reaction is blocked.
3. Show which nucleotide synthesis stops and how DNA/RNA synthesis is affected.
4. Explain why rapidly dividing cells are damaged more.

5. Name the adverse effects (anemia, leukopenia, mucositis) and the protective measures (with methotrexate — folinic acid, leucovorin).

Review questions

1. List the functions of nucleotides in the body.
2. How are nucleoproteins digested?
3. Which substances supply the atoms of the purine ring?
4. Name the main steps of de novo purine nucleotide synthesis.
5. How is the synthesis of purine nucleotides regulated?
6. What is the salvage pathway and what is its importance?
7. What is the biochemical basis of Lesch–Nyhan syndrome?
8. How are purines broken down? What is the role of xanthine oxidase?
9. What is the normal uric acid level in blood?
10. List the causes of primary and secondary hyperuricemia.
11. Explain the biochemical basis of the treatment of gout.
12. How are pyrimidine nucleotides synthesized?
13. What is orotic aciduria and how is it treated?
14. How are deoxyribonucleotides and dTMP formed?
15. Which drugs affect nucleotide synthesis?

Test questions

1. Which is the key enzyme of purine nucleotide synthesis?
   A) PRPP synthetase
   B) Amidophosphoribosyltransferase
   C) Xanthine oxidase
   D) Carbamoyl phosphate synthetase II
   E) Ribonucleotide reductase

2. From which source are the C2 and C8 atoms of the purine ring taken?
   A) From N10-formyl-tetrahydrofolate
   B) From glycine
   C) From aspartate
   D) From CO_2
   E) From glutamine

3. Which nucleotide is formed first in de novo synthesis of purine nucleotides?
   A) AMP
   B) GMP
   C) IMP
   D) UMP
   E) XMP

4. Which enzyme is deficient in Lesch–Nyhan syndrome?
   A) Hypoxanthine-guanine phosphoribosyltransferase
   B) Xanthine oxidase
   C) Adenosine deaminase
   D) UMP synthase
   E) Uricase

5. What is the final product of purine metabolism in humans?
   A) Allantoin
   B) Uric acid
   C) Urea
   D) Xanthine
   E) β-Alanine

6. What is the mechanism of action of allopurinol?
   A) It reduces urate reabsorption
   B) It breaks uric acid down to allantoin
   C) It inhibits xanthine oxidase
   D) It activates PRPP synthetase
   E) It reduces neutrophil migration

7. What is the normal level of uric acid in blood serum in men?
   A) 2.5–8.3 mmol/l
   B) 50–100 µmol/l
   C) 600–800 µmol/l
   D) 11–35 µmol/l
   E) 200–420 µmol/l

8. Which condition is not a cause of secondary hyperuricemia?
   A) Chemotherapy of tumors
   B) Renal failure
   C) Thiazide diuretics
   D) A purine-free diet
   E) Leukemias

9. Which is the key enzyme of pyrimidine nucleotide synthesis?
   A) Carbamoyl phosphate synthetase I
   B) Amidophosphoribosyltransferase
   C) Ornithine carbamoyltransferase
   D) Carbamoyl phosphate synthetase II
   E) Thymidylate synthase

10. How is hereditary orotic aciduria treated?
    A) With allopurinol
    B) With folic acid
    C) With vitamin B_{12}
    D) With protein restriction
    E) With uridine

11. Which substrates does ribonucleotide reductase reduce?
    A) Nucleoside monophosphates
    B) Nucleoside triphosphates
    C) Free bases
    D) Ribonucleoside diphosphates
    E) Nucleosides

12. Which enzyme does 5-fluorouracil inhibit?
    A) Thymidylate synthase
    B) Dihydrofolate reductase
    C) Ribonucleotide reductase
    D) Xanthine oxidase
    E) IMP dehydrogenase

13. Which enzyme does methotrexate inhibit?
    A) Thymidylate synthase
    B) Dihydrofolate reductase
    C) Ribonucleotide reductase
    D) DNA polymerase
    E) Reverse transcriptase

14. Which qualitative reaction is specific for uric acid?
    A) The biuret reaction
    B) The ninhydrin reaction
    C) The murexide reaction
    D) Trommer's reaction
    E) Legal's reaction

15. Which disease does adenosine deaminase deficiency lead to?
    A) Gout
    B) Orotic aciduria
    C) Megaloblastic anemia
    D) Lesch–Nyhan syndrome
    E) Severe combined immunodeficiency

Clinical cases

Case 1

A 50-year-old man ate a lot of meat and drank beer at a party, and at night woke up with severe pain at the base of his big toe: the joint is swollen and red, and cannot be touched. Uric acid in blood is 610 µmol/l. Needle-shaped crystals were found in the joint fluid.

Questions: What is the diagnosis? Explain the causes of the hyperuricemia and of the attack. State the principles of treatment.

Answer: Acute gouty arthritis. Purine-rich food (meat) and beer (purines + alcohol) increased uric acid formation; in the metabolism of alcohol lactate accumulates and reduces urate secretion in the kidney. When uric acid exceeds the saturation limit, monosodium urate crystals precipitate in cooler peripheral joints; neutrophils engulf them and release inflammatory mediators. During an attack — colchicine, NSAIDs or glucocorticoids; afterwards, long-term — allopurinol (a xanthine oxidase inhibitor), diet, restriction of alcohol, plenty of fluid.

Case 2

In a patient with lymphoma, 2 days after chemotherapy was started the volume of urine fell sharply. Uric acid in blood is 950 µmol/l, potassium and phosphate are increased, calcium is decreased, creatinine is increased.

Questions: What condition is this? Explain its mechanism and the measures for preventing it.

Answer: Tumor lysis syndrome. Under the action of chemotherapy many tumor cells break down and the purines of their nucleic acids are converted into uric acid; intracellular K^+ and phosphate enter the blood, and phosphate binds calcium and causes hypocalcemia. Uric acid crystallizes in the renal tubules and causes acute renal failure. Prevention: allopurinol or rasburicase before chemotherapy (it breaks uric acid down to the much more soluble allantoin), plenty of fluid (hydration), monitoring of electrolytes.

Case 3

A 1-year-old child was found to have megaloblastic anemia and growth retardation. Treatment with folate and B_{12} gave no effect. Orotic acid in the urine is sharply increased, ammonia in blood is normal.

Questions: What disease is this? Why does the anemia not respond to folate and B_{12}? What is the treatment?

Answer: Hereditary orotic aciduria — deficiency of UMP synthase. Orotic acid is not converted into UMP and is excreted in large amounts in the urine; pyrimidine nucleotides (UTP, CTP, dTTP) are lacking, and DNA synthesis is disturbed in rapidly dividing cells (megaloblastic anemia). The disease is caused not by a deficiency of folate or B_{12} but by an enzyme defect, and for that reason vitamins have no effect. The normal ammonia level rules out a disturbance of the ornithine cycle. Treatment — uridine (a source of UMP and, through UTP, a feedback inhibitor of carbamoyl phosphate synthetase II).

Quick review

Table 14.4. *Key concepts of topic 14*

Concept	Brief definition
Functions of nucleotides	Monomers of nucleic acids, ATP, coenzymes, cAMP, activated compounds
The purine ring	Glycine, glutamine, aspartate, CO_2, 2 formyl-THF
Purine synthesis	PRPP → (amidophosphoribosyltransferase — key) → IMP → AMP, GMP
Salvage pathway	HGPRT, APRT; its deficiency — Lesch–Nyhan syndrome
Breakdown	Hypoxanthine → xanthine → uric acid (xanthine oxidase)
Uric acid	Men 200–420, women 140–360 µmol/l
Gout	Monosodium urate crystals; allopurinol, uricosurics, colchicine
Pyrimidine synthesis	CPS-II (key; cytosol) → orotate → UMP → UTP → CTP
Orotic aciduria	UMP synthase; megaloblastic anemia; uridine
Ribonucleotide reductase	NDP → dNDP; thioredoxin; hydroxyurea
Thymidylate synthase	dUMP → dTMP; 5-fluorouracil
Dihydrofolate reductase	DHF → THF; methotrexate, trimethoprim

TOPIC 15

The interrelationship of carbohydrate, lipid and amino acid metabolism

Learning objectives

To give the student an integrated understanding of the general scheme of the catabolism of proteins, fats and carbohydrates, of the metabolism of the nitrogen-free residue of amino acids, of glucogenic and ketogenic amino acids, of the synthesis of glucose from amino acids and glycerol, of the formation of amino acids from carbohydrates and fats, of the role of insulin and cortisol in the regulation of metabolism, of the changes in metabolism in starvation and diabetes mellitus, and of the biochemical basis of the metabolic syndrome.

By the end of the class the student should know:

- the common intermediate products of the catabolism of carbohydrates, lipids and proteins (pyruvate, acetyl-CoA, metabolites of the Krebs cycle);
- glucogenic and ketogenic amino acids;
- which substances can be converted into one another and which cannot (that glucose is not formed from fatty acids);
- the metabolic cooperation between organs in the fed and the fasting state;
- the main metabolic effects of insulin, glucagon, adrenaline and cortisol;
- the changes in metabolism in diabetes mellitus and the metabolic syndrome.

The student should be able to:

- depict the connections between metabolites in the form of a scheme;
- evaluate several blood indices at once (glucose, ketone bodies, urea, lipids) and determine the metabolic state (fed, fasting, diabetes);
- analyze the results of a glucose tolerance test.

1. The general scheme of catabolism

The catabolism of carbohydrates, lipids and proteins proceeds in three stages (topic 5): 1) breakdown of polymers into monomers; 2) conversion of the monomers into a few common intermediate products — **pyruvate, acetyl-CoA**, metabolites of the Krebs cycle; 3) the Krebs cycle and oxidative phosphorylation. **Acetyl-CoA** and the **Krebs cycle** are the “crossroads” of all metabolic pathways.

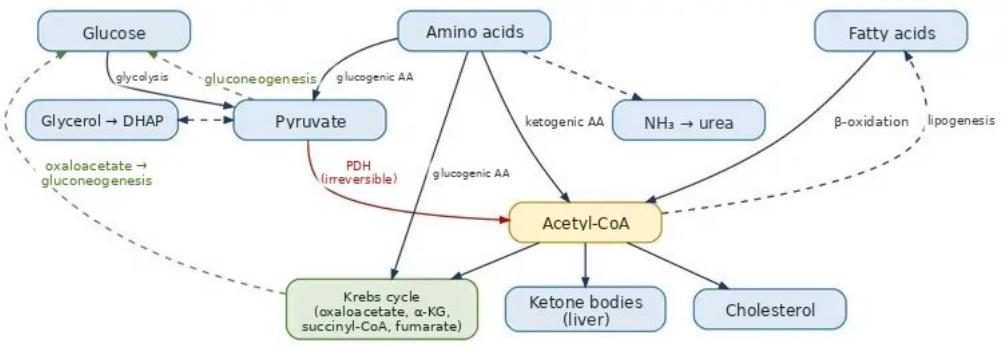


Figure 15.1. *The main points of convergence of carbohydrate, lipid and amino acid metabolism (drawn after Severin E.S.)*

The main points of convergence:

- **glucose-6-phosphate** — glycolysis, synthesis and breakdown of glycogen, the pentose phosphate pathway;
- **dihydroxyacetone phosphate (DHAP)** — the link between glycolysis and the synthesis of triacylglycerols (glycerol-3-phosphate);
- **pyruvate** — glycolysis, gluconeogenesis, alanine metabolism (ALT), formation of acetyl-CoA and oxaloacetate;
- **acetyl-CoA** — synthesis of fatty acids, ketone bodies and cholesterol, and the Krebs cycle;
- **oxaloacetate, α-ketoglutarate** — transamination with aspartate and glutamate, gluconeogenesis. **An important rule: the pyruvate dehydrogenase reaction is irreversible**, and therefore **glucose is not synthesized** from acetyl-CoA (and hence from even-chain **fatty acids**): the 2 carbons of acetyl-CoA leave the Krebs cycle as 2 CO_2, and the net amount of oxaloacetate does not increase. The propionyl-CoA part of odd-chain fatty acids (through succinyl-CoA) and the **glycerol** part of triacylglycerols are converted into glucose.

2. Metabolism of the nitrogen-free residue of amino acids

After deamination the carbon skeleton of amino acids (α-keto acids) is converted into one of seven products: pyruvate, acetyl-CoA, acetoacetyl-CoA, α-ketoglutarate, succinyl-CoA, fumarate, oxaloacetate.

Table 15.1. *Classification of amino acids by their carbon skeleton*

Group	Amino acids	Product formed
Glucogenic	Ala, Gly, Ser, Cys, Thr, Asp, Asn, Glu, Gln, Arg, His, Pro, Met, Val	Pyruvate or metabolites of the Krebs cycle → glucose
Ketogenic (only)	Leucine, lysine	Acetyl-CoA, acetoacetyl-CoA → ketone bodies, fatty acids
Mixed (glucogenic and ketogenic)	Phenylalanine, tyrosine, tryptophan, isoleucine	Products of both types

Glucogenic amino acids serve as substrates of gluconeogenesis in starvation; **ketogenic amino acids** are used for the synthesis of ketone bodies and lipids.

3. Synthesis of glucose from amino acids and glycerol

In starvation and physical exertion **gluconeogenesis** (topic 7) maintains glucose in the blood. The main substrates:

- **amino acids** — muscle proteins break down and reach the liver and the kidney in the form of **alanine** (the glucose-alanine cycle) and **glutamine**; alanine → pyruvate → glucose;
- **glycerol** — from lipolysis in adipose tissue; in the liver glycerol kinase → glycerol-3-phosphate → DHAP → glucose;
- **lactate** — from muscle and erythrocytes (the Cori cycle). Thus, although fatty acids are not converted into glucose, their oxidation supplies the liver with **ATP and NADH** for gluconeogenesis.

4. Biosynthesis of amino acids from carbohydrates and fats

The body can synthesize **nonessential amino acids** from the intermediate products of carbohydrates:

- pyruvate + transamination → **alanine**;
- oxaloacetate → **aspartate** → asparagine;
- α-ketoglutarate → **glutamate** → glutamine, proline, arginine;
- 3-phosphoglycerate → **serine** → glycine, cysteine (with the sulfur of methionine). The nitrogen is taken from other amino acids or from NH_3 (glutamate dehydrogenase). **Essential amino acids** (valine, leucine, isoleucine, threonine, methionine, phenylalanine, tryptophan, lysine; in children also arginine and histidine) are not formed from carbohydrates and fats — they come only from food.

Synthesis of fats from carbohydrates: excess glucose → pyruvate → acetyl-CoA (mitochondria) → into the cytosol in the form of citrate → synthesis of fatty acids; NADPH from the pentose phosphate pathway and from malic enzyme; glycerol-3-phosphate from DHAP. For this reason food rich in carbohydrates leads to obesity.

5. Hormonal regulation of metabolism

Table 15.2. *The main metabolic effects of hormones*

Process	Insulin	Glucagon	Adrenaline	Cortisol
Glycogen synthesis	↑	↓	↓	↑ (in the liver)
Glycogenolysis	↓	↑ (liver)	↑ (liver, muscle)	—
Glycolysis	↑	↓	↑ (muscle)	↓
Gluconeogenesis	↓	↑	↑	↑↑ (synthesis of enzymes)
Lipogenesis	↑	↓	↓	↑ (central)
Lipolysis	↓	↑	↑	↑ (permissive)
Protein synthesis	↑	—	—	↓ (in muscle)
Protein breakdown	↓	↑ (in the liver)	—	↑ (in muscle)
Blood glucose	↓	↑	↑	↑

Insulin is the only hypoglycemic, **anabolic** hormone: through GLUT4 it increases the entry of glucose into muscle and adipose tissue; through phosphoprotein phosphatases it activates glycogen synthase, pyruvate dehydrogenase and acetyl-CoA carboxylase; it induces the synthesis of glucokinase, PFK-1 and pyruvate kinase.

Cortisol is a **catabolic** hormone (muscle proteins are broken down) that induces the synthesis of the enzymes of gluconeogenesis (PEPCK, glucose-6-phosphatase); it is important in prolonged stress and starvation. The ratio of insulin to the contrainsular hormones (the insulin/glucagon index) determines the direction of metabolism.

6. Metabolism in the fed and the fasting state

The absorptive period (2–4 hours after a meal): the insulin/glucagon index is high. The liver converts glucose into glycogen and fatty acids (which it releases in VLDL); muscle stores glucose as glycogen and synthesizes protein; adipose tissue takes the triacylglycerols of chylomicrons and VLDL (lipoprotein lipase) into its stores.

The postabsorptive period and starvation: insulin decreases, and glucagon, adrenaline and cortisol increase.

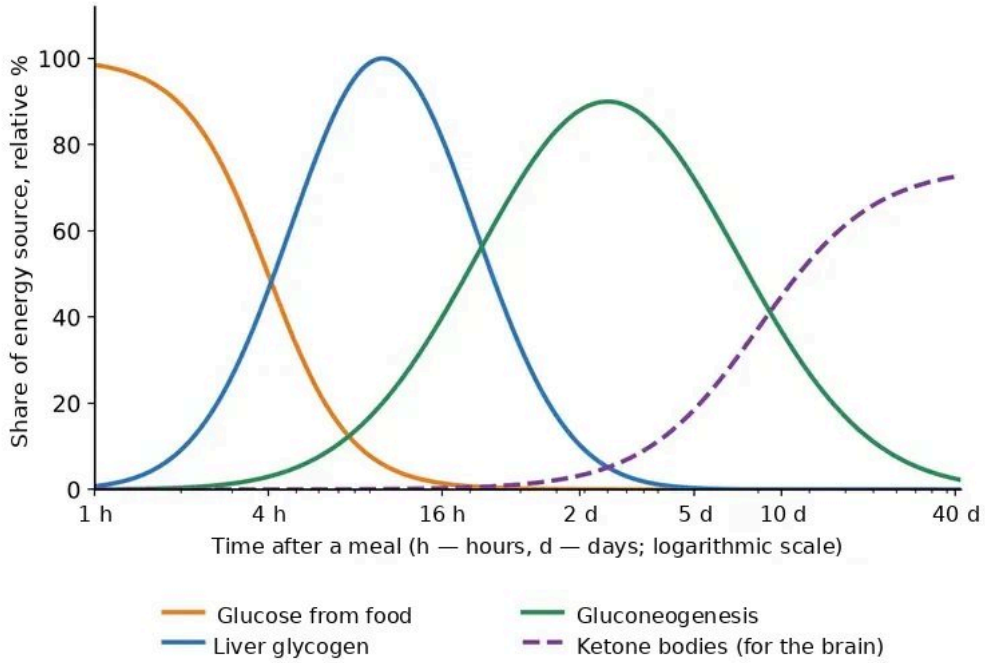


Figure 15.2. *The change of energy sources at the different stages of starvation (drawn after Harper's Illustrated Biochemistry)*

Table 15.3. *Changes in metabolism at the stages of starvation*

Period	Main processes	Blood indices
Day 1	Liver glycogen (used up in ≈ 12–24 hours), gluconeogenesis begins, lipolysis increases	Glucose at the lower limit of normal, free fatty acids ↑
Days 2–5	Gluconeogenesis is maximal (muscle proteins → alanine, glutamine), ketone bodies ↑	Excretion of urea ↑, ketone bodies ↑
After 1 week (prolonged starvation)	The brain obtains ≈ 2/3 of its energy from ketone bodies; protein breakdown decreases (proteins are spared); renal gluconeogenesis ↑	Ketone bodies 5–7 mmol/l, urea decreases, excretion of NH_4^+ ↑

In prolonged starvation life expectancy depends mainly on the **fat store**; death occurs when ≈ 1/3–1/2 of the proteins (muscle, and then the vital organs) have been broken down.

7. Disturbance of metabolism in diabetes mellitus

Diabetes mellitus is chronic hyperglycemia resulting from insulin deficiency (type 1 — absolute, autoimmune damage of the β-cells) or from relative deficiency with resistance to insulin (type 2). In the absence of insulin the body works “in a state of starvation”, even though there is plenty of glucose in the blood.

Table 15.4. *Changes in metabolism in diabetes mellitus*

Metabolism	Change	Result
Carbohydrates	Entry of glucose into tissues (GLUT4) ↓, synthesis of glycogen ↓, glycogenolysis and gluconeogenesis ↑	Hyperglycemia, glucosuria (when the renal threshold of ≈ 10 mmol/l is exceeded), polyuria, polydipsia
Lipids	Lipolysis ↑, free fatty acids ↑, β-oxidation ↑, ketone bodies ↑; VLDL ↑ (in type 2)	Ketonemia, ketonuria, ketoacidosis; dyslipidemia, atherosclerosis
Proteins	Protein synthesis ↓, breakdown ↑	Negative nitrogen balance, urea ↑, wasting, slow healing of wounds

Metabolism	Change	Result
Water and electrolytes	Osmotic diuresis	Dehydration, loss of K^+

Mechanisms of the late complications: nonenzymatic glycosylation of proteins (HbA_{1c}, collagen, the basement membrane — microangiopathy, nephropathy, retinopathy); the **polyol pathway** — aldose reductase converts glucose into sorbitol (the lens of the eye, the nerves — cataract, neuropathy); oxidative stress.

> **CLINICAL SIGNIFICANCE**
> **HbA_{1c}** (glycated hemoglobin) reflects the mean glycemia of the last 2–3 months (erythrocyte lifespan ≈ 120 days): normal < 6.0%; ≥ 6.5% — diagnostic criterion of diabetes mellitus. **Fructosamine** (glycated albumin) reflects the last 2–3 weeks.

8. Metabolic syndrome

Metabolic syndrome is a cluster of interrelated factors that develops on the basis of insulin resistance: **abdominal obesity** (waist circumference: men > 94 cm, women > 80 cm), **arterial hypertension** (≥ 130/85 mm Hg), **hyperglycemia** (fasting ≥ 5.6 mmol/l), **hypertriglyceridemia** (≥ 1.7 mmol/l), **decreased HDL cholesterol** (men < 1.0, women < 1.3 mmol/l). The diagnosis is made if three of them are present.

Biochemical basis: visceral adipose tissue releases many free fatty acids and inflammatory cytokines (TNF-α, IL-6), and adiponectin decreases → insulin signaling in muscle and liver is impaired (IRS-1 phosphorylation) → compensatory hyperinsulinemia. In the liver VLDL synthesis increases (hypertriglyceridemia, small dense LDL) and HDL decreases. Hyperinsulinemia increases renal Na^+ reabsorption and sympathetic activity (hypertension). Complications: type 2 diabetes mellitus, atherosclerosis, coronary heart disease, nonalcoholic fatty liver disease, gout, polycystic ovary syndrome.

9. Laboratory work

Experiment 1: glucose tolerance test (GTT)

Principle of the method: glucose is measured in blood in the fasting state and after a glucose load, which allows the body's ability to utilize glucose to be assessed.

1. After the subject has fasted for 8–14 hours, determine glucose in capillary or venous blood (glucose oxidase method, topic 7).
2. Dissolve 75 g of anhydrous glucose in 250–300 ml of water and drink it within 5 minutes (for children 1.75 g/kg, at most 75 g).
3. After 2 hours (for teaching purposes — every 30 minutes) determine glucose again; plot the glycemic curve.

Table 15.5. *Interpretation of GTT results (venous plasma, mmol/l)*

Condition	Fasting	After 2 hours
Normal value	< 6,1	< 7,8
Impaired fasting glycemia	6,1–6,9	< 7,8
Impaired glucose tolerance	< 7,0	7,8–11,0

Condition	Fasting	After 2 hours
Diabetes mellitus	≥ 7,0	≥ 11,1

Analysis: normally glucose peaks at 30–60 minutes (1.5–1.7 times the initial level) and returns to the initial level by 2 hours. In diabetes the peak is high and the decline is slow.

Experiment 2: determination of glucose and ketone bodies in urine (test strips)

1. Dip the test strip into fresh urine for 1–2 seconds and blot off the excess urine against the edge of the container.
2. After 30–60 seconds compare the color with the scale on the container. **Results and analysis:** the glucose zone — glucose oxidase–peroxidase reaction; the ketone zone — acetoacetate and acetone give a violet color with sodium nitroprusside (based on Legal's reaction; it does not detect β-hydroxybutyrate). Simultaneous glucosuria and ketonuria is a sign of decompensated diabetes; ketonuria alone — starvation, vomiting, ketogenic diet.

10. Practical skills

Skill: determining the metabolic state from laboratory parameters

1. Assess blood glucose, ketone bodies, free fatty acids, triacylglycerols, urea and insulin.
2. Fed state: glucose ↑ (≤ 7.8), insulin ↑, ketone bodies and free fatty acids ↓.
3. Starvation: glucose at the lower limit of normal, insulin ↓, ketone bodies ↑ (moderately), urea ↑ (days 2–5).
4. Decompensated type 1 diabetes: glucose ↑↑, insulin ↓↓, ketone bodies ↑↑, pH ↓.
5. Type 2 diabetes / metabolic syndrome: glucose ↑, insulin ↑ (resistance), triacylglycerols ↑, HDL ↓.

Skill: showing the further fate of a metabolite in a scheme

1. Choose a substance (for example, alanine) and write down its first transformation (ALT → pyruvate).
2. Show the fate of pyruvate in the fed state (acetyl-CoA → fatty acids, Krebs cycle) and in starvation (pyruvate carboxylase → gluconeogenesis).
3. Mark the key enzymes and the hormones that regulate them.
4. Mark the irreversible reactions (PDH) separately.

Review questions

1. Name the three stages of catabolism and the common intermediate products.
2. Through which metabolites are carbohydrate and lipid metabolism linked?
3. Why is glucose not synthesized from fatty acids?
4. Which part of triacylglycerols is converted to glucose?
5. Give examples of glucogenic and ketogenic amino acids.
6. Which amino acids are exclusively ketogenic?
7. How does the glucose–alanine cycle work?
8. Which amino acids are synthesized from carbohydrates?
9. How are fats synthesized from carbohydrates?

10. List the metabolic effects of insulin.
11. How does cortisol affect metabolism?
12. What processes take place in the liver, muscle and adipose tissue in the absorptive period?
13. How does metabolism change at different stages of starvation?
14. How are carbohydrate, lipid and protein metabolism disturbed in diabetes mellitus?
15. What are the criteria and the biochemical basis of the metabolic syndrome?

Test questions

1. What is the common intermediate product of the catabolism of carbohydrates, lipids and amino acids?
   A) Acetyl-CoA
   B) Glucose-6-phosphate
   C) Malonyl-CoA
   D) Carbamoyl phosphate
   E) Lactate

2. Why is glucose not synthesized from even-chain fatty acids?
   A) Fatty acids do not enter the liver
   B) The pyruvate dehydrogenase reaction is irreversible
   C) Acetyl-CoA does not leave the mitochondrion
   D) There is not enough ATP for gluconeogenesis
   E) Insulin stops this process

3. The amino acids that are purely ketogenic:
   A) Leucine and lysine
   B) Alanine and serine
   C) Glutamate and aspartate
   D) Valine and threonine
   E) Histidine and proline

4. Which part of triacylglycerols is a substrate of gluconeogenesis?
   A) Palmitic acid
   B) Oleic acid
   C) Acetyl-CoA
   D) Malonyl-CoA
   E) Glycerol

5. In the glucose–alanine cycle, to which organ is alanine transported from muscle?
   A) To the kidney
   B) To the brain
   C) To the liver
   D) To adipose tissue
   E) To the intestine

6. Which amino acid is not synthesized from intermediates of carbohydrate metabolism?
    A) Alanine
    B) Aspartate
    C) Glutamate
    D) Serine
    E) Leucine

7. Which process does insulin enhance?
    A) Glycogenolysis
    B) Gluconeogenesis
    C) Lipolysis
    D) Glycogen synthesis
    E) Ketogenesis

8. What is the main metabolic effect of cortisol?
    A) Induces the synthesis of gluconeogenic enzymes
    B) Enhances protein synthesis in muscle
    C) Brings GLUT4 to the membrane
    D) Stops lipogenesis
    E) Lowers blood glucose

9. After about how long is liver glycogen depleted in starvation?
    A) After 1 hour
    B) After 3–4 hours
    C) After 5 days
    D) After 2 weeks
    E) After 12–24 hours

10. In prolonged starvation (after 1 week), what supplies most of the brain's energy?
    A) Glucose
    B) Fatty acids
    C) Amino acids
    D) Ketone bodies
    E) Lactate

11. Which change is not observed in decompensated type 1 diabetes?
    A) Hyperglycemia
    B) Ketonemia
    C) Positive nitrogen balance
    D) Glucosuria
    E) Metabolic acidosis

12. HbA_{1c} reflects the mean glycemia of which period?
    A) The last 1 day
    B) The last 1 week
    C) The last 2–3 months
    D) The last 1 year
    E) Only at the time of the test

13. In the glucose tolerance test, which result is consistent with diabetes mellitus?
    A) 6.5 mmol/l after 2 hours
    B) 12.0 mmol/l after 2 hours
    C) 5.0 mmol/l fasting
    D) 4.2 mmol/l fasting
    E) 7.0 mmol/l after 2 hours

14. Which parameter is not included in the criteria of the metabolic syndrome?
    A) Abdominal obesity
    B) Hyperuricemia
    C) Hypertriglyceridemia
    D) Decreased HDL
    E) Arterial hypertension

15. In diabetes, under the action of which enzyme does sorbitol accumulate in the lens of the eye?
    A) Hexokinase
    B) Glucokinase
    C) Sorbitol dehydrogenase
    D) Aldose reductase
    E) Glucose-6-phosphatase

Clinical cases

Case 1

A healthy male volunteer, under a physician's supervision, underwent therapeutic fasting and drank only water. Blood was drawn on day 3 and in week 3. Day 3: glucose 3.9 mmol/l, ketone bodies 2.5 mmol/l, a lot of urea in the urine. Week 3: glucose 3.6 mmol/l, ketone bodies 6.5 mmol/l, urinary urea decreased, ammonium salts increased.

Questions: Why is there a lot of urea in the urine on day 3 and little in week 3? What are the main sources of glucose? Why did ammonium excretion increase?

Answer: On day 3 liver glycogen is depleted and glucose comes mainly from gluconeogenesis — muscle proteins are broken down, and alanine and glutamine go to the liver; their nitrogen is converted to urea (a lot of urea). In week 3 the brain obtains a large part of its energy from ketone bodies, the need for glucose decreases, and protein breakdown slows down (proteins are spared) — urea decreases. Sources of glucose: amino acids, glycerol (lipolysis), lactate. To compensate for ketoacidosis, glutaminase is activated in the kidney: NH_3 excretes H^+ as NH_4^+, and the carbon skeleton of glutamine is used in renal gluconeogenesis.

Case 2

A 16-year-old girl has been drinking a lot of water and urinating a lot for a month and has lost weight; today she has abdominal pain, vomiting, deep noisy breathing (Kussmaul) and an acetone smell on the breath. Blood glucose 24 mmol/l, pH 7.15, ketone bodies 9 mmol/l; the urine contains a lot of glucose and ketone bodies.

Questions: What is the diagnosis? How are carbohydrate, lipid and protein metabolism disturbed? Why has the patient lost weight?

Answer: Type 1 diabetes mellitus, diabetic ketoacidosis. Without insulin, glucose does not enter muscle and adipose tissue, and glycogenolysis and gluconeogenesis are enhanced in the liver

(hyperglycemia, glucosuria, osmotic diuresis — polyuria, polydipsia, dehydration). Lipolysis is enhanced and free fatty acids undergo β-oxidation in the liver; because oxaloacetate is used for gluconeogenesis, acetyl-CoA is converted to ketone bodies — ketoacidosis (Kussmaul breathing — compensation by exhaling CO_2). Protein synthesis decreases and breakdown increases (negative nitrogen balance). The weight loss is due to the breakdown of adipose and muscle tissue and the loss of glucose in the urine (loss of energy).

Case 3

A 52-year-old man: height 172 cm, weight 98 kg, waist circumference 112 cm, BP 150/95 mm Hg. Fasting glucose 6.4 mmol/l, triacylglycerols 3.1 mmol/l, HDL cholesterol 0.8 mmol/l, insulin elevated, uric acid 480 µmol/l.

Questions: What condition is this? Explain the biochemical mechanism. What complications are at risk?

Answer: Metabolic syndrome (abdominal obesity, hypertension, impaired fasting glycemia, hypertriglyceridemia, decreased HDL — 5 criteria). Visceral adipose tissue releases many free fatty acids and inflammatory cytokines, and insulin signaling in muscle and liver is impaired — insulin resistance and compensatory hyperinsulinemia. In the liver VLDL synthesis increases (triacylglycerols ↑) and HDL decreases. Hyperinsulinemia enhances renal reabsorption of Na^+ and urate (hypertension, hyperuricemia). Complications: type 2 diabetes mellitus, atherosclerosis, myocardial infarction, stroke, fatty liver disease, gout. The basis of treatment is weight loss, physical activity and diet.

Quick review

Table 15.6. *Key concepts of topic 15*

Concept	Brief definition
Junction points	Glucose-6-phosphate, DHAP, pyruvate, acetyl-CoA, metabolites of the Krebs cycle
Fatty acid → glucose	No: PDH is irreversible; glycerol and propionyl-CoA — yes
Ketogenic amino acids	Leucine, lysine (only); Phe, Tyr, Trp, Ile — mixed
Substrates of gluconeogenesis	Amino acids (alanine, glutamine), glycerol, lactate
Amino acids from carbohydrates	Alanine, aspartate, glutamate, serine, glycine (nonessential ones)
Insulin	Anabolic: glycogen, lipogenesis, protein synthesis ↑; glucose ↓
Cortisol	Gluconeogenic enzymes ↑, breakdown of muscle protein ↑
Starvation, day 1	Glycogen is depleted, gluconeogenesis begins
Prolonged starvation	The brain uses ketone bodies, proteins are spared
Diabetes	Hyperglycemia, ketoacidosis, negative nitrogen balance
HbA_{1c}	Glycemia over 2–3 months; ≥ 6.5% — diabetes
Metabolic syndrome	Insulin resistance + obesity, HTN, TAG ↑, HDL ↓, glucose ↑

TOPIC 16

Molecular biology. DNA structure, replication, transcription

Learning objectives

To give the student knowledge of the structure of nucleic acids (the Watson–Crick model), the role of DNA in heredity, the genetic system of mitochondria, DNA hybridization, replication and its relation to the cell cycle, post-replicative modifications (methylation, telomeres), transcription and post-transcriptional modifications of RNA, the organization of prokaryotic and eukaryotic genes, recombination, transposons and plasmids, and the use of PCR and recombinant DNA in medicine.

By the end of the class the student should know:

- the primary, secondary and tertiary structure of DNA and RNA and Chargaff's rules;
- the structure of chromatin (nucleosome, histones);
- the principles of replication (semiconservative, antiparallel, discontinuous) and its enzymes;
- the importance of telomerase and DNA methylation;
- the stages of transcription and the eukaryotic RNA polymerases;
- the post-transcriptional modifications of mRNA, rRNA and tRNA (capping, polyadenylation, splicing);
- the principle of PCR and the basics of molecular diagnostics;
- the drugs that block replication and transcription.

The student should be able to:

- isolate nucleoproteins from tissue and perform qualitative reactions for their components;
- write the complementary strand and the mRNA from a DNA sequence;
- interpret PCR results (gel electropherogram).

1. Structure of nucleic acids

Nucleic acids are polymers formed by the joining of mononucleotides through 3’,5’-**phosphodiester bonds**. The 5’-end of the chain carries a phosphate and the 3’-end a free OH group; the sequence is written in the 5’ → 3’ direction.

Table 16.1. *Differences between DNA and RNA*

Characteristic	DNA	RNA
Pentose	2-Deoxyribose	Ribose
Pyrimidine bases	Cytosine, thymine	Cytosine, uracil
Structure	Double helix	Usually single-stranded (internal “hairpins”)
Location	Nucleus, mitochondria	Nucleus, cytoplasm, ribosomes
Function	Storage and transmission of genetic information	Implementation of information (mRNA, tRNA, rRNA), regulation (microRNA)
Alkali resistance	Resistant	Hydrolyzed in alkali because of the 2’-OH

Chargaff's rules: in DNA A = T and G = C; the sum of purines equals the sum of pyrimidines (A + G = T + C); the (G + C)/(A + T) ratio is species-specific.

Watson–Crick model (1953, B-form): two **antiparallel** polynucleotide chains form a right-handed double helix twisted around a common axis; the bases are inside and the sugar-phosphate backbone is outside. The bases pair **complementarily**: A=T (2 hydrogen bonds), G≡C (3 hydrogen bonds). One turn of the helix is ≈ 10 base pairs (3.4 nm), the diameter is 2 nm; the major and minor grooves are important for the binding of proteins to DNA. The helix is stabilized by hydrogen bonds and by the **stacking interactions** of the bases.

Tertiary structure and chromatin: DNA winds 1.7 times around a histone octamer (H2A, H2B, H3, H4 — 2 of each) with ≈ 146 base pairs to form a **nucleosome**; the linker histone H_1 assembles nucleosomes into a 30 nm fibril; then come loops and metaphase chromosomes. **Euchromatin** is loosely packed and actively transcribed; **heterochromatin** is dense and inactive.

Denaturation and hybridization: heating (or alkali) breaks the hydrogen bonds and separates the double helix — the **melting temperature (T_m)** is higher when there are many G≡C pairs; on denaturation the absorbance at 260 nm increases (**hyperchromic effect**). On cooling, complementary strands reassociate (renaturation). The joining of complementary strands from different sources is **hybridization** (Southern blot, FISH, DNA microarrays, PCR-based methods).

Genetic system of mitochondria: mtDNA is circular, 16,569 base pairs, 37 genes (13 respiratory chain proteins, 22 tRNAs, 2 rRNAs); it is inherited only **through the mother**; repair is weak and mutations are frequent — mitochondrial diseases (MELAS, Leber optic atrophy), aging.

2. Organization of the gene and the genome

A **gene** is a segment of DNA that carries information about a functional product (a polypeptide or an RNA). A eukaryotic gene consists of a **promoter** (TATA box, CAAT box), **exons** (coding regions) and **introns** (noncoding regions removed in splicing), **enhancers** and silencers.

Table 16.2. *Differences between prokaryotic and eukaryotic genomes*

Characteristic	Prokaryotes	Eukaryotes
DNA	Circular, one chromosome, no histones	Linear, many chromosomes, with histones
Genes	No introns, operons (polycistronic mRNA)	Intron–exon structure, monocistronic mRNA
Noncoding DNA	Little	A lot (≈ 98%), repetitive sequences
Transcription and translation	Simultaneous, in the cytoplasm	Transcription in the nucleus, translation in the cytoplasm
Extrachromosomal DNA	Plasmids	Mitochondrial DNA

The human genome is ≈ 3.1×10^9 base pairs (haploid), with ≈ 20,000 protein-coding genes.

Elements of genetic variability:

- **recombination** — exchange of DNA segments (crossing-over in meiosis, V(D)J recombination of immunoglobulin genes);
- **transposons** (“jumping genes”) — DNA segments that change their position within the genome; in humans, LINE and Alu elements;
- **plasmids** — extrachromosomal circular DNA of bacteria; they carry **antibiotic resistance genes**; vectors in genetic engineering;
- **bacteriophages** — viruses of bacteria; transduction, vectors in genetic engineering.

3. DNA replication

Replication is the copying of DNA; it takes place in the **S phase** of the cell cycle.

Principles:

- **semiconservative** — each new molecule contains one old and one new strand (the Meselson–Stahl experiment);
- **template-directed and complementary**;
- **antiparallel** — the new strand is elongated only in the 5’→3’ direction;
- **discontinuous (semidiscontinuous)** — the leading strand is synthesized continuously, the lagging strand as **Okazaki fragments**;
- **requires a primer** — DNA polymerases cannot start synthesis; they only add nucleotides to an existing 3’-OH;
- in eukaryotes there are many **origins of replication (ori)** — replicons.

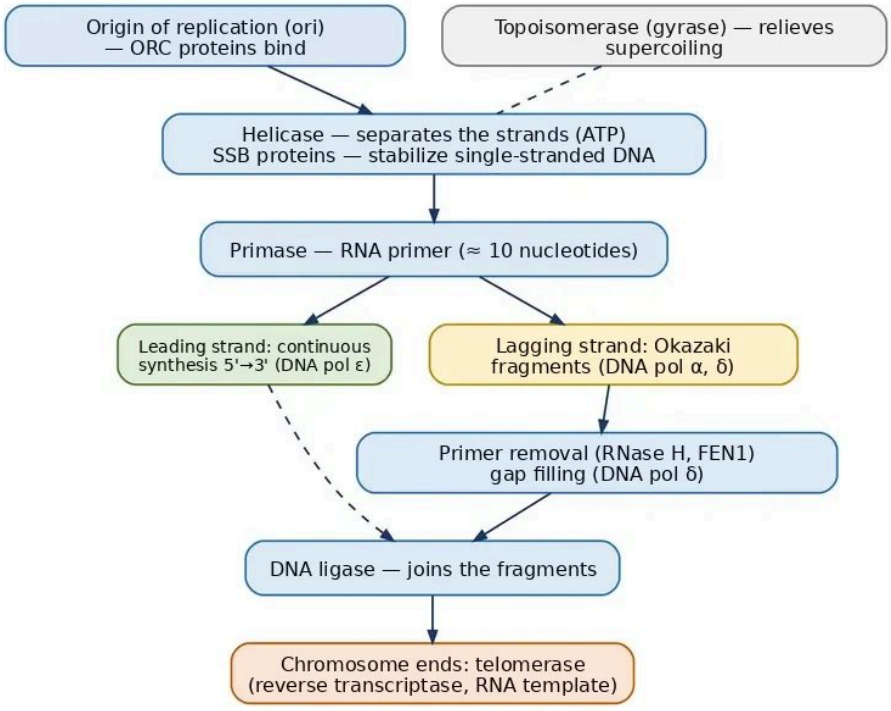


Figure 16.1. *The main enzymes and processes at the replication fork (drawn after Severin E.S.)*

The substrates are **dNTPs** (dATP, dGTP, dCTP, dTTP); pyrophosphate is released and then hydrolyzed, which makes the process irreversible.

Table 16.3. *Eukaryotic DNA polymerases*

Polymerase	Function
α (with primase)	Primer synthesis, initiation of Okazaki fragments
δ	Synthesis of the lagging strand; 3’→5’ exonuclease (proofreading)
ε	Synthesis of the leading strand; 3’→5’ exonuclease (proofreading)
β	Repair (base excision pathway)
γ	Replication of mitochondrial DNA

Fidelity: the **3’→5’ exonuclease (proofreading)** activity of DNA polymerases immediately removes a wrongly inserted nucleotide; together with mismatch repair, the error rate is ≈ 10^{-9}–10^{-10}.

Relation to the cell cycle: G_1 → S → G_2 → M; the transitions are controlled by **cyclins** and **cyclin-dependent kinases (CDKs)**; at the checkpoints the **p53** and **Rb** proteins arrest the cycle when DNA is damaged (topics 18–19). Replication licensing is granted only once per cycle.

4. Post-replicative modifications: methylation, telomeres

DNA methylation: DNA methyltransferases (donor — S-adenosylmethionine) convert cytosine in **CpG** dinucleotides to 5-methylcytosine. Methylation of a promoter **silences** the gene — epigenetic regulation (X-chromosome inactivation, genomic imprinting; silencing of tumor suppressor genes by methylation in tumors).

Telomeres and telomerase: the primer site at the 3’-end of linear DNA is not filled in, so the chromosome shortens with every division. The ends carry **telomeres** — (TTAGGG)n repeats. **Telomerase** is a **reverse transcriptase** with its own **RNA template**; it lengthens telomeres. It is active in germ cells, stem cells and **≈ 85–90% of tumor cells** (immortality), while its activity in somatic cells is low — telomere shortening is associated with **aging** (the Hayflick limit).

5. Transcription

Transcription is the synthesis of RNA on a DNA template. One DNA strand (antisense, noncoding) serves as the template; the RNA is complementary to it and similar to the coding strand (with U instead of T); synthesis proceeds in the 5’→3’ direction, **no primer is required**, and the substrates are NTPs.

Table 16.4. *Eukaryotic RNA polymerases*

Enzyme	Location	Products	Sensitivity to α-amanitin
RNA polymerase I	Nucleolus	28S, 18S, 5.8S rRNA	Not sensitive
RNA polymerase II	Nucleoplasm	mRNA (pre-mRNA), most microRNAs	Highly sensitive
RNA polymerase III	Nucleoplasm	tRNA, 5S rRNA	Moderate

Stages:

1. **Initiation** — the general transcription factors (TFIID binds to the TATA box) and RNA polymerase II bind to the **promoter** and open the DNA over ≈ 14 base pairs; TFIIH phosphorylates the CTD domain of the polymerase.
2. **Elongation** — the polymerase moves along the DNA and elongates the RNA chain (≈ 40 nucleotides/s).
3. **Termination** — synthesis stops at termination signals (in prokaryotes, the rho factor or a GC-rich hairpin). Bacteria have a single RNA polymerase (the σ subunit recognizes the promoter) — **rifampicin** inhibits this enzyme.

6. Post-transcriptional modifications of RNA (processing)

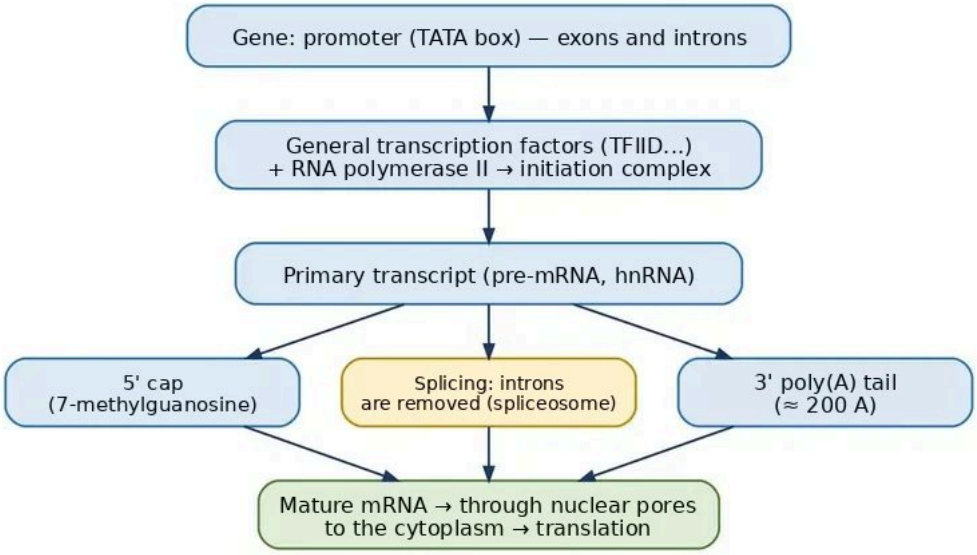


Figure 16.2. *Formation and maturation of eukaryotic mRNA (drawn after Lippincott's Illustrated Reviews: Biochemistry)*

Maturation of mRNA:

- **5'-cap** — 7-methylguanosine attached by a 5'-5'-triphosphate bond; it protects the mRNA from exonucleases and is required for ribosome binding;
- **3'-polyadenylation** — after cleavage downstream of the AAUAAA signal, poly(A) polymerase adds ≈ 200 adenylate residues; stability and export;
- **splicing** — removal of introns and joining of exons; the **spliceosome** (small nuclear RNAs — snRNAs — and proteins) recognizes the GU…AG boundaries. **Alternative splicing** — several proteins are produced from one gene (for example, calcitonin and CGRP);
- **RNA editing** — changes of individual nucleotides (apo B-100 in the liver, apo B-48 in the intestine — C→U editing, topic 8). **Maturation of rRNA:** the 45S precursor is cleaved in the nucleolus to form 28S, 18S and 5.8S rRNAs; nucleotides are methylated.

Maturation of tRNA: the 5'- and 3'-ends are trimmed, **CCA** is added to the 3'-end (the amino acid attachment site), bases are modified (pseudouridine, dihydrouridine, inosine), and in some an intron is removed.

> **CLINICAL SIGNIFICANCE**
> Splicing defects: many forms of β-thalassemia — mutations in an intron of the β-globin gene lead to aberrant splicing. In systemic lupus erythematosus (SLE), autoantibodies against snRNPs (anti-Sm) are produced.

7. Inhibitors of replication and transcription

Table 16.5. *Substances that affect template-directed syntheses*

Substance	Mechanism	Use
Fluoroquinolones (ciprofloxacin)	Inhibitor of bacterial DNA gyrase (topoisomerase II)	Antibacterial
Doxorubicin, etoposide	Topoisomerase II inhibitor, intercalation	Antitumor

Substance	Mechanism	Use
Irinotecan, topotecan	Topoisomerase I inhibitor	Antitumor
Cisplatin, alkylating agents	Cross-links between DNA strands	Antitumor
Acyclovir, zidovudine	Chain terminators (no 3’-OH)	Antiviral
Rifampicin	Bacterial RNA polymerase	Tuberculosis, antibacterial
Actinomycin D	Intercalates into DNA, stops transcription	Antitumor
α-Amanitin	Eukaryotic RNA polymerase II	Death cap toxin — liver necrosis

8. Use of recombinant DNA and PCR in medicine

Recombinant DNA technology: restriction enzymes (bacterial endonucleases that cut DNA at specific palindromic sequences) are used to cut out a gene, which is joined into a **vector** (plasmid, virus) with **DNA ligase** and introduced into bacteria, yeast or animal cells. This is how the following are obtained: human **insulin**, growth hormone, erythropoietin, interferons, clotting factors (VIII), hepatitis B vaccine, monoclonal antibodies.

Polymerase chain reaction (PCR): in vitro amplification of a specific DNA segment. Each cycle: **denaturation** (94–95 °C) → **primer binding (annealing)** (50–65 °C) → **elongation** (72 °C, thermostable **Taq polymerase**). In 30–35 cycles the number of copies increases ≈ 2^n times (billions). The result is detected by agarose gel electrophoresis or in real time (fluorescent probes).

PCR-based diagnosis: infections (SARS-CoV-2, HIV, hepatitis B and C viral load, mycobacteria of tuberculosis), hereditary diseases (mutations in cystic fibrosis and phenylketonuria), tumors (BCR-ABL), personal identification and paternity testing (forensic medicine), prenatal diagnosis. **Reverse transcription PCR (RT-PCR)** detects RNA viruses.

> **REMEMBER**
> Replication and transcription are “DNA → DNA” and “DNA → RNA”. Main difference: replication requires a **primer** and its substrate is **dNTP**; transcription does not need a primer and its substrate is **NTP**.

9. Laboratory work

Experiment 1: isolation and hydrolysis of nucleoproteins from yeast

Principle of the method: nucleoproteins dissolve in weak alkali and precipitate when acid is added; on acid hydrolysis they break down into protein, purine bases, pentose and phosphoric acid.

1. Grind 5 g of yeast with 0.4% NaOH solution, stir for 20 minutes and centrifuge.
2. Add acetic acid to the supernatant — the nucleoproteins precipitate as white flakes.
3. Hydrolyze the precipitate by boiling with 10% sulfuric acid for 1 hour, cool and filter.

Experiment 2: qualitative reactions for the components of the hydrolysate

1. **Protein part** — the biuret reaction: a violet color when NaOH and $CuSO_4$ are added to the hydrolysate.
2. **Purine bases** — the silver test: when ammoniacal $AgNO_3$ is added, the silver salts of the purine bases form a light brown precipitate.

3. **Pentose** — the orcinol (Bial) reaction: a **green** color on heating with HCl and $FeCl_3$ (ribose); the **diphenylamine** reaction — a **blue** color (deoxyribose, DNA).
4. **Phosphoric acid** — the molybdate reagent: on heating with ammonium molybdate and HNO_3, a **yellow** crystalline precipitate (ammonium phosphomolybdate). **Results and analysis:** positive results of all the reactions confirm that nucleoproteins contain protein, nitrogenous bases, pentose and phosphate. Yeast contains mainly RNA, so the orcinol reaction is clearly positive.

Experiment 3: interpretation of a PCR electropherogram (case-based)

1. Identify the lanes on the gel image: marker (lengths of DNA fragments), positive control, negative control, patient samples.
2. Compare the band in the patient sample with the positive control (the length must be the same).
3. There must be no band in the negative control (if there is one — contamination, the result is invalid). **Results and analysis:** a band of the expected length in the patient lane is a positive result (pathogen DNA or a mutant allele is present).

10. Practical skills

Skill: writing the complementary strand and the mRNA from a DNA sequence

1. Mark the direction of the given strand (5’ → 3’).
2. Write the complementary strand antiparallel: A ↔ T, G ↔ C (3’ → 5’).
3. Write the mRNA from the template strand: U instead of T; the mRNA is identical to the coding strand (with U).
4. Example: coding 5’-ATG GCT TAA-3’ → template 3’-TAC CGA ATT-5’ → mRNA 5’-AUG GCU UAA-3’.

Skill: clinical interpretation of a PCR result

1. Determine the purpose of the test (infection, mutation, viral load).
2. Check that the controls worked correctly.
3. Assess a qualitative (present/absent) or quantitative result (real-time PCR: the Ct value — the smaller it is, the more starting DNA there is).
4. Consider the causes of false positives (contamination) and false negatives (inhibitors, sample quality).

Review questions

1. State the differences in the chemical composition and structure of DNA and RNA.
2. State Chargaff's rules.
3. List the main features of the Watson–Crick model.
4. How are the nucleosome and chromatin organized?
5. What are DNA denaturation and hybridization? What is their practical importance?
6. What are the features of the mitochondrial genome?
7. State the differences between prokaryotic and eukaryotic genes.
8. List the main principles of replication.
9. Which enzymes take part in replication?
10. How are Okazaki fragments formed and joined?

11. What is telomerase? What is its importance in aging and in tumors?
12. What is the biological role of DNA methylation?
13. Describe the stages of transcription and the eukaryotic RNA polymerases.
14. State the post-transcriptional modifications of mRNA.
15. Explain the principle of PCR and its use in medicine.

Test questions

1. Which nitrogenous base is not found in DNA?
   A) Adenine
   B) Guanine
   C) Cytosine
   D) Thymine
   E) Uracil

2. According to Chargaff's rule, which equality is true for DNA?
   A) A = G
   B) A = C
   C) G = C
   D) T = G
   E) A + T = G + C

3. How is the protein core of the nucleosome organized?
   A) A tetramer of histone H_1
   B) 2 each of H2A, H2B, H3, H4 — an octamer
   C) Only an H3 and H4 dimer
   D) Nonhistone proteins
   E) Protamines

4. What does it mean that replication is semiconservative?
   A) Only one strand is copied
   B) Each new molecule has one old and one new strand
   C) The new molecules consist entirely of new strands
   D) Only half of the gene is copied
   E) Replication takes place only in half of the S phase

5. Which enzyme joins Okazaki fragments?
   A) Helicase
   B) Primase
   C) DNA ligase
   D) Topoisomerase
   E) Telomerase

6. What kind of enzyme is telomerase?
   A) A reverse transcriptase with an RNA template
   B) DNA ligase
   C) A restriction enzyme
   D) An RNA polymerase
   E) Helicase

7. Which enzyme carries out mRNA synthesis in eukaryotes?
   A) RNA polymerase II
   B) RNA polymerase I
   C) RNA polymerase III
   D) DNA polymerase α
   E) Primase

8. Which feature distinguishes replication from transcription?
   A) DNA is used as the template
   B) Synthesis proceeds in the 5’ → 3’ direction
   C) The principle of complementarity
   D) A primer is required
   E) Phosphodiester bonds are formed

9. What is splicing?
   A) Addition of a cap to the 5’-end of mRNA
   B) Removal of introns and joining of exons
   C) Addition of a poly(A) tail
   D) Addition of CCA to tRNA
   E) DNA methylation

10. Which enzyme is inhibited by α-amanitin, the death cap toxin?
    A) RNA polymerase II
    B) DNA gyrase
    C) Bacterial RNA polymerase
    D) Reverse transcriptase
    E) Topoisomerase I

11. What is the target of rifampicin?
    A) DNA gyrase
    B) The 30S ribosomal subunit
    C) Reverse transcriptase
    D) Topoisomerase II
    E) Bacterial RNA polymerase

12. On which factor does the DNA melting temperature (Tm) depend?
    A) The adenine content
    B) The thymine content
    C) The proportion of G≡C pairs
    D) The type of pentose
    E) The phosphate content

13. Which enzyme carries out the elongation step in PCR?
    A) DNA ligase
    B) A restriction enzyme
    C) Reverse transcriptase
    D) Taq polymerase
    E) An RNA polymerase

14. Which is a feature of mitochondrial DNA?
    A) It has a linear structure
    B) It is inherited through the father
    C) It is bound to histones
    D) It contains many introns
    E) It is inherited only through the mother

15. What color does the diphenylamine reaction give in identifying the type of pentose?
    A) Green (ribose)
    B) Violet
    C) Yellow
    D) Blue (deoxyribose)
    E) Red

Clinical cases

Case 1

A family ate mushrooms gathered in the forest. After 10–12 hours vomiting, diarrhea and abdominal pain began; on day 3 jaundice appeared, blood ALT was 3500 U/l and the prothrombin index fell sharply.

Questions: Which toxin is responsible? What is its molecular mechanism of action? Why is the liver specifically affected?

Answer: The toxin of the death cap (Amanita phalloides) — α-amanitin. It inhibits eukaryotic RNA polymerase II: mRNA synthesis stops, proteins (including clotting factors) are not synthesized, and the cell dies. The toxin is absorbed from the intestine and reaches the liver first via the portal blood, actively enters hepatocytes through OATP transporters and acts again and again through the enterohepatic circulation — liver necrosis. The fall in the prothrombin index shows that protein synthesis in the liver is impaired.

Case 2

A patient was diagnosed with tuberculosis and treatment with rifampicin was started. After a few weeks his urine and tears turned orange-red. The patient asked whether the drug also stops RNA synthesis in human cells.

Questions: What is the mechanism of action of rifampicin? Why does it have almost no effect on human cells?

Answer: Rifampicin binds to the β subunit of the bacterial DNA-dependent RNA polymerase and blocks the initiation of transcription (the start of chain elongation). Eukaryotic RNA polymerases differ in structure and are not sensitive to rifampicin — hence its selective action. The orange color is the drug's own color (harmless). In addition, rifampicin induces cytochrome P450 in the liver — interactions with other drugs must be taken into account.

Case 3

A nasopharyngeal swab was taken from a patient and tested for a viral infection by real-time RT-PCR. Result: Ct = 18 (positive control Ct = 25, no signal in the negative control).

Questions: How do you interpret the result? Why is the “RT” step needed to detect an RNA virus? What does the Ct value mean?

Answer: The result is positive: the sample contains viral RNA and the controls worked correctly. Taq polymerase cannot amplify the genome of RNA viruses — first, complementary DNA (cDNA) is synthesized from the RNA by reverse transcriptase and then amplified by PCR. Ct is the number of the cycle at which the fluorescent signal crosses the threshold; the smaller the Ct, the more starting copies there are (Ct 18 — a high viral load, the patient is likely to be infectious).

Quick review

Table 16.6. *Key concepts of topic 16*

Concept	Brief definition
Chargaff's rules	A = T, G = C; A + G = T + C
The Watson–Crick model	Antiparallel right-handed double helix, A=T (2), G≡C (3 hydrogen bonds)
Nucleosome	Histone octamer + ≈ 146 base pairs of DNA
Hybridization	Joining of complementary strands; Tm depends on G≡C
mtDNA	Circular, 37 genes, inherited through the mother
Replication	Semiconservative, primer required, dNTPs, Okazaki fragments
Enzymes	Helicase, topoisomerase, primase, DNA pol α/δ/ε, ligase
Telomerase	Reverse transcriptase with an RNA template; active in tumors
Methylation	5-Methylcytosine in CpG — silences the gene (epigenetics)
Transcription	No primer, NTPs; RNA pol II — mRNA (α-amanitin)
Processing	Cap, poly(A), splicing (snRNA), alternative splicing
PCR	95 → 55–65 → 72 °C; Taq polymerase; 2^n copies

TOPIC 17

Translation. Regulation of gene expression

Learning objectives

To give the student knowledge of the genetic code and its properties, the components of the protein-synthesizing system, the stages of polypeptide chain synthesis on ribosomes, post-translational modification, protein folding and targeting, inhibitors of protein synthesis (antibiotics, toxins), the regulation of gene expression in prokaryotes and eukaryotes, and cell differentiation.

By the end of the class the student should know:

- the properties of the genetic code;
- the structure of tRNA and the role of aminoacyl-tRNA synthetases;
- the structure of ribosomes (prokaryotic and eukaryotic);
- the initiation, elongation and termination stages of translation;
- the main post-translational modifications;
- the antibiotics and toxins that block protein synthesis;
- the operon model (the lactose operon) and the levels of regulation of gene expression in eukaryotes.

The student should be able to:

- determine a peptide sequence from an mRNA sequence using the genetic code table;
- predict the effect of a mutation on a protein;
- explain the basis of the selective action of antibiotics.

1. The genetic code and its properties

The **genetic code** is the system that matches the sequence of mRNA nucleotides to the sequence of amino acids in a protein. The 4 kinds of nucleotides form 64 **codons** (triplets): 61 of them code for amino acids and 3 are **stop codons** (UAA, UAG, UGA). **AUG** is the start codon (methionine).

Table 17.1. *Properties of the genetic code*

Property	Meaning
Triplet	Each amino acid is coded by three nucleotides
Degenerate	One amino acid is coded by several codons (leucine, serine, arginine — 6 each); only Met and Trp — one each
Unambiguous (specific)	Each codon codes for only one amino acid
Nonoverlapping	Each nucleotide belongs to only one codon
Commaless (continuous)	There are no separators between codons — the reading frame is set from the start codon
Colinear	The sequences in the gene and in the protein are in the same order
Universal	Almost the same in all organisms (small differences in mitochondria: UGA — tryptophan)

Wobble hypothesis: pairing between the 3rd nucleotide of the codon and the 1st nucleotide of the anticodon is less strict (for example, inosine pairs with U, C and A) — therefore the 61 codons can be

read by ≈ 40 different tRNAs.

2. Components of the protein-synthesizing system

1. **Amino acids** (20 standard ones; selenocysteine — the UGA codon in a special context).
2. **mRNA** — the template.
3. **tRNA** — the “adapter”: a “cloverleaf” secondary structure and an L-shaped tertiary structure; the **anticodon loop** (recognizes the codon) and the **CCA at the 3’-end** (the amino acid is attached).
4. **Aminoacyl-tRNA synthetases** — a separate enzyme for each amino acid; they activate the amino acid and attach it to the corresponding tRNA (2 high-energy bonds are consumed). They are the **main guarantee of the fidelity** of translation — they hydrolyze a wrongly attached amino acid (proofreading).
5. **Ribosomes.**
6. **Protein factors:** initiation (eIF), elongation (eEF), termination (eRF).
7. **Energy sources:** ATP (activation), GTP (initiation, elongation, termination).
8. Mg^{2+} and K^{+} ions.

Table 17.2. *Structure of ribosomes*

Characteristic	Prokaryotes	Eukaryotes
Complete ribosome	70S	80S
Small subunit	30S (16S rRNA)	40S (18S rRNA)
Large subunit	50S (23S, 5S rRNA)	60S (28S, 5.8S, 5S rRNA)
Active sites	A (aminoacyl), P (peptidyl), E (exit)	A, P, E

3. Stages of translation

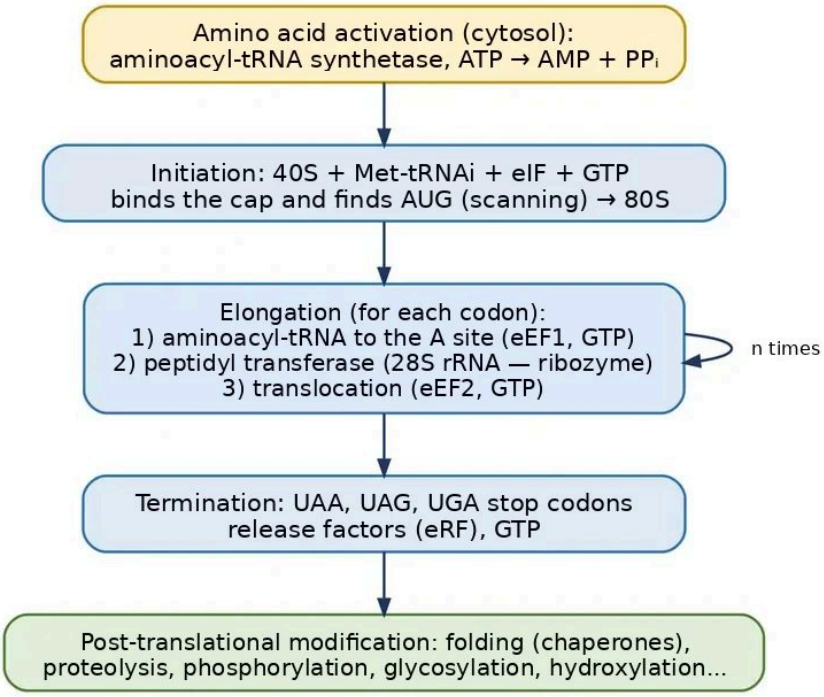


Figure 17.1. *Stages of translation (drawn after Severin E.S.)*

Activation of amino acids: amino acid + ATP → aminoacyl-AMP + PP_i; aminoacyl-AMP + tRNA → aminoacyl-tRNA + AMP.

Initiation (in eukaryotes): the 40S subunit, the initiator **Met-tRNA$_i^{Met}$**, eIF2-GTP and other factors bind together to the **5'-cap of the mRNA** (eIF4E recognizes the cap) and move along the mRNA to the first **AUG** codon (**scanning**, the Kozak sequence). GTP is hydrolyzed and the 60S subunit joins — the **80S** initiation complex is formed, with Met-tRNAi in the **P site**. In prokaryotes the initiator is **formylmethionine-tRNA**, and 16S rRNA recognizes the **Shine–Dalgarno** sequence in the mRNA.

Elongation — three steps repeated for each codon:

1. The next aminoacyl-tRNA arrives at the **A site** with eEF1α-GTP (if the anticodon matches the codon, GTP is hydrolyzed).
2. **Peptidyl transferase** transfers the peptide in the P site to the amino group of the amino acid in the A site — a peptide bond is formed. This activity belongs to the **28S rRNA** of the 60S subunit (a **ribozyme**; in prokaryotes, 23S rRNA).
3. **Translocation** — under the action of eEF2-GTP the ribosome moves one codon along the mRNA; the peptidyl-tRNA moves to the P site, and the empty tRNA moves to the E site and is released. **4 high-energy bonds** are consumed for each peptide bond (2 in activation, 2 GTP in elongation). Many ribosomes read one mRNA at the same time — a **polysome**.

Termination: when a stop codon enters the A site, **release factors** (eRF1, eRF3-GTP) bind instead of a tRNA; peptidyl transferase transfers the peptide to water (hydrolysis), the polypeptide is released and the ribosome dissociates into subunits.

4. Post-translational modification and protein targeting

Folding: a polypeptide often reaches its correct conformation spontaneously; **chaperones** (heat shock proteins — Hsp70, Hsp60/chaperonins) prevent misfolding and aggregation. Misfolded proteins are degraded by the **ubiquitin–proteasome** system. Folding disorders — **prion diseases**, Alzheimer disease (β-amyloid), Parkinson disease (α-synuclein), cystic fibrosis (CFTR ΔF508).

Targeting: secretory and membrane proteins are directed to the **rough endoplasmic reticulum** by an N-terminal **signal peptide** (SRP — signal recognition particle) and are then sorted in the Golgi apparatus. Lysosomal enzymes are directed to lysosomes by the **mannose-6-phosphate** tag (I-cell disease — absence of this tag). Mitochondrial and nuclear proteins have special signal sequences.

Table 17.3. *Main post-translational modifications*

Modification	Example
Partial proteolysis	Proinsulin → insulin + C-peptide; activation of proenzymes (trypsinogen); cleavage of the signal peptide
Phosphorylation (Ser, Thr, Tyr)	Glycogen phosphorylase, receptor tyrosine kinases
Glycosylation (N- and O-)	Immunoglobulins, hormones, membrane proteins
Hydroxylation (Pro, Lys)	Collagen — ascorbic acid is required (scurvy, topic 22)
γ-Carboxylation (Glu)	Prothrombin, factors VII, IX, X — vitamin K is required
Acetylation, methylation	Histones (chromatin activity)
Lipidation (prenylation, myristoylation)	Ras proteins — membrane attachment
Formation of disulfide bonds	Insulin, immunoglobulins

Modification	Example
Ubiquitination	A tag for degradation in the proteasome

5. Inhibitors of protein synthesis

Table 17.4. *Antibiotics and toxins that block protein synthesis*

Substance	Target	Mechanism
Streptomycin, gentamicin (aminoglycosides)	30S	Disrupt initiation, cause misreading of the code
Tetracyclines	30S	Block the binding of aminoacyl-tRNA to the A site
Chloramphenicol	50S	Inhibits peptidyl transferase (also affects mitochondrial ribosomes)
Erythromycin, azithromycin (macrolides)	50S	Block translocation
Linezolid	50S	Blocks formation of the initiation complex
Puromycin	Prokaryotic and eukaryotic	Aminoacyl-tRNA analog — causes premature chain termination
Cycloheximide	80S	Eukaryotic peptidyl transferase
Diphtheria toxin	eEF2	Inactivates it by ADP-ribosylation — translocation stops
Ricin (castor bean), Shiga toxin	60S (28S rRNA)	Removes an adenine (N-glycosidase) — the ribosome is inactive

The **selective action** of antibiotics is based on the difference in structure between prokaryotic (70S) and eukaryotic (80S) ribosomes. Because mitochondria contain 70S-type ribosomes, chloramphenicol can damage the bone marrow (aplastic anemia; in newborns, the “gray baby syndrome” — due to insufficient glucuronidation).

6. Regulation of gene expression in prokaryotes

An **operon** (Jacob and Monod, 1961) is a group of structural genes under the control of one promoter and operator that is transcribed as a single polycistronic mRNA.

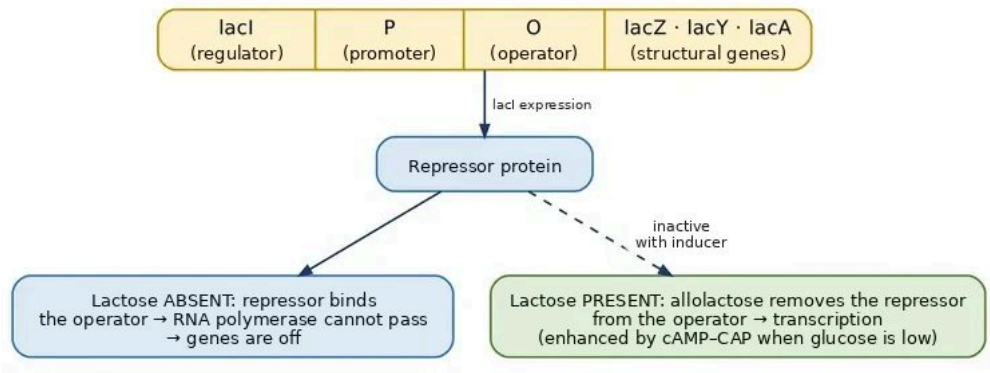


Figure 17.2. *Regulation of the lactose operon — induction (drawn after Lippincott’s Illustrated Reviews: Biochemistry)*

Lactose operon (inducible): structural genes — lacZ (β-galactosidase), lacY (permease), lacA (transacetylase). In the absence of lactose the **repressor** binds to the operator and blocks transcription. Lactose (more precisely, allolactose — the **inducer**) binds to the repressor and releases it from the operator — the genes are expressed. **Positive control:** when glucose is low, cAMP increases, and the **cAMP–CAP** complex binds to the promoter and enhances transcription (catabolite repression — when glucose is present, the lactose operon works weakly).

Tryptophan operon (repressible): when tryptophan is abundant it acts as a **corepressor**, activating the repressor, and the genes for the synthetic enzymes are switched off; an additional mechanism is **attenuation**.

7. Regulation of gene expression in eukaryotes

In eukaryotes, regulation operates at many levels:

1. **Chromatin level** — histone **acetylation** (histone acetyltransferases) loosens chromatin and activates transcription, while deacetylation silences it; **DNA methylation** (CpG) silences the gene — **epigenetic** regulation; genomic imprinting (Prader–Willi and Angelman syndromes).
2. **Transcription level** — the main level: **transcription factors** bind to the promoter, **enhancers** (which work even thousands of base pairs away) and silencers. Examples: **steroid hormone receptors** — bind the hormone and then bind to hormone response elements in DNA (GRE); cAMP → PKA → **CREB**; metals → the metallothionein gene; hypoxia → HIF-1 → the erythropoietin gene.
3. **Gene amplification** — an increase in the number of gene copies (HER2, MYC in tumors; the dihydrofolate reductase gene in methotrexate resistance).
4. **Gene rearrangement** — V(D)J recombination of immunoglobulin genes.
5. **Post-transcriptional level** — alternative splicing, RNA editing, mRNA stability (the transferrin receptor mRNA is regulated by iron, topic 21), **microRNAs** and siRNAs (RNA interference) degrade mRNA or stop translation.
6. **Translation level** — phosphorylation of eIF2 (under stress, and in heme deficiency globin synthesis stops in reticulocytes); iron-dependent translation of ferritin mRNA (IRE/IRP).
7. **Post-translational level** — phosphorylation, proteolysis, degradation by ubiquitination. **Cell differentiation:** the genome is the same in all somatic cells, but in each cell type only a certain set of genes is expressed (globin in erythrocyte precursors, insulin in β-cells). This **selective expression** is provided by tissue-specific transcription factors and epigenetic marks; in ontogenesis the **homeotic (HOX) genes** determine the body plan. For example, the switch from fetal hemoglobin (HbF) to adult hemoglobin (HbA) is the silencing of the γ-globin gene and the activation of the β-globin gene.

> **CLINICAL SIGNIFICANCE**
> **Hydroxycarbamide** (hydroxyurea) in sickle cell anemia reactivates expression of the γ-globin gene and increases the amount of HbF — HbF reduces the polymerization of HbS.

8. Laboratory work

Experiment 1: solving translation problems using the genetic code table

1. Find the first AUG codon in the given mRNA sequence — the reading frame starts there.
2. Divide the sequence into codons of three and determine the amino acids from the table.

3. Continue up to the stop codon. **Example:** 5’-GCC **AUG** UUU GGC UGG AAA UAG CC-3’ → Met–Phe–Gly–Trp–Lys (UAG — stop).

Experiment 2: determining the effect of a mutation on a protein

1. Compare the normal and mutant sequences.
2. Determine the type of mutation: substitution (missense, nonsense, synonymous), insertion or deletion (frameshift).
3. Predict its effect on the protein structure. **Example:** codon 6 of the β-globin gene GAG → GUG: glutamate → valine (missense) — HbS (sickle cell anemia). Deletion of a single nucleotide shifts the reading frame — all subsequent amino acids change, and a premature stop codon often appears.

Experiment 3: comparing the effects of protein synthesis inhibitors (working with a table)

1. Write the target of each antibiotic (30S, 50S, 80S, eEF2) in the table.
2. Mark its toxicity to humans and the reason (effect on mitochondrial ribosomes, on eukaryotic ribosomes). **Results and analysis:** substances that act selectively on prokaryotic ribosomes are used as antibiotics; those that block eukaryotic ribosomes (cycloheximide, ricin, diphtheria toxin) are toxic to humans.

9. Practical skills

Skill: writing the flow of information from DNA to protein step by step

1. Write the template strand from the DNA coding strand.
2. Write the mRNA from the template (T → U).
3. Find the start codon in the mRNA and divide it into codons.
4. Write the peptide sequence using the genetic code table.
5. Show the tRNA anticodons (complementary and antiparallel to the codon).

Skill: assessing the effect of an antibiotic on bacterial and human cells

1. Determine which process (replication, transcription, translation, cell wall) the antibiotic blocks.
2. Show the difference in the target structure between prokaryotes and eukaryotes.
3. State the likelihood of an effect on mitochondria and the adverse effects (ototoxicity, aplastic anemia, tooth discoloration).

Review questions

1. What is the genetic code? List its properties.
2. Name the start and stop codons.
3. What is the structure of tRNA and what is its function?
4. What is the role of aminoacyl-tRNA synthetases?
5. How do prokaryotic and eukaryotic ribosomes differ?
6. Describe the initiation stage of translation.
7. What steps make up elongation?
8. Which enzyme catalyzes the formation of the peptide bond?
9. How does translation termination take place?

10. How much energy is consumed for the formation of one peptide bond?
11. Name the main post-translational modifications with examples.
12. What are the functions of chaperones and the proteasome?
13. List the antibiotics and toxins that block protein synthesis.
14. How is the lactose operon regulated?
15. At what levels is gene expression regulated in eukaryotes?

Test questions

1. What does the “degenerate” property of the genetic code mean?
   A) One codon codes for several amino acids
   B) Codons overlap
   C) One amino acid is coded by several codons
   D) The code is the same in all organisms
   E) There are commas between codons

2. Which codon is the start codon?
   A) UAA
   B) AUG
   C) UGA
   D) UAG
   E) GUG

3. To which part of tRNA is the amino acid attached?
   A) To the anticodon loop
   B) To the D loop
   C) To the TψC loop
   D) To the 5’-end
   E) To the CCA at the 3’-end

4. Which is the small subunit of the eukaryotic ribosome?
   A) 30S
   B) 50S
   C) 40S
   D) 60S
   E) 70S

5. In eukaryotes, peptidyl transferase activity belongs to what?
   A) The eEF1 protein
   B) The eEF2 protein
   C) 18S rRNA
   D) 28S rRNA
   E) Aminoacyl-tRNA synthetase

6. Which factor takes part in translocation?
   A) eIF4E
   B) eRF1
   C) eEF1α
   D) eEF2
   E) CAP

7. What is the mechanism of action of diphtheria toxin?
    A) It ADP-ribosylates eEF2
    B) It removes an adenine from 28S rRNA
    C) It binds to the 30S subunit
    D) It inhibits peptidyl transferase
    E) It blocks RNA polymerase II

8. How do tetracyclines act?
    A) It inhibits peptidyl transferase
    B) Block the binding of aminoacyl-tRNA to the A site
    C) Block translocation
    D) They cause premature chain termination
    E) They inactivate eEF2

9. Which vitamin is required for the hydroxylation of proline and lysine in collagen?
    A) Vitamin K
    B) Vitamin B_{12}
    C) Vitamin A
    D) Ascorbic acid
    E) Vitamin D

10. What acts as the inducer in the lactose operon?
    A) Allolactose
    B) Glucose
    C) Tryptophan
    D) cAMP
    E) The repressor

11. What is the weak activity of the lactose operon in the presence of abundant glucose called?
    A) Attenuation
    B) Catabolite repression
    C) Induction
    D) Amplification
    E) Imprinting

12. How does histone acetylation affect gene expression?
    A) It completely silences the gene
    B) It has no effect
    C) It methylates DNA
    D) It stops splicing
    E) It loosens chromatin and activates transcription

13. Which antibiotic can cause aplastic anemia by acting on mitochondrial ribosomes?
    A) Penicillin
    B) Rifampicin
    C) Ciprofloxacin
    D) Cycloheximide
    E) Chloramphenicol

14. Which small RNAs degrade mRNA or stop its translation in eukaryotes?
    A) MicroRNAs
    B) tRNA
    C) rRNA
    D) snRNA
    E) Telomerase RNA

15. What type of mutation is the GAG → GUG substitution in the β-globin gene?
    A) Nonsense
    B) Frameshift
    C) Missense
    D) Synonymous (silent)
    E) Deletion

Clinical cases

Case 1

An unvaccinated 6-year-old child has fever, sore throat, a gray, hard-to-remove membrane on the tonsils and swelling of the neck. A week later a heart rhythm disturbance developed (myocarditis).

Questions: What is the disease? What is the molecular mechanism of action of the toxin? Why is the heart affected?

Answer: Diphtheria. The toxin of Corynebacterium diphtheriae enters the cell by endocytosis, and its A fragment transfers an ADP-ribose residue from NAD^+ to the elongation factor eEF2 (to the diphthamide residue) — eEF2 is inactivated, translocation and protein synthesis stop, and the cell dies. A single toxin molecule is enough to kill a cell. The toxin spreads through the blood and damages especially the myocardium and nerve tissue, where protein synthesis is active. Treatment — antitoxic serum (neutralizes the toxin before it enters the cells) and antibiotics; prevention — vaccine (toxoid).

Case 2

A patient has pneumonia; the physician wants to choose one of the antibiotics that act on the 70S ribosome. The patient is a pregnant woman (second trimester).

Questions: Which antibiotics that block protein synthesis are not used in pregnancy, and why? Which group is relatively safe?

Answer: Tetracyclines (30S — block the binding of aminoacyl-tRNA) form chelates with calcium and accumulate in fetal bones and teeth (tooth discoloration, slowed bone growth). Aminoglycosides (30S — misreading of the code) are ototoxic to the fetus (deafness). Chloramphenicol (50S) — the “gray baby syndrome” in newborns (insufficient glucuronidation in the liver) and bone marrow suppression through mitochondrial ribosomes. Macrolides (azithromycin — 50S, blocks translocation) are considered relatively safe. The selective action is based on the structural difference between 70S and 80S ribosomes.

Case 3

A patient has been diagnosed with sickle cell anemia. In the coding strand of the β-globin gene, codon 6 is GTG instead of GAG.

Questions: What kind of mutation is this? What change occurs in the protein and how does it lead to the disease? What is the basis of treatment with hydroxycarbamide?

Answer: A point missense mutation: GAG → GUG in the mRNA; in the protein, glutamate at position 6 (negatively charged, hydrophilic) is replaced by valine (hydrophobic) — HbS. Deoxygenated HbS molecules form long polymers through hydrophobic interactions; the erythrocytes become sickle-shaped, occlude small vessels (painful crises) and are rapidly destroyed (hemolytic anemia). Hydroxycarbamide reactivates expression of the γ-globin gene and increases the proportion of fetal hemoglobin (HbF); HbF blocks the polymerization of HbS.

Quick review

Table 17.5. *Key concepts of topic 17*

Concept	Brief definition
Genetic code	Triplet, degenerate, unambiguous, nonoverlapping, universal
Codons	64; AUG — start (Met); UAA, UAG, UGA — stop
Aminoacyl-tRNA synthetase	Activates the amino acid, ensures fidelity
Ribosome	80S (40S + 60S); 70S (30S + 50S)
Initiation	Cap, scanning, Met-tRNAi in the P site
Elongation	Arrival at the A site → peptidyl transferase (28S rRNA) → translocation (eEF2)
Energy	4 high-energy bonds / peptide bond
Post-translational	Proteolysis, phosphorylation, glycosylation, hydroxylation (vit. C), carboxylation (vit. K)
Antibiotics	Tetracycline, aminoglycoside (30S); chloramphenicol, macrolide (50S)
Toxins	Diphtheria (eEF2), ricin (28S rRNA)
Lactose operon	Repressor + inducer (allolactose); cAMP–CAP
Eukaryotic regulation	Chromatin, transcription factors, splicing, microRNA, translation

TOPIC 18

DNA damage and repair. Apoptosis. Basics of gene therapy

Learning objectives

To give the student knowledge of the causes and types of DNA damage, molecular mutations (substitution, deletion, insertion), repair systems and their defects, genetic variability and protein polymorphism, the molecular basis of hereditary diseases, the use of DNA technologies in medicine, the differences between apoptosis and necrosis, the regulation of apoptosis, and the basics of cloning and gene therapy.

By the end of the class the student should know:

- the physical, chemical and biological factors that damage DNA;
- the classification of mutations and their effect on proteins;
- the main repair systems (BER, NER, MMR, repair of double-strand breaks);
- the diseases associated with repair defects;
- protein polymorphism and its clinical importance;
- the extrinsic and intrinsic pathways of apoptosis and the proteins that regulate it;
- the principles and vectors of gene therapy.

The student should be able to:

- identify the type of mutation from a sequence;
- distinguish apoptosis from necrosis by biochemical features;
- choose a method of molecular diagnosis for a hereditary disease.

1. Causes of DNA damage

Tens of thousands of DNA lesions occur in each cell per day; most of them are corrected by the repair systems.

Table 18.1. *Factors that damage DNA*

Factors	Examples	Type of damage
Spontaneous (internal)	Water, reactive oxygen species, replication errors	Depurination, deamination of cytosine to uracil, 8-oxoguanine, mispairing
Physical	Ultraviolet radiation	Thymine (pyrimidine) dimers
	Ionizing radiation (X-rays, γ-rays)	Single- and double-strand breaks, free radicals
Chemical	Alkylating agents (nitrosamines, sulfur mustard, cyclophosphamide)	Alkylation of bases (O^6-methylguanine)
	Benzo(a)pyrene, aflatoxin B_1	Bulky adducts (with guanine)
	Intercalators (acridines, doxorubicin)	Frameshift
	Base analogs (5-bromouracil)	Mispairing
Biological	Viruses (HPV, hepatitis B, retroviruses), transposons	Integration into DNA, gene disruption

2. Mutations

A **mutation** is a heritable change in the nucleotide sequence of DNA. Mutations in germ cells are passed on to offspring (hereditary diseases), while somatic mutations lead to tumors and aging.

By level: gene (point), chromosomal (deletion, duplication, inversion, translocation — the Philadelphia chromosome t(9;22)), genomic (aneuploidy — Down syndrome, trisomy 21).

Table 18.2. *Molecular (gene) mutations and their effect on proteins*

Mutation	Meaning	Effect on the protein	Example
Synonymous (silent)	The codon changes, the amino acid does not	None	Because of the degeneracy of the code
Missense	A different amino acid is coded	Activity is changed or preserved	HbS (Glu6Val)
Nonsense	The codon becomes a stop codon	A shortened, often inactive protein	Some forms of β^0-thalassemia
Frameshift	Insertion or deletion of a number of nucleotides that is not a multiple of 3	All subsequent amino acids change	Tay–Sachs disease, many forms of Duchenne muscular dystrophy
In-frame deletion	Loss of 3 nucleotides	One amino acid is missing	Cystic fibrosis (ΔF508)
Trinucleotide repeat expansion	Increase of (CAG)n, (CGG)n	Toxic protein or gene silencing	Huntington disease, fragile X syndrome
Splicing mutations	At the intron–exon boundary	Abnormal mRNA	β^+-thalassemia

Transition — a purine is replaced by a purine or a pyrimidine by a pyrimidine; **transversion** — a purine is replaced by a pyrimidine.

3. DNA repair

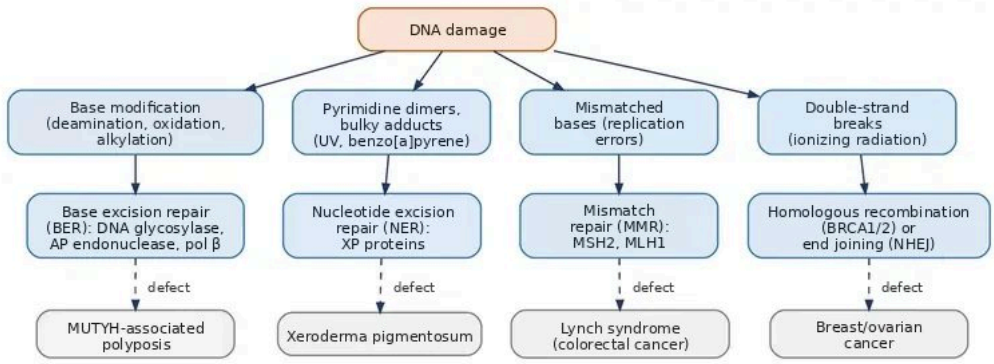


Figure 18.1. *Types of DNA damage, repair systems and their defects (drawn after Lippincott's Illustrated Reviews: Biochemistry)*

Direct repair: O^6-methylguanine-DNA methyltransferase transfers the methyl group to itself (a "suicide" enzyme); in bacteria **photolyase** splits thymine dimers under the action of light (absent in humans).

Base excision repair (BER): a damaged base (uracil, 8-oxoguanine, an alkylated base) is recognized by a specific **DNA glycosylase**, which cleaves the N-glycosidic bond; the resulting apurinic/apyrimidinic (AP) site is cut by **AP endonuclease**; DNA polymerase β fills the gap and

DNA ligase seals it. This is also why DNA contains thymine instead of uracil: uracil formed by deamination of cytosine is recognized as “foreign”.

Nucleotide excision repair (NER): bulky lesions (thymine dimers, benzo(a)pyrene adducts) — a complex of XP proteins cuts the damaged strand on both sides and removes a fragment of ≈ 24–32 nucleotides; DNA polymerase δ/ε and ligase fill it in.

CLINICAL SIGNIFICANCE

Xeroderma pigmentosum is a hereditary defect of the NER enzymes (XP-A … XP-G) (autosomal recessive). Thymine dimers formed under sunlight are not repaired: the skin is very sensitive, there are freckles, and multiple **skin cancers** (basal cell carcinoma, melanoma) appear at an early age (before 10 years). Prevention — complete protection from the sun.

Mismatch repair (MMR): after replication, mispaired bases and small loops are recognized by **MSH2/MSH6**, and **MLH1/PMS2** nick the new strand; the new strand is removed by an exonuclease and resynthesized. Its defect causes **microsatellite instability** and **Lynch syndrome** (hereditary nonpolyposis colorectal cancer, endometrial cancer).

Repair of double-strand breaks:

- **homologous recombination** — uses the sister chromatid as a template (in the S and G_2 phases) and is accurate; the **BRCA1 and BRCA2** proteins take part — their defect sharply increases the risk of breast and ovarian cancer; the ATM kinase senses the break (ataxia-telangiectasia);
- **nonhomologous end joining (NHEJ)** — Ku proteins and DNA-PK; fast, but nucleotides are sometimes lost (error-prone). **SOS repair (translesion synthesis):** special low-fidelity DNA polymerases “bypass” the damaged site — the probability of mutations is high.

REMEMBER

Diseases associated with repair defects: **xeroderma pigmentosum** (NER), **Lynch syndrome** (MMR), **BRCA1/2** (homologous recombination), **ataxia-telangiectasia** (ATM), **Fanconi anemia** (cross-link repair), **Bloom syndrome** (helicase). The common feature is a predisposition to tumors.

4. Genetic variability. The human genome. Protein polymorphism

Mechanisms of genetic variability: mutations, recombination in meiosis (crossing-over), independent assortment of chromosomes, transposons.

The human genome: ≈ 3.1×10^9 base pairs; the protein-coding part is ≈ 1.5%; the rest consists of introns, regulatory elements, repetitive sequences (Alu, LINE, microsatellites) and pseudogenes. The genomes of two people are ≈ 99.9% identical; the differences consist mainly of **single nucleotide polymorphisms (SNPs)** and copy number variations.

Polymorphism means gene variants that occur in a population with a frequency of ≥ 1%. **Protein polymorphism** means several variants (allelic forms) of one protein:

- **AB0 blood groups** — glycosyltransferase variants;
- the **HLA system** — very high polymorphism (transplantation, susceptibility to autoimmune diseases);
- **hemoglobin variants** — HbS, HbC, HbE (more than 300);
- **glucose-6-phosphate dehydrogenase** variants (topic 7);
- **cytochrome P450** polymorphism (CYP2D6, CYP2C19) — “fast” or “slow” drug metabolizers (**pharmacogenetics**);

- **N-acetyltransferase** — slow and fast acetylators of isoniazid;
- **pseudocholinesterase** — with the atypical variant suxamethonium acts for a long time (apnea);
- **α_1-antitrypsin** variants (PiZZ — emphysema, liver cirrhosis).

5. Hereditary diseases and DNA technologies

Monogenic diseases: autosomal dominant (familial hypercholesterolemia, Marfan syndrome, Huntington disease), autosomal recessive (phenylketonuria, cystic fibrosis, sickle cell anemia, glycogenoses), X-linked (hemophilia A and B, Duchenne muscular dystrophy, G6PD deficiency, Lesch–Nyhan syndrome). **Mitochondrial** — through the mother. **Multifactorial** — type 2 diabetes, atherosclerosis, hypertension.

DNA technologies in medicine:

- **PCR** and real-time PCR — detection of mutations and infections (topic 16);
- **restriction fragment length polymorphism (RFLP)** — if a mutation changes a restriction enzyme cutting site, the fragment lengths differ (diagnosis of sickle cell anemia);
- **Southern blot** (DNA), **Northern blot** (RNA), **Western blot** (protein);
- **sequencing** — the Sanger method, next-generation sequencing (whole exome/genome);
- **FISH** — detection of chromosomal changes with fluorescent probes;
- **DNA microarrays** — expression of thousands of genes or SNPs;
- **genetic fingerprinting** (microsatellites) — forensic medicine, paternity testing;
- **prenatal diagnosis** — chorionic villi, amniocytes, cell-free fetal DNA in maternal blood.

6. Apoptosis and necrosis

Apoptosis is an energy-requiring, genetically programmed cell death. Its importance: embryogenesis (tissue between the fingers), the immune system (elimination of autoreactive lymphocytes), elimination of damaged and tumor cells, tissue homeostasis.

Table 18.3. *Differences between apoptosis and necrosis*

Characteristic	Apoptosis	Necrosis
Cause	Physiological and pathological signals, DNA damage	Severe injury: ischemia, toxins, temperature
Extent	Individual cells	Groups of cells
Cell volume	Decreases (shrinks)	Swells
Membrane	Remains intact; phosphatidylserine moves to the outer leaflet	Is destroyed
Chromatin	Condenses; DNA is cut into fragments of ≈ 180–200 base pairs (a “ladder”)	Is degraded randomly
Energy	Required	Not required (ATP is depleted)
(ATP)		
Result	Phagocytosis of apoptotic bodies, no inflammation	Cell contents are released — inflammation
Enzymes	Caspases	Lysosomal enzymes, calpains

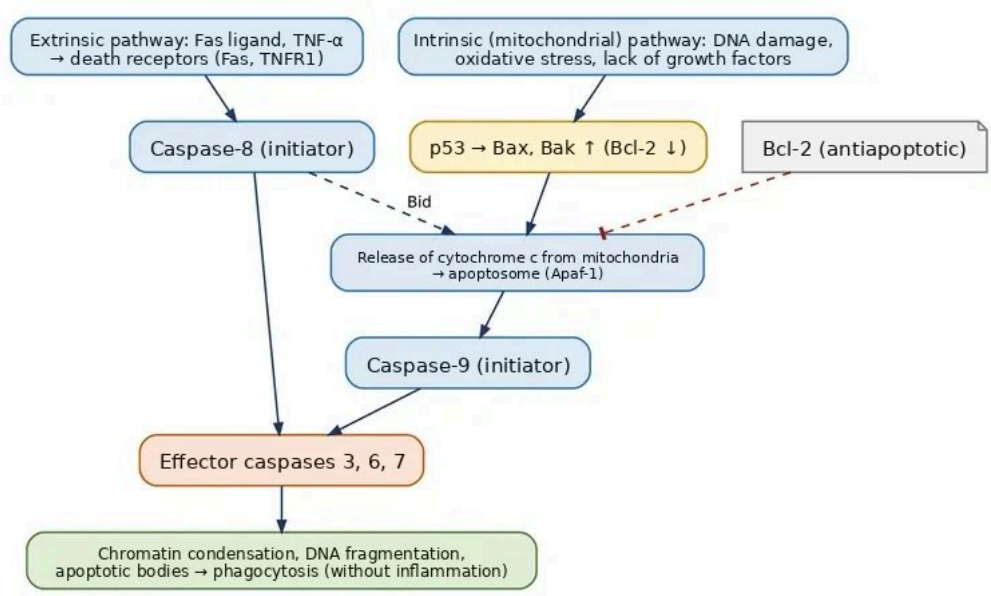


Figure 18.2. *The extrinsic and intrinsic pathways of apoptosis (drawn after Severin E.S.)*

Caspases are proteases with cysteine in the active site that cleave after aspartate; they are synthesized as procaspases. **Initiator** (8, 9) and **effector** (3, 6, 7) caspases are distinguished. Effector caspases cleave cytoskeletal proteins, nuclear lamins and DNA repair enzymes (PARP) and activate **caspase-activated DNase**.

Regulation:

- The **Bcl-2 family** — antiapoptotic (Bcl-2, Bcl-xL) and proapoptotic (Bax, Bak, Bid, Bad) proteins regulate the permeability of the outer mitochondrial membrane;
- **p53** ("guardian of the genome") accumulates when DNA is damaged, arrests the cell cycle in G_1 (p21) and activates repair; if the damage cannot be repaired, it triggers apoptosis through Bax, PUMA and NOXA;
- **IAP** proteins (inhibitors of apoptosis) block caspases;
- growth factors and the PI3K/Akt pathway — survival signals. **Disorders of apoptosis:** decreased apoptosis — tumors (overexpression of Bcl-2 in follicular lymphoma, t(14;18); p53 mutations) and autoimmune diseases; increased apoptosis — neurodegenerative diseases (Alzheimer, Parkinson), the penumbra zone in myocardial infarction and stroke, death of CD4 lymphocytes in AIDS.

7. Basics of cloning and gene therapy

Cloning: gene (molecular) cloning — inserting a gene into a vector and propagating it in bacteria or cell culture (production of recombinant proteins); **cell clones** — genetically identical cells derived from a single cell (monoclonal antibodies — hybridomas); **reproductive cloning** — somatic cell nuclear transfer ("Dolly" the sheep); **therapeutic cloning** — obtaining stem cells matched to the patient. The clinical importance of clones: monoclonal gammopathies (myeloma) and the **clonal** origin of tumors.

Gene therapy is the introduction of genetic material into cells in order to treat a disease.

Table 18.4. *Strategies and vectors of gene therapy*

Approach	Meaning
Gene addition (replacement)	Introduction of a normal copy in place of the defective gene (adenosine deaminase deficiency — the first gene therapy, 1990; spinal muscular atrophy; hemophilia B)
Gene editing	CRISPR-Cas9 — cutting DNA at a precise site and correcting it (increasing HbF via BCL11A in sickle cell anemia and β-thalassemia)
Silencing of gene expression	Antisense oligonucleotides, siRNA (transthyretin amyloidosis)
Cell therapy	CAR-T cells — a gene for a receptor against a tumor antigen is introduced into the patient's T lymphocytes (B-cell leukemias)
Viral vectors	Retroviruses and lentiviruses (integration into the genome — for stem cells; risk of insertional mutagenesis), adenoviruses, adeno-associated viruses (AAV — no integration, safer)
Nonviral methods	Liposomes, lipid nanoparticles, electroporation, “naked” DNA
Methods	Ex vivo (cells are taken from the body, the gene is introduced and the cells are returned) and in vivo (the vector is delivered directly into the body)

Problems: immune response, insertional mutagenesis (leukemia), short duration of expression, high cost, ethical issues (modification of germ cells is prohibited).

8. Laboratory work

Experiment 1: identifying the type of mutation from a sequence

1. Compare the coding strand sequences of normal and patient DNA.
2. Identify the differing nucleotide and its position.
3. Convert the codon to mRNA and find the amino acid in the genetic code table.
4. Name the type of mutation (synonymous, missense, nonsense, frameshift). **Example:** normal … TGG… (Trp) → mutant …TGA… (stop) — a nonsense mutation: the protein is shortened.

Experiment 2: diagnosis of sickle cell anemia by the RFLP method (case-based)

Principle of the method: the A→T mutation in the β-globin gene (GAG→GTG) destroys the cutting site of the restriction enzyme MstII.

1. Amplify the relevant part of the β-globin gene by PCR.
2. Treat with MstII and perform electrophoresis.
3. Compare the fragments. **Results and analysis:** the normal allele gives two short fragments (1.15 and 0.2 kb), and the mutant allele gives one long fragment (1.35 kb); in a heterozygote (HbAS) all three fragments are seen.

Experiment 3: introduction to methods for detecting signs of apoptosis (demonstration)

1. **DNA electrophoresis** — a “ladder” in apoptosis (fragments that are multiples of ≈ 180–200 base pairs) and a “smear” in necrosis.
2. Staining with **annexin V** — detects phosphatidylserine in the outer leaflet of the membrane (flow cytometry).
3. The **TUNEL method** — labels the ends of DNA breaks.

4. **Caspase-3 activity** — with a fluorogenic substrate. **Results and analysis:** using the methods together reliably distinguishes apoptosis from necrosis.

9. Practical skills

Skill: choosing a method of molecular diagnosis for a hereditary disease

1. Determine the mode of inheritance of the disease and the known mutations.
2. If there is a single known mutation — PCR, RFLP or allele-specific PCR.
3. If there are many different mutations — gene sequencing.
4. Chromosomal changes — karyotype, FISH.
5. Trinucleotide expansion — special PCR or Southern blot.

Skill: distinguishing apoptosis from necrosis

1. Assess the cell volume and membrane integrity.
2. Determine whether there is an inflammatory reaction.
3. Assess the pattern of DNA fragmentation (ladder or smear).
4. Take caspase activity and the annexin V result into account.

Review questions

1. List the factors that damage DNA.
2. How do ultraviolet and ionizing radiation damage DNA?
3. How are mutations classified?
4. What is the difference between missense, nonsense and frameshift mutations?
5. How does base excision repair proceed?
6. How are nucleotide excision repair and xeroderma pigmentosum related?
7. Explain the mismatch repair system and Lynch syndrome.
8. How are double-strand breaks repaired? What is the role of BRCA1/2?
9. What is protein polymorphism? Give examples.
10. What does pharmacogenetics study?
11. Which DNA technologies are used to diagnose hereditary diseases?
12. State the differences between apoptosis and necrosis.
13. Describe the extrinsic and intrinsic pathways of apoptosis.
14. What is the role of the p53 and Bcl-2 proteins?
15. List the principles and vectors of gene therapy.

Test questions

1. What damage do ultraviolet rays cause in DNA?
   A) Double-strand breaks
   B) Deamination of cytosine
   C) Alkylation of guanine
   D) Thymine dimers
   E) Depurination

2. Which repair system is defective in xeroderma pigmentosum?
   A) Base excision
   B) Mismatch repair
   C) Homologous recombination
   D) Nucleotide excision
   E) Direct repair

3. Which repair system is defective in Lynch syndrome?
   A) Nucleotide excision
   B) Mismatch repair
   C) Base excision
   D) Nonhomologous end joining
   E) Photoreactivation

4. What kind of mutation converts a codon into a stop codon?
   A) Nonsense
   B) Missense
   C) Synonymous
   D) Frameshift
   E) Transversion

5. What kind of mutation is ΔF508 in cystic fibrosis?
   A) In-frame deletion (3 nucleotides)
   B) Nonsense mutation
   C) Trinucleotide expansion
   D) Splicing mutation
   E) Synonymous mutation

6. Which enzyme initiates base excision repair?
   A) DNA ligase
   B) DNA glycosylase
   C) Telomerase
   D) Photolyase
   E) Topoisomerase

7. Which feature is characteristic of apoptosis?
   A) Cell swelling
   B) Membrane destruction
   C) Inflammatory reaction
   D) No ATP requirement
   E) Cleavage of DNA into fragments of ≈ 180–200 base pairs

8. What is released from mitochondria in the intrinsic pathway of apoptosis?
   A) ATP
   B) Caspase-8
   C) Fas ligand
   D) Bcl-2
   E) Cytochrome c

9. Which protein is antiapoptotic?
    A) Bax
    B) Bak
    C) p53
    D) Bcl-2
    E) Caspase-3

10. Which protein is called the “guardian of the genome”?
    A) p53
    B) Rb
    C) Ras
    D) Myc
    E) Bcl-2

11. Which proteins take part in repair by homologous recombination?
    A) XP-A, XP-C
    B) MSH2, MLH1
    C) BRCA1, BRCA2
    D) Caspases
    E) Ku proteins and photolyase

12. Slow and fast acetylators of isoniazid are associated with polymorphism of which enzyme?
    A) Cytochrome P450 2D6
    B) Pseudocholinesterase
    C) N-acetyltransferase
    D) G6PD
    E) Glutathione transferase

13. In which disease was the first gene therapy used?
    A) Hemophilia A
    B) Adenosine deaminase deficiency
    C) Cystic fibrosis
    D) Phenylketonuria
    E) Duchenne muscular dystrophy

14. Which viral vector does not integrate into the genome and is relatively safe?
    A) Retrovirus
    B) Lentivirus
    C) HIV
    D) Bacteriophage
    E) Adeno-associated virus

15. Which lipid moves to the outer leaflet of the membrane in apoptosis and is detected with annexin V?
    A) Phosphatidylcholine
    B) Sphingomyelin
    C) Phosphatidylserine
    D) Cholesterol
    E) Cardiolipin

Clinical cases

Case 1

An 8-year-old child has many freckles, dryness and ulcers on sun-exposed areas of the skin; a basal cell carcinoma was found on the face. The parents are related and healthy.

Questions: What is the disease? What is its molecular basis? What is the mode of inheritance?

Answer: Xeroderma pigmentosum. Ultraviolet rays form covalent bonds between adjacent pyrimidines in DNA — thymine dimers. They are repaired by the nucleotide excision repair (NER) system; in this disease the XP proteins (enzymes that recognize and excise the damage) are defective — the dimers accumulate and lead to mutations during replication, including in p53 and other genes, so skin cancer develops at an early age. The inheritance is autosomal recessive (consanguineous marriage increases the risk). Treatment and prevention — complete protection from sunlight and early removal of tumors.

Case 2

A 35-year-old woman's mother and maternal aunt died of breast cancer before the age of 45. Genetic testing of the woman found a frameshift mutation in the BRCA1 gene (heterozygous).

Questions: What is the function of the BRCA1 protein? Why does even a single mutant allele increase the risk of cancer? Which group of drugs is used in treatment?

Answer: BRCA1 takes part in the accurate repair of DNA double-strand breaks by homologous recombination (a tumor suppressor). In an inherited mutation one allele is inactive in all cells; if the second allele is lost by a somatic mutation during life (Knudson's “two-hit” hypothesis), homologous recombination does not work, the genome becomes unstable and a tumor develops. Therefore the inheritance appears dominant (risk ≈ 60–70%). PARP inhibitors (olaparib) are used in treatment: by blocking the repair of single-strand breaks, PARP inhibition selectively kills tumor cells with defective homologous recombination (synthetic lethality).

Case 3

In a patient with chronic lymphocytic leukemia, an excess of the Bcl-2 protein was found in the tumor cells. The effect of chemotherapy is poor. The drug venetoclax was recommended.

Questions: How does Bcl-2 excess contribute to tumor growth? Why is chemotherapy ineffective? What is the mechanism of action of venetoclax?

Answer: Bcl-2 is an antiapoptotic protein: it blocks Bax and Bak and prevents the increase in mitochondrial membrane permeability and the release of cytochrome c. As a result, damaged tumor cells do not undergo apoptosis and accumulate. Many chemotherapy drugs damage DNA and trigger the intrinsic apoptotic pathway through p53 — when Bcl-2 is in excess, this pathway is blocked. Venetoclax is a Bcl-2 inhibitor (a BH3 mimetic): it releases the proapoptotic proteins and restores apoptosis in tumor cells.

Quick review

Table 18.5. *Key concepts of topic 18*

Concept	Brief definition
UV rays	Thymine dimers → NER
Ionizing radiation	Double-strand breaks → homologous recombination, NHEJ

Concept	Brief definition
Mutations	Synonymous, missense, nonsense, frameshift, expansion
BER	DNA glycosylase → AP endonuclease → pol β → ligase
NER defect	Xeroderma pigmentosum
MMR defect	Lynch syndrome, microsatellite instability
BRCA1/2	Homologous recombination; breast and ovarian cancer
Polymorphism	AB0, HLA, Hb variants, CYP450, N-acetyltransferase
Apoptosis	Caspases, DNA “ladder”, no inflammation
Pathways	Extrinsic (Fas → caspase-8), intrinsic (cytochrome c → caspase-9)
Regulators	p53, Bax/Bak (pro), Bcl-2 (anti)
Gene therapy	Gene addition, CRISPR, siRNA, CAR-T; AAV and lentivirus vectors

TOPIC 19

Biochemistry of oncogenesis

Learning objectives

To give the student knowledge of chemical, physical and viral carcinogenesis, the biochemical features of tumor cells, proto-oncogenes, oncogenes and tumor suppressor genes, the mechanisms of neoplastic transformation and the multistage theory of carcinogenesis, invasion and metastasis, tumor markers, and the biochemical principles of tumor diagnosis and treatment.

By the end of the class the student should know:

- the types of carcinogenic factors and the metabolic activation of procarcinogens;
- the stages of carcinogenesis (initiation, promotion, progression);
- the mechanisms by which proto-oncogenes are converted into oncogenes;
- the main tumor suppressor genes (TP53, RB1, APC, BRCA);
- the metabolic features of tumor cells (the Warburg effect);
- the biochemical basis of invasion and metastasis;
- tumor markers and their clinical importance;
- the mechanism of multidrug resistance.

The student should be able to:

- interpret tumor marker results;
- show the activation pathway of a carcinogen in a scheme;
- explain the mechanism of targeted therapy.

1. Carcinogenic factors

Table 19.1. *Classification of carcinogenic factors*

Group	Examples	Tumor
Chemical: polycyclic aromatic hydrocarbons	Benzo(a)pyrene (tobacco smoke, charred meat)	Lung, larynx, urinary bladder
Aromatic amines	2-Naphthylamine, benzidine (dye industry)	Urinary bladder
N-Nitroso compounds	Nitrosamines (formed from nitrites in the stomach)	Stomach, esophagus
Mycotoxins	Aflatoxin B_1 (moldy nuts, grain)	Liver (TP53 codon 249)
Inorganic substances	Asbestos, arsenic, chromium, nickel, cadmium, benzene	Mesothelioma, skin, lung, leukemia
Physical	Ultraviolet radiation	Skin (melanoma, basal cell carcinoma)
	Ionizing radiation	Leukemia, thyroid cancer
Biological: DNA viruses	Human papillomavirus (HPV 16, 18)	Cervical cancer
	Hepatitis B virus; Epstein–Barr virus	Liver cancer; Burkitt lymphoma, nasopharyngeal cancer
RNA viruses	Hepatitis C virus, HTLV-1	Liver cancer, T-cell leukemia

Group	Examples	Tumor
Bacteria	Helicobacter pylori	Gastric cancer, MALT lymphoma
Hormones	Estrogens (long-term use)	Endometrium, breast

2. Chemical carcinogenesis

Most chemical carcinogens are **procarcinogens**: they are not active themselves but are converted in the body by the **cytochrome P450** system (topic 26) into electrophilic **ultimate carcinogens**, which form covalent **adducts** with nucleophilic centers of DNA (most often the N7 or N2 atom of guanine). Direct-acting carcinogens (alkylating agents) do not require activation.

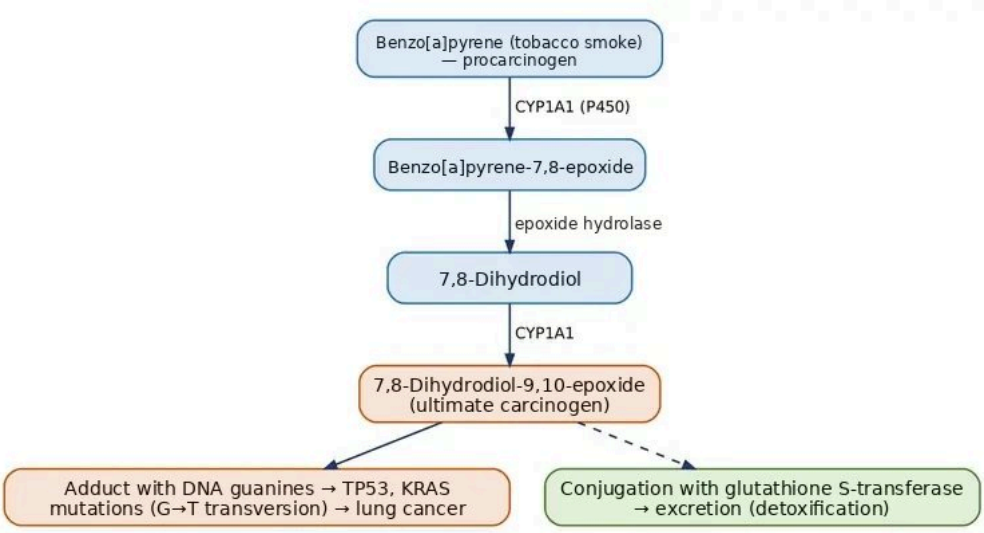


Figure 19.1. *Conversion of benzo(a)pyrene into an ultimate carcinogen and its detoxification (drawn after Severin E.S.)*

Cytochrome P450 polymorphism and carcinogenesis: smokers with highly inducible variants of CYP1A1 have a higher risk of lung cancer; gene variants of detoxifying enzymes (glutathione S-transferase GSTM1, N-acetyltransferase NAT2) affect the risk: for example, the risk is higher in smokers who lack the GSTM1 gene (null genotype); in slow acetylators, bladder cancer caused by aromatic amines is more common.

> **CLINICAL SIGNIFICANCE**
> Tobacco smoke contains more than 60 carcinogens. Benzo(a)pyrene diol epoxide causes G→T transversions at certain “hot spots” of the TP53 gene (codons 157, 248, 273) — the very same mutations are found in lung cancer in smokers. This is evidence of a direct molecular link between smoking and cancer.

3. Proto-oncogenes, oncogenes and tumor suppressor genes

Proto-oncogenes are normal genes that stimulate cell growth, division and differentiation. As a result of an activating mutation they become **oncogenes** (a **dominant** effect — one allele is enough; “a stuck gas pedal”).

Mechanisms of proto-oncogene activation:

1. **Point mutation** — the RAS protein loses its GTPase activity and remains permanently active (codons 12, 13, 61; in ≈ 30% of tumors);
2. **Gene amplification** — MYC (N-MYC in neuroblastoma), HER2/neu (≈ 20% of breast cancers);

3. **Chromosomal translocation** — the gene comes under a strong promoter (Burkitt lymphoma: t(8;14) — MYC next to an immunoglobulin gene) or a hybrid protein is formed (chronic myeloid leukemia: t(9;22) — **BCR-ABL**, a constitutively active tyrosine kinase);
4. **Integration next to a viral promoter** (insertional mutagenesis).

Table 19.2. *Functions of proto-oncogene products*

Group	Examples
Growth factors	SIS (platelet-derived growth factor)
Growth factor receptors (tyrosine kinases)	EGFR (ERBB1), HER2 (ERBB2), RET, KIT
Signal transduction proteins	RAS (G protein), RAF, BRAF (kinases), ABL
Transcription factors	MYC, FOS, JUN
Cell cycle regulators	Cyclin D1, CDK4
Apoptosis inhibitors	BCL-2

Tumor suppressor genes are genes that stop cell division and ensure DNA repair and apoptosis (the "brakes"). For a tumor to develop, **both alleles** must be inactive (a **recessive** effect) — **Knudson's "two-hit" hypothesis**: in the hereditary form the first mutation is inborn and the second is somatic.

Table 19.3. *Main tumor suppressor genes*

Gene	Function	Tumors associated with its defect
TP53	"Guardian of the genome": cell cycle arrest when DNA is damaged, repair, apoptosis	Mutated in ≈ 50% of tumors; Li–Fraumeni syndrome
RB1	Binds the transcription factor E2F and stops the $G_1 \rightarrow S$ transition	Retinoblastoma, osteosarcoma
APC	Ensures the degradation of β-catenin (the Wnt pathway)	Familial adenomatous polyposis, colorectal cancer
BRCA1, BRCA2	Homologous recombination repair	Breast, ovarian and prostate cancer
PTEN	A phosphatase that stops the PI3K/Akt pathway	Glioblastoma, endometrial cancer
NF1	A protein that activates RAS GTPase	Neurofibromatosis
VHL	Targets HIF for degradation	Kidney cancer, hemangioblastoma
CDKN2A (p16)	Inhibitor of CDK4/6	Melanoma, pancreatic cancer

4. The multistage theory of carcinogenesis

A tumor develops not as a result of a single mutation but as a result of successive changes in several (usually 4–7) genes over years (in the colon: APC → KRAS → SMAD4 → TP53 — the **Vogelstein model**: normal epithelium → adenoma → carcinoma).

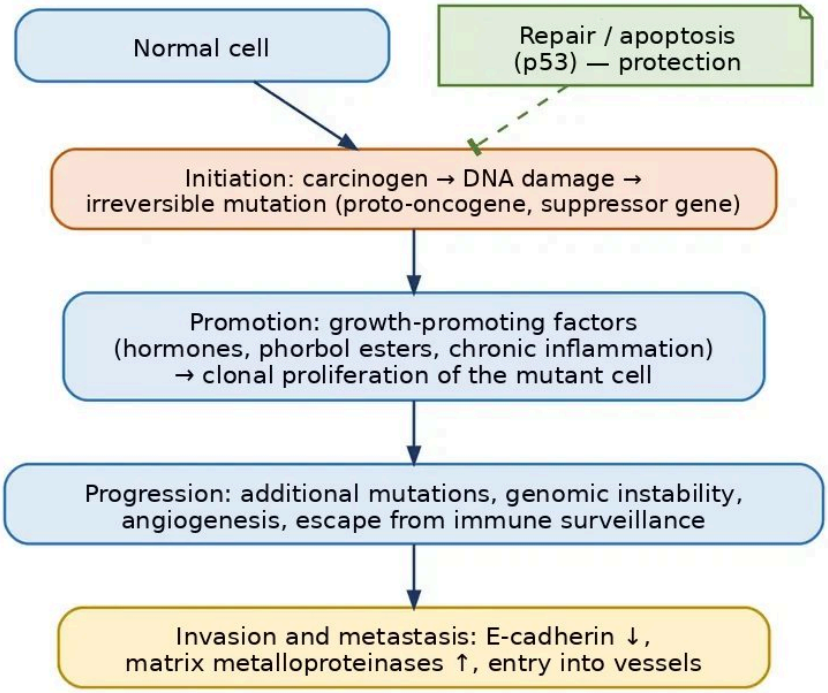


Figure 19.2. *The multistage model of carcinogenesis (drawn after Harper's Illustrated Biochemistry)*

As a result of **neoplastic transformation**, the cell acquires tumor features (the "hallmarks of cancer" according to Hanahan and Weinberg): self-sufficiency in growth signals, insensitivity to growth-inhibitory signals, evasion of apoptosis, unlimited division (**telomerase** activity), **angiogenesis** (VEGF), invasion and metastasis, reprogramming of energy metabolism, evasion of immune surveillance (PD-L1), genome instability, tumor-promoting inflammation.

5. Biochemical features of tumor cells

- **The Warburg effect (aerobic glycolysis):** tumor cells break down glucose mainly to lactate even when oxygen is sufficient; they take up a lot of glucose (GLUT1, hexokinase II ↑) — this is the basis of diagnosis by **PET-CT**, which relies on the accumulation of ^{18}F-fluorodeoxyglucose. Glycolytic intermediates are used for the synthesis of nucleotides, lipids and amino acids; the M2 isoform of pyruvate kinase is expressed.
- **Glutamine "addiction"** — glutamine is a source of nitrogen and carbon (glutaminolysis).
- **Anabolic processes are intense:** synthesis of nucleotides, DNA, protein and fatty acids (fatty acid synthase ↑); the pentose phosphate pathway is active (NADPH, ribose-5-phosphate).
- **Shift of the isoenzyme spectrum to the embryonic type** ("dedifferentiation"): LDH-5 ↑, synthesis of embryonic proteins (α-fetoprotein, carcinoembryonic antigen).
- **Membrane changes:** loss of contact inhibition, adhesion molecules (E-cadherin) ↓, changes in the composition of glycoproteins and gangliosides.
- **Mutant IDH1/2** (gliomas, leukemias) produces the "oncometabolite" 2-hydroxyglutarate, which disrupts the demethylation of DNA and histones.
- **Cachexia:** the tumor "draws in" glucose and amino acids; cytokines (TNF-α, IL-6) enhance lipolysis and muscle proteolysis; lactate is used for gluconeogenesis in the liver (the Cori cycle — a waste of energy).

6. Invasion and metastasis

1. **Detachment** — decrease in E-cadherin, epithelial–mesenchymal transition (EMT).
2. **Degradation of the basement membrane and the extracellular matrix — matrix metalloproteinases** (MMP-2, MMP-9 — type IV collagenases), plasminogen activator (uPA), cathepsins.
3. **Migration** — integrins, cytoskeletal rearrangement.
4. **Entry into blood and lymphatic vessels (intravasation)**, survival in the blood (a "coat" of platelets).
5. **Extravasation** and settling in a new site, **angiogenesis** (VEGF) — a metastatic focus.

7. Tumor markers

Table 19.4. *Main tumor markers*

Marker	Tumor	Comment
α-Fetoprotein (AFP)	Hepatocellular carcinoma, testicular tumors	Physiological ↑ in pregnancy
Carcinoembryonic antigen (CEA)	Colorectal cancer	Also ↑ in smokers; monitoring of treatment effect
PSA (prostate-specific antigen)	Prostate cancer	Also ↑ in adenoma; free/total PSA ratio
CA-125	Ovarian cancer	Also ↑ in endometriosis and peritonitis
CA 19-9	Pancreatic cancer	Also ↑ in cholestasis
CA 15-3	Breast cancer	For monitoring
β-hCG	Choriocarcinoma, testicular tumors	Same as the pregnancy test
Calcitonin	Medullary thyroid cancer	—
Paraprotein (M spike), Bence Jones protein	Multiple myeloma	Detected by electrophoresis (topic 20)
Neuron-specific enolase	Small cell lung cancer, neuroblastoma	—

Tumor markers are used mainly for **monitoring the effect of treatment and early detection of recurrence**; most of them are not specific enough for screening.

8. Biochemical principles of tumor treatment

- **Cytotoxic chemotherapy:** antimetabolites (methotrexate, 5-fluorouracil, 6-mercaptopurine — topic 14), alkylating agents and platinum drugs (DNA cross-links), topoisomerase inhibitors (doxorubicin, etoposide), mitotic poisons (vincristine, paclitaxel — microtubules).
- **Targeted therapy: imatinib** — an inhibitor of the BCR-ABL tyrosine kinase (chronic myeloid leukemia); **trastuzumab** — a monoclonal antibody against HER2; gefitinib, osimertinib — EGFR inhibitors; vemurafenib — a BRAF V600E inhibitor; PARP inhibitors (in BRCA mutation); bevacizumab — against VEGF (an angiogenesis inhibitor).

- **Hormonal therapy:** tamoxifen (an estrogen receptor modulator), aromatase inhibitors, androgen blockade.
- **Immunotherapy:** immune checkpoint inhibitors (antibodies against PD-1/PD-L1, CTLA-4), CAR-T cells (topic 18).
- **Enzymes:** L-asparaginase — leukemia cells cannot synthesize asparagine (asparagine synthetase is low), and asparagine in the blood is broken down. **Multidrug resistance:** in the membrane of tumor cells **P-glycoprotein** (the product of the MDR1 gene, an ABC transporter) increases and, using the energy of ATP, pumps drugs of various structures (doxorubicin, vincristine, paclitaxel) out of the cell. Other mechanisms: amplification of the target enzyme (dihydrofolate reductase with methotrexate), mutation of the target (imatinib-resistant BCR-ABL), increased glutathione S-transferase activity, enhanced repair, disruption of apoptotic pathways.

9. Laboratory work

Experiment 1: determination of α-fetoprotein in blood serum by enzyme immunoassay (demonstration)

Principle of the method: “sandwich” ELISA: the wells of the plate are coated with antibodies against AFP; AFP in the sample binds to them, and then a second, peroxidase-labeled antibody is added; when the substrate (TMB) is added, the color intensity is proportional to the amount of AFP (450 nm).

1. Pipette 50 µl of the standards and serum samples into the wells.
2. Add the conjugate, incubate at 37 °C for 60 minutes and wash 5 times.
3. Add the substrate, after 15 minutes add the stop solution and measure at 450 nm.
4. Find the concentration from the calibration curve. **Interpretation:** in adults the normal value is < 10 IU/ml; > 400 is highly characteristic of hepatocellular carcinoma (moderate increases are possible in chronic hepatitis and cirrhosis).

Experiment 2: assessment of lactate dehydrogenase isoenzymes in tumor tissue and blood serum (case-based)

1. Compare the results for total LDH activity and the isoenzyme spectrum (electrophoresis) with the normal values.
2. Pay attention to an increase in the proportion of LDH-4 and LDH-5. **Results and analysis:** in tumor cells the M-type isoenzymes adapted to anaerobic glycolysis (LDH-4, LDH-5) increase; in lymphomas and leukemias total serum LDH activity is an indicator of tumor mass and prognosis.

10. Practical skills

Skill: clinical interpretation of tumor marker results

1. Determine which tumor the marker is characteristic of.
2. Compare the result with the normal value and assess the degree of increase.
3. Exclude causes of false positives (pregnancy — AFP, hCG; smoking — CEA; inflammation — CA-125).
4. Assess the dynamics: a decrease after treatment — effect; a renewed increase — recurrence or metastasis.

Skill: identifying the target of a targeted therapy drug

1. Identify the molecular abnormality in the tumor (oncogene, amplification, translocation).
2. Choose a drug against the product of this abnormality (a kinase inhibitor or a monoclonal antibody).
3. Explain why a molecular test (HER2, EGFR, BCR-ABL, BRAF) is needed before the drug is used.

Review questions

1. Classify the carcinogenic factors into groups.
2. Explain the concepts of procarcinogen and ultimate carcinogen.
3. How is benzo(a)pyrene converted into a carcinogen?
4. How does cytochrome P450 polymorphism affect carcinogenesis?
5. What are a proto-oncogene and an oncogene?
6. State the mechanisms of proto-oncogene activation with examples.
7. Give examples of tumor suppressor genes.
8. What does Knudson's “two-hit” hypothesis explain?
9. Describe the stages of carcinogenesis.
10. What is the Warburg effect?
11. List the other biochemical features of tumor cells.
12. Which enzymes take part in invasion and metastasis?
13. Name the main tumor markers and their importance.
14. Give examples of targeted therapy.
15. What is the mechanism of multidrug resistance?

Test questions

1. What is the main enzyme system that converts benzo(a)pyrene into an ultimate carcinogen?
   A) Glutathione S-transferase
   B) UDP-glucuronosyltransferase
   C) Monoamine oxidase
   D) Cytochrome P450 (CYP1A1)
   E) Catalase
2. Which hybrid oncogene is formed in chronic myeloid leukemia?
   A) RAS
   B) MYC
   C) HER2
   D) BCL-2
   E) BCR-ABL
3. Which gene is the “guardian of the genome” and is mutated in ≈ 50% of tumors?
   A) RB1
   B) APC
   C) RAS
   D) MYC
   E) TP53

4. How does the RAS protein change in an oncogenic mutation?
   A) It is not synthesized
   B) It loses its GTPase activity and becomes permanently active
   C) It detaches from the membrane
   D) It is rapidly degraded in the proteasome
   E) It becomes a transcription factor

5. What does the Warburg effect mean?
   A) Enhanced glycolysis to lactate in tumor cells even in the presence of oxygen
   B) Enhanced oxidative phosphorylation
   C) Decreased glucose uptake
   D) Enhanced β-oxidation of fatty acids
   E) Accumulation of glycogen

6. What is the main marker of hepatocellular carcinoma?
   A) PSA
   B) CA-125
   C) α-Fetoprotein
   D) Calcitonin
   E) CA 15-3

7. With a tumor of which organ is aflatoxin B_1 associated?
   A) Lung
   B) Urinary bladder
   C) Liver
   D) Skin
   E) Thyroid gland

8. How do tumor suppressor genes act?
   A) Dominantly — one allele is enough
   B) Recessively — both alleles must be inactive
   C) Only on the X chromosome
   D) Only in the mitochondria
   E) Only through a virus

9. In multidrug resistance, which protein pumps drugs out of the cell?
   A) GLUT1
   B) P-glycoprotein (MDR1)
   C) E-cadherin
   D) Telomerase
   E) Caspase-3

10. Which molecule does imatinib inhibit?
   A) BCR-ABL tyrosine kinase
   B) The HER2 receptor
   C) VEGF
   D) PARP
   E) Topoisomerase II

11. Which enzymes break down the basement membrane during invasion?
    A) Caspases
    B) Kinases
    C) Telomerases
    D) Phosphatases
    E) Matrix metalloproteinases

12. Which virus is associated with cervical cancer?
    A) Human papillomavirus 16, 18
    B) Epstein–Barr virus
    C) Hepatitis C virus
    D) HTLV-1
    E) Herpes simplex virus type 1

13. Which translocation is observed in Burkitt lymphoma?
    A) t(9;22)
    B) t(14;18)
    C) t(8;14)
    D) t(15;17)
    E) t(11;22)

14. Which gene is associated with retinoblastoma?
    A) TP53
    B) APC
    C) BRCA1
    D) RB1
    E) VHL

15. Which marker is used to monitor prostate cancer?
    A) CEA
    B) CA 19-9
    C) AFP
    D) PSA
    E) β-hCG

Clinical cases

Case 1

A 58-year-old man has smoked a pack of cigarettes a day for 35 years. Prolonged cough, hemoptysis, weight loss. Bronchoscopy shows a tumor, biopsy — squamous cell carcinoma; a G→T transversion at codon 248 of the TP53 gene was found in the tumor DNA.

Questions: What is the molecular mechanism of action of the carcinogen? Why is the TP53 mutation in particular important? Which stages of carcinogenesis can be identified?

Answer: Benzo(a)pyrene from tobacco smoke is converted in lung cells by CYP1A1 and epoxide hydrolase into the ultimate carcinogen — a diol epoxide — which forms an adduct with DNA guanine; during replication a G→T transversion occurs. TP53 is a tumor suppressor: it arrests cells with damaged DNA or sends them into apoptosis; its inactivation allows damaged cells to proliferate and new mutations to accumulate (genome instability). Initiation — DNA adducts and mutation; promotion — the chronic irritant and inflammatory effect of smoke; progression — additional

mutations (KRAS and others), invasion. The long duration (decades) is consistent with the multistage theory.

Case 2

A 45-year-old patient has a leukocyte count of 120 × 10^9/l and an enlarged spleen. Cytogenetic testing revealed the Philadelphia chromosome. Treatment with imatinib was started; after 3 years the effect decreased.

Questions: What is the molecular basis of the disease? How does imatinib act? What could be the cause of resistance?

Answer: Chronic myeloid leukemia. As a result of the t(9;22) translocation, the BCR and ABL genes fuse to form the constitutively active BCR-ABL tyrosine kinase; it activates signaling pathways (RAS-MAPK, PI3K/Akt, STAT), leading to uncontrolled proliferation of myeloid cells and evasion of apoptosis. Imatinib occupies the ATP-binding site of the ABL kinase and selectively inhibits it (targeted therapy). Resistance is most often due to point mutations in the kinase domain (for example, T315I), so the drug cannot bind; also BCR-ABL amplification or an increase in P-glycoprotein. In such cases next-generation inhibitors (dasatinib, nilotinib, ponatinib) are used.

Case 3

A 62-year-old patient was operated on for colorectal cancer. Before surgery CEA was 42 ng/ml, 1 month after surgery 3 ng/ml (normal < 5). After 8 months CEA rose to 28 ng/ml.

Questions: How do you interpret the CEA dynamics? Why is CEA unsuitable for screening? Which genes are involved in the multistage development of this tumor?

Answer: The fall of CEA to normal after surgery indicates that the tumor was completely removed; the renewed rise after 8 months is a sign of recurrence or metastasis (most often to the liver) — investigation (CT, PET-CT) is needed. CEA is not specific: it is also raised in smokers and in liver cirrhosis, inflammatory bowel disease and pancreatitis, and it is often normal at an early stage — so it is unsuitable for screening and is used mainly for monitoring treatment. Colorectal cancer develops according to the Vogelstein model: inactivation of APC (start of the adenoma) → activation of KRAS → loss of SMAD4 → inactivation of TP53 (carcinoma).

Quick review

Table 19.5. *Key concepts of topic 19*

Concept	Brief definition
Procarcinogen	Activated by P450 → electrophile → DNA adduct
Benzo(a)pyrene	Diol epoxide; TP53 G→T; lung cancer
Oncogenes	RAS, MYC, HER2, BCR-ABL; dominant; “gas pedal”
Activation	Point mutation, amplification, translocation
Suppressors	TP53, RB1, APC, BRCA1/2, PTEN; recessive; “brakes”
Knudson	The “two-hit” hypothesis
Stages	Initiation → promotion → progression → metastasis
Warburg effect	Aerobic glycolysis; PET-CT (FDG)
Invasion	E-cadherin ↓, MMP ↑, angiogenesis (VEGF)

Concept	Brief definition
Markers	AFP, CEA, PSA, CA-125, CA 19-9, β-hCG
Targeted therapy	Imatinib, trastuzumab, EGFR inhibitors, PARP inhibitors
Multidrug resistance	P-glycoprotein (MDR1) — pumps drugs out

TOPIC 20

Blood composition, plasma proteins. Biochemistry of the lymphoreticular system

Learning objectives

To give the student knowledge of the functions of blood and the chemical composition of plasma, plasma proteins and enzymes, “acute phase” proteins, the role of albumin in water distribution and the mechanisms of edema, hematopoiesis and erythrocyte maturation, heme synthesis and its disorders (porphyrias), erythrocyte metabolism, hemoglobin and its derivatives, the buffer systems of blood, gas transport, the lymphoreticular and immune systems, and the metabolism of phagocytes.

By the end of the class the student should know:

- the chemical composition and reference values of blood plasma and serum;
- the fractions of plasma proteins and their functions;
- acute phase proteins and their diagnostic importance;
- oncotic pressure and the mechanisms of edema;
- the stages and regulation of heme synthesis, and the porphyrias;
- the features of erythrocyte metabolism;
- hemoglobin derivatives (oxy-, carboxy-, methemoglobin) and Hb variants;
- the buffer systems of blood and the transport of oxygen and CO_2;
- the mechanism of the “respiratory burst” in phagocytes.

The student should be able to:

- determine total protein and albumin in blood serum;
- interpret proteinogram (electrophoregram) results;
- identify the types of dysproteinemia.

1. Functions and chemical composition of blood

Functions of blood: transport (gases, nutrients, metabolites, hormones), homeostatic (pH, osmotic pressure, temperature), protective (immunity, hemostasis), regulatory (transport of hormones and cytokines), excretory.

The blood volume is ≈ 6–8% of body weight (≈ 5 l); the **hematocrit** is 40–48% in men and 36–42% in women. **Plasma** contains fibrinogen; **serum** is the fluid after clotting, without fibrinogen.

Table 20.1. *Main biochemical parameters of blood plasma (serum)*

Index	Normal value
Total protein	65–85 g/l
Albumin	35–50 g/l
Glucose (fasting)	3.3–5.5 mmol/l
Urea	2.5–8.3 mmol/l
Creatinine	53–115 µmol/l
Total bilirubin	3.4–20.5 µmol/l
Total cholesterol	< 5.2 mmol/l

Index	Normal value
Sodium / potassium	135–145 / 3.5–5.1 mmol/l
Total calcium	2.15–2.55 mmol/l
Chlorides	98–107 mmol/l
Osmolality	280–295 mOsm/kg
pH (arterial blood)	7,35–7,45

2. Blood plasma proteins

Most plasma proteins (albumin, fibrinogen, most of the globulins) are synthesized in the **liver**, and γ-globulins (immunoglobulins) in **plasma cells**.

Functions: maintenance of oncotic pressure, transport, buffering, protection (immunoglobulins, complement), hemostasis (clotting factors), enzymatic, protein reserve.

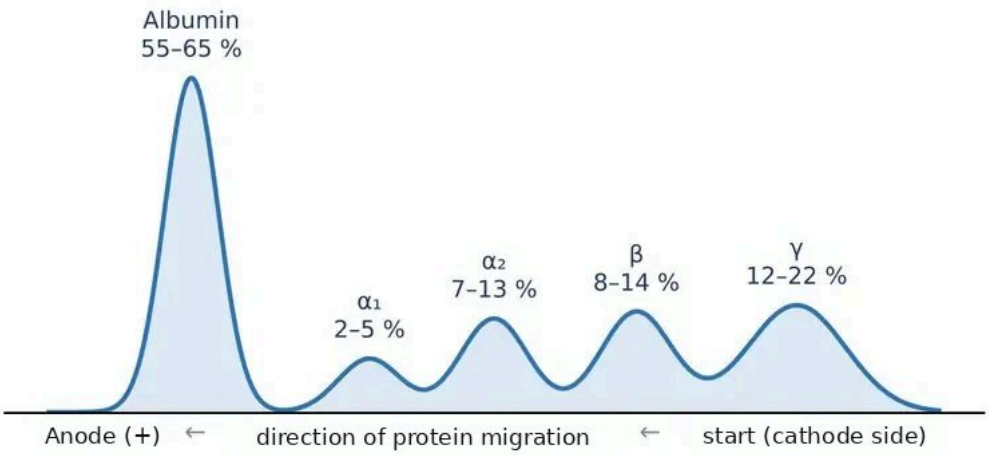


Figure 20.1. *Electrophoregram of serum proteins (normal)*

Table 20.2. *Plasma protein fractions and their main representatives*

Fraction	Main proteins	Functions
Albumin (55–65%)	Albumin, transthyretin (prealbumin)	Oncotic pressure (≈ 80%), transport of fatty acids, bilirubin, drugs, Ca^{2+}, hormones
α_1-Globulins (2–5%)	α_1-antitrypsin, α_1-acid glycoprotein, HDL	Protease inhibitor, transport
α_2-Globulins (7–13%)	α_2-macroglobulin, haptoglobin, ceruloplasmin	Protease inhibitor, binding of free Hb, Cu transport (ferroxidase)
β-Globulins (8–14%)	Transferrin, LDL, hemopexin, C3, fibrinogen (in plasma)	Fe transport, lipids, heme binding, complement
γ-Globulins (12–22%)	Immunoglobulins (IgG, IgA, IgM, IgD, IgE), C-reactive protein	Humoral immunity

Albumin is a small (66 kDa), very abundant protein of 585 amino acids; ≈ 12 g is synthesized per day, and its half-life is ≈ 20 days. **Transthyretin** (half-life ≈ 2 days) is a sensitive indicator of nutritional status; it transports thyroxine and retinol-binding protein.

Plasma enzymes:

- **secretory** (act in plasma) — clotting factors, plasmin, lipoprotein lipase (on the endothelium), lecithin–cholesterol acyltransferase, pseudocholinesterase;
- **excretory** — amylase, lipase (pancreas), alkaline phosphatase (liver, bone);
- **intracellular (indicator)** — ALT, AST, LDH, CK; their increase in serum indicates cell damage (topics 2, 12, 23, 25).

3. Acute phase proteins

In inflammation, trauma, infection and tumors, **IL-6**, IL-1 and TNF-α released from macrophages change the synthesis of acute phase proteins in the liver.

Table 20.3. *Acute phase proteins*

Group	Proteins	Degree of increase
Positive — "major"	C-reactive protein, serum amyloid A	100–1000 times, within 6–8 hours
Positive — moderate	α_1-antitrypsin, α_1-acid glycoprotein, haptoglobin, fibrinogen	2–5 times, within 24 hours
Positive — weak	Ceruloplasmin, complement C3, C4	30–60 %
Negative	Albumin, transferrin, transthyretin	Decrease

C-reactive protein (CRP) reacts with the C-polysaccharide of pneumococci; it activates complement and is an opsonin. Normal < 5 mg/l; in bacterial infection > 100 mg/l; **high-sensitivity CRP** (1–3 mg/l) is used to assess cardiovascular risk. **Procalcitonin** is a specific marker of bacterial sepsis. **Ferritin** is also an acute phase protein (it can mask iron deficiency, topic 21). An increase in acute phase proteins (especially fibrinogen) causes an increase in the **erythrocyte sedimentation rate (ESR)**.

4. The role of albumin in the distribution of body water. Edema

Water exchange in capillaries is determined by **Starling forces**: at the arterial end the hydrostatic pressure (≈ 30–35 mm Hg) is greater than the oncotic pressure (≈ 25 mm Hg) — fluid moves out into the tissue; at the venous end the hydrostatic pressure (≈ 15) is lower than the oncotic pressure — fluid returns. **Albumin** provides ≈ 80% of the oncotic pressure.

Mechanisms of edema:

- **decreased oncotic pressure** (hypoalbuminemia < 25–30 g/l): nephrotic syndrome (proteinuria), liver cirrhosis (synthesis ↓), starvation and kwashiorkor (protein deficiency), protein-losing enteropathy, burns;
- increased hydrostatic pressure (heart failure, venous obstruction);
- increased capillary permeability (inflammation, allergy);
- impaired lymph flow (lymphedema).

5. Dysproteinemias

- **Hypoproteinemia** (< 60 g/l): malnutrition, liver diseases, nephrotic syndrome, blood loss, burns.
- **Hyperproteinemia** (> 85 g/l): relative — dehydration; absolute — paraproteinemia (multiple myeloma), chronic infections (γ-globulins ↑).

- **Dysproteinemia** is a change in the ratio of the fractions.

Table 20.4. *Main types of proteinogram changes*

Condition	Albumin	α_1	α_2	β	γ
Acute inflammation	↓	↑	↑	N	N
Chronic inflammation	↓	↑	↑	N	↑
Nephrotic syndrome	↓↓	N/↑	↑↑ (α_2-macroglobulin)	↑ (LDL)	↓
Liver cirrhosis	↓↓	↓	↓	β–γ “bridge” (IgA)	↑↑
Multiple myeloma	↓	N	N	N or M spike	Narrow M spike (monoclonal Ig)
α_1-Antitrypsin deficiency	N	↓↓	N	N	N

CLINICAL SIGNIFICANCE

Multiple myeloma is a tumor of a single clone of plasma cells: a **paraprotein** in serum (M spike), free light chains in urine — **Bence Jones protein** (precipitates at 60 °C and dissolves on boiling), osteolysis in bones (hypercalcemia), renal failure, anemia (the “CRAB” features).

6. Hematopoiesis and erythrocyte maturation. Heme synthesis

Hematopoiesis begins in the bone marrow from a pluripotent stem cell and is controlled by growth factors: **erythropoietin** (produced in the kidney in hypoxia via HIF-1), thrombopoietin (liver), G-CSF, GM-CSF and interleukins. Erythropoiesis requires iron, **B_{12} and folate** (DNA synthesis), B_6 (heme synthesis), copper and protein.

Erythroid precursors → reticulocyte (anucleate, remnants of ribosomes and mitochondria, 1–2 days) → erythrocyte (lives ≈ 120 days). In erythroid cells the synthesis of heme and globin is coordinated: heme activates globin synthesis (it blocks the eIF2 kinase).

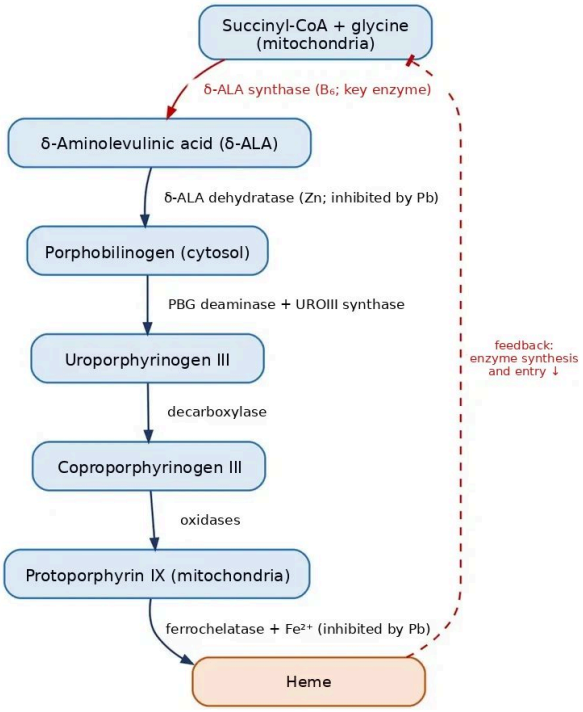


Figure 20.2. *Stages and regulation of heme synthesis (drawn after Severin E.S.)*

Heme is synthesized mainly in the **bone marrow** (≈ 85%) and the **liver** (for cytochromes). The first and the last three reactions take place in the mitochondria, and the intermediate reactions in the cytosol. **The key enzyme is δ-ALA synthase** (coenzyme — pyridoxal phosphate); heme inhibits it by feedback (it decreases its synthesis and its entry into the mitochondria). Hepatic ALA synthase-1 is induced by barbiturates, alcohol and some drugs (heme is needed for P450 synthesis).

Porphyrias are hereditary or acquired defects of the enzymes of heme synthesis; intermediate products accumulate.

- **Acute intermittent porphyria** — PBG deaminase deficiency (autosomal dominant); severe abdominal pain, neurological and psychiatric disturbances; a lot of **δ-ALA and porphobilinogen** in the urine (the urine darkens in light); attacks are provoked by barbiturates, sulfonamides, alcohol and fasting. Treatment — hemin (inhibits ALA synthase), glucose.
- **Porphyria cutanea tarda** — uroporphyrinogen decarboxylase (the most common): skin photosensitivity, blisters; provoked by alcohol, hepatitis C, iron overload.
- **Erythropoietic protoporphyria**, congenital erythropoietic porphyria (Günther disease) — photosensitivity.
- **Lead poisoning** — δ-ALA dehydratase and ferrochelatase are inhibited: δ-ALA increases in the urine and protoporphyrin (zinc protoporphyrin) in erythrocytes; microcytic anemia, basophilic stippling of erythrocytes.

7. Features of erythrocyte metabolism

The mature erythrocyte has no nucleus, mitochondria or ribosomes — protein synthesis, the Krebs cycle and oxidative phosphorylation do not take place.

- **Glucose** (GLUT1, insulin-independent) is the only energy source; ≈ 90% goes through **anaerobic glycolysis** → lactate (ATP for Na^+,K^+-ATPase and the membrane).
- **The 2,3-bisphosphoglycerate (2,3-BPG) shunt** (Rapoport–Luebering) decreases the affinity of hemoglobin for oxygen (facilitates the delivery of oxygen to tissues); it increases at high altitude

and in anemia.

- **Pentose phosphate pathway** (≈ 10%) — **NADPH** → glutathione reductase → reduced **glutathione** → detoxification of H_2O_2 and lipid peroxides (glutathione peroxidase, Se). **In G6PD deficiency**, oxidizing drugs (primaquine, sulfonamides) and legumes (favism) cause a hemolytic crisis (topic 7).
- **Methemoglobin reductase** (NADH–cytochrome b5 reductase) reduces methemoglobin (Fe^{3+}) back to hemoglobin (Fe^{2+}).
- Antioxidant enzymes: superoxide dismutase, catalase, glutathione peroxidase.
- **Pyruvate kinase deficiency** — ATP decreases and erythrocytes undergo hemolysis (2,3-BPG increases).

8. Hemoglobin and its derivatives

Hemoglobin is a hemoprotein of 4 subunits (HbA: $\alpha_2\beta_2$), each of which contains one heme (Fe^{2+} + protoporphyrin IX); it binds 4 molecules of O_2. **Cooperativity** — the dissociation curve is S-shaped (hyperbolic for myoglobin). **Bohr effect** — an increase in H^+ and CO_2 decreases the affinity for O_2 (facilitating O_2 delivery in tissues); 2,3-BPG has the same effect.

Table 20.5. *Types and derivatives of hemoglobin*

Form	Description
HbA ($\alpha_2\beta_2$)	≈ 97% in adults
HbA_2 ($\alpha_2\delta_2$)	2–3%; increases in β-thalassemia
HbF ($\alpha_2\gamma_2$)	Fetal; binds 2,3-BPG weakly — higher affinity for O_2; < 1% in adults
HbA_{1c}	Nonenzymatically glycated HbA; normal < 6.0% (topic 15)
Oxyhemoglobin (HbO_2)	Reversible compound with O_2 (Fe^{2+})
Carbaminohemoglobin ($HbCO_2$)	CO_2 bound to the amino groups of globin
Carboxyhemoglobin (HbCO)	With CO; the affinity is ≈ 200–250 times higher than for O_2; cherry-red color; poisoning — hyperbaric oxygenation
Methemoglobin (MetHb)	Fe^{3+}; does not carry O_2; normal < 1–2%; nitrates, nitrites, aniline, dapsone, benzocaine; cyanosis, “chocolate-colored” blood; treatment — methylene blue, ascorbic acid
HbS	β6 Glu → Val; sickle cell anemia

Hemoglobinopathies: qualitative — changes in the structure of Hb (HbS, HbC, HbM); quantitative — **thalassemias** (reduced synthesis of α or β chains, microcytic anemia).

9. Buffer systems of blood. Gas transport

Table 20.6. *Buffer systems of blood*

System	Composition	Contribution
Bicarbonate	H_2CO_3 / HCO_3^- (1:20)	≈ 53%; an open system (the lungs regulate CO_2, the kidneys regulate HCO_3^-)
Hemoglobin	HHb / KHb, HHbO2 / $KHbO_2$	≈ 35%; imidazole groups of histidine
Protein	Plasma proteins	≈ 7 %

System	Composition	Contribution
Phosphate	$H_2PO_4^-$ / HPO_4^{2-}	≈ 5%; important inside cells and in urine

System Composition Contribution

Henderson–Hasselbalch equation: pH = 6.1 + lg ([HCO_3^-] / 0.03 × pCO_2). Normally HCO_3^- ≈ 24 mmol/l, pCO_2 ≈ 40 mm Hg → pH ≈ 7.4.

Disorders of acid–base balance: metabolic acidosis (HCO_3^- ↓: ketoacidosis, lactic acidosis, renal failure, diarrhea), **metabolic alkalosis** (HCO_3^- ↑: vomiting, diuretics), **respiratory acidosis** (pCO_2 ↑: hypoventilation), **respiratory alkalosis** (pCO_2 ↓: hyperventilation). **Anion gap** = Na^+ − (Cl^- + HCO_3^-), normal 8–16 mmol/l; it increases in ketoacidosis, lactic acidosis and methanol poisoning.

Oxygen transport: 1 g of Hb binds 1.34 ml of O_2; 100 ml of arterial blood carries ≈ 20 ml of O_2; very little is dissolved (0.3 ml).

CO_2 transport: ≈ 70% as **bicarbonate** (in the erythrocyte, **carbonic anhydrase**: $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$; HCO_3^- moves out into the plasma and Cl^- enters the erythrocyte — the **chloride shift**, Hamburger); ≈ 20–23% as **carbaminohemoglobin**; ≈ 7–10% dissolved. Deoxyhemoglobin binds H^+ and CO_2 better (the **Haldane effect**).

10. The lymphoreticular system. Metabolism of phagocytes

The lymphoreticular (immune) system: central organs — bone marrow and thymus; peripheral — lymph nodes, spleen and mucosa-associated lymphoid tissue. Functions: recognition and elimination of foreign antigens, immune memory, surveillance of tumor cells. **Lymph** is formed from tissue fluid; its composition is close to that of plasma but it contains little protein (20–40 g/l; intestinal lymph contains many chylomicrons — it is “milky”) and many lymphocytes. Age-related features: in children the thymus and lymphoid tissue are active; in old age the thymus involutes and the immune response weakens.

Biochemical tools of the immune system: immunoglobulins (IgG — the most abundant, crosses the placenta; IgM — primary response, a pentamer; IgA — mucous membranes, milk; IgE — allergy), the complement system, cytokines, interferons.

Metabolism of phagocytes (neutrophils, macrophages): they obtain energy mainly from **glycolysis** (there is little oxygen at the site of inflammation); the pentose phosphate pathway is active. During phagocytosis there is a **“respiratory (oxidative) burst”**: O_2 consumption rises sharply —

1. **NADPH oxidase** (in the membrane): $2O_2$ + NADPH → $2O_2$ •- (superoxide) + $NADP^+$ + H^+;
2. **superoxide dismutase**: $2O_2$ •- + $2H^+$ → H_2O_2 + O_2;
3. **myeloperoxidase** (neutrophil granules): H_2O_2 + Cl^- → HOCl (hypochlorite) — a powerful bactericide;
4. NO synthase (in macrophages) — NO and peroxynitrite.

> **CLINICAL SIGNIFICANCE**
> **Chronic granulomatous disease** is a hereditary defect of NADPH oxidase (most often X-linked): phagocytes can engulf bacteria but cannot kill them — severe recurrent infections with catalase-positive microorganisms (staphylococci, Aspergillus) and granulomas. Diagnosis — the nitroblue tetrazolium test (negative) or the dihydrorhodamine test.

11. Laboratory work

Experiment 1: determination of total protein in blood serum by the biuret method

Principle of the method: in an alkaline medium peptide bonds form a violet complex with Cu^{2+} ions; the color intensity is proportional to the amount of protein (540 nm).

1. Add 1 ml of biuret reagent and 20 µl of serum to a test tube, 20 µl of standard (70 g/l) to a second tube and 20 µl of water to a third.
2. Keep for 30 minutes at room temperature.
3. Measure at 540 nm against the blank. **Calculation:** C (g/l) = (Esample / Estandard) × 70. **Normal:** 65–85 g/l.

Experiment 2: determination of albumin in blood plasma on a biochemical analyzer (with bromocresol green)

Principle of the method: at pH 4.2 albumin binds the dye bromocresol green (BCG) and changes its color from yellow to green-blue (628 nm).

1. Select the “Albumin (BCG)” test on the analyzer; measure the calibrator and the control serum (the control values must be within the permitted range).
2. Place the tube with the patient's serum in the rack and enter its identifier.
3. The analyzer automatically mixes 5 µl of sample with 1 ml of reagent, measures after 1 minute and reports the result. **Interpretation:** normal 35–50 g/l. **Albumin/globulin ratio** = albumin / (total protein − albumin), normal 1.2–2.0; it decreases in inflammation, cirrhosis and myeloma.

Experiment 3: fractionation of serum proteins by electrophoresis (demonstration)

1. Apply serum to a cellulose acetate strip and run electrophoresis in veronal buffer (pH 8.6) at 200 V for 20 minutes.
2. Stain (amido black or Ponceau S) and read on a densitometer. **Results and analysis:** at pH 8.6 the proteins are negatively charged and move toward the anode; albumin is the fastest. Compare the proportions of the fractions with the normal values and determine the type of dysproteinemia.

12. Practical skills

Skill: interpreting a proteinogram

1. Assess the amounts of total protein and albumin.
2. Compare the proportion of each fraction with the normal values.
3. Look for characteristic features: α_2 ↑ (acute inflammation, nephrotic syndrome), β–γ bridge (cirrhosis), narrow M spike (myeloma), γ ↓ (immunodeficiency), α_1 ↓ (α_1-antitrypsin deficiency).
4. Draw a conclusion by comparing with the clinical picture.

Skill: distinguishing hemoglobin derivatives

1. Assess the color of the blood: cherry-red (HbCO), chocolate-brown (MetHb), dark red (deoxy-Hb).
2. Compare the pulse oximetry reading with arterial blood gases (with HbCO and MetHb the pulse oximeter may read falsely high).
3. Assess the CO-oximetry result and choose the principle of treatment (O_2 or hyperbaric oxygenation; methylene blue).

Review questions

1. List the main functions of blood.
2. What is the difference between plasma and serum?
3. Name the functions and fractions of plasma proteins.
4. What are the functions of albumin?
5. Give examples of acute phase proteins.
6. What is the diagnostic importance of C-reactive protein?
7. Explain the biochemical mechanisms of edema formation.
8. What are the main types of dysproteinemia?
9. Name the stages of heme synthesis and its key enzyme.
10. What are porphyrias? How is heme synthesis disturbed in lead poisoning?
11. List the features of erythrocyte metabolism.
12. What derivatives of hemoglobin exist? What is the danger of carboxyhemoglobin and methemoglobin?
13. Describe the buffer systems of blood.
14. How is CO_2 transported in the blood?
15. How does the “respiratory burst” proceed in phagocytes?

Test questions

1. Which protein mainly maintains the oncotic pressure of plasma?
   A) Fibrinogen
   B) Albumin
   C) Immunoglobulin G
   D) Transferrin
   E) Haptoglobin

2. Which protein is a “negative” acute phase protein?
   A) C-reactive protein
   B) Albumin
   C) Fibrinogen
   D) Haptoglobin
   E) α_1-Antitrypsin

3. Which acute phase protein increases the most (100–1000 times) in inflammation?
   A) C-reactive protein
   B) Ceruloplasmin
   C) Fibrinogen
   D) α_1-Antitrypsin
   E) Transferrin

4. Which change in the proteinogram is characteristic of multiple myeloma?
   A) β–γ bridge
   B) A sharp decrease in α_1-globulins
   C) A sharp increase in α_2-globulins
   D) A narrow M spike in the γ zone
   E) An increase in albumin

5. What is the key enzyme of heme synthesis?
   A) Ferrochelatase
   B) δ-ALA dehydratase
   C) PBG deaminase
   D) Uroporphyrinogen decarboxylase
   E) δ-ALA synthase

6. What increases in the urine in acute intermittent porphyria?
   A) Bilirubin
   B) Uric acid
   C) Hemoglobin
   D) δ-ALA and porphobilinogen
   E) Urobilinogen

7. Which enzymes of heme synthesis does lead inhibit?
   A) δ-ALA synthase and PBG deaminase
   B) Only uroporphyrinogen III synthase
   C) Coproporphyrinogen oxidase
   D) Only δ-ALA synthase
   E) δ-ALA dehydratase and ferrochelatase

8. Which process is the main source of energy in erythrocytes?
   A) Oxidative phosphorylation
   B) β-Oxidation
   C) Anaerobic glycolysis
   D) Krebs cycle
   E) Oxidation of ketone bodies

9. How does 2,3-bisphosphoglycerate affect hemoglobin?
   A) It decreases its affinity for oxygen
   B) It increases its affinity for oxygen
   C) It forms methemoglobin
   D) It enhances its binding to CO
   E) It has no effect

10. What is the valence of iron in methemoglobin?
    A) Fe^{2+}
    B) Fe^{3+}
    C) Fe^{0}
    D) Fe^{4+}
    E) There is no iron

11. Which is the most powerful buffer system of blood?
    A) Bicarbonate
    B) Phosphate
    C) Protein
    D) Ammonia
    E) Acetate

12. In what form is CO_2 mainly transported?
    A) Dissolved
    B) As carbaminohemoglobin
    C) As carboxyhemoglobin
    D) As bicarbonate
    E) As carbamoyl phosphate

13. Which enzyme produces superoxide in phagocytes?
    A) Myeloperoxidase
    B) Superoxide dismutase
    C) NADPH oxidase
    D) Catalase
    E) Glutathione peroxidase

14. What is the normal value of total protein in serum?
    A) 20–40 g/l
    B) 35–50 g/l
    C) 100–120 g/l
    D) 3.3–5.5 g/l
    E) 65–85 g/l

15. What is the determination of albumin by the bromocresol green method based on?
    A) On the reaction of peptide bonds with Cu^{2+}
    B) On the ninhydrin reaction
    C) On albumin binding the dye and changing its color
    D) On the absorbance of albumin at 280 nm
    E) On the precipitation of albumin by heat

Clinical cases

Case 1

A 6-year-old child has swelling of the face, legs and abdomen. Urine protein is 8 g/day. Blood total protein 42 g/l, albumin 15 g/l, cholesterol 11 mmol/l. The proteinogram shows α_2-globulins 28%.

Questions: What syndrome is this? What is the mechanism of the edema? Why are α_2-globulins and cholesterol increased?

Answer: Nephrotic syndrome. The glomerular filter is damaged, and albumin (a small protein) is lost in large amounts in the urine — hypoalbuminemia; the oncotic pressure falls and fluid leaves the vessels for the tissues (edema); the decrease in blood volume activates the renin–angiotensin–aldosterone system and enhances retention of Na^+ and water. α_2-Macroglobulin (a large molecule, 720 kDa) is not filtered, while the liver increases the synthesis of all proteins in response to the fall in oncotic pressure — a relative and absolute increase. For the same reason the synthesis of lipoproteins (LDL) in the liver increases and LPL activity decreases — hypercholesterolemia.

Case 2

In winter a coal stove was burning in a closed room. In the morning the family members woke up with headache, nausea and weakness; one was unconscious. The skin was pinkish red. The pulse oximeter showed SpO_2 98%; CO-oximetry showed HbCO 35%.

Questions: What is the diagnosis? Why did the pulse oximeter show a normal value? What are the mechanism of poisoning and the principles of treatment?

Answer: Carbon monoxide (CO) poisoning. CO binds to the iron of hemoglobin ≈ 200–250 times more strongly than O_2, forming carboxyhemoglobin; the remaining hemes release O_2 less readily (the dissociation curve shifts to the left) — tissue hypoxia. CO also inhibits cytochrome oxidase. An ordinary pulse oximeter cannot distinguish HbCO from HbO_2 (both are red) — a falsely normal value; CO-oximetry is needed. Treatment: removal from the source of poisoning, 100% oxygen (the half-life of HbCO shortens from 4–5 hours to ≈ 1 hour), hyperbaric oxygenation in severe cases.

Case 3

A 30-year-old woman has severe abdominal pain, vomiting, tachycardia and weakness in the legs. The attack began after she took a barbiturate for insomnia. Her urine turned red-brown on standing in the light. Urinary porphobilinogen is sharply increased.

Questions: What is the disease? What is its biochemical basis? Why did the barbiturate provoke the attack? What is the principle of treatment?

Answer: Acute intermittent porphyria — PBG deaminase deficiency (autosomal dominant). Barbiturates induce the synthesis of cytochrome P450 in the liver; heme consumption increases and its amount decreases — the feedback is lost and the synthesis of δ-ALA synthase rises sharply; because of the PBG deaminase deficiency, δ-ALA and porphobilinogen accumulate. They are neurotoxic (autonomic and peripheral nervous system) — abdominal pain, polyneuropathy. In light, porphobilinogen is converted into porphyrins and porphobilin, which turn the urine red. Treatment: stopping the provoking drugs, intravenous hemin (inhibits ALA synthase by feedback) and glucose (reduces the induction of ALA synthase).

Quick review

Table 20.7. *Key concepts of topic 20*

Concept	Brief definition
Plasma proteins	65–85 g/l; albumin 35–50 g/l
Fractions	Albumin, α_1, α_2, β, γ
Acute phase (+)	CRP, serum amyloid A, fibrinogen, haptoglobin, α_1-antitrypsin
Acute phase (−)	Albumin, transferrin, transthyretin
Edema	Oncotic pressure ↓ (albumin < 25–30 g/l)
Myeloma	M spike, Bence Jones protein
Heme synthesis	Succinyl-CoA + glycine → δ-ALA (synthase — the key enzyme, B_6) → … → heme
Porphyrias	AIP: PBG deaminase; lead: ALA dehydratase, ferrochelatase
Erythrocyte	Glycolysis, 2,3-BPG, PPP (NADPH, glutathione), MetHb reductase
Hb derivatives	HbO_2, HbCO (CO), MetHb (Fe^{3+}), HbA_{1c}
Buffers	Bicarbonate (≈ 53%), hemoglobin, protein, phosphate
Respiratory burst	NADPH oxidase → $O_2^{\bullet-}$ → H_2O_2 → HOCl (MPO)

TOPIC 21

Iron metabolism. Biochemistry of hemostasis

Learning objectives

To give the student knowledge of the distribution, absorption, transport, storage and regulation of iron in the body, states of iron deficiency and iron overload, the vascular–platelet and coagulation components of hemostasis, the modern cell-based theory of hemostasis, the anticoagulant and fibrinolytic systems, disorders of hemostasis and the mechanisms of action of antithrombotic drugs.

By the end of the class the student should know:

- the amount and distribution of iron in the body;
- the absorption, transport (transferrin) and storage of iron (ferritin, hemosiderin);
- the regulation of iron metabolism through hepcidin and the IRE/IRP system;
- the biochemical features of iron deficiency anemia, anemia of chronic disease and hemochromatosis;
- the mechanisms of platelet adhesion and aggregation;
- the clotting factors and the role of vitamin K;
- the stages of coagulation according to the cell-based model;
- the anticoagulant and fibrinolytic systems.

The student should be able to:

- determine iron in blood serum and assess the parameters of iron metabolism;
- interpret coagulogram results (PT/INR, aPTT, fibrinogen, D-dimer).

1. Distribution and functions of iron in the body

The adult body contains ≈ 3–5 g of iron (≈ 50 mg/kg in men, ≈ 35–40 mg/kg in women).

Table 21.1. *Distribution of iron in the body*

Pool	Amount	Form
Hemoglobin	≈ 2.5 g (≈ 65%)	Heme
Stores (macrophages of the liver, spleen and bone marrow)	≈ 1 g (men), ≈ 0.3–0.5 g (women)	Ferritin, hemosiderin
Myoglobin	≈ 0.3 g	Heme
Enzymes	≈ 0.1 g	Heme (cytochromes, catalase, peroxidase), Fe-S proteins, ribonucleotide reductase
Transport pool	≈ 3–4 mg	Transferrin

Functions: transport and storage of oxygen (Hb, myoglobin), electron transport (cytochromes, Fe-S centers), redox reactions (P450, catalase), DNA synthesis (ribonucleotide reductase). Free iron is toxic — it forms hydroxyl radicals through the **Fenton reaction** ($Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH^- + OH^\bullet$); therefore iron in the body is always bound to proteins.

2. Absorption, transport and storage of iron

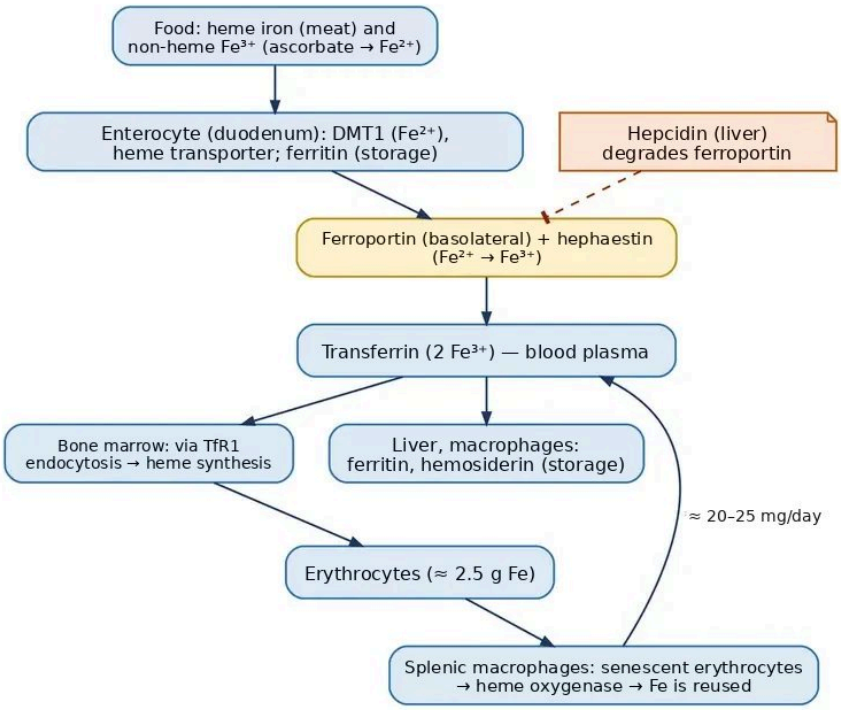


Figure 21.1. *Iron metabolism and its regulation by hepcidin (drawn after Lippincott's Illustrated Reviews: Biochemistry)*

Absorption: the daily diet supplies 10–20 mg of iron, of which only **1–2 mg** (10%) is absorbed (3–5 times more in deficiency) — mainly in the **duodenum and the initial part of the jejunum**.

- **Heme iron** (meat, liver, fish) is well absorbed (20–30%);
- **nonheme iron** (plants) — Fe^{3+} is reduced to Fe^{2+} by **duodenal cytochrome b** (a ferrireductase) on the surface of the intestinal cell and by **ascorbic acid**, and enters through **DMT1**. Gastric HCl, ascorbate, fructose and amino acids improve absorption; phytates (grain), polyphenols (tea, coffee), phosphates, calcium and antacids reduce it. In the enterocyte iron is stored as **ferritin** (the intestinal cell is shed in 2–3 days — the iron is lost) or released into the blood through **ferroportin** in the basolateral membrane; **hephaestin** (a Cu-containing protein) oxidizes it to Fe^{3+}.

Transport: transferrin (a β-globulin synthesized in the liver) binds 2 Fe^{3+} ions. Normally ≈ 1/3 of transferrin is saturated (**saturation coefficient** 20–45%). Cells take up iron by endocytosis through the **transferrin receptor (TfR1)**; in the acidic medium of the endosome the iron is released, and apotransferrin and the receptor return to the membrane.

Storage: ferritin — up to 4500 Fe^{3+} atoms (as hydroxyphosphate) inside a shell of 24 subunits; it is water-soluble and rapidly mobilized. Serum ferritin reflects storage iron (1 μg/l ≈ 8–10 mg of storage iron). **Hemosiderin** is an insoluble aggregate formed by the breakdown of ferritin in lysosomes; it accumulates in iron overload (stains with Prussian blue).

Recycling: erythropoiesis requires ≈ 20–25 mg of iron per day — almost all of it is obtained from the heme of senescent erythrocytes in the macrophages of the spleen and liver (**heme oxygenase**). The body has **no** mechanism for active iron excretion: the daily loss is ≈ 1 mg (intestinal and skin epithelium), in women an additional ≈ 0.5–1 mg with menstruation, and in pregnancy ≈ 3–4 mg/day.

3. Regulation of iron metabolism

Systemic regulation — hepcidin: a peptide hormone of 25 amino acids synthesized in the liver. It binds to ferroportin and causes its endocytosis and degradation — iron absorption from the intestine and iron release from macrophages decrease.

- Hepcidin **increases** in iron overload and **in inflammation (IL-6)** — the cause of anemia of chronic disease;
- hepcidin **decreases** in iron deficiency, hypoxia, enhanced erythropoiesis (erythroferrone) and hemochromatosis (HFE, hemojuvelin defect). **Intracellular regulation — the IRE/IRP system:** when iron is low, **iron regulatory proteins (IRP)** bind to the **iron-responsive elements (IRE)** in mRNA:
- binding to the IRE at the 5'-end of **ferritin** mRNA, they **stop translation** (stores decrease);
- binding to the IRE at the 3'-end of **transferrin receptor** mRNA, they **stabilize** the mRNA (iron uptake increases). When iron is abundant, IRP dissociates — the opposite result.

4. Disorders of iron metabolism

Table 21.2. *Parameters of iron metabolism in various conditions*

Parameter (normal)	Iron deficiency anemia	Anemia of chronic disease	Hemochromatosis
Serum iron (men 11–30, women 9–27 µmol/l)	↓	↓	↑↑
Total iron-binding capacity (45–77 µmol/l)	↑	↓ or N	↓
Transferrin saturation (20–45%)	↓ (< 16 %)	↓ or N	↑↑ (> 45 %)
Ferritin (men 20–250, women 10–120 µg/l)	↓ (< 15–30)	N or ↑	↑↑
Soluble transferrin receptor	↑	N	N
Hepcidin	↓	↑	↓

Iron deficiency anemia is the most common anemia: chronic blood loss (menstruation, gastrointestinal bleeding — in adults a tumor must be excluded), pregnancy, periods of growth, malabsorption (celiac disease, gastric resection), diet. A **microcytic, hypochromic** anemia (MCV, MCH ↓); sideropenic signs — dry skin, hair loss, brittle nails (koilonychia), altered taste (eating chalk or ice — pica). Stages: prelatent (stores ↓ — ferritin ↓), latent (transport iron ↓), overt anemia. Treatment — oral Fe^{2+} preparations (with ascorbic acid), intravenous if necessary.

Anemia of chronic disease (inflammation) — IL-6 → hepcidin ↑ → iron is "trapped" in macrophages (ferritin N or ↑) and does not reach erythropoiesis.

Hemochromatosis: hereditary — mutation of the HFE gene (C282Y, autosomal recessive): low hepcidin → persistently high iron absorption → liver (cirrhosis, hepatocellular carcinoma), pancreas ("bronze diabetes"), heart (cardiomyopathy), skin pigmentation, joints, hypogonadism. **Secondary** — multiple blood transfusions (thalassemia), ineffective erythropoiesis. Treatment — phlebotomy (bloodletting), deferoxamine, deferasirox (chelators).

Acute iron poisoning (iron tablets in children): gastrointestinal injury, metabolic acidosis, liver necrosis; antidote — deferoxamine.

5. Hemostasis. The vascular–platelet component

Hemostasis is the system that stops bleeding when a vessel is damaged and keeps the blood in a fluid state. Its components: the vessel wall, platelets, coagulation factors, anticoagulants, fibrinolysis.

Vascular–platelet (primary) hemostasis:

1. **Vasoconstriction** (thromboxane A_2, serotonin, endothelin).
2. **Adhesion** — platelets adhere to exposed subendothelial collagen through **von Willebrand factor** (vWF) and the GPIb-IX-V receptor, and directly to collagen through GPVI and integrin $\alpha_2\beta_1$.
3. **Activation and secretion** — platelets change shape and release ADP, serotonin, Ca^{2+}, vWF, fibrinogen and factor V from their granules; **thromboxane A_2** is synthesized (from arachidonic acid by **cyclooxygenase-1**); phosphatidylserine of the inner leaflet of the membrane moves to the outer leaflet — a surface for the coagulation complexes.
4. **Aggregation** — the **GPIIb/IIIa** receptor (integrin $\alpha IIb\beta_3$) is activated and links platelets to one another through **fibrinogen** bridges — the **platelet plug**. Normally the **endothelium** is antithrombotic: **prostacyclin (PGI_2)** and **nitric oxide (NO)** stop aggregation and vasoconstriction (through cAMP and cGMP), ecto-ADPase breaks down ADP, and thrombomodulin, heparan sulfate, t-PA and tissue factor pathway inhibitor (TFPI) are produced.

6. Coagulation hemostasis

Table 21.3. *Clotting factors*

Factor	Name	Feature
I	Fibrinogen	Fibrin precursor; acute phase protein
II	Prothrombin	Vitamin K-dependent; precursor of thrombin
III	Tissue factor (TF)	Membrane protein; initiation
IV	Ca^{2+}	Binds γ-carboxyglutamate residues to phospholipids
V	Proaccelerin	Cofactor of prothrombinase
VII	Proconvertin	Vitamin K-dependent; the shortest half-life (≈ 6 hours)
VIII	Antihemophilic globulin A	Cofactor of tenase; carried by vWF; deficiency — hemophilia A
IX	Christmas factor	Vitamin K-dependent; deficiency — hemophilia B
X	Stuart–Prower factor	Vitamin K-dependent; the center of prothrombinase
XI	Plasma thromboplastin antecedent	—
XII	Hageman factor	Contact activation (in vitro)
XIII	Fibrin-stabilizing factor	Transglutaminase

The role of vitamin K: in factors II, VII, IX, X and proteins C and S, glutamate residues are converted to **γ-carboxyglutamate** in the liver by **γ-glutamyl carboxylase** (cofactor — reduced vitamin K, the hydroquinone form). γ-Carboxyglutamate binds the factors to the phospholipid membrane via Ca^{2+}. The vitamin K epoxide formed is regenerated by **vitamin K epoxide reductase (VKORC1)** — **warfarin** inhibits this enzyme.

The cell-based model (modern theory) — clotting takes place on phospholipid surfaces (cells bearing tissue factor and activated platelets) in three stages:

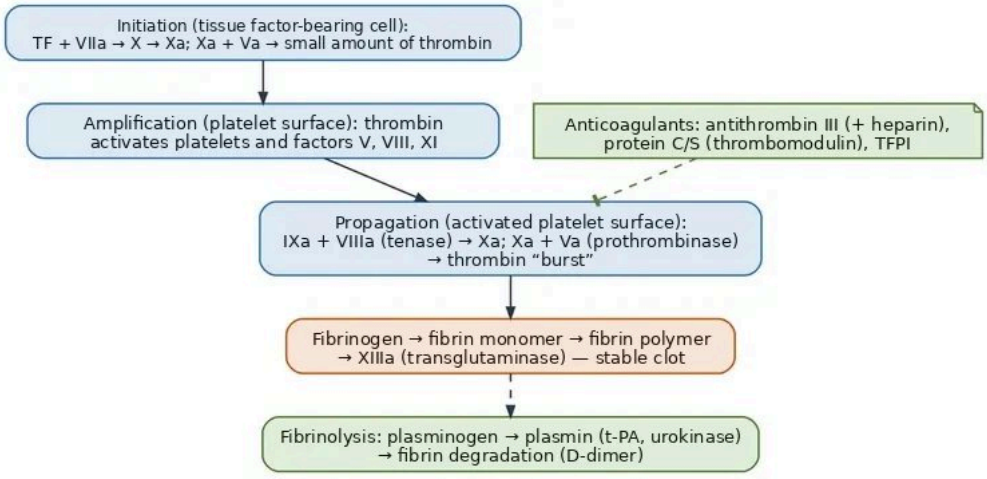


Figure 21.2. *The cell-based model of hemostasis: initiation, amplification, propagation (drawn after Harper's Illustrated Biochemistry)*

1. **Initiation** — when a vessel is damaged, **tissue factor** on fibroblasts and subendothelial cells forms a complex with VIIa and activates X and IX; Xa + Va generate a small amount of thrombin.
2. **Amplification** — this small amount of thrombin activates platelets and activates factors V, VIII (releasing it from vWF) and XI.
3. **Propagation** — on the surface of activated platelets **tenase** (IXa + VIIIa) generates a lot of Xa, and **prothrombinase** (Xa + Va + Ca^{2+} + phospholipid) provides the thrombin "burst". **Fibrin formation:** thrombin cleaves fibrinopeptides A and B from fibrinogen → fibrin monomers polymerize spontaneously (soluble fibrin) → **XIIIa** (activated by thrombin) forms isopeptide bonds between glutamine and lysine residues, creating a stable, insoluble fibrin mesh. Actomyosin inside the platelets (thrombosthenin) contracts the clot (**retraction**).

The classical "extrinsic" (TF, VII) and "intrinsic" (XII, XI, IX, VIII) pathways have been retained to explain laboratory tests: **prothrombin time (PT/INR)** — the extrinsic and common pathways (VII, X, V, II, I); **activated partial thromboplastin time (aPTT)** — the intrinsic and common pathways.

7. Anticoagulant and fibrinolytic systems

Physiological anticoagulants:

- **antithrombin III** — a serpin; it inhibits thrombin, Xa, IXa and XIa; **heparin** (and heparan sulfate on the endothelium) increases its activity ≈ 1000 times;
- **the protein C – protein S system** — thrombin binds to **thrombomodulin** on the endothelium and activates protein C; activated protein C (cofactor — protein S) proteolyzes Va and VIIIa. **Factor V Leiden** (Arg506Gln) is a factor V resistant to activated protein C, the most common hereditary thrombophilia;
- **tissue factor pathway inhibitor (TFPI)** — blocks the TF–VIIa–Xa complex;
- α_2-macroglobulin, heparin cofactor II.

Fibrinolysis: plasminogen (synthesized in the liver) binds to fibrin and is converted into **plasmin** by **tissue plasminogen activator (t-PA)** (endothelium) or **urokinase**; plasmin breaks down fibrin into **fibrin degradation products**, including **D-dimer**. **D-dimer** is a marker of the breakdown of stabilized (cross-linked by XIIIa) fibrin; it increases in deep vein thrombosis, pulmonary embolism and DIC.

Antifibrinolytic system: α_2-antiplasmin (inhibits plasmin), **plasminogen activator inhibitor-1 (PAI-1)**, thrombin-activatable fibrinolysis inhibitor. Drug — **tranexamic acid** (a lysine analog that blocks the binding of plasminogen to fibrin).

8. Disorders of hemostasis and antithrombotic drugs

- **Hemophilia A** (factor VIII) and **B** (factor IX) — X-linked recessive; bleeding into joints (hemarthrosis) and muscles; aPTT prolonged, PT normal.
- **Von Willebrand disease** is the most common hereditary bleeding disorder (autosomal dominant): mucosal bleeding; bleeding time and aPTT prolonged.
- **Vitamin K deficiency** — hemorrhagic disease of the newborn (no intestinal microflora, little vitamin K in breast milk — vitamin K is given in the maternity ward), obstructive jaundice (impaired fat absorption), prolonged antibiotic use, warfarin overdose; PT prolonged.
- **Thrombocytopenias and thrombocytopathies** — hereditary (Glanzmann thrombasthenia — GPIIb/IIIa; Bernard–Soulier syndrome — GPIb) and acquired (aspirin, uremia, immune thrombocytopenia).
- **DIC** (disseminated intravascular coagulation) — sepsis, obstetric complications, tumors: first hypercoagulation, then consumption of factors and platelets and bleeding; fibrinogen ↓, platelets ↓, PT and aPTT prolonged, D-dimer ↑↑.
- **Thrombophilias** — factor V Leiden, prothrombin G20210A, deficiency of antithrombin III, protein C and S, antiphospholipid syndrome, hyperhomocysteinemia.

Table 21.4. *Mechanisms of action of antithrombotic drugs*

Drug	Target
Aspirin	Irreversibly acetylates platelet cyclooxygenase-1 — thromboxane A_2 ↓ (for the lifetime of the platelet, 7–10 days)
Clopidogrel, ticagrelor	Blockers of the ADP receptor $P2Y_{12}$
Abciximab, eptifibatide	GPIIb/IIIa blockers
Heparin (unfractionated)	Activates antithrombin III (thrombin and Xa); monitoring — aPTT; antidote — protamine sulfate
Low-molecular-weight heparins (enoxaparin)	Mainly Xa (via antithrombin)
Warfarin	VKORC1 inhibitor — vitamin K-dependent factors are not carboxylated; monitoring — INR (2.0–3.0); antidote — vitamin K
Rivaroxaban, apixaban	Direct inhibitors of Xa
Dabigatran	Direct thrombin inhibitor
Alteplase (recombinant t-PA), streptokinase	Fibrinolytics — activate plasminogen (infarction, stroke)

9. Laboratory work

Experiment 1: determination of iron in blood serum by the ferrozine method

Principle of the method: in an acidic medium iron is released from transferrin, and a reducing agent (ascorbic acid) converts Fe^{3+} to Fe^{2+}; Fe^{2+} forms a violet complex with ferrozine (562 nm).

1. Add 1 ml of working reagent to the tubes, then 200 µl of serum, standard (17.9 µmol/l) or water.

2. Keep for 10 minutes at room temperature and measure at 562 nm. **Calculation:** C (µmol/l) = (Esample / Estandard) × 17.9.

Interpretation: 11–30 µmol/l in men, 9–27 µmol/l in women. A hemolyzed sample is unsuitable (iron is released from erythrocytes). The diurnal variation of iron is large (higher in the morning) — therefore ferritin and transferrin saturation are also determined for diagnosis.

Experiment 2: determination of coagulogram parameters (prothrombin time and aPTT, demonstration)

Principle of the method: the time to clot formation after adding thromboplastin (tissue factor + phospholipid) and $CaCl_2$ to citrated plasma (PT); with a kaolin–phospholipid activator and $CaCl_2$ — aPTT.

1. Collect blood with 3.2% sodium citrate (9:1) and separate the plasma by centrifugation.
2. At 37 °C add 100 µl of thromboplastin–calcium reagent to 50 µl of plasma and record the clotting time on a coagulometer.
3. For aPTT add 50 µl of activator to 50 µl of plasma, incubate for 3 minutes, then add 50 µl of $CaCl_2$ and record the time.

Table 21.5. *Reference values of the coagulogram*

Index	Normal value	Prolongation
Prothrombin time	11–15 s	Warfarin, vitamin K deficiency, liver diseases, factor VII
INR (international normalized ratio)	0.8–1.2 (2.0–3.0 on warfarin)	The same causes
aPTT	25–35 s	Heparin, hemophilia A and B, von Willebrand disease, lupus anticoagulant
Fibrinogen	2–4 g/l	↓ DIC, liver; ↑ inflammation, pregnancy
D-dimer	< 0.5 mg/l (FEU)	↑ thrombosis, PE, DIC

10. Practical skills

Skill: assessing iron metabolism parameters in anemia

1. Assess hemoglobin, MCV and MCH (microcytic hypochromic?).
2. Determine ferritin: < 30 µg/l — iron deficiency (in the absence of inflammation).
3. Assess serum iron, TIBC and transferrin saturation.
4. Determine CRP: in inflammation ferritin may be falsely normal.
5. Distinguish iron deficiency from anemia of chronic disease; look for the source of blood loss.

Skill: determining the cause of bleeding from the coagulogram

1. Assess the platelet count and bleeding time (primary hemostasis).
2. PT prolonged, aPTT normal — factor VII, warfarin, early vitamin K deficiency.
3. aPTT prolonged, PT normal — factors VIII, IX, XI (hemophilia), heparin, von Willebrand disease.
4. Both prolonged, fibrinogen and platelets low, D-dimer high — DIC.

Review questions

1. How much iron is in the body and how is it distributed?
2. How is iron absorbed in the intestine? What factors affect it?
3. What are the functions of transferrin and ferritin?
4. How does hepcidin regulate iron metabolism?
5. How does the IRE/IRP system work?
6. Name the causes and biochemical features of iron deficiency anemia.
7. What is hemochromatosis?
8. Describe the stages of vascular–platelet hemostasis.
9. List the antithrombotic factors of the endothelium.
10. Name the clotting factors. Which of them are vitamin K-dependent?
11. What role does vitamin K play in blood clotting? How does warfarin act?
12. Explain the stages of the cell-based model of hemostasis.
13. List the physiological anticoagulants.
14. How does fibrinolysis proceed? What does D-dimer indicate?
15. Name the mechanisms of action of antithrombotic drugs.

Test questions

1. Which protein exports iron from the intestine into the blood?
   A) Transferrin
   B) Ferritin
   C) DMT1
   D) Hepcidin
   E) Ferroportin

2. What is the effect of hepcidin?
   A) It enhances iron absorption
   B) It increases transferrin synthesis
   C) It stops ferritin translation
   D) It degrades ferroportin and reduces iron release
   E) It activates heme oxygenase

3. Which parameter best reflects storage iron?
   A) Serum iron
   B) Hemoglobin
   C) Serum ferritin
   D) Transferrin
   E) Bilirubin

4. Which parameter increases in iron deficiency anemia?
   A) Ferritin
   B) Transferrin saturation
   C) Total iron-binding capacity
   D) Serum iron
   E) Hepcidin

5. Hereditary hemochromatosis is associated with a mutation in which gene?
   A) HFE
   B) TfR1
   C) DMT1
   D) VKORC1
   E) F8

6. Which receptor mediates platelet aggregation through fibrinogen?
   A) GPIb-IX-V
   B) GPVI
   C) $P2Y_{12}$
   D) GPIIb/IIIa
   E) Thrombomodulin

7. Which clotting factor is not vitamin K-dependent?
   A) II
   B) VII
   C) IX
   D) VIII
   E) X

8. Which enzyme does warfarin inhibit?
   A) Cyclooxygenase-1
   B) Thrombin
   C) Factor Xa
   D) Plasmin
   E) Vitamin K epoxide reductase

9. Through which protein is the anticoagulant effect of heparin exerted?
   A) Protein C
   B) Antithrombin III
   C) Protein S
   D) Thrombomodulin
   E) α_2-Antiplasmin

10. Which is a marker of the breakdown of stabilized fibrin?
    A) Fibrinopeptide A
    B) D-dimer
    C) Plasminogen
    D) Fibrinogen
    E) Prothrombin

11. What is the antiplatelet effect of aspirin based on?
    A) Blocking ADP receptors
    B) Blocking GPIIb/IIIa
    C) Irreversible acetylation of cyclooxygenase-1
    D) Activation of antithrombin III
    E) Inhibition of factor Xa

12. Which factor is deficient in hemophilia A?
    A) VIII
    B) IX
    C) VII
    D) XI
    E) von Willebrand factor

13. Which factor stabilizes the fibrin mesh with isopeptide bonds?
    A) Thrombin
    B) Plasmin
    C) Va
    D) IXa
    E) XIIIa

14. Which is the most common hereditary thrombophilia?
    A) Factor V Leiden
    B) Hemophilia B
    C) Glanzmann thrombasthenia
    D) Von Willebrand disease
    E) Afibrinogenemia

15. Which factors does the prothrombin time mainly reflect?
    A) VIII, IX, XI, XII
    B) VII, X, V, II, I
    C) Only platelets
    D) Only factor XIII
    E) Only von Willebrand factor

Clinical cases

Case 1

A 32-year-old woman complains of weakness, fatigue, hair loss and brittle nails; she craves chalk. Her menstrual periods are heavy and prolonged. Hb 92 g/l, MCV 68 fl, serum iron 5 µmol/l, TIBC 88 µmol/l, transferrin saturation 6%, ferritin 6 µg/l.

Questions: What kind of anemia is this? Explain the parameters biochemically. What are the principles of treatment?

Answer: Iron deficiency anemia (chronic blood loss — menorrhagia). Storage iron is exhausted (ferritin very low), transport iron is low (serum iron and transferrin saturation low); in iron deficiency the liver increases transferrin synthesis — TIBC is high. There is not enough iron for heme synthesis — the erythrocytes are small and contain little hemoglobin (microcytosis, hypochromia). Deficiency of iron-containing enzymes (in the epithelium) causes changes in the skin, hair and nails and pica. Treatment: identify and eliminate the cause of the menorrhagia; Fe^{2+} preparations with ascorbic acid, taken separately from tea and dairy products; continue for 2–3 months after Hb normalizes to replenish the stores (under ferritin control).

Case 2

A 68-year-old man takes warfarin for atrial fibrillation. A week after he started taking an antibiotic for pneumonia, his gums began to bleed and his urine turned red. INR 7.5 (target 2–3).

Questions: What is the mechanism of action of warfarin? Why did the INR rise? How should help be provided?

Answer: Warfarin inhibits vitamin K epoxide reductase and decreases the reduced form of vitamin K; factors II, VII, IX, X and proteins C and S are not carboxylated in the liver — they cannot bind Ca^{2+} and phospholipids, and clotting slows down. Antibiotics eliminate the intestinal microflora and reduce the source of vitamin K_2; some of them (metronidazole, fluoroquinolones, macrolides) inhibit CYP2C9 in the liver and slow the breakdown of warfarin — its effect increases. Help: stop warfarin and give vitamin K (phytomenadione); in serious bleeding, prothrombin complex concentrate or fresh frozen plasma (ready-made factors).

Case 3

A 5-year-old boy has prolonged bleeding after minor injuries and bleeding into the knee joint. His mother's brother has the same disease. Platelets 280 × 10^9/l, PT 13 s, aPTT 78 s, factor VIII activity 2%.

Questions: What is the diagnosis? Explain the laboratory results on the basis of the hemostasis model. What is the mode of inheritance?

Answer: Hemophilia A (severe form). Factor VIII is the cofactor of the tenase complex (IXa + VIIIa); without it, the “burst” of Xa and thrombin on the platelet surface does not occur in the propagation stage — a stable fibrin clot is not formed; primary (platelet) hemostasis is preserved, so what is characteristic is not bleeding from minor wounds but deep bleeding into joints and muscles. aPTT (intrinsic pathway: XII, XI, IX, VIII) is prolonged, PT (VII, X, V, II) is normal. The inheritance is X-linked recessive: the mother is a carrier, her brother is affected. Treatment — recombinant factor VIII or emicizumab (a bispecific antibody that binds IXa and X).

Quick review

Table 21.6. *Key concepts of topic 21*

Concept	Brief definition
Iron	3–5 g; ≈ 65% in hemoglobin; absorption 1–2 mg/day
Absorption	DMT1 (Fe^{2+}), ascorbate improves, phytates and tea reduce
Transport and storage	Transferrin (2 Fe^{3+}); ferritin, hemosiderin
Regulation	Hepcidin → ferroportin ↓; IRE/IRP (ferritin, TfR1)
IDA	Ferritin ↓, TIBC ↑, saturation ↓, microcytosis
Hemochromatosis	HFE; saturation > 45%, ferritin ↑↑; phlebotomy
Primary hemostasis	vWF + GPIb (adhesion), TxA_2, ADP, GPIIb/IIIa (aggregation)
Vitamin K-dependent	II, VII, IX, X, proteins C, S; warfarin — VKORC1
Cell-based model	Initiation (TF-VIIa) → amplification → propagation (tenase, prothrombinase)
Anticoagulants	Antithrombin III (+ heparin), protein C/S, TFPI
Fibrinolysis	t-PA → plasmin → D-dimer; α_2-antiplasmin, PAI-1
Tests	PT/INR (extrinsic), aPTT (intrinsic), fibrinogen, D-dimer

TOPIC 22

Biochemistry of connective tissue

Learning objectives

To give the student knowledge of the composition and functions of the extracellular matrix, the amino acid composition, structure, types, synthesis and breakdown of collagen, the importance of ascorbic acid, diseases associated with disorders of collagen synthesis, the structure and metabolism of elastin, the structure, functions, synthesis and breakdown of glycosaminoglycans and proteoglycans (mucopolysaccharidoses), and adhesive proteins.

By the end of the class the student should know:

- the main components of the extracellular matrix;
- the specific features of the amino acid composition and structure of collagen;
- the types of collagen and their location;
- the intracellular and extracellular stages of collagen synthesis and the role of vitamin C and copper;
- the breakdown of collagen and the diagnostic importance of hydroxyproline determination;
- the properties of elastin and the role of desmosine;
- the structure and functions of glycosaminoglycans and proteoglycans;
- mucopolysaccharidoses and collagenopathies.

The student should be able to:

- determine hydroxyproline in urine and interpret the result;
- determine glycosaminoglycans (sialic acids) in blood serum;
- explain the biochemical features of connective tissue diseases.

1. The extracellular matrix

Connective tissue consists of cells (fibroblasts, chondrocytes, osteoblasts, macrophages, mast cells) and a large amount of **extracellular matrix**. Composition of the matrix:

- **fibers** — collagen, elastin, reticulin;
- **ground (amorphous) substance** — proteoglycans, glycosaminoglycans, water;
- **adhesive glycoproteins** — fibronectin, laminin, nidogen, tenascin. **Functions:** supportive-mechanical (tendon, bone, cartilage), protective (barrier, phagocytosis), trophic (exchange of substances), water–salt balance, reparative (wound healing), signaling (storage of growth factors, cell migration and differentiation).

2. Collagen: composition and structure

Collagen is the most abundant protein in the body (≈ 25–33% of total protein).

Specific features of the amino acid composition:

- every third amino acid is **glycine** (≈ 33%): the (Gly–X–Y)n repeat; because glycine is the smallest, the three chains pack tightly along the axis of the helix;
- there is a lot of **proline** and **hydroxyproline** (≈ 21%; proline is most often in position X and hydroxyproline in position Y);

- **hydroxylysine** — for glycosylation and cross-links;
- **no tryptophan**, little cysteine and tyrosine — collagen is not a nutritionally complete protein.

Structure:

- **primary** — (Gly–X–Y)n;
- **secondary** — each α-chain is a left-handed **polyproline helix** (not an α-helix; the proline ring does not allow an α-helix);
- **tertiary** — three α-chains wind around one another to form a right-handed **triple helix** (superhelix) — **tropocollagen** (length 300 nm, diameter 1.5 nm); hydrogen bonds (with the participation of the OH groups of hydroxyproline) stabilize it;
- **quaternary** — tropocollagen molecules are arranged in a staggered manner (shifted by 1/4 of their length) to form **fibrils** (periodic cross-striations of 67 nm in the electron microscope); fibrils combine into fibers.

Table 22.1. *Main types of collagen*

Type	Location	Feature
I	Skin, bone, tendon, cornea, dentin	The most abundant (≈ 90%); fibril-forming
II	Hyaline cartilage, vitreous body	Fibril-forming
III	Blood vessels, skin (embryonic), internal organs, reticular fibers	Fibril-forming; early stage of wound healing
IV	Basement membranes	Network-forming; the propeptides are retained
VII	Fibrils anchoring the epithelium to the basement membrane	Defect — epidermolysis bullosa
IX, XII	Fibril surface (FACIT)	Link fibrils to other molecules

3. Collagen synthesis

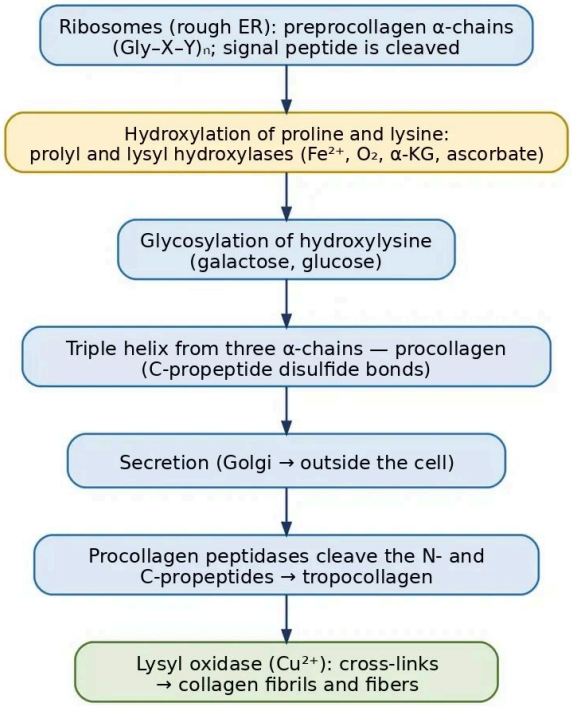


Figure 22.1. *Stages of collagen biosynthesis (drawn after Severin E.S.)*

Intracellular stages (fibroblast):

1. Synthesis of the **preprocollagen** α-chains in the rough endoplasmic reticulum; the signal peptide is cleaved off.
2. **Hydroxylation — prolyl 4-hydroxylase**, prolyl 3-hydroxylase and **lysyl hydroxylase**; cofactors: **Fe^{2+}, O_2, α-ketoglutarate and ascorbic acid** (keeps the iron in the reduced state). Hydroxylation is possible only until the helix is formed.
3. **Glycosylation** of hydroxylysine residues (galactose, glucosyl-galactose).
4. The C-propeptides join the three chains by disulfide bonds — the **triple helix** forms like a "zipper" (from the C-end to the N-end) — **procollagen**.
5. **Secretion** via the Golgi apparatus. **Extracellular stages:**

1. **Procollagen N- and C-peptidases** cleave off the propeptides → **tropocollagen** (the propeptides enter the blood — **markers of collagen synthesis**: PINP, PICP).
2. The molecules assemble spontaneously into fibrils.
3. **Lysyl oxidase** (a copper-containing enzyme; cofactor — Cu^{2+}) oxidizes the ε-amino groups of lysine and hydroxylysine to aldehydes (allysine); the aldehydes form **cross-links** with amino groups on neighboring molecules — the fibers are strengthened. With age the number of cross-links increases (skin elasticity decreases).

CLINICAL SIGNIFICANCE

Scurvy (vitamin C deficiency) — prolyl and lysyl hydroxylases do not work; unhydroxylated chains do not form a stable triple helix and are degraded in the cell. Collagen is scarce and fragile: swollen and bleeding gums, loosening and loss of teeth, perifollicular hemorrhages, slow wound healing, reopening of old scars, bone fragility (Möller–Barlow disease in children). Treatment — ascorbic acid.

Regulation: collagen synthesis is enhanced by **TGF-β**, insulin, somatotropin (IGF-1) and ascorbate; **glucocorticoids** reduce it (with long-term use the skin thins, wounds heal slowly, osteoporosis develops). Factors that chronically activate fibroblasts cause **fibrosis** (liver cirrhosis, pulmonary fibrosis, keloid).

4. Breakdown and turnover of collagen

Collagen turns over slowly (half-life from weeks to years), faster in bone and during growth. **Collagenases** (matrix metalloproteinases — MMP-1, MMP-8, MMP-13; Zn^{2+}-containing) cut the triple helix at a specific site (into 3/4 and 1/4 fragments); then gelatinases (MMP-2, MMP-9) and lysosomal cathepsins break it down. MMPs are regulated by tissue inhibitors (TIMPs).

Because **hydroxyproline** is found almost only in collagen and is not reused, its **amount in the urine** is an indicator of collagen breakdown: normal in adults 15–50 mg/day (more in children because of growth). **It increases** with enhanced bone resorption (hyperparathyroidism, Paget disease, bone metastases, osteoporosis), collagenoses (rheumatoid arthritis, systemic lupus erythematosus, scleroderma), burns, thyrotoxicosis. **It decreases** in hypothyroidism, scurvy and hypopituitarism. Modern markers of bone resorption — **C-telopeptide (CTX)**, deoxypyridinoline.

5. Disorders of collagen synthesis and maturation

Table 22.2. *Diseases associated with collagen*

Disease	Defect	Main features
Osteogenesis imperfecta (“glass man”)	Type I collagen genes (COL1A1, COL1A2) — most often substitution of glycine	Multiple fractures, blue sclerae, deafness, dental defects
Ehlers–Danlos syndrome	Various types: genes of type III or V collagen, lysyl hydroxylase, procollagen peptidase	Excessive skin extensibility, joint hypermobility, fragile vessels (type III — arterial rupture)
Alport syndrome	Type IV collagen (basement membrane)	Nephritis (hematuria), deafness, eye defects
Epidermolysis bullosa	Type VII collagen, keratins	The skin blisters on friction
Menkes disease	Copper transporter ATP7A (X-linked) — lysyl oxidase is inactive	“Kinky hair”, neurological disorders, vascular defects
Lathyrism	β-Aminopropionitrile (from the sweet pea plant) inhibits lysyl oxidase	Bone deformities, aortic aneurysm
Scurvy	Vitamin C deficiency — hydroxylation is impaired	Bleeding gums, hemorrhages
Collagenoses (autoimmune)	Systemic lupus erythematosus, scleroderma, dermatomyositis, rheumatoid arthritis	Systemic inflammation of connective tissue; excessive collagen synthesis in scleroderma

Biochemical diagnosis of collagenoses: acute phase proteins (CRP, fibrinogen, seromucoid, sialic acids), proteinogram (α_2 and γ ↑), urinary hydroxyproline ↑, autoantibodies (antinuclear antibodies, anti-double-stranded DNA antibodies — SLE, rheumatoid factor and anti-cyclic citrullinated peptide antibodies — rheumatoid arthritis, Scl-70 — scleroderma), complement ↓.

6. Elastin

Elastin is the protein that gives elasticity to the large vessels (aorta), lungs, skin and ligaments (ligamentum flavum); it stretches and then returns to its original state.

- It contains a lot of glycine, alanine, valine and proline (hydrophobic); little hydroxyproline, no hydroxylysine and no carbohydrates.
- The monomer is **tropoelastin**; outside the cell it assembles on **fibrillin** microfibrils.
- **Lysyl oxidase** forms from lysine residues the characteristic cross-links that join four lysine residues — **desmosine and isodesmosine** — a three-dimensional network of randomly coiled chains ("rubber").
- It turns over very slowly (over a lifetime); it is broken down by **neutrophil elastase**; α_1-antitrypsin inhibits elastase.

CLINICAL SIGNIFICANCE

α_1-Antitrypsin deficiency — in the lungs neutrophil elastase acts uncontrolled and breaks down the elastin in the alveolar walls — early **panacinar emphysema** (smoking oxidizes and inactivates α_1-antitrypsin and sharply increases the risk); in the PiZZ variant the misfolded protein accumulates in hepatocytes and leads to liver cirrhosis. **Marfan syndrome** — mutation of the **fibrillin-1** gene: tall stature, long fingers (arachnodactyly), dislocation of the lens, aortic aneurysm and dissection.

7. Glycosaminoglycans and proteoglycans

Glycosaminoglycans (GAGs) are long unbranched polysaccharides made of repeating disaccharide units: a **uronic acid** (glucuronic or iduronic) + an **amino sugar** (N-acetylglucosamine or N-acetylgalactosamine); most of them are sulfated. They carry many negative charges ($–COO^-$, $–SO_3^-$) — they strongly bind Na^+ and **water** and form a gel.

Table 22.3. *Main glycosaminoglycans*

GAG	Disaccharide unit	Location and function
Hyaluronic acid	Glucuronic acid + N-acetylglucosamine; unsulfated, not covalently bound to protein	Synovial fluid (lubrication), vitreous body, cartilage, umbilical cord; cell migration
Chondroitin sulfates	Glucuronic acid + N-acetylgalactosamine sulfate	Cartilage, bone, vessels — the most abundant GAG
Dermatan sulfate	Iduronic acid + N-acetylgalactosamine sulfate	Skin, vessels, heart valves
Keratan sulfate	Galactose + N-acetylglucosamine sulfate (no uronic acid)	Cornea (transparency), cartilage
Heparan sulfate	Glucuronic/iduronic acid + N-acetylglucosamine	Basement membranes, cell surface (syndecan) — growth factor receptor, antithrombin
Heparin	Iduronic acid + glucosamine (the most highly sulfated)	Mast cell granules — anticoagulant (topic 21)

Proteoglycans are GAG chains covalently bound to a core protein (protein ≈ 5%, carbohydrate ≈ 95%). In cartilage, **aggrecan** molecules attach to a hyaluronic acid chain through a link protein and form huge aggregates.

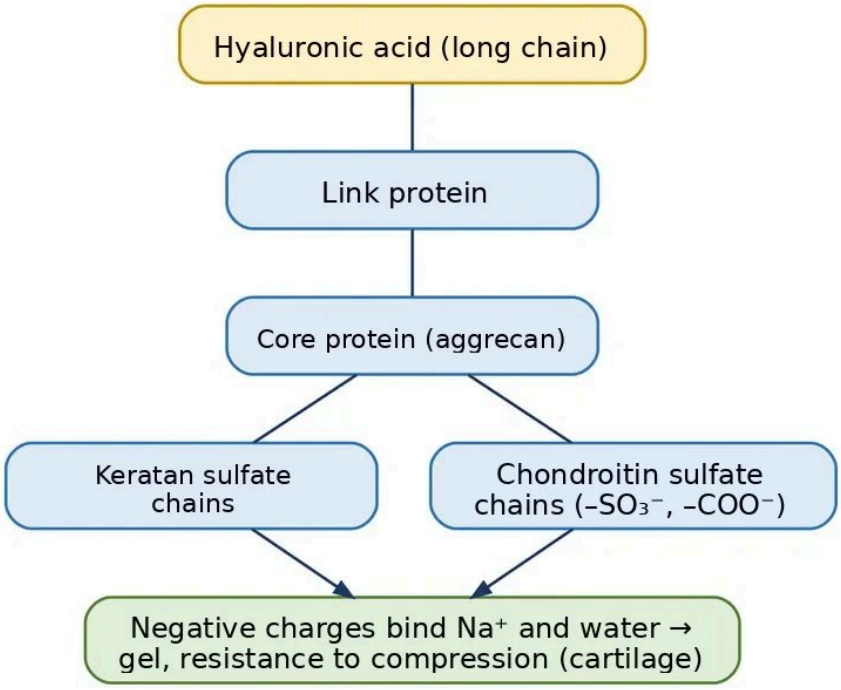


Figure 22.2. *Structure of the cartilage proteoglycan aggregate (drawn after Lippincott's Illustrated Reviews: Biochemistry)*

Biological functions: supportive (resistance of cartilage to compression), protective (a barrier to the spread of bacteria — hyaluronic acid; bacterial **hyaluronidase** is a "spreading factor"), mechanical (lubrication of joints), binding (ordering of collagen fibrils — decorin), ion exchange (binding of Na^+, K^+, Ca^{2+} — bone mineralization), filtration (heparan sulfate in the glomerular basement membrane repels albumin by its negative charge), signaling (binding of growth factors).

Synthesis: the core protein is synthesized in the RER; in the Golgi apparatus the chains are elongated from UDP-sugars (UDP-glucuronic acid, UDP-N-acetylglucosamine) by **glycosyltransferases**; the sulfate donor is **PAPS** (phosphoadenosine phosphosulfate). Hyaluronic acid is synthesized at the plasma membrane.

Breakdown: in lysosomes, exoglycosidases and sulfatases degrade them step by step.

> **CLINICAL SIGNIFICANCE**
>
> **Mucopolysaccharidoses** are hereditary defects of lysosomal enzymes: incompletely degraded GAGs accumulate in tissues and are excreted in the urine. **Hurler syndrome** (α-L-iduronidase; autosomal recessive) — coarse facial features ("gargoylism"), corneal clouding, intellectual disability, enlarged liver and spleen, skeletal deformities. **Hunter syndrome** (iduronate-2-sulfatase; X-linked) — the cornea is clear. **Sanfilippo syndrome** — mainly severe neurological disorders. Diagnosis — increased dermatan and heparan sulfate in the urine, determination of the enzyme in leukocytes; treatment — enzyme replacement therapy, stem cell transplantation.

Use of GAGs in medicine: heparin (anticoagulant), chondroitin sulfate and glucosamine (chondroprotectors in osteoarthritis), hyaluronic acid (intra-articular — "liquid prosthesis", ophthalmic surgery, cosmetology), hyaluronidase (softening of scars, enhancing the spread of drugs).

8. Adhesive and antiadhesive proteins

- **Fibronectin** is a dimeric glycoprotein; it binds collagen, heparin, fibrin and cell **integrins** (through the RGD — Arg-Gly-Asp sequence); cell migration (embryogenesis, wound healing),

opsonin.

- **Laminin** is the main glycoprotein of the basement membrane (cross-shaped); it binds type IV collagen, nidogen and cells.
- **Integrins** are receptors for matrix proteins (α and β subunits); they link the matrix to the cytoskeleton and transmit signals.
- **Cadherins** (Ca^{2+}-dependent), selectins and adhesion molecules of the immunoglobulin family (ICAM, VCAM) — cell–cell adhesion; attachment of leukocytes to the endothelium.
- **Antiadhesive proteins** — **tenascin**, **thrombospondin**, **SPARC (osteonectin)** — enable cells to detach from the matrix and migrate (embryogenesis, wound healing, tumor invasion).

9. Laboratory work

Experiment 1: determination of hydroxyproline in urine

Principle of the method: urine is subjected to acid hydrolysis to convert bound hydroxyproline into the free form; it is oxidized by chloramine T to form a pyrrole, which gives a pink-red color with **Ehrlich's reagent** (p-dimethylaminobenzaldehyde) (558 nm).

1. Add 1 ml of concentrated HCl to 1 ml of daily urine, hydrolyze at 110 °C for 3 hours and neutralize.
2. Add chloramine T solution to the hydrolysate and keep for 20 minutes at room temperature.
3. Add Ehrlich's reagent, heat at 60 °C for 15 minutes, cool and measure at 558 nm.
4. Calculate by comparison with a standard hydroxyproline solution.

Interpretation: 15–50 mg/day in adults. An increase indicates enhanced bone resorption and collagen breakdown (hyperparathyroidism, metastases, collagenoses, burns). Collagen-rich foods (jelly, aspic) should not be eaten for 2 days before the test.

Experiment 2: determination of sialic acids in blood serum (Hess method)

Principle of the method: sialic (N-acetylneuraminic) acids are the terminal residues of the oligosaccharide chains of glycoproteins and GAGs; after the serum proteins are precipitated, the sialic acids in the hydrolysate give a colored compound when heated with a mixture of acetic and sulfuric acids.

1. Add 1 ml of 5% trichloroacetic acid to 0.1 ml of serum, boil for 5 minutes and centrifuge.
2. Add 5 ml of Hess reagent to 0.4 ml of supernatant and keep in a boiling water bath for 30 minutes.
3. Cool and measure at 540 nm against the blank; find the amount from the calibration graph. **Interpretation:** normal 2.0–2.33 mmol/l (≈ 180–200 conventional units). An increase is seen in rheumatic fever, rheumatoid arthritis, collagenoses, acute inflammation and tumors (a nonspecific indicator of inflammatory activity).

10. Practical skills

Skill: assessing the biochemical features of a connective tissue disease

1. Inflammatory activity: CRP, ESR, fibrinogen, sialic acids, seromucoid, proteinogram (α_2, γ).
2. Collagen turnover: urinary hydroxyproline, serum PINP (synthesis), CTX (resorption).
3. Immunological markers: rheumatoid factor, antinuclear antibodies, complement.

4. Compare with the clinical diagnosis.

Skill: identifying a hereditary connective tissue disease by its molecular defect

1. Clinical features: bone fractures and blue sclerae (type I collagen), skin extensibility and hypermobility (Ehlers–Danlos), arachnodactyly and aortic aneurysm (fibrillin-1), corneal clouding and gargoylism (Hurler).
2. Biochemical confirmation: urinary GAGs, enzyme activity, gene analysis.
3. Determine at which stage the defect lies: synthesis, hydroxylation, propeptide cleavage, cross-linking, breakdown.

Review questions

1. Name the composition and functions of the extracellular matrix.
2. What is special about the amino acid composition of collagen?
3. Describe the primary, secondary, tertiary and quaternary structure of collagen.
4. Where are the main types of collagen located?
5. Name the intracellular stages of collagen synthesis.
6. What is the role of ascorbic acid in collagen synthesis? What are the signs of scurvy?
7. What are the role and the cofactor of lysyl oxidase?
8. How is collagen broken down? What is the diagnostic importance of hydroxyproline determination?
9. List the hereditary diseases associated with disorders of collagen synthesis.
10. How does the structure of elastin differ from that of collagen? What is desmosine?
11. What is the biochemical basis of α_1-antitrypsin deficiency and Marfan syndrome?
12. Name the structure and types of glycosaminoglycans.
13. What are the structure and biological functions of proteoglycans?
14. What are mucopolysaccharidoses?
15. Give examples of adhesive and antiadhesive proteins.

Test questions

1. Which amino acid occupies every third position in the collagen polypeptide chain?
   A) Proline
   B) Lysine
   C) Glycine
   D) Alanine
   E) Cysteine

2. Which vitamin is required for the activity of prolyl 4-hydroxylase?
   A) B_1
   B) K
   C) Ascorbic acid
   D) A
   E) D

3. What is the cofactor of lysyl oxidase?
    A) Zn^{2+}
    B) Fe^{3+}
    C) Mg^{2+}
    D) Cu^{2+}
    E) Mn^{2+}

4. Which type of collagen forms a network in basement membranes?
    A) I
    B) IV
    C) II
    D) III
    E) VII

5. The amount of which substance in the urine reflects collagen breakdown?
    A) Hydroxyproline
    B) Creatinine
    C) Urea
    D) Uric acid
    E) Indican

6. Osteogenesis imperfecta is associated with a defect of which protein?
    A) Elastin
    B) Fibrillin-1
    C) Type IV collagen
    D) Fibronectin
    E) Type I collagen

7. Which cross-links are characteristic of elastin?
    A) Disulfide bonds
    B) Isopeptide bonds
    C) Hydroxyproline bonds
    D) Phosphodiester bonds
    E) Desmosine and isodesmosine

8. Which gene is mutated in Marfan syndrome?
    A) Fibrillin-1
    B) COL1A1
    C) α-L-Iduronidase
    D) ATP7A
    E) Laminin

9. Which glycosaminoglycan is unsulfated and not covalently bound to protein?
    A) Hyaluronic acid
    B) Chondroitin sulfate
    C) Heparin
    D) Keratan sulfate
    E) Dermatan sulfate

10. Which enzyme is deficient in Hurler syndrome?
    A) Iduronate-2-sulfatase
    B) Hyaluronidase
    C) α-L-Iduronidase
    D) Lysyl hydroxylase
    E) Collagenase

11. What is the donor of the sulfate group in glycosaminoglycan synthesis?
    A) S-adenosylmethionine
    B) UDP-glucuronic acid
    C) Cysteine
    D) Phosphoadenosine phosphosulfate (PAPS)
    E) Taurine

12. Through which sequence does fibronectin bind to cell integrins?
    A) Gly–X–Y
    B) Arg–Gly–Asp (RGD)
    C) Lys–Asp–Glu–Leu
    D) TTAGGG
    E) AUG

13. Which enzyme acts uncontrolled in the lungs in α_1-antitrypsin deficiency?
    A) Collagenase
    B) Hyaluronidase
    C) Lysyl oxidase
    D) Neutrophil elastase
    E) Trypsin

14. How do glucocorticoids affect collagen metabolism?
    A) They enhance its synthesis
    B) They reduce its synthesis (the skin thins, wounds heal slowly)
    C) They enhance hydroxylation
    D) They increase cross-links
    E) It has no effect

15. Collagen is not a nutritionally complete protein because it:
    A) Lacks glycine
    B) Lacks proline
    C) Lacks lysine
    D) Lacks alanine
    E) Lacks tryptophan

Clinical cases

Case 1

A 45-year-old sailor returning from a long sea voyage (for several months he ate mainly dried bread and canned food) complains of swollen, bleeding gums, loose teeth and small hemorrhages on his legs. An old wound scar has reopened.

Questions: What is the diagnosis? Explain the biochemical mechanism of the symptoms.

Answer: Scurvy (vitamin C deficiency). Ascorbic acid is the cofactor of prolyl and lysyl hydroxylases: it keeps the iron in the enzyme in the Fe^{2+} state. When it is deficient, proline and lysine are not hydroxylated; without hydroxyproline the triple helix is not stabilized by hydrogen bonds and is degraded in the cell, and without hydroxylysine cross-links are not formed. Collagen is scarce and fragile: capillary walls weaken (hemorrhages, bleeding gums), the periodontal ligaments break down (loose teeth), and because scar collagen is constantly renewed, old scars reopen. Treatment — ascorbic acid, fresh fruit and vegetables.

Case 2

A 3-year-old child breaks bones after minor falls (6 times in two years). The sclerae are blue, and the teeth are translucent and yellowish. One of the parents also has blue sclerae and reduced hearing.

Questions: What is the disease? What is its molecular basis? Why does a single mutant allele cause the disease?

Answer: Osteogenesis imperfecta (autosomal dominant). A mutation in one of the type I collagen genes (COL1A1 or COL1A2) — most often substitution of glycine in the triple helix by a larger amino acid: such a chain cannot pack tightly into the triple helix. Because the triple helix consists of three chains, the mutant chain combines with normal chains and makes the whole molecule defective (a dominant-negative effect) — most of the molecules are abnormal and are degraded. As a result, bone, the sclera (it thins and the vascular layer shows through — blue color), dentin and the auditory ossicles are weakened.

Case 3

A 2-year-old child has coarse facial features, corneal clouding, an enlarged liver and spleen, delayed mental development and spinal deformity. The urine contains a lot of dermatan sulfate and heparan sulfate.

Questions: What is the disease? What is its biochemical basis? What treatment is possible?

Answer: Mucopolysaccharidosis type I — Hurler syndrome. The lysosomal enzyme α-L-iduronidase is deficient: iduronic acid residues at the ends of dermatan sulfate and heparan sulfate chains are not removed, the GAGs accumulate in lysosomes (cells enlarge, organ function is impaired), and the excess is excreted in the urine. Accumulation in cartilage, bone (skeletal deformities), the cornea (clouding), liver, spleen and brain (intellectual disability) produces the symptoms. Treatment: enzyme replacement therapy with recombinant α-L-iduronidase (it does not reach the brain), hematopoietic stem cell transplantation at an early age; prenatal diagnosis and genetic counseling (autosomal recessive).

Quick review

Table 22.4. *Key concepts of topic 22*

Concept	Brief definition
Composition of collagen	(Gly–X–Y)n; proline, hydroxyproline, hydroxylysine; no tryptophan
Structure	Polyproline helix → triple helix (tropocollagen) → fibrils
Types	I (skin, bone), II (cartilage), III (vessels), IV (basement membrane)
Hydroxylation	Prolyl and lysyl hydroxylases: Fe^{2+}, O_2, α-KG, ascorbate
Cross-links	Lysyl oxidase (Cu^{2+}); Menkes, lathyrism

Concept	Brief definition
Breakdown	Collagenases (MMP); urinary hydroxyproline 15–50 mg/day
Diseases	Scurvy, osteogenesis imperfecta, Ehlers–Danlos, Alport
Elastin	Hydrophobic; desmosine; elastase ↔ α_1-antitrypsin; fibrillin-1 (Marfan)
GAG	Uronic acid + amino sugar; sulfated; binds water
Hyaluronic acid	Unsulfated, protein-free; synovial fluid
Proteoglycans	Core protein + GAG (aggrecan); PAPS in synthesis
Mucopolysaccharidoses	Hurler (α-L-iduronidase), Hunter (iduronate-2-sulfatase)

TOPIC 23

Biochemistry of muscle and the cardiovascular system

Learning objectives

To give the student knowledge of the myofibrillar and sarcoplasmic proteins of muscle, the biochemical mechanisms of muscle contraction and relaxation, the extractive substances of muscle, the specific features of energy metabolism in muscle and the role of creatine phosphate, changes in muscular dystrophy and denervation, the metabolism and secretory function of the heart muscle, the diagnostic importance of cardiospecific proteins and enzymes, and endothelial factors.

By the end of the class the student should know:

- the structure and functions of muscle proteins (myosin, actin, tropomyosin, troponin, titin, dystrophin);
- the role of Ca^{2+} and ATP in muscle contraction and relaxation;
- the pathways of ATP regeneration in muscle (creatine phosphate, glycolysis, oxidative phosphorylation);
- creatine synthesis and creatinine formation;
- the differences in the metabolism of skeletal, cardiac and smooth muscle;
- the biochemical markers of myocardial infarction and their dynamics;
- the importance of natriuretic peptides;
- the role of endothelial factors in cardiovascular pathology.

The student should be able to:

- determine creatine kinase activity in blood serum;
- interpret the results of myocardial infarction markers;
- assess the biochemical features of myopathies.

1. Muscle proteins

Muscle mass is ≈ 40% of body weight; ≈ 80% of its dry residue is protein.

Table 23.1. *Muscle proteins*

Group	Protein	Function
Myofibrillar — contractile	Myosin (thick filaments)	2 heavy chains (head — ATPase and actin-binding) + 4 light chains
	Actin (thin filaments)	Double helix of F-actin made of G-actin globules
Myofibrillar — regulatory	Tropomyosin	Lies in the groove of actin and covers the myosin-binding sites at rest
	Troponin complex	TnC (binds Ca^{2+}), TnI (inhibits actomyosin ATPase), TnT (binds to tropomyosin)
Supportive (cytoskeletal)	Titin (the largest protein), nebulin, α-actinin, desmin	Elasticity and order of the sarcomere
	Dystrophin	Links actin to the extracellular matrix through sarcolemmal glycoproteins
Sarcoplasmic	Myoglobin	O_2 reserve (abundant in red muscles)

Group	Protein	Function
	Glycolytic enzymes, creatine kinase, myokinase	Energy metabolism
	Calsequestrin, SERCA, ryanodine receptor	Storage and transport of Ca^{2+}

The I and T isoforms of cardiac and skeletal muscle troponins differ in amino acid composition — this is the basis of the high diagnostic specificity of **cardiac troponins**.

2. Mechanism of muscle contraction and relaxation

Sliding filament theory (Huxley): during contraction the length of the filaments does not change — the thin (actin) filaments slide between the thick (myosin) filaments, and the sarcomere shortens.

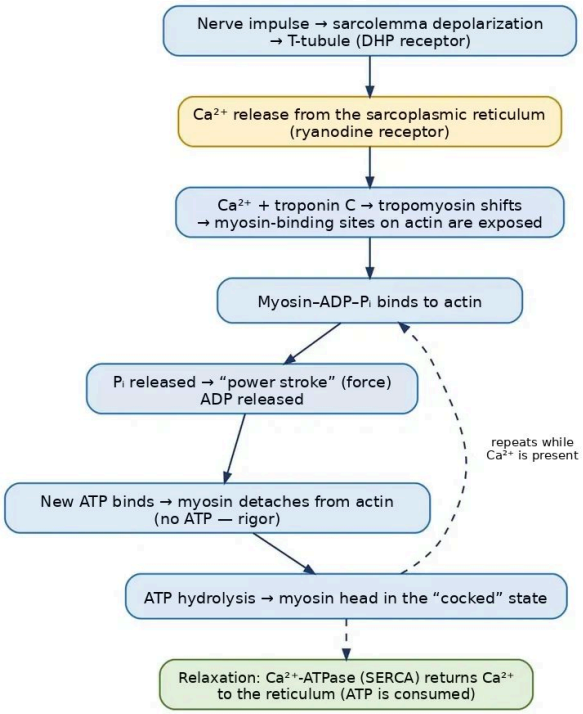


Figure 23.1. *Biochemical cycle of skeletal muscle contraction and relaxation (drawn after Severin E.S.)*

Excitation–contraction coupling: the action potential spreads through the T tubules → the **dihydropyridine receptor** (voltage-sensitive) opens the **ryanodine receptor** → **Ca^{2+}** is released from the sarcoplasmic reticulum (10^{-7} → 10^{-5} M) → Ca^{2+} binds to **troponin C** → tropomyosin shifts → the myosin heads bind to actin.

Cross-bridge cycle: myosin–ADP–P_i binds to actin → P_i is released and the "power stroke" occurs (the filament moves ≈ 10 nm) → ADP is released → a new **ATP** binds and detaches myosin from actin → ATP hydrolysis returns the myosin head to its initial position. Without ATP, myosin remains bound to actin — **rigor mortis**.

Relaxation: Ca^{2+}-ATPase (SERCA) returns Ca^{2+} to the sarcoplasmic reticulum (ATP is consumed); Ca^{2+} dissociates from troponin C, and tropomyosin covers the binding sites. Thus, ATP is required for both contraction and relaxation.

Smooth muscle: there is no troponin; Ca^{2+} combines with **calmodulin** and activates **myosin light chain kinase** — myosin is phosphorylated and interacts with actin. Relaxation — myosin

phosphatase; **NO** → cGMP → PKG causes relaxation (vasodilation).

3. Extractive substances of muscle

Nonprotein substances extracted from muscle with water (they give meat broth its taste and appetite-stimulating effect):

- **nitrogenous:** ATP, ADP, AMP, **creatine phosphate and creatine**, **carnosine and anserine** (histidine dipeptides — buffer, antioxidant, activate the Ca^{2+} pump), carnitine, glutamine, free amino acids, urea, purine bases;
- **non-nitrogenous:** glycogen, glucose, lactate, pyruvate, Krebs cycle acids, inositol;
- **minerals:** K^+, Mg^{2+}, phosphates (abundant inside cells), Na^+, Cl^- (little).

4. Energy metabolism in muscle. Creatine phosphate

The ATP reserve in muscle (≈ 5 mmol/kg) is enough for only 1–2 seconds of contraction. Pathways of ATP regeneration:

Table 23.2. *Pathways of ATP resynthesis in muscle*

Pathway	Rate	Duration	Feature
Creatine phosphate + ADP → creatine + ATP (creatine kinase)	The fastest	10–15 seconds (sprint)	Oxygen-independent
Myokinase reaction: 2ADP → ATP + AMP	Fast	Short	AMP activates PFK-1 and glycogen phosphorylase
Anaerobic glycolysis (from glycogen)	Fast	30 s – 2 minutes	Lactate accumulates — fatigue, acidosis
Aerobic oxidation (glucose, fatty acids, ketone bodies)	Slower	For hours	The most efficient; requires O_2 and mitochondria

Creatine phosphate shuttle: mitochondrial creatine kinase forms creatine phosphate using ATP from oxidative phosphorylation; it diffuses through the cytosol to the myofibrils, where creatine kinase regenerates ATP — an energy carrier.

Creatine synthesis: 1) in the kidney: arginine + glycine → guanidinoacetate (amidinotransferase); 2) in the liver: guanidinoacetate + S-adenosylmethionine → **creatine** (methyltransferase). Creatine is carried by the blood to muscle and brain. Creatine and creatine phosphate are converted nonenzymatically at a rate of ≈ 2% per day into **creatinine**, which is excreted by the kidneys; the daily amount of creatinine is proportional to **muscle mass** and constant (8.8–17.7 mmol/day in men, 7.1–15.9 mmol/day in women). Therefore blood creatinine is an indicator of **glomerular filtration** (topic 29).

Types of muscle fibers: type I (red, slow) — rich in myoglobin and mitochondria, aerobic, fatigue-resistant (postural muscles, marathon runners); **type II (white, fast)** — rich in glycogen and glycolytic enzymes, anaerobic, fatigue quickly (sprinters).

Causes of **fatigue**: depletion of creatine phosphate and glycogen, accumulation of lactate and H^+ (acidosis weakens the binding of Ca^{2+} to troponin and PFK-1), accumulation of P_i.

5. Biochemical changes in muscular dystrophy and denervation

Duchenne muscular dystrophy — X-linked recessive; a deletion or frameshift in the **dystrophin** gene (the largest gene, ≈ 2.4 million base pairs) — dystrophin is absent. The sarcolemma tears during

contraction, Ca^{2+} enters and activates proteases, and the muscle fibers undergo necrosis and are replaced by fat and connective tissue. Signs at 3–5 years: weakness of proximal muscles, the **Gowers sign**, pseudohypertrophy of the calf muscles; respiratory and heart failure at about 20 years. **Becker dystrophy** — in-frame mutations, partially functional dystrophin, a milder course.

Biochemical signs of dystrophies: serum **CK (MM) is sharply increased** (10–100 times, even before clinical signs), ALT, AST, LDH, aldolase and myoglobin are increased; muscle does not retain creatine — urinary **creatinuria** increases and **creatinine** excretion decreases (muscle mass is reduced); in muscle, myofibrillar proteins decrease and stromal (collagen) proteins increase; ATP and creatine phosphate decrease.

Denervation (nerve injury, poliomyelitis, amyotrophic lateral sclerosis): muscle atrophy, decreased protein synthesis, enhanced proteolysis; CK is usually slightly increased; creatinuria; acetylcholine receptors spread over the entire sarcolemma.

Other myopathies: McArdle disease (glycogenosis type V — muscle glycogen phosphorylase is absent; pain and cramps on exertion, lactate does not rise), **carnitine and carnitine palmitoyltransferase** deficiency (myoglobinuria on prolonged exertion or fasting), mitochondrial myopathies (lactic acidosis), **statin myopathy**, **rhabdomyolysis** (trauma — "crush syndrome", heavy exertion, alcohol, drugs): CK > 10,000 U/l, myoglobinuria (brown urine), hyperkalemia, acute kidney injury. **Myasthenia gravis** — autoantibodies against acetylcholine receptors.

6. Metabolism and secretory function of the heart muscle

Myocardial metabolism: the heart works continuously and consumes ≈ 6 kg of ATP per day (the ATP reserve lasts a few seconds). Mitochondria occupy ≈ 35% of the cell volume; > 95% of ATP comes from **oxidative phosphorylation**. Energy sources: **fatty acids** (60–80% at rest), glucose, **lactate** (during exercise — to pyruvate via LDH-1), ketone bodies (in fasting and diabetes). In oxygen deficiency (ischemia), β-oxidation stops and anaerobic glycolysis cannot meet the ATP demand — lactate and H^+ accumulate (anginal pain), ATP is depleted and the membrane pumps fail, and Ca^{2+} overload and free radicals (on reperfusion) lead to cardiomyocyte necrosis.

Regulation of myocardial contraction: β_1-adrenoceptors → cAMP → PKA: L-type Ca^{2+} channels and **phospholamban** are phosphorylated (inhibition of SERCA is removed — rapid relaxation), and troponin I is phosphorylated. **Cardiac glycosides** (digoxin) inhibit Na^+,K^+-ATPase → Na^+ ↑ in the cell → Na^+/Ca^{2+} exchange decreases → Ca^{2+} ↑ → contraction is enhanced.

Endocrine function — natriuretic peptides: atrial natriuretic peptide (ANP) is released when the atria are stretched, and **brain natriuretic peptide (BNP)** mainly when the ventricular walls are stretched. Action: guanylate cyclase receptor → cGMP; natriuresis and diuresis, vasodilation, decreased secretion of renin, aldosterone and vasopressin — lowering blood volume and pressure. **BNP and NT-proBNP** are the main laboratory markers of **heart failure** (diagnosis, severity and prognosis): NT-proBNP < 125 pg/ml practically excludes (chronic) heart failure.

7. Diagnostic importance of cardiospecific proteins and enzymes

In myocardial necrosis the contents of cardiomyocytes are released into the blood.

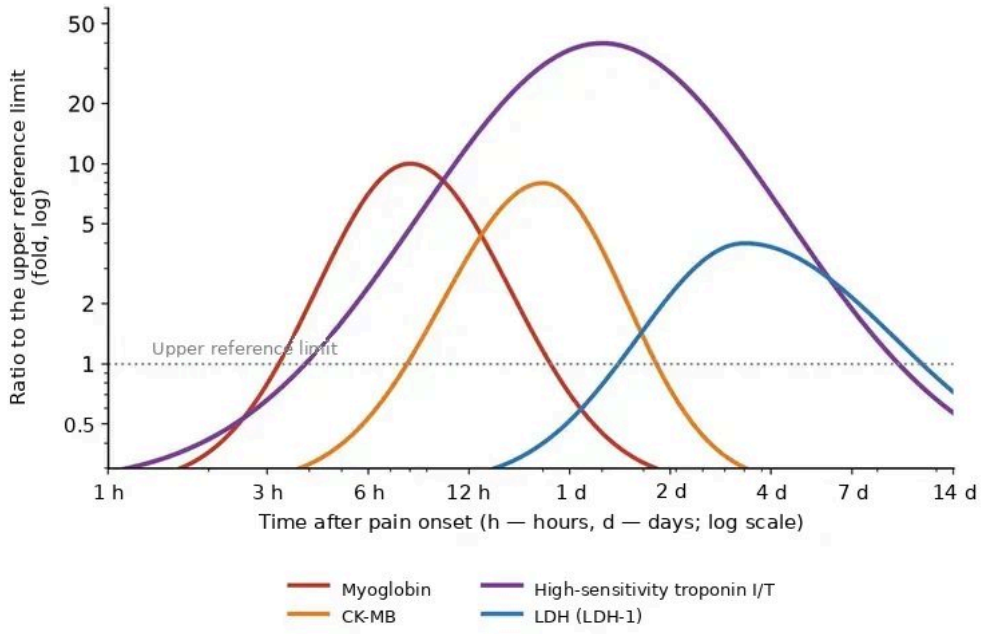


Figure 23.2. *Dynamics of biochemical markers in myocardial infarction (drawn after Harper's Illustrated Biochemistry)*

Table 23.3. *Markers of myocardial injury*

Marker	Onset of rise	Peak	Return to normal	Comment
High-sensitivity troponin I or T	1–3 hours	24–48 hours	7–14 days	The "gold standard" — the most sensitive and specific; 0/1- or 0/2-hour algorithm
Myoglobin	1–2 hours	6–10 hours	24 hours	Early but nonspecific (also in skeletal muscle)
CK-MB	3–6 hours	12–24 hours	48–72 hours	Convenient for detecting reinfarction
CK (total)	4–6 hours	24 hours	3–4 days	Also in skeletal muscle injury
AST	6–12 hours	24–48 hours	4–7 days	Now almost never used
LDH (LDH-1, LDH-2)	8–12 hours	48–72 hours	10–14 days	LDH-1 > LDH-2 ("flipped ratio"); late diagnosis

Creatine kinase isoenzymes (dimers of M and B subunits): **CK-MM** — skeletal muscle (≈ 95% of total activity), **CK-MB** — myocardium (15–40% of myocardial CK), **CK-BB** — brain, smooth muscle; there is also a mitochondrial CK. Normal total CK is < 190 U/l in men and < 170 U/l in women; CK-MB < 25 U/l or < 6% of the total.

Modern diagnosis of myocardial infarction (universal definition): troponin above the 99th percentile of normal with a **rise-and-fall pattern** + clinical signs of ischemia or ECG changes. Troponin can also rise in myocarditis, PE, sepsis, renal failure and heart failure (myocardial injury without infarction).

8. Secretory function of the endothelium

The endothelium (≈ 1.5 kg, ≈ 1000 m^2) is an active endocrine organ; it regulates vascular tone, hemostasis, inflammation and permeability.

Table 23.4. *Endothelial factors*

Action	Factors
Vasodilators	Nitric oxide (NO) — synthesized by eNOS from arginine (BH_4, NADPH) → guanylate cyclase in smooth muscle → cGMP; prostacyclin (PGI_2); endothelium-derived hyperpolarizing factor; bradykinin
Vasoconstrictors	Endothelin-1 (the most potent), angiotensin II (ACE on the endothelium), thromboxane A_2
Antithrombotic	NO, PGI_2, thrombomodulin, heparan sulfate, t-PA, TFPI
Prothrombotic	von Willebrand factor, tissue factor, PAI-1
Inflammation and adhesion	Selectins (E, P), ICAM-1, VCAM-1, IL-6, MCP-1
Growth factors	VEGF, PDGF (smooth muscle proliferation)

Endothelial dysfunction is a decrease in the formation and biological activity of NO (oxidative stress converts NO to peroxynitrite, asymmetric dimethylarginine inhibits eNOS) with a predominance of vasoconstrictor, prothrombotic and inflammatory factors. Risk factors: hypercholesterolemia (oxidized LDL), arterial hypertension, diabetes mellitus (glycation products), smoking, **hyperhomocysteinemia** (deficiency of B_6, B_{12}, folate), obesity. Endothelial dysfunction is the first stage of **atherosclerosis** (topic 10) and the basis for the development of hypertension, thrombosis and coronary heart disease.

> **REMEMBER**
> **Nitroglycerin** and other nitrates release NO in the body → cGMP → vasodilation (in angina). **Sildenafil** is a phosphodiesterase-5 inhibitor that slows the breakdown of cGMP. Using them together leads to a dangerous fall in blood pressure.

9. Laboratory work

Experiment 1: determination of creatine kinase activity in blood serum ("Biotest" kit, kinetic method)

Principle of the method (coupled reactions): creatine phosphate + ADP → (CK) → creatine + ATP; ATP + glucose → (hexokinase) → ADP + glucose-6-phosphate; glucose-6-phosphate + $NADP^+$ → (G6PD) → 6-phosphogluconate + NADPH. The rate of NADPH formation at 340 nm is proportional to CK activity. N-acetylcysteine activates CK, while AMP and diadenosine pentaphosphate inhibit erythrocyte myokinase.

1. Warm the working reagent to 37 °C; add 1 ml of reagent and 20 µl of serum to the cuvette and mix.
2. After 2 minutes of incubation record the absorbance at 340 nm, then measure every minute for 3 minutes.
3. Calculate the mean ΔE/min. **Calculation:** activity (U/l) = ΔE/min × 8095 (based on the millimolar extinction coefficient of NADPH, 6.22, and the sample dilution).

Interpretation: < 190 U/l in men, < 170 U/l in women. Hemolysis falsely increases the result (adenylate kinase and ATP from erythrocytes). Physiological increases occur after intramuscular injections and heavy physical exertion.

Experiment 2: determination of CK-MB by the immunoinhibition method (demonstration)

Principle of the method: antibodies against the M subunit completely inhibit CK-MM and half of CK-MB (its M part); the remaining activity (B subunit) is multiplied by 2.

1. Incubate the serum with the antibody-containing reagent.
2. Measure the residual activity as in Experiment 1; × 2. **Results and analysis:** CK-MB > 25 U/l and > 6% of total CK are characteristic of myocardial injury; today this method has been replaced by troponin determination.

10. Practical skills

Skill: interpreting markers in chest pain

1. Determine the time elapsed since the onset of pain.
2. Determine high-sensitivity troponin on arrival and after 1–3 hours; assess the dynamics.
3. With early presentation (< 3 hours), a first negative result does not exclude infarction.
4. With late presentation (> 3–4 days), troponin is still high, while CK-MB may have returned to normal.
5. Detect reinfarction by a renewed rise in CK-MB.
6. Exclude causes of troponin elevation without infarction.

Skill: assessing the biochemical features of a myopathy

1. Determine serum CK (MM), AST, ALT, LDH, aldolase and myoglobin.
2. Assess creatine and creatinine in the urine.
3. Very high CK (> 10,000 U/l) and brown urine — rhabdomyolysis (monitor potassium and creatinine).
4. Very high CK and proximal weakness in a young boy — dystrophin gene analysis.

Review questions

1. List the main muscle proteins and their functions.
2. What are the composition and function of the troponin complex?
3. Explain the sliding filament theory of muscle contraction.
4. What is the role of Ca^{2+} and ATP in muscle contraction and relaxation?
5. How does the regulation of smooth muscle contraction differ from that of skeletal muscle?
6. List the extractive substances of muscle.
7. By what pathways is ATP regenerated in muscle?
8. How is creatine synthesized? What is the diagnostic importance of creatinine?
9. What are the biochemical basis and features of Duchenne muscular dystrophy?
10. What is special about the metabolism of the heart muscle?
11. What are the effects and diagnostic importance of natriuretic peptides?
12. Name the creatine kinase isoenzymes.
13. Describe the dynamics of the biochemical markers of myocardial infarction.
14. What factors does the endothelium produce?
15. What is endothelial dysfunction and what are its causes?

Test questions

1. Which protein binds Ca^{2+} in skeletal muscle?
   A) Tropomyosin
   B) Troponin I
   C) Troponin T
   D) Calmodulin
   E) Troponin C

2. Which enzyme returns Ca^{2+} to the sarcoplasmic reticulum during muscle relaxation?
   A) Na^{+},K^{+}-ATPase
   B) Ca^{2+}-ATPase (SERCA)
   C) Creatine kinase
   D) Myosin ATPase
   E) Adenylate kinase

3. Which pathway regenerates ATP the fastest?
   A) Aerobic oxidation
   B) The creatine kinase reaction
   C) Anaerobic glycolysis
   D) β-Oxidation
   E) Krebs cycle

4. In which organs does creatine synthesis take place?
   A) In muscle and brain
   B) In the heart and lungs
   C) In the kidney and liver
   D) In the intestine and spleen
   E) Only in muscle

5. Which protein is absent in Duchenne muscular dystrophy?
   A) Titin
   B) Myosin
   C) Dystrophin
   D) Actin
   E) Myoglobin

6. What is rigor mortis due to?
   A) Lack of Ca^{2+}
   B) Breakdown of myosin
   C) Absence of troponin
   D) Failure of myosin to detach from actin in the absence of ATP
   E) Accumulation of lactate

7. What is the main energy source of the heart muscle at rest?
   A) Glucose
   B) Amino acids
   C) Fatty acids
   D) Glucogenic
   E) Creatine

8. Which is the most sensitive and specific marker of myocardial infarction?
   A) Myoglobin
   B) AST
   C) LDH
   D) High-sensitivity troponin
   E) Total CK

9. Which marker rises the earliest after myocardial infarction but is not specific?
   A) LDH
   B) Myoglobin
   C) AST
   D) CK-MB
   E) Troponin

10. What is the main laboratory marker of heart failure?
    A) NT-proBNP
    B) Troponin C
    C) CK-BB
    D) Aldolase
    E) Amylase

11. From which amino acid is nitric oxide synthesized in the endothelium?
    A) Glycine
    B) Glutamine
    C) Lysine
    D) Arginine
    E) Histidine

12. Which creatine kinase isoenzyme is specific to the myocardium?
    A) CK-MB
    B) CK-MM
    C) CK-BB
    D) Mitochondrial CK
    E) LDH-1

13. In smooth muscle, with which protein does Ca^{2+} combine to activate myosin light chain kinase?
    A) Troponin C
    B) Tropomyosin
    C) Calsequestrin
    D) Actin
    E) Calmodulin

14. Which is the most potent endothelial vasoconstrictor?
    A) Endothelin-1
    B) Prostacyclin
    C) Nitric oxide
    D) Bradykinin
    E) Thrombomodulin

15. What change is observed in the urine in muscular dystrophies?
    A) Creatinine increases, there is no creatine
    B) Urea decreases sharply
    C) Glucosuria
    D) Uric acid increases
    E) Creatinuria, decreased creatinine excretion

Clinical cases

Case 1

A 58-year-old man presented with severe crushing chest pain lasting for 2 hours. On arrival high-sensitivity troponin T was 18 ng/l (normal < 14), and 74 ng/l 1 hour later; myoglobin was increased; CK-MB 28 U/l.

Questions: What is the diagnosis? Explain the dynamics of the markers. Why do myocardial cells die?

Answer: Acute myocardial infarction: troponin above the 99th percentile with a sharp rise within 1 hour (dynamics) together with clinical signs of ischemia. Myoglobin is a small molecule and is released the earliest, but it is also present in skeletal muscle; CK-MB rises at 3–6 hours; troponin becomes detectable at 1–3 hours and persists for 7–14 days (the sarcomere-bound fraction is released slowly). Coronary thrombosis cuts off oxygen: β-oxidation and oxidative phosphorylation stop, and anaerobic glycolysis cannot meet the ATP demand; lactate and H^+ accumulate; Na^+,K^+-ATPase and Ca^{2+}-ATPases fail, Na^+ and Ca^{2+} increase inside the cell, phospholipases and proteases are activated, and the membranes are destroyed — necrosis, and the enzymes are released into the blood.

Case 2

A 5-year-old boy tires quickly, has difficulty climbing stairs and pushes on his legs with his hands when getting up from the floor; his calf muscles are enlarged. Serum CK 15,000 U/l, ALT and AST increased.

Questions: What is the disease? Explain the biochemical changes. How can the diagnosis be confirmed?

Answer: Duchenne muscular dystrophy (X-linked recessive). Because dystrophin is absent, the cytoskeleton is not linked to the sarcolemma and the extracellular matrix; the membrane tears during contraction, Ca^{2+} enters and activates calpains, and the fibers undergo necrosis — CK (MM), aminotransferases, LDH and aldolase are released into the blood (the increase in ALT/AST must not be mistaken for liver disease). The enlarged calves are a pseudohypertrophy due to fat and connective tissue; the way of getting up from the floor is the Gowers sign. Confirmation: molecular analysis of the DMD gene (deletions — MLPA, sequencing), immunohistochemical demonstration of the absence of dystrophin in a muscle biopsy.

Case 3

A 25-year-old man presented after a marathon with severe muscle pain and weakness; his urine is dark brown. CK 85,000 U/l, potassium 6.2 mmol/l, creatinine 280 μmol/l. The urine test strip is positive for “blood”, but there are no erythrocytes on microscopy.

Questions: What condition is this? Explain the laboratory results. Why are the kidneys damaged?

Answer: Rhabdomyolysis after heavy physical exertion. As a result of ATP depletion and Ca^{2+} accumulation in muscle cells, the fibers break down and CK, myoglobin and K^+ are released into the blood (hyperkalemia — dangerous for the heart). The test strip detects the peroxidase activity of heme — myoglobin also gives a positive result, while erythrocytes are absent (myoglobinuria). Myoglobin is filtered in the glomeruli, precipitates in the tubules in acidic urine and causes oxidative injury through heme iron; together with dehydration and vasoconstriction this leads to acute kidney injury (creatinine increased). Treatment: abundant intravenous fluids, correction of hyperkalemia, alkalinization of the urine, dialysis if necessary.

Quick review

Table 23.5. *Key concepts of topic 23*

Concept	Brief definition
Contractile proteins	Myosin (ATPase), actin
Regulatory proteins	Tropomyosin, troponin C, I, T
Contraction	Ca^{2+} → TnC → cross-bridge cycle; ATP for detachment
Relaxation	SERCA (ATP is consumed)
Smooth muscle	Ca^{2+}–calmodulin → myosin light chain kinase
ATP resynthesis	Creatine phosphate (the fastest) → glycolysis → aerobic oxidation
Creatine	Kidney (guanidinoacetate) + liver (SAM) → creatine → creatinine
Duchenne	Dystrophin absent; CK ↑↑, creatinuria
Myocardium	Aerobic; fatty acids are the main source
Infarction markers	Troponin (gold standard), myoglobin (early), CK-MB
Natriuretic peptides	ANP, BNP, NT-proBNP — heart failure
Endothelium	NO, PGI_2 (dilation); endothelin-1 (constriction)

TOPIC 24

Biochemistry of the gastrointestinal system

Learning objectives

To give the student knowledge of the composition and functions of saliva, gastric juice, pancreatic juice, bile and intestinal juice, the biochemical regulation of taste and appetite, the regulation of digestive juice secretion, the breakdown products of organic substances in the intestine, the absorption of water and salts and its disorders, and the biochemical basis of pancreatitis, celiac disease and inflammatory bowel diseases.

By the end of the class the student should know:

- the composition and functions of saliva;
- taste receptors and the hormones that regulate appetite (ghrelin, leptin);
- the mechanism of HCl secretion and its regulation;
- the enzymes of pancreatic juice and their activation;
- the composition of bile and the enterohepatic circulation of bile acids;
- intestinal juice and membrane digestion;
- the mechanisms of absorption of monosaccharides, amino acids, lipids, water and electrolytes;
- the biochemical basis of pancreatitis, celiac disease, lactose intolerance, Crohn disease and ulcerative colitis.

The student should be able to:

- perform qualitative reactions for the components of saliva and bile;
- interpret the results of amylase in blood and urine and of serum lipase;
- identify the biochemical features of the malabsorption syndrome.

1. Saliva

Saliva consists of the secretions of three pairs of major glands (parotid — serous; submandibular and sublingual — mixed) and many minor glands; 1–1.5 l is secreted per day; pH 6.4–7.4.

Composition: water (≈ 99.4%); organic substances — **mucin** (a glycoprotein that makes the food bolus slippery), **α-amylase** (ptyalin — breaks down starch into dextrins; activated by Cl^-), lingual lipase (important in infants), **lysozyme** (breaks down the bacterial wall), lactoferrin, **secretory IgA**, histatins, kallikrein, growth factors (epidermal growth factor); inorganic — Na^+, K^+ (more than in plasma), Ca^{2+}, phosphates, HCO_3^-, Cl^-, **thiocyanates** (rhodanides; abundant in smokers), F-.

Functions: digestive (the start of carbohydrate digestion), protective (antibacterial, buffering, remineralization of enamel — saturated with Ca^{2+} and phosphate), moistening and swallowing of food, taste perception (dissolves taste substances), excretory (urea, drugs, heavy metals).

> **CLINICAL SIGNIFICANCE**
> Reduced salivary secretion (xerostomia — Sjögren syndrome, radiation, anticholinergic drugs) increases dental caries and oral infections. Caries — **lactic acid**, formed when bacteria (Streptococcus mutans) ferment sucrose, dissolves the hydroxyapatite of enamel; fluoride forms the more resistant **fluorapatite** in enamel.

2. Taste perception. Regulation of appetite and hunger

Biochemical basis of taste perception: there are five basic tastes. **Sweet** (T1R2/T1R3), **umami** (T1R1/T1R3 — L-glutamate) and **bitter** (the T2R family, ≈ 25 types) are G protein-coupled receptors; the G protein **gustducin** → phospholipase Cβ2 → IP_3 → Ca^{2+} → TRPM5 channel → depolarization. **Salty** — direct entry of Na^+ through ENaC sodium channels; **sour** — H^+ ions (the OTOP1 proton channel). Zinc (gustin) is required for taste perception — in Zn deficiency taste is impaired (hypogeusia).

Regulation of appetite and hunger — two groups of neurons in the **arcuate nucleus** of the hypothalamus:

- **orexigenic** (increase appetite): **neuropeptide Y** and AgRP;
- **anorexigenic** (satiety): **POMC** → **α-MSH** (melanocortin-4 receptor) and CART.

Table 24.1. *Peripheral signals that regulate appetite*

Signal	Source	Action
Ghrelin	Stomach (increases in fasting)	Increases appetite (NPY/AgRP ↑) — the “hunger hormone”
Leptin	Adipose tissue (proportional to fat mass)	Long-term satiety signal (POMC ↑, NPY ↓), increases energy expenditure
Insulin	β-cells	Central anorexigenic effect
Cholecystokinin	Duodenum	Short-term satiety (vagus nerve)
GLP-1, PYY3-36	L cells of the small and large intestine	Satiety, slows gastric emptying
Glucose, amino acids, gastric distension	—	Satiety

In **obesity** leptin is usually high, but **leptin resistance** develops; defects of the leptin or MC4R genes lead to severe monogenic obesity. GLP-1 receptor agonists (semaglutide, liraglutide) are used to treat obesity and type 2 diabetes.

3. Gastric juice

2–2.5 l per day; pH 1.5–2.0 (during digestion).

Table 24.2. *Composition of gastric juice*

Component	Cells	Function
HCl	Parietal (oxyntic) cells	Activates pepsinogen, denatures proteins, bactericidal, converts iron into a soluble form, stimulates the release of secretin
Intrinsic factor (Castle factor)	Parietal cells	Binds vitamin B_{12} and ensures its absorption in the ileum
Pepsinogen I, II	Chief cells	Converted into pepsin by HCl and pepsin (pH optimum 1.5–2.0)
Gastricsin, rennin (in infants — chymosin)	Chief cells	Break down proteins; curdle milk (caseinogen → casein)
Gastric lipase	Chief cells	Triacylglycerols of short-chain fatty acids (milk fat)

Component	Cells	Function
Mucin, HCO_3^-	Accessory (neck) and surface epithelial cells	The mucus–bicarbonate barrier — protects the gastric wall from self-digestion
Gastrin	Antral G cells (into the blood)	Stimulates the secretion of HCl and pepsinogen

Mechanism of HCl secretion:

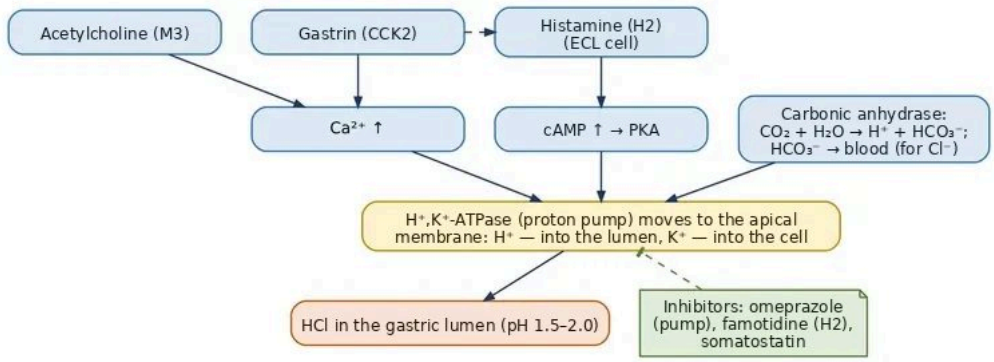


Figure 24.1. *HCl secretion in the gastric parietal cell and its regulation (drawn after Lippincott's Illustrated Reviews: Biochemistry)*

In the parietal cell, **carbonic anhydrase** forms H_2CO_3 from CO_2 and H_2O; H^+ is pumped into the lumen by the **H^+,K^+-ATPase** (proton pump) in the apical membrane (against a ≈ 106-fold concentration gradient), and HCO_3^- passes into the blood across the basolateral membrane in exchange for Cl^- (the postprandial **"alkaline tide"** — the urine becomes alkaline). Cl^- and K^+ leave into the lumen through apical channels. **Stimulators:** acetylcholine (vagus nerve, M_3), gastrin, **histamine** (from ECL cells, H_2 receptor — cAMP); **inhibitors:** somatostatin, prostaglandin E_2, secretin, GIP, pH < 2 (feedback).

CLINICAL SIGNIFICANCE

Gastric and duodenal ulcer — disturbance of the balance between aggression (HCl, pepsin, Helicobacter pylori, NSAIDs) and protection (mucus, HCO_3^-, prostaglandins, blood flow). The H. pylori enzyme **urease** breaks down urea into NH_3 and CO_2, creating an alkaline "cloud" around the bacterium; NH_3 damages the epithelium. Diagnosis — the ^{13}C-urea breath test; treatment — proton pump inhibitors + antibiotics. NSAIDs inhibit cyclooxygenase and reduce protective prostaglandins. **Zollinger–Ellison syndrome** — a gastrin-producing tumor (gastrinoma): multiple recurrent ulcers.

4. Pancreatic juice

1.5–2 l per day; pH 7.8–8.4 (HCO_3^- ≈ 120–140 mmol/l — neutralizes the acidic chyme).

Table 24.3. *Enzymes of pancreatic juice*

Enzyme	Form	Activation	Substrate
Trypsin	Trypsinogen	Enteropeptidase (intestine), then autocatalysis	Proteins (after Arg, Lys residues)
Chymotrypsin	Chymotrypsinogen	Trypsin	Proteins (after aromatic amino acids)
Elastase	Proelastase	Trypsin	Elastin, proteins
Carboxypeptidase A, B	Procarboxypeptidases	Trypsin	C-terminus of peptides

Enzyme	Form	Activation	Substrate
α-Amylase	Active	—	Starch, glycogen
Lipase + colipase	Lipase is active; procolipase	Trypsin (colipase)	Triacylglycerols (positions 1 and 3)
Phospholipase A_2	Prophospholipase	Trypsin	Phospholipids (position 2)
Cholesterol esterase	Active	Bile acids	Cholesterol esters
Ribonuclease, deoxyribonuclease	Active	—	Nucleic acids

Protective mechanisms: proteases are synthesized as inactive **zymogens** and stored in membrane-bound granules; the granules and the juice contain a **trypsin inhibitor** (SPINK1); activation occurs only in the intestine (enteropeptidase).

Regulation of secretion:

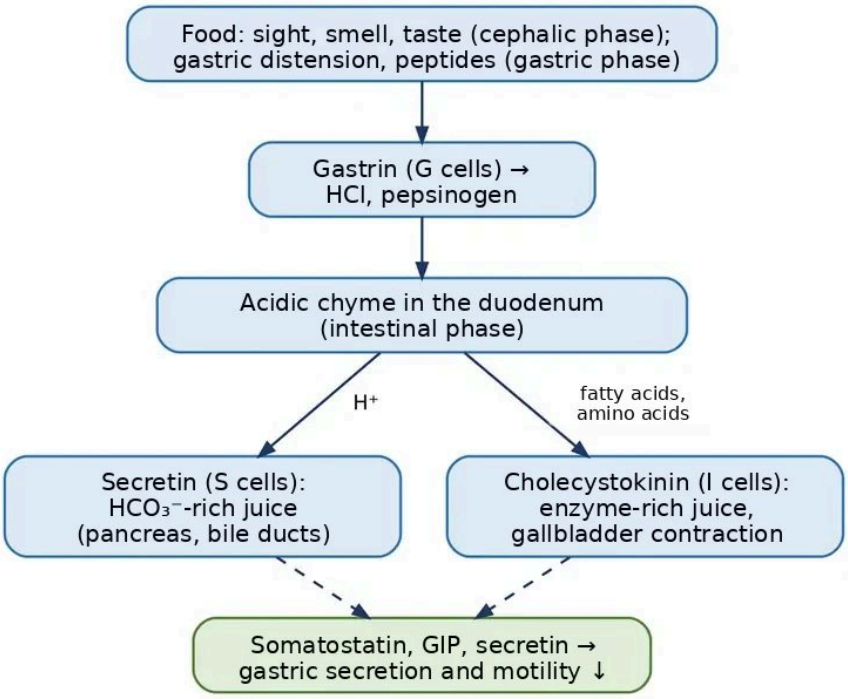


Figure 24.2. *Hormonal regulation of digestive juice secretion (drawn after Severin E.S.)*

5. Bile

0.6–1 l is formed in the liver per day; in the gallbladder water is reabsorbed and bile is concentrated 5–10 times.

Composition: bile acids (bile salts — ≈ 67% of the dry residue), **phospholipids** (lecithin), **cholesterol**, **bilirubin diglucuronide** (conjugated bilirubin — the color of bile), proteins, electrolytes (HCO_3^-), metabolites of drugs and toxins.

Bile acids are synthesized in the liver **from cholesterol** (key enzyme — **7α-hydroxylase**, a cytochrome P450; topic 10): **primary** — cholic and chenodeoxycholic acids; with glycine or taurine they **are conjugated** (glycocholic and taurocholic acids — highly soluble, amphipathic). In the intestine, bacteria deconjugate them and 7-dehydroxylate them into **secondary** bile acids (deoxycholic, lithocholic).

Functions: emulsification of fats, enhancement of lipase activity, formation of **micelles** for the absorption of lipids and fat-soluble vitamins (A, D, E, K), keeping cholesterol in solution, stimulation of intestinal peristalsis, bacteriostatic action, excretion of cholesterol and bilirubin.

Enterohepatic circulation: ≈ 95% of bile acids are reabsorbed in the **ileum** (active transport together with Na^+) and return to the liver via the portal vein; they circulate 6–10 times a day; only ≈ 0.5 g/day is lost with the feces — this is the main route of cholesterol elimination from the body. Sequestrants (colestyramine) bind bile acids and lower blood cholesterol.

CLINICAL SIGNIFICANCE

Cholesterol gallstones — an excess of cholesterol in bile relative to bile acids and phospholipids (lithogenic bile): obesity, estrogens, pregnancy, rapid weight loss, ileal resection (bile acids are lost). **Ursodeoxycholic acid** is used to dissolve small stones. **Pigment stones** — calcium bilirubinate (hemolysis, infection).

6. Intestinal juice and membrane digestion

Intestinal juice (2–3 l/day, pH 7.5–8.0) consists of the secretion of the Brunner and Lieberkühn glands and shed epithelial cells. Digestion takes place mainly as **membrane (parietal) digestion** — by enzymes located in the brush border of the enterocyte microvilli:

- **disaccharidases:** sucrase-isomaltase, **lactase** (β-galactosidase), maltase-glucoamylase, trehalase;
- **peptidases:** aminopeptidases, dipeptidases, tripeptidases;
- **enteropeptidase (enterokinase)** — activates trypsinogen;
- alkaline phosphatase, nucleotidases.

7. Intestinal absorption and its disorders

Table 24.4. *Mechanisms of absorption of digestion products*

Substance	Mechanism
Glucose, galactose	SGLT1 (symport with Na^+ — secondary active transport; the Na^+ gradient is maintained by Na^+,K^+-ATPase); basolateral — GLUT2
Fructose	GLUT5 (facilitated diffusion)
Amino acids	Many Na^+-dependent and Na^+-independent transporters (for neutral, basic and acidic amino acids and imino acids); di- and tripeptides — PEPT1 (symport with H^+)
Monoacylglycerols, fatty acids, cholesterol	Delivered within micelles; in the enterocyte TAG is resynthesized and released into the lymph as chylomicrons; short- and medium-chain fatty acids go directly into the portal vein
Water	Osmotic (aquaporins) — mainly in the jejunum and ileum; the colon absorbs the rest
Na^+	ENaC, Na^+/H^+ exchanger (NHE3), symport with nutrients
Cl^-	Cl^-/HCO_3^- exchange; secretion — CFTR channel (via cAMP)
Ca^{2+}	Under the action of calcitriol (calbindin) — in the duodenum
Fe	DMT1, heme transporter (topic 21)
Vitamin B_{12}	Complex with intrinsic factor — the cubilin receptor in the ileum

Malabsorption syndrome — weight loss, diarrhea, steatorrhea, anemia, osteoporosis, hypoproteinemic edema, vitamin deficiencies.

- **Lactose intolerance** — lactase deficiency (in adults a genetically programmed decline — widespread in many populations; secondary — in intestinal diseases): after drinking milk, lactose passes into the colon, where bacteria ferment it to gases (H_2, CO_2) and short-chain acids — **osmotic diarrhea**, bloating. Diagnosis — the hydrogen breath test, the lactose tolerance test (blood glucose does not rise).
- **Celiac disease (gluten enteropathy)** — an autoimmune response to **gliadin** peptides of wheat, barley and rye (rich in proline and glutamine, resistant to pepsin and trypsin): **tissue transglutaminase** deamidates gliadin, and T cells are activated through HLA-DQ2/DQ8; atrophy of the intestinal villi — the absorption of all substances is impaired (iron deficiency anemia, osteoporosis). Diagnosis — IgA antibodies against tissue transglutaminase, biopsy; treatment — a **lifelong gluten-free diet**.
- **Exocrine pancreatic insufficiency** (chronic pancreatitis, **cystic fibrosis** — a CFTR defect, thick secretions block the ducts): steatorrhea, decreased fecal **elastase-1**; treatment — enzyme preparations (pancreatin).
- **Bile acid deficiency** (obstructive jaundice, ileal resection) — the absorption of fats and vitamins A, D, E, K is impaired (vitamin K deficiency — bleeding).
- **Crohn disease and ulcerative colitis** are chronic inflammatory bowel diseases: genetic predisposition (NOD2 — a receptor that recognizes bacterial muramyl dipeptide), disturbed microbiota (dysbiosis), disturbed epithelial barrier and immune response (excess TNF-α, IL-12/23); protein loss, anemia, malabsorption (in Crohn disease, the ileum — B_{12}, bile acids). Activity marker — fecal **calprotectin**. Treatment includes monoclonal antibodies against TNF-α (infliximab) and 5-aminosalicylates.
- **Familial adenomatous polyposis** — mutation of the APC gene: hundreds of polyps in the colon, with a ≈ 100% chance of malignant transformation (topic 19). **Functions of the colon:** absorption of water and electrolytes; the **microbiota** (10^{14} cells) — fermentation of undigested polysaccharides (fiber) to form **short-chain fatty acids** (acetate, propionate, **butyrate** — the main energy source of colonocytes, anti-inflammatory), synthesis of vitamin K_2 and B vitamins, transformation of bile acids and bilirubin (urobilinogen, stercobilin), **putrefaction** of amino acids (indole, skatole, phenol, cresol, putrescine, cadaverine — topic 26), a competitive barrier to pathogens.

8. Mechanisms of action of drugs in diarrhea and constipation

- **Secretory diarrhea** — cholera toxin ADP-ribosylates the G_s protein and permanently activates adenylate cyclase → cAMP ↑ → a lot of Cl^- and water is secreted through CFTR (up to 10–20 l per day).

Oral rehydration solution (glucose + NaCl) — based on the fact that the absorption of Na^+ and water together with glucose through SGLT1 is preserved.

- **Osmotic diarrhea** — unabsorbed substances (lactose, sorbitol, magnesium salts).
- **Antidiarrheal drugs:** loperamide (a μ-opioid receptor agonist — peristalsis ↓, secretion ↓), adsorbents (diosmectite, activated charcoal), racecadotril (an enkephalinase inhibitor — secretion ↓), probiotics.
- **Laxatives:** osmotic (lactulose — bacteria ferment it to form acids; macrogol; magnesium sulfate), bulk-forming (fiber, psyllium), stimulant (bisacodyl, sennosides — enhance epithelial

secretion and peristalsis), softeners (liquid paraffin).

9. Laboratory work

Experiment 1: determination of the components of saliva

1. **Mucin:** add 3–4 drops of acetic acid to 2 ml of saliva — a white stringy precipitate (mucin); filter the precipitate and perform the biuret reaction (the protein part).
2. **Amylase:** add 1 ml of diluted (1:10) saliva to 1 ml of 1% starch solution and keep at 37 °C for 10 minutes; reaction with iodine — no color or reddish (starch has been broken down); the Trommer reaction is positive (maltose).
3. **Thiocyanates (rhodanides):** add 2–3 drops of HCl and 2 drops of 5% $FeCl_3$ to 1 ml of saliva — a red color (Fe(SCN)3). **Results and analysis:** the saliva of smokers contains a lot of thiocyanates (a product of the detoxification of cyanides in tobacco smoke). Preserved amylase activity indicates the digestive function of saliva.

Experiment 2: qualitative reactions for the components of bile

1. **Bile acids — the Pettenkofer reaction:** place 2 drops of bile and 1 drop of 20% sucrose solution in a dry test tube and carefully add 1 ml of concentrated sulfuric acid down the wall — a red-violet ring at the interface (condensation of the hydroxymethylfurfural formed from sucrose with bile acids).
2. **Bile pigments — the Gmelin reaction:** pour 1 ml of concentrated nitric acid into a test tube and layer 1 ml of bile over it down the wall — colored rings at the interface (green, blue, violet, red, yellow) — products of successive oxidation of bilirubin (biliverdin and others).
3. **Reduction of surface tension by bile (Hay's test):** sprinkle sulfur powder on the surface of two test tubes containing water and diluted bile — in the bile the powder sinks (the surface tension is reduced). **Results and analysis:** the reactions confirm the presence of bile acids and pigments in bile; a positive Gmelin reaction in urine indicates conjugated bilirubinuria (obstructive and hepatocellular jaundice).

Experiment 3: interpretation of serum amylase and lipase results (case-based)

1. Compare the results of serum α-amylase (normal < 100 U/l; pancreatic isoenzyme < 53 U/l) and lipase (< 60 U/l) with the normal values.
2. Take into account the result of urinary diastase (amylase). **Results and analysis:** in acute pancreatitis amylase rises within 2–12 hours and returns to normal in 3–5 days; urinary amylase persists longer; **lipase** is more specific and stays high longer (7–14 days) — an increase of **3 times** the upper limit of normal is the diagnostic criterion. Amylase also rises in mumps, intestinal obstruction and macroamylasemia.

10. Practical skills

Skill: biochemical diagnosis of acute pancreatitis

1. Determine lipase and amylase activity (≥ 3-fold increase).
2. Assess severity: glucose ↑, calcium ↓ (calcium soaps in fat necrosis), LDH ↑, leukocytosis, CRP > 150 mg/l (at 48 hours), urea ↑.
3. Determine the cause: gallstones (bilirubin, alkaline phosphatase, ALT ↑), alcohol, hypertriglyceridemia (> 11 mmol/l), hypercalcemia.

4. Explain the mechanism: premature activation of trypsinogen inside the gland → activation of the other proenzymes → self-digestion of the gland (autolysis), elastase — vascular damage, phospholipase A_2 — destruction of membranes.

Skill: determining the cause of malabsorption

1. Confirm steatorrhea (fat in the feces — coprogram).
2. Determine fecal elastase-1: low — pancreatic insufficiency.
3. Antibodies against tissue transglutaminase — celiac disease.
4. Lactose tolerance or hydrogen breath test — lactase deficiency.
5. Assess blood albumin, iron, ferritin, B_{12}, folate, calcium, vitamin D and PT.

Review questions

1. Name the composition and functions of saliva.
2. What are the biochemical mechanisms of taste perception?
3. Which hormones and neuropeptides regulate appetite?
4. What is the composition of gastric juice?
5. How is HCl formed in the parietal cell?
6. How is HCl secretion regulated? Name the drugs that reduce it.
7. What protects the gastric wall from self-digestion?
8. List the enzymes of pancreatic juice and their activation.
9. Which intestinal hormones regulate the secretion of digestive juices?
10. What is the composition of bile and what are the functions of bile acids?
11. What is the enterohepatic circulation of bile acids?
12. What is membrane digestion?
13. How are monosaccharides and amino acids absorbed?
14. What is the biochemical basis of lactose intolerance and celiac disease?
15. What are the mechanism of development and the diagnosis of acute pancreatitis?

Test questions

1. Which component of saliva breaks down the bacterial cell wall?
   A) Mucin
   B) Amylase
   C) Thiocyanate
   D) Kallikrein
   E) Lysozyme

2. Which hormone is called the “hunger hormone”?
   A) Ghrelin
   B) Leptin
   C) Cholecystokinin
   D) Insulin
   E) GLP-1

3. Which protein pumps H^+ into the gastric lumen in the parietal cell?
   A) Na^+,K^+-ATPase
   B) Ca^{2+}-ATPase
   C) Carbonic anhydrase
   D) CFTR
   E) H^+,K^+-ATPase

4. Through which receptor does histamine enhance HCl secretion?
   A) H_1
   B) M_3
   C) CCK_2
   D) H_2
   E) β_2

5. Which enzyme activates trypsinogen in the intestine?
   A) Enteropeptidase
   B) Pepsin
   C) Chymotrypsin
   D) Carboxypeptidase
   E) Lipase

6. Which factor produced in the stomach is required for the absorption of vitamin B_{12}?
   A) Pepsinogen
   B) Gastrin
   C) Mucin
   D) Rennin
   E) Intrinsic factor (Castle factor)

7. Where are ≈ 95% of bile acids reabsorbed?
   A) In the duodenum
   B) In the stomach
   C) In the ileum
   D) In the large intestine
   E) In the gallbladder

8. Which transporter is responsible for the intestinal absorption of glucose?
   A) GLUT4
   B) SGLT1
   C) GLUT5
   D) PEPT1
   E) GLUT1

9. Which hormone stimulates the secretion of bicarbonate-rich juice by the pancreas?
   A) Gastrin
   B) Cholecystokinin
   C) Secretin
   D) Ghrelin
   E) Somatostatin

10. Which protein triggers the autoimmune response in celiac disease?
    A) Gliadin
    B) Casein
    C) Albumin
    D) Collagen
    E) Lactoglobulin

11. What is the mechanism of diarrhea in lactose intolerance?
    A) Secretory (cAMP ↑)
    B) Osmotic (unabsorbed lactose and fermentation products)
    C) Inflammatory
    D) Motor (peristalsis ↑)
    E) Exudative

12. What is the most specific laboratory marker of acute pancreatitis?
    A) Amylase
    B) Lipase
    C) ALT
    D) CK
    E) Bilirubin

13. What is the key enzyme of bile acid synthesis?
    A) HMG-CoA reductase
    B) Cholesterol esterase
    C) Lipase
    D) 7α-Hydroxylase
    E) Phospholipase A_2

14. How cholera toxin acts:
    A) Inhibits Na^+,K^+-ATPase
    B) It blocks SGLT1
    C) It ADP-ribosylates the G_s protein and increases cAMP
    D) It breaks down lactase
    E) It releases histamine

15. What is the main energy source of the colonic epithelium?
    A) Glucose
    B) Glutamine
    C) Ketone bodies
    D) Butyrate
    E) Lactate

Clinical cases

Case 1

A 45-year-old man presented after eating a large fatty meal and drinking alcohol with severe girdle-like pain in the upper abdomen and repeated vomiting. Serum lipase 1250 U/l (normal < 60), amylase 820 U/l, glucose 9.8 mmol/l, calcium 1.9 mmol/l.

Questions: What is the diagnosis? Explain the biochemical mechanism of the disease. Why is calcium decreased and glucose increased?

Answer: Acute pancreatitis. Alcohol and fatty food enhance secretion, raise the pressure in the ducts and damage the acinar cells; trypsinogen is converted to trypsin prematurely inside the gland and activates the other proenzymes (chymotrypsinogen, proelastase, prophospholipase A_2) — the gland digests itself (autolysis): elastase destroys the vessels (hemorrhage), phospholipase A_2 destroys membranes, and lipase breaks down adipose tissue. The enzymes are released into the blood (increased lipase and amylase). In foci of fat necrosis, free fatty acids form soaps with calcium, leading to hypocalcemia (a sign of a severe course). Damage to the islets of Langerhans and stress hormones cause hyperglycemia.

Case 2

A 25-year-old woman complains of bloating, rumbling and loose stools 1–2 hours after drinking milk or eating dairy products. After 50 g of lactose, blood glucose did not rise (+ 0.6 mmol/l), and breath hydrogen rose sharply.

Questions: What is the diagnosis? Explain the biochemical mechanism of the symptoms. What recommendations would you give?

Answer: Lactase deficiency (lactose intolerance). Without lactase, lactose is not broken down into glucose and galactose in the small intestine — blood glucose does not rise. Lactose passes into the colon, raises the osmotic pressure and retains water (osmotic diarrhea); bacteria ferment it to H_2, CO_2, methane and short-chain acids (bloating, rumbling; H_2 is absorbed into the blood and exhaled). Recommendations: limit fresh milk; fermented dairy products (yogurt, cheese — little lactose, contain bacterial β-galactosidase), lactose-free milk or lactase preparations; prevention of calcium deficiency.

Case 3

A 7-year-old child is growth-retarded and thin, with a large abdomen and frequent, large, fatty, foul-smelling stools. Blood Hb 88 g/l, ferritin 5 µg/l, albumin 30 g/l. IgA antibodies against tissue transglutaminase are sharply increased. Intestinal biopsy shows villous atrophy.

Questions: What is the diagnosis? Explain the mechanism of the disease. Why did iron deficiency and hypoalbuminemia develop? What is the treatment?

Answer: Celiac disease (gluten enteropathy). The gliadin peptides of wheat, barley and rye are resistant to proteases; tissue transglutaminase deamidates them, and the deamidated peptides are presented to T lymphocytes with HLA-DQ2/DQ8 — inflammation; cytotoxic lymphocytes destroy the enterocytes, and the villi become atrophic. The absorptive surface and membrane enzymes decrease: the absorption of fats (steatorrhea), iron (in the duodenum — iron deficiency anemia), amino acids (hypoalbuminemia), calcium and vitamin D is impaired — growth retardation. Treatment — a lifelong strict gluten-free diet (rice, corn and buckwheat are allowed); iron and vitamin supplementation; a decrease in antibodies shows that the diet is being followed.

Quick review

Table 24.5. *Key concepts of topic 24*

Concept	Brief definition
Saliva	Amylase, mucin, lysozyme, IgA, thiocyanates; pH 6.4–7.4
Appetite	Ghrelin (↑), leptin, CCK, GLP-1, PYY (↓); NPY ↔ POMC
HCl	Carbonic anhydrase + H^+,K^+-ATPase; ACh, gastrin, histamine (H_2)

Concept	Brief definition
Gastric protection	Mucin, HCO_3^-, prostaglandins
Pancreatic juice	Zymogens (trypsinogen → enteropeptidase), amylase, lipase + colipase
Regulation	Secretin (HCO_3^-), cholecystokinin (enzymes, bile)
Bile	Bile acids (from cholesterol, 7α-hydroxylase), phospholipids, bilirubin
Enterohepatic circulation	≈ 95% are reabsorbed in the ileum
Absorption	SGLT1 (glucose), GLUT5 (fructose), PEPT1, chylomicrons
Lactose intolerance	Lactase ↓ → osmotic diarrhea, H_2 breath test
Celiac disease	Gliadin, transglutaminase antibodies; gluten-free diet
Pancreatitis	Activation of trypsinogen, autolysis; lipase ≥ 3 × normal

TOPIC 25

Biochemistry of the liver. Bilirubin metabolism and jaundice

Learning objectives

To give the student knowledge of the role of the liver in the metabolism of carbohydrates, lipids, proteins and amino acids and its age-related features, heme catabolism and bilirubin metabolism, the types of jaundice and their differential diagnosis, physiological and pathological jaundice in newborns, the syndromes of liver damage (cytolysis, cholestasis, mesenchymal-inflammatory, hepatocellular insufficiency), and the biochemical mechanisms of the development of hepatic coma.

By the end of the class the student should know:

- the main functions of the liver in metabolism;
- the breakdown of heme and the formation, transport, conjugation and excretion of bilirubin;
- the properties of conjugated and unconjugated bilirubin;
- the biochemical differences between hemolytic, hepatocellular and obstructive jaundice;
- the causes of neonatal jaundice and the hereditary hyperbilirubinemias;
- the laboratory parameters of the syndromes of liver damage;
- the biochemical mechanisms of hepatic coma.

The student should be able to:

- determine bilirubin and its fractions in blood on a biochemical analyzer;
- determine bilirubin and urobilinogen in urine;
- distinguish the type of jaundice by laboratory parameters.

1. The role of the liver in metabolism

The liver (≈ 1.5 kg) is the body's main “biochemical laboratory”; blood arrives through the portal vein (≈ 75%) and the hepatic artery; hepatocytes make up ≈ 80% of the cells.

Table 25.1. *Metabolic functions of the liver*

Metabolism	Functions
Carbohydrates	Maintaining the blood glucose level (the “glucostat”): glycogen synthesis and breakdown (glucose-6-phosphatase only in the liver and kidney), gluconeogenesis; conversion of fructose and galactose to glucose; pentose phosphate pathway
Lipids	Synthesis of fatty acids and TAG, formation of VLDL and HDL, synthesis of cholesterol and its conversion to bile acids, synthesis of ketone bodies (the liver does not use them itself), phospholipid synthesis
Proteins	Synthesis of plasma proteins (albumin, fibrinogen, prothrombin and other clotting factors, transferrin, ceruloplasmin, acute phase proteins, apolipoproteins); almost all of them except γ-globulins
Amino acids	Transamination, deamination, urea synthesis (the only site), creatine synthesis (the last stage), synthesis of nonessential amino acids
Pigments	Conjugation of bilirubin and its excretion with bile
Detoxification	Xenobiotics, drugs, hormones, ammonia, ethanol (topic 26)
Vitamins and minerals	A, D (25-hydroxylation), B_{12} store, synthesis of K-dependent factors; store of iron and copper

Metabolism	Functions
Excretory	Bile formation

Age-related features: in newborns the activity of UDP-glucuronosyltransferase, some P450 isoforms and gluconeogenic enzymes is low (physiological jaundice, sensitivity to drugs, tendency to hypoglycemia); the synthesis of albumin and clotting factors is lower. In old age liver mass and blood flow decrease, and drug metabolism slows down.

2. Heme catabolism and bilirubin metabolism

≈ 250–300 mg of bilirubin is formed per day: ≈ 80% from the hemoglobin of senescent erythrocytes, the rest from myoglobin, cytochromes and ineffective erythropoiesis.

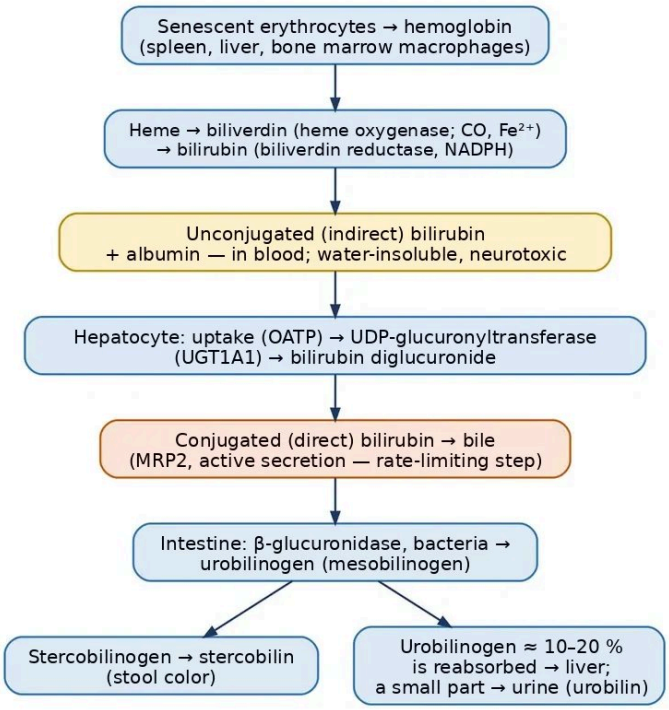


Figure 25.1. *Bilirubin metabolism (drawn after Severin E.S.)*

1. **In macrophages** (spleen, Kupffer cells of the liver, bone marrow): **heme oxygenase** (microsomal, requires O_2 and NADPH) cleaves the α-methene bridge of heme to form **biliverdin**, **CO** and Fe^{2+}; **biliverdin reductase** (NADPH) reduces biliverdin to **bilirubin** (from green to yellow — the color change of a bruise).
2. **Transport in the blood:** bilirubin is insoluble in water (lipophilic) and is carried to the liver bound to **albumin** — **unconjugated (indirect, free)** bilirubin. It does not pass the renal filter — it is not found in urine. Its free (non-albumin-bound) fraction crosses the blood–brain barrier and is **neurotoxic**. Sulfonamides, salicylates and free fatty acids displace it from albumin.
3. **In the liver:** hepatocytes take up bilirubin with membrane transporters (OATP) and bind it inside the cell to ligandin (glutathione S-transferase); in the endoplasmic reticulum, **UDP-glucuronosyltransferase (UGT1A1)** attaches two glucuronic acid residues from UDP-glucuronic acid — **bilirubin diglucuronide** (**conjugated, direct** bilirubin) — which is water-soluble.
4. **Secretion into bile** — active transport (ATP) through the **MRP2** transporter; this is the **slowest (rate-limiting) step** of the process.

5. **In the intestine:** bacterial β-glucuronidase removes the glucuronic acid; bacteria reduce bilirubin to **urobilinogens** (mesobilinogen, stercobilinogen). Most of it is oxidized to **stercobilin**, which colors the feces (200–300 mg per day). Part of the urobilinogen is reabsorbed and returns to the liver (enterohepatic circulation), and a small part is excreted in the urine and oxidized to **urobilin** (the color of urine).

Table 25.2. *Properties of the bilirubin fractions*

Characteristic	Unconjugated (indirect)	Conjugated (direct)
Chemical nature	Free bilirubin + albumin	Bilirubin diglucuronide
Water solubility	Insoluble (lipophilic)	Soluble
Reaction with the diazo reagent	Only after an “accelerator” (caffeine, alcohol) is added	Direct
Passage through the renal filter	Does not pass	Passes (appears in the urine)
Toxicity	Neurotoxic (soluble in lipids)	Low toxicity
Normal in serum	≈ 75% (≤ 17 µmol/l)	≤ 3.4–5 µmol/l

Normal: total bilirubin 3.4–20.5 µmol/l. **Jaundice** is yellowing of the skin, mucous membranes and sclerae; it becomes visible when bilirubin is > 34 µmol/l (earliest in the sclerae).

3. Types of jaundice and their differential diagnosis

Table 25.3. *Biochemical differences between the types of jaundice*

Index	Hemolytic (prehepatic)	Hepatocellular (hepatic)	Obstructive (posthepatic, cholestatic)
Cause	Excessive breakdown of erythrocytes (hemolytic anemias, Rh incompatibility, malaria, poisons)	Damage to hepatocytes (viral hepatitis, toxic, alcohol, cirrhosis)	Obstruction of the bile ducts (stones, tumor of the head of the pancreas, stricture)
Unconjugated bilirubin	↑↑	↑	N or slightly ↑
Conjugated bilirubin	N	↑	↑↑
Urinary bilirubin	None	Present	A lot (urine the color of “beer”)
Urinary urobilinogen	↑↑	↑ (the liver cannot reprocess it)	Absent (in complete obstruction)
Stool color	Dark (stercobilin ↑)	Lighter	Colorless (acholic)
ALT, AST	N	↑↑	↑
Alkaline phosphatase, GGT	N	N or ↑	↑↑
Other features	Reticulocytosis, LDH ↑, haptoglobin ↓	Albumin, PT ↓ (in severe cases)	Pruritus (bile acids), cholesterol ↑, vitamin K deficiency

4. Neonatal jaundice. Hereditary hyperbilirubinemias

Physiological jaundice is observed in ≈ 60% of full-term and ≈ 80% of preterm newborns. Causes: the large number and short lifespan (≈ 80 days) of fetal erythrocytes (HbF) — bilirubin production is 2–3 times higher than in adults; the low activity of **UDP-glucuronosyltransferase** (≈ 1% of the adult level, matures by 1–3 months); little ligandin; the intestine is sterile — urobilinogen is not formed, and bilirubin is reabsorbed under the action of β-glucuronidase. Features: it begins on **days 2–3**, peaks on days 3–5 (< 205–220 μmol/l) and resolves by days 10–14; only unconjugated bilirubin increases; the general condition is good.

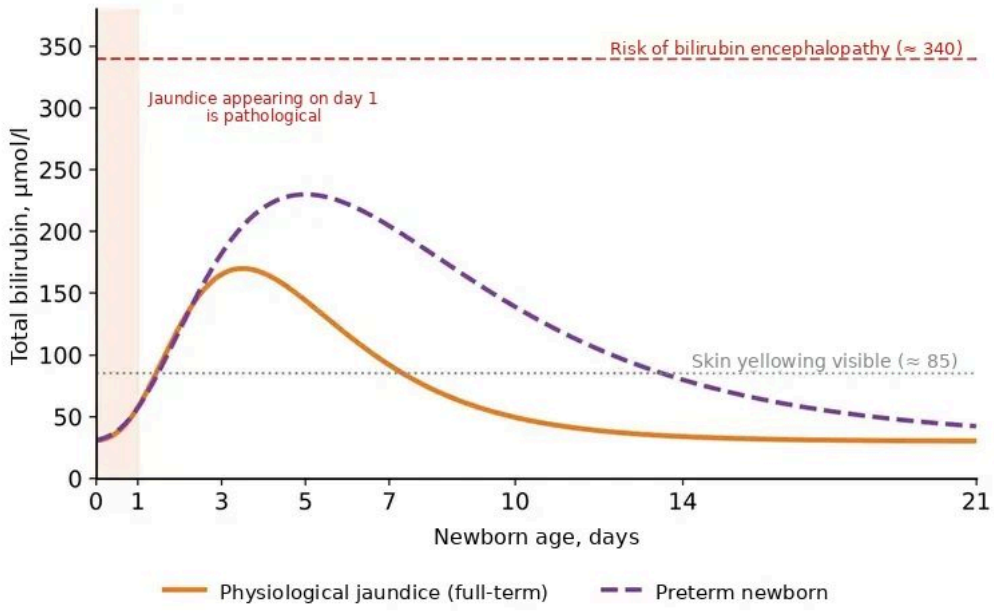


Figure 25.2. *Dynamics of bilirubin in newborns (drawn after Harper's Illustrated Biochemistry)*

Signs of pathological jaundice: appearance on the first day, a rapid rise in bilirubin (> 85 μmol/l/day), duration of more than 2 weeks in full-term and more than 3 weeks in preterm infants, an increase in conjugated bilirubin (> 25–34 μmol/l), an undulating course. Causes: **hemolytic disease** (Rh or AB0 incompatibility), G6PD deficiency, spherocytosis, infections, hypothyroidism, biliary atresia, hereditary hyperbilirubinemias.

CLINICAL SIGNIFICANCE

Bilirubin encephalopathy (kernicterus) — unconjugated bilirubin (> 340 μmol/l in full-term infants, lower in preterm infants) crosses the blood–brain barrier and accumulates in the basal ganglia: it uncouples oxidative phosphorylation in the mitochondria — neurons die (deafness, cerebral palsy, intellectual disability). **Phototherapy** (blue light, 460–490 nm) converts bilirubin into water-soluble photoisomers (lumirubin), which are excreted in bile and urine without conjugation. In severe cases — exchange transfusion. Phenobarbital induces UGT1A1.

Table 25.4. *Hereditary hyperbilirubinemias*

Disease	Defect	Bilirubin	Course
Gilbert syndrome	UGT1A1 promoter (TA repeat) — activity ≈ 30%	Unconjugated, slightly ↑ (in fasting, stress, infection)	Mild, harmless; 5–10% of the population
Crigler–Najjar type I	UGT1A1 completely absent	Unconjugated ↑↑ (> 340)	Severe, kernicterus; liver transplantation
Crigler–Najjar type II	UGT1A1 partial (< 10%)	Unconjugated ↑	Responds to phenobarbital

Disease	Defect	Bilirubin	Course
Dubin–Johnson syndrome	MRP2 (secretion)	Conjugated ↑	Black liver (pigment); mild
Rotor syndrome	OATP1B1/1B3 (reuptake)	Conjugated ↑	Mild

5. Syndromes of liver damage

Table 25.5. *Biochemical syndromes of liver damage*

Syndrome	Essence	Laboratory parameters
Cytolysis	Damage to the hepatocyte membrane, release of the cell contents into the blood	ALT, AST ↑ (AST/ALT — the de Ritis ratio: < 1 in viral hepatitis, > 2 in alcoholic damage), LDH-5, GLDH (necrosis), direct bilirubin, serum iron ↑
Cholestasis	Impaired bile formation or flow	Alkaline phosphatase, GGT, 5'-nucleotidase ↑; conjugated bilirubin, bile acids, cholesterol ↑; pruritus
Mesenchymal-inflammatory	Inflammation of the liver stroma and the reticuloendothelial system, immune response	γ-Globulins, IgG, IgA (alcohol), IgM (primary biliary cholangitis) ↑; thymol test ↑; autoantibodies
Hepatocellular insufficiency (hepatodepressive)	Decreased synthetic function	Albumin ↓, prothrombin index ↓ (PT, INR prolonged) — not corrected by vitamin K administration; cholinesterase ↓, cholesterol ↓, urea ↓, glucose ↓ (in severe cases), ammonia ↑
Shunting (portal hypertension)	Portal venous blood bypasses the liver	Ammonia, phenols, aromatic amino acids ↑

Liver fibrosis and cirrhosis — in chronic damage (hepatitis B, C, alcohol, fatty liver disease) **stellate (Ito) cells** are activated, transform into myofibroblasts and synthesize a lot of collagen (types I, III); nodules form. Fibrosis markers — hyaluronic acid, type III procollagen peptide, elastography.

6. Biochemical mechanisms of hepatic coma

Hepatic encephalopathy and coma is a disturbance of central nervous system function in acute or chronic liver failure.

1. **The ammonia hypothesis (the main one):** urea synthesis in the liver decreases, and ammonia from the intestine reaches the brain directly through portocaval shunts. In the brain, astrocytes convert NH_3 into **glutamine** (glutamine synthetase) — glutamine accumulates and the astrocytes swell (osmotically), brain edema; **α-ketoglutarate** is consumed for glutamate — the Krebs cycle and ATP formation decrease; inhibitory and excitatory neurotransmission are disturbed (topic 13).
2. **False neurotransmitters:** aromatic amino acids (phenylalanine, tyrosine, tryptophan) increase in the blood (the liver does not break them down), while branched-chain amino acids (valine, leucine, isoleucine — used in muscle) decrease — the **Fischer ratio** (BCAA/AAA) falls from 3–3.5 to < 1. Many aromatic amino acids enter the brain and form octopamine, phenylethanolamine (false transmitters) and serotonin; dopamine and norepinephrine decrease.
3. **Enhanced GABAergic inhibition** (endogenous benzodiazepines).
4. **Intestinal toxins:** mercaptans (the liver odor — "foetor hepaticus"), phenols, short-chain fatty acids, indole, skatole — the liver does not detoxify them.

5. Hypoglycemia, electrolyte disturbances (hypokalemia, alkalosis — enhances the passage of NH_3 into the brain), acidosis, renal failure (hepatorenal syndrome). **Principles of treatment:** temporary restriction of protein (but not complete), **lactulose** (in the colon bacteria ferment it and acidify the medium: $NH_3 \rightarrow NH_4^+$ — not absorbed, excreted from the intestine; laxative effect), bowel cleansing, nonabsorbable antibiotics (rifaximin — reduces the ammonia-producing bacteria), L-ornithine L-aspartate (enhances the synthesis of urea and glutamine), branched-chain amino acids, correction of hypokalemia; in severe cases liver transplantation.

7. Laboratory work

Experiment 1: determination of total and direct bilirubin in blood on a biochemical analyzer (Jendrassik–Grof method)

Principle of the method: bilirubin forms red-violet azobilirubin with diazotized sulfanilic acid (the diazo reagent). Conjugated bilirubin reacts directly (**direct bilirubin**); unconjugated bilirubin requires an **accelerator** (caffeine benzoate) to be released from albumin — with it **total bilirubin** is determined. Indirect bilirubin = total − direct. Measurement — 546 nm (in an alkaline medium — 578 nm, blue color).

1. Select the “Total bilirubin” and “Direct bilirubin” tests on the analyzer; perform quality control with calibrators and control sera.
2. Place the serum sample in the rack (the sample must be protected from light — bilirubin breaks down in light).
3. The analyzer mixes 10–20 µl of sample with the appropriate reagents, measures after the set time and reports the result. **Interpretation:** total bilirubin 3.4–20.5 µmol/l; direct ≤ 3.4–5 µmol/l (≤ 25% of the total). Hemolysis and lipemia alter the result.

Experiment 2: determination of bilirubin and urobilinogen in urine

1. **Bilirubin — the Rosin test:** carefully layer a 1% alcoholic iodine solution over 3 ml of urine — a **green ring** at the interface (bilirubin is oxidized to biliverdin). Test strips (diazo reaction) are also used.
2. **Urobilinogen — the Ehrlich reaction:** add 0.5 ml of Ehrlich's reagent (p-dimethylaminobenzaldehyde + HCl) to 2 ml of urine — a red color at room temperature (a lot of urobilinogen); normally the color is faint and appears only on heating. **Results and analysis:** bilirubin in urine — conjugated hyperbilirubinemia (hepatocellular or obstructive jaundice); a lot of urobilinogen — hemolytic or hepatocellular jaundice; no urobilinogen and a lot of bilirubin — complete obstruction of the bile ducts.

8. Practical skills

Skill: determining the type of jaundice by laboratory parameters

1. Assess total bilirubin and its fractions: which fraction predominates?
2. Assess urinary bilirubin and urobilinogen and the stool color.
3. Assess the enzymes: ALT/AST (cytolysis) or alkaline phosphatase/GGT (cholestasis).
4. Check for signs of hemolysis: reticulocytes, LDH, haptoglobin.
5. Conclusion: hemolytic, hepatocellular or obstructive jaundice; recommend a confirmatory investigation (ultrasound, markers).

Skill: grouping liver function tests by syndrome

1. Identify the parameters of cytolysis (ALT, AST, GLDH).
2. Identify the parameters of cholestasis (ALP, GGT, direct bilirubin, bile acids).
3. Assess the synthetic function (albumin, PT/INR, cholinesterase).
4. Assess the inflammatory and immune parameters (γ-globulins).
5. Assess the severity (Child–Pugh: bilirubin, albumin, INR, ascites, encephalopathy).

Review questions

1. Explain the role of the liver in carbohydrate metabolism.
2. Which processes of lipid metabolism take place in the liver?
3. Which plasma proteins does the liver synthesize?
4. How is heme broken down? Name the products of the heme oxygenase reaction.
5. How is unconjugated bilirubin transported in the blood and why is it toxic?
6. How is bilirubin conjugated in the liver?
7. How is bilirubin changed in the intestine? How are stercobilin and urobilin formed?
8. State the differences between the bilirubin fractions.
9. How can hemolytic, hepatocellular and obstructive jaundice be distinguished?
10. What are the causes of physiological jaundice in newborns?
11. What are the signs of pathological jaundice and the mechanism of kernicterus?
12. List the hereditary hyperbilirubinemias.
13. Describe the main biochemical syndromes of liver damage.
14. Which parameters reflect hepatocellular insufficiency?
15. Explain the biochemical mechanisms of hepatic coma.

Test questions

1. Which enzyme converts heme into biliverdin?
   A) Heme oxygenase
   B) Biliverdin reductase
   C) UDP-glucuronosyltransferase
   D) β-Glucuronidase
   E) Ferrochelatase

2. With what is unconjugated bilirubin transported in the blood?
   A) With transferrin
   B) With albumin
   C) With haptoglobin
   D) With hemopexin
   E) With VLDL

3. Which substance is used in the conjugation of bilirubin?
   A) Glycine
   B) Taurine
   C) PAPS
   D) UDP-glucuronic acid
   E) S-adenosylmethionine

4. Which is the slowest (rate-limiting) step of bilirubin metabolism in the liver?
   A) Secretion of conjugated bilirubin into bile
   B) Uptake
   C) Binding to ligandin
   D) Glucuronidation
   E) Release from albumin

5. Which pigment colors the feces?
   A) Bilirubin
   B) Stercobilin
   C) Biliverdin
   D) Urobilinogen
   E) Porphobilinogen

6. Which change is characteristic of hemolytic jaundice?
   A) A lot of bilirubin in the urine
   B) Colorless stools
   C) A sharp increase in unconjugated bilirubin in the blood
   D) A sharp increase in alkaline phosphatase
   E) No urobilinogen in the urine

7. Which feature is characteristic of obstructive jaundice?
   A) A lot of urobilinogen in the urine
   B) Reticulocytosis
   C) Decreased haptoglobin
   D) Acholic stools, a lot of bilirubin in the urine
   E) Only unconjugated bilirubin is increased

8. What is the main cause of physiological jaundice in newborns?
   A) Absence of heme oxygenase
   B) Excess of albumin
   C) Biliary atresia
   D) Abundance of intestinal bacteria
   E) Low activity of UDP-glucuronosyltransferase

9. What is the mechanism of action of phototherapy?
   A) It induces UGT1A1
   B) It releases bilirubin from albumin
   C) It converts bilirubin into water-soluble photoisomers
   D) It stops hemolysis
   E) It inhibits biliverdin reductase

10. Gilbert syndrome is associated with decreased activity of which enzyme?
    A) MRP2
    B) UGT1A1
    C) Heme oxygenase
    D) β-Glucuronidase
    E) OATP1B1

11. What is the main parameter of the cytolysis syndrome?
    A) Albumin
    B) Alkaline phosphatase
    C) γ-Globulins
    D) Cholinesterase
    E) ALT

12. Which parameter decreases in hepatocellular insufficiency?
    A) Albumin and the prothrombin index
    B) Ammonia
    C) ALT
    D) GGT
    E) Unconjugated bilirubin

13. What are the main enzymes of the cholestasis syndrome?
    A) ALT and AST
    B) CK and LDH
    C) Amylase and lipase
    D) Alkaline phosphatase and GGT
    E) Cholinesterase and lipase

14. Which amino acids decrease in the blood in hepatic coma?
    A) Phenylalanine and tyrosine
    B) Tryptophan
    C) Methionine
    D) Glutamine
    E) Valine, leucine, isoleucine

15. What is the mechanism of action of lactulose in hepatic encephalopathy?
    A) It enhances urea synthesis
    B) It activates glutamine synthetase
    C) It acidifies the intestine, converting NH_3 to NH_4^+ and eliminating it
    D) It binds aromatic amino acids
    E) It enhances bilirubin conjugation

Clinical cases

Case 1

A 62-year-old patient complains of gradually increasing yellowing of the skin and sclerae, severe itching and weight loss; there is no pain. The urine is dark and the stools are colorless. Total bilirubin 285 µmol/l (direct 230), alkaline phosphatase 980 U/l, GGT 650 U/l, ALT 95 U/l, no urobilinogen in the urine. PT is prolonged and normalized after vitamin K was given. CA 19-9 is sharply increased.

Questions: Determine the type of jaundice. Explain the laboratory results. Why was the PT corrected by vitamin K?

Answer: Obstructive (cholestatic) jaundice; the painless progression, weight loss and CA 19-9 point to a tumor of the head of the pancreas. Because bile flow is blocked, conjugated bilirubin returns to the blood (the direct fraction predominates), is filtered by the kidneys and darkens the urine; no bile reaches the intestine — urobilinogen and stercobilin are not formed (acholic stools, no urobilinogen in the urine). In cholestasis the synthesis of alkaline phosphatase and GGT and their release into the blood increase sharply; bile acids accumulate and cause itching. Because there are no bile acids in the intestine, fat-soluble vitamin K is not absorbed — prothrombin and other factors are not carboxylated; since the synthetic function of the liver is preserved, parenteral vitamin K corrected the PT (this does not happen with hepatocellular damage).

Case 2

A 3-day-old newborn (from the second pregnancy of an Rh-negative mother) developed jaundice 10 hours after birth. Total bilirubin was 150 µmol/l on day 1 and 310 µmol/l on day 2 (almost all unconjugated), Hb 105 g/l, many reticulocytes, direct Coombs test positive.

Questions: What is the diagnosis? Why is the jaundice not physiological? What is the danger and how should help be provided?

Answer: Hemolytic disease of the newborn (Rh incompatibility): the mother's IgG antibodies against the Rh antigen cross the placenta and destroy the fetal erythrocytes (Coombs positive). The jaundice began on the first day, bilirubin is rising by > 85 µmol/l per day, and there is anemia — it is pathological. A lot of heme is broken down and unconjugated bilirubin is formed; the newborn's low UGT1A1 activity cannot conjugate it in time. The danger is bilirubin encephalopathy (> 340 µmol/l; free bilirubin exceeding the albumin capacity enters the brain). Help: intensive phototherapy, intravenous immunoglobulin, exchange transfusion (removal of antibodies and bilirubin, replacement of erythrocytes); avoidance of drugs that occupy albumin binding sites (sulfonamides, ceftriaxone). Prevention — anti-D immunoglobulin for the Rh-negative mother.

Case 3

A patient with liver cirrhosis developed drowsiness, confusion, hand tremor (“flapping”) and a peculiar sweetish-musty breath odor after bleeding from esophageal varices. Blood ammonia 145 µmol/l (normal < 50), albumin 26 g/l, PT prolonged.

Questions: What condition is this? Why did the condition worsen after the bleeding? What are the biochemical mechanism and the principles of treatment?

Answer: Hepatic encephalopathy (precoma). Blood in the intestine is a large protein load: bacteria break it down and form a lot of ammonia; in cirrhosis urea synthesis in the liver is reduced, and NH_3 reaches the brain through portocaval shunts. In the brain, astrocytes convert NH_3 into glutamine and swell; α-ketoglutarate is consumed and ATP formation decreases; aromatic amino acids increase and false neurotransmitters are formed; GABAergic inhibition is enhanced. Mercaptans — the liver odor. Albumin and PT show the decline in the synthetic function of the liver. Treatment: stopping the bleeding, bowel cleansing, lactulose (elimination as NH_4^+), rifaximin, L-ornithine L-aspartate, correction of electrolytes and glucose, temporary protein restriction.

Quick review

Table 25.6. *Key concepts of topic 25*

Concept	Brief definition
Liver functions	Glucostat, ketone bodies, plasma proteins, urea, detoxification
Heme breakdown	Heme oxygenase → biliverdin (+ CO, Fe) → bilirubin
Unconjugated bilirubin	With albumin, water-insoluble, neurotoxic, absent from urine
Conjugation	UGT1A1 + UDP-glucuronic acid → diglucuronide
In the intestine	Urobilinogen → stercobilin (feces), urobilin (urine)
Hemolytic jaundice	Indirect ↑, urobilinogen ↑
Hepatocellular	Both fractions ↑, ALT ↑↑
Obstructive	Direct ↑↑, ALP and GGT ↑↑, acholic stools
Newborns	Physiological: days 2–3, < 220 µmol/l; phototherapy
Hereditary	Gilbert, Crigler–Najjar (UGT1A1); Dubin–Johnson (MRP2)
Syndromes	Cytolysis (ALT), cholestasis (ALP, GGT), insufficiency (albumin, PT)
Hepatic coma	NH_3 ↑, false transmitters, Fischer ratio ↓; lactulose

TOPIC 26

Detoxification of toxic substances in the liver

Learning objectives

To give the student knowledge of the mechanisms of detoxification (biotransformation) of xenobiotics — microsomal oxidation and conjugation reactions, the detoxification of toxic products formed by the putrefaction of amino acids in the colon, the inactivation of normal metabolites and hormones, the biotransformation of drugs and the factors that affect it, inducers and inhibitors of cytochrome P450, P450 polymorphism and personalized treatment, and ethanol metabolism.

By the end of the class the student should know:

- phases I and II of biotransformation;
- the composition and mechanism of the microsomal oxidation system;
- the main conjugation reactions and their donors;
- the detoxification of putrefaction products such as indole, skatole, phenol and cresol;
- the inactivation of hormones and bilirubin in the liver;
- the clinical importance of P450 inducers and inhibitors;
- pharmacogenetic polymorphisms (CYP2D6, CYP2C19, CYP2C9, NAT2, TPMT);
- ethanol metabolism and the metabolic effects of alcohol.

The student should be able to:

- determine hippuric acid in urine (based on the Quick test);
- predict drug–drug interactions on the basis of P450;
- explain the biochemical mechanism of paracetamol poisoning.

1. General principles of biotransformation

Xenobiotics are substances foreign to the body: drugs, industrial poisons, pesticides, food additives, carcinogens. Most of them are **lipophilic** — even when filtered in the kidneys they are reabsorbed in the tubules, and they accumulate in membranes and adipose tissue. The aim of biotransformation is to convert them into **hydrophilic** (water-soluble) compounds and excrete them in the urine or bile.

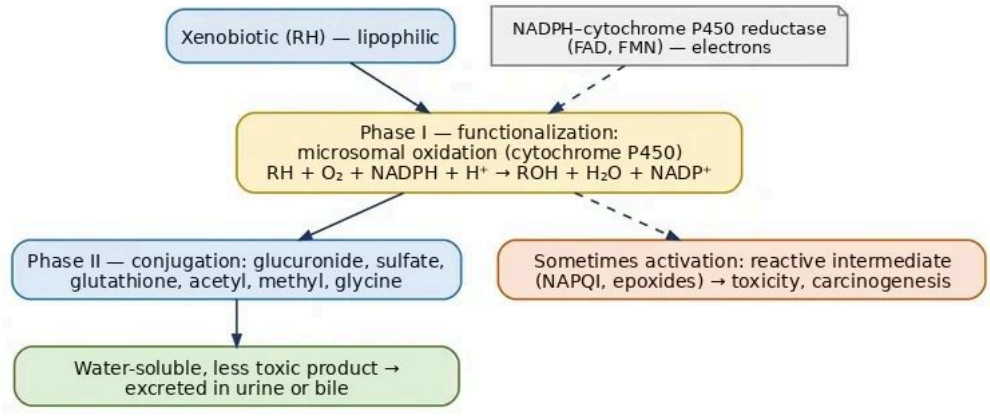


Figure 26.1. *The two phases of xenobiotic detoxification (drawn after Severin E.S.)*

- **Phase I (functionalization)** — polar groups (–OH, –NH_2, –COOH, –SH) are introduced into the molecule or unmasked: **oxidation** (mainly cytochrome P450), reduction (nitro and azo

compounds), **hydrolysis** (esterases, amidases, epoxide hydrolase).

- **Phase II (conjugation)** — a small endogenous molecule is attached to the polar group; the product is highly water-soluble and usually inactive.
- **Phase III** — export of the conjugates from the cell by membrane transporters (P-glycoprotein, MRP). Besides the liver, detoxification also takes place in the intestine, kidneys, lungs and skin.

2. Microsomal oxidation

Microsomes are fragments of the smooth endoplasmic reticulum isolated from a cell homogenate. The **monooxygenase** (mixed-function oxidase) reaction:

$RH + O_2 + NADPH + H^+ \rightarrow ROH + H_2O + NADP^+$

One oxygen atom goes to the substrate and the other to water.

Electron transport chain: NADPH → **NADPH–cytochrome P450 reductase** (a flavoprotein: FAD, FMN) → **cytochrome P450** (a hemoprotein). The second electron can also be donated by cytochrome b5 (via NADH–cytochrome b5 reductase).

The cytochrome P450 cycle: the substrate binds to Fe^{3+}-P450 → first electron (Fe^{2+}) → O_2 binds → second electron → the O–O bond is broken, one O atom forms water → the active iron–oxo complex hydroxylates the substrate → the product is released. The name “P450”: its reduced form bound to CO gives an absorption peak at 450 nm.

The P450 family: 57 genes in humans; mainly the **CYP1, 2, 3** families take part in xenobiotic metabolism: **CYP3A4** (≈ 50% of drugs), CYP2D6 (≈ 25%), CYP2C9, CYP2C19, CYP1A2, CYP2E1 (ethanol, paracetamol, organic solvents). Other P450s take part in the synthesis of endogenous substances: steroid hormones (adrenal glands, gonads), bile acids (7α-hydroxylase), hydroxylation of vitamin D, eicosanoids.

Types of reactions: aliphatic and aromatic hydroxylation, epoxidation, N-, O- and S-dealkylation, deamination, N- and S-oxidation, dehalogenation.

Toxic activation (lethal synthesis): some substances become more toxic under the action of P450: benzo(a)pyrene → diol epoxide (a carcinogen, topic 19), paracetamol → NAPQI, CCl_4 → trichloromethyl radical (lipid peroxidation, liver necrosis), methanol → formaldehyde → formic acid (blindness, acidosis), ethylene glycol → oxalate (kidney).

Difference from mitochondrial oxidation: microsomal oxidation does not produce ATP (energy is consumed — NADPH), has low substrate specificity and is inducible.

3. Conjugation reactions

Table 26.1. *Phase II — conjugation reactions*

Reaction	Donor (activated form)	Enzyme	Substrates (examples)
Glucuronidation (the most common)	UDP-glucuronic acid	UDP-glucuronosyltransferases (UGT)	Bilirubin, steroid hormones, paracetamol, morphine, phenols
Sulfation	PAPS (3’-phosphoadenosine-5’-phosphosulfate)	Sulfotransferases	Phenol, cresol, indoxyl, steroids, paracetamol, catecholamines

Reaction	Donor (activated form)	Enzyme	Substrates (examples)
With glutathione	Glutathione (GSH)	Glutathione S-transferases	Electrophilic substances, epoxides, NAPQI, heavy metals → mercapturic acids (in the urine)
Acetylation	Acetyl-CoA	N-acetyltransferases (NAT2)	Isoniazid, sulfonamides, procainamide, hydralazine, aromatic amines
Methylation	S-adenosylmethionine	Methyltransferases (COMT, TPMT)	Catecholamines, histamine, 6-mercaptopurine, nicotinamide
With amino acids	Glycine, taurine, glutamine (the substrate is activated as acyl-CoA)	Acyltransferases	Benzoic acid → hippuric acid; bile acids; phenylacetate → phenylacetylglutamine

4. Detoxification of intestinal putrefaction products

In the colon, undigested proteins and unabsorbed amino acids **putrefy** under the action of microorganisms (decarboxylation, deamination, breakdown of the side chain). The toxic substances formed are absorbed and reach the liver via the portal vein.

Table 26.2. *Putrefaction products and their detoxification*

Amino acid	Toxic product	Detoxification in the liver
Tryptophan	Indole, skatole (methylindole)	Oxidation (P450) → indoxyl, skatoxyl → conjugation with PAPS or UDP-glucuronic acid → indoxyl sulfate (potassium salt — animal indican), skatoxyl sulfate → urine
Tyrosine	Phenol, cresol	Sulfate or glucuronide conjugates → urine
Lysine, ornithine	Cadaverine, putrescine	Oxidation by amine oxidases
Cysteine, methionine	Hydrogen sulfide, methyl mercaptan	Oxidation (sulfate)
All amino acids	Ammonia	Urea synthesis (topic 13)
Histidine	Histamine	Methylation, oxidation

Urinary indican (the Obermayer reaction, topic 11) reflects the intensity of putrefaction in the intestine and the detoxifying activity of the liver: it increases in constipation, intestinal obstruction, achylia and dysbiosis.

5. Detoxification of normal metabolites and hormones

- **Bilirubin** — glucuronidation (topic 25).
- **Steroid hormones:** the cortisol ring is reduced and conjugated with glucuronic acid (17-hydroxycorticosteroids in the urine); androgens — 17-ketosteroids (sulfate and glucuronide conjugates); estrogens — hydroxylation, methylation, glucuronidation (enterohepatic circulation).
- **Catecholamines** — catechol-O-methyltransferase (COMT) and monoamine oxidase (MAO) → vanillylmandelic acid (VMA).

- **Peptide hormones** (insulin, glucagon, somatotropin) — proteolysis (insulinase); ≈ 50% of insulin is degraded in the liver on first pass.
- **Thyroid hormones** — deiodination, glucuronidation, sulfation. **Clinical importance:** in liver cirrhosis the inactivation of estrogens decreases — gynecomastia in men, "spider angiomas", palmar erythema; the inactivation of aldosterone decreases — retention of Na^+ and water (ascites); the breakdown of insulin decreases.

Age-related features: in newborns P450 and UGT activity is low (chloramphenicol — the "gray baby syndrome", physiological jaundice); in children some P450s are more active; in the elderly liver mass, blood flow and phase I reactions decrease — drug doses must be reduced.

6. Biotransformation of drugs and the factors that affect it

Influencing factors: age, sex (pregnancy), genetic polymorphism, liver and kidney diseases, nutrition (protein deficiency decreases P450; aromatic hydrocarbons in grilled meat induce CYP1A2), smoking (CYP1A2 induction), alcohol (CYP2E1 induction), other drugs (induction and inhibition), circadian rhythm.

The "first-pass" effect: when an orally administered drug passes through the intestinal wall and the liver, a large part of it is broken down (nitroglycerin, propranolol, morphine) — therefore it is given sublingually or parenterally.

Prodrugs are inactive drugs that are activated in the body: codeine → morphine (CYP2D6), clopidogrel → active thiol metabolite (CYP2C19), enalapril → enalaprilat (esterases), cyclophosphamide (CYP2B6).

Table 26.3. *Inducers and inhibitors of cytochrome P450*

Group	Substances	Clinical consequence
Inducers (enzyme synthesis ↑, 1–2 weeks)	Rifampicin, phenobarbital, carbamazepine, phenytoin, St John's wort, chronic alcohol use (CYP2E1), smoking (CYP1A2)	The effect of other drugs decreases: oral contraceptives fail, the warfarin dose must be increased; the activity of prodrugs increases; paracetamol toxicity increases
Inhibitors (immediate)	Ketoconazole, itraconazole, clarithromycin, erythromycin, HIV protease inhibitors (ritonavir), grapefruit juice (intestinal CYP3A4), cimetidine, fluconazole, fluoroquinolones (CYP1A2), amiodarone	The concentration and toxicity of other drugs increase: statins (myopathy), warfarin (bleeding), tacrolimus, ciclosporin

Phenobarbital is used as an inducer in neonatal jaundice and Crigler–Najjar type II (induction of UGT1A1).

7. P450 polymorphism and personalized treatment

Because of pharmacogenetic polymorphism, people are divided into **poor**, **intermediate**, **normal** and **ultrarapid** metabolizers of drugs.

- **CYP2D6** — poor metabolizers (≈ 7% in Europe): toxicity of β-blockers and antidepressants, codeine is ineffective (not converted to morphine); in **ultrarapid** metabolizers (many gene copies), codeine is converted to large amounts of morphine and depresses the respiratory center (deaths in children).

- **CYP2C19** — in poor metabolizers **clopidogrel** is not activated (risk of thrombosis); proton pump inhibitors act more strongly.
- **CYP2C9** and **VKORC1** — the dose of **warfarin** is chosen according to the genotype.
- **NAT2** — in slow acetylators, isoniazid polyneuropathy (protection with B_6), and lupus-like syndrome with hydralazine and procainamide.
- **TPMT** — in its deficiency 6-mercaptopurine and azathioprine cause severe bone marrow suppression — genotyping before treatment is recommended.
- **UGT1A1*28** (Gilbert) — irinotecan toxicity.
- **Pseudocholinesterase** — in the atypical variant suxamethonium causes prolonged apnea.
- **G6PD** — hemolysis with oxidizing drugs (topic 7). **Personalized medicine** — choosing the drug and the dose on the basis of pharmacogenetic tests and preventing adverse effects.

8. Detoxification of ethanol in the liver

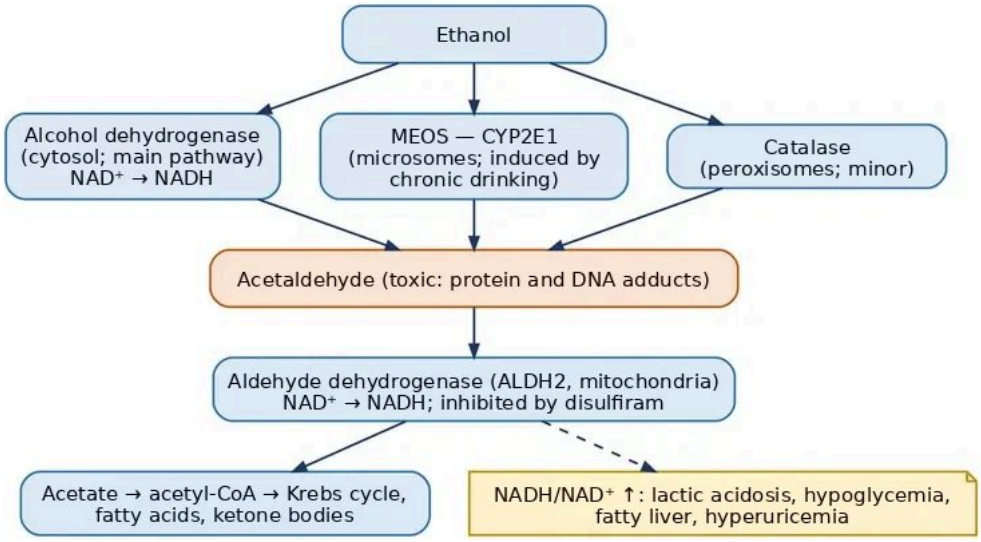


Figure 26.2. *Pathways of ethanol metabolism and its metabolic consequences (drawn after Lippincott's Illustrated Reviews: Biochemistry)*

≈ 90% of ethanol is oxidized in the liver (2–10% is excreted unchanged by the lungs and in the urine); the rate is ≈ 7–10 g/hour (0.1–0.15‰ per hour).

1. **Alcohol dehydrogenase (ADH)** — cytosol, NAD^+; the main pathway (also a small amount in the stomach; women have less gastric ADH).
2. **The microsomal ethanol-oxidizing system (MEOS) — CYP2E1** — NADPH, O_2; it is induced 5–10 times by chronic drinking ("tolerance"); it generates free radicals and alters the metabolism of paracetamol and other drugs.
3. **Catalase** — in peroxisomes, with H_2O_2 (of little importance). **Acetaldehyde** is the most toxic intermediate: adducts with proteins, DNA and phospholipids, damage to microtubules, depletion of glutathione. **Aldehyde dehydrogenase (ALDH)** oxidizes it to acetate: **ALDH2** — mitochondrial, low K_m, the main one; ALDH1 — cytosolic.

CLINICAL SIGNIFICANCE

The **ALDH2*2** (Glu487Lys) variant occurs in ≈ 40% of the East Asian population: the enzyme is almost inactive, and after alcohol acetaldehyde accumulates and causes facial flushing, tachycardia and nausea (a protective factor, but the risk of esophageal cancer increases). **Disulfiram**, by inhibiting ALDH, causes the same reaction — it is used in the treatment of alcoholism; metronidazole and some cephalosporins also have a disulfiram-like effect.

Acetate is converted to acetyl-CoA in various tissues.

The metabolic consequences of alcohol are mainly due to an **increase in the NADH/NAD$^+$ ratio** in the cytosol and mitochondria:

- pyruvate is converted to lactate — **lactic acidosis**; lactate competes with urate secretion in the kidney — **hyperuricemia** (gout attack);
- gluconeogenesis stops (pyruvate and oxaloacetate are reduced) — **hypoglycemia** when drinking on an empty stomach (dangerous, especially in children and diabetic patients);
- β-oxidation and the Krebs cycle slow down, and the synthesis of glycerol-3-phosphate and fatty acids increases — triacylglycerols accumulate: **fatty liver**, hyperlipidemia;
- a lot of acetyl-CoA and little oxaloacetate — **alcoholic ketoacidosis**;
- in chronic drinking: CYP2E1 induction, oxidative stress, activation of stellate cells — **alcoholic hepatitis** and **cirrhosis**; deficiency of thiamine (Wernicke–Korsakoff), folate and B_6; pancreatitis; in methanol and ethylene glycol poisoning ethanol acts as an **antidote** as a competitive substrate for ADH (fomepizole — an ADH inhibitor). **Markers of chronic alcoholism:** GGT ↑, AST/ALT > 2, MCV ↑ (macrocytosis), **carbohydrate-deficient transferrin** (CDT) ↑, triacylglycerols and uric acid ↑.

9. Laboratory work

Experiment 1: determination of hippuric acid in urine (assessment of the detoxifying function of the liver — based on the Quick–Pytel test)

Principle of the method: the liver conjugates benzoic acid with glycine and converts it into **hippuric acid**. The amount of hippuric acid excreted in the urine after a sodium benzoate load reflects the conjugating activity of the liver. Hippuric acid crystallizes in an acidic medium.

1. The subject drinks 4 g of sodium benzoate dissolved in 30 ml of water on an empty stomach (for teaching purposes a prepared urine sample is used).
2. Urine is collected for 4 hours and its volume is measured.
3. Add 1 g of NaCl and acetic acid (or HCl) to 20 ml of urine and keep in an ice bath for 30 minutes — hippuric acid precipitates as needle-shaped crystals.
4. Filter the crystals, wash with a small amount of cold water, dry and weigh (or titrate with 0.1 M NaOH in the presence of phenolphthalein: 1 ml of 0.1 M NaOH = 17.9 mg of hippuric acid). **Interpretation:** in a healthy person ≈ 65–85% of the administered sodium benzoate (≈ 3–3.5 g calculated as benzoic acid) is excreted as hippuric acid within 4 hours. A decrease is seen with damage to the liver parenchyma (hepatitis, cirrhosis) or with glycine deficiency; kidney function must be preserved.

Experiment 2: analysis of paracetamol metabolism (case-based)

1. Write the metabolic pathways of paracetamol at a therapeutic dose: glucuronidation (≈ 55%), sulfation (≈ 30%), NAPQI via CYP2E1 (≈ 5–10%) → conjugation with glutathione → mercapturic acid.
2. Show which pathways become saturated at a toxic dose (> 150 mg/kg) and the fate of NAPQI.
3. State the mechanism of action of N-acetylcysteine.

Results and analysis: the conjugation pathways become saturated, a lot of NAPQI is formed and the glutathione store is exhausted (< 30%) — NAPQI binds covalently to the cysteine residues of hepatocyte proteins — centrilobular liver necrosis. N-acetylcysteine supplies cysteine for glutathione synthesis and binds NAPQI directly; it is very effective if given within 8–10 hours. Chronic alcoholism (CYP2E1 induction, decreased glutathione) and fasting increase the risk.

10. Practical skills

Skill: predicting drug–drug interactions on the basis of P450

1. Determine which P450 isoform metabolizes each drug.
2. Check whether one of the drugs is an inducer or an inhibitor.
3. Inhibitor — the concentration and toxicity of the substrate increase; inducer — the effect decreases (for a prodrug — it increases).
4. Recommend clinical measures: dose adjustment, choice of another drug, therapeutic monitoring (INR, drug concentration).

Skill: assessing the metabolic effects of alcohol by laboratory parameters

1. Assess GGT, the AST/ALT ratio, MCV and CDT.
2. Check lactate, glucose, ketone bodies, uric acid and triacylglycerols.
3. Link each change to the increase in the NADH/NAD^+ ratio.
4. State the risk of vitamin deficiencies (B_1, folate) and their prevention.

Review questions

1. What are xenobiotics and what is the aim of their detoxification?
2. Describe the phases of biotransformation.
3. What is the composition of the microsomal oxidation system?
4. Write the equation of the monooxygenase reaction and explain its mechanism.
5. Name the main isoforms of cytochrome P450 and their substrates.
6. Give examples of toxic activation.
7. List the main conjugation reactions and their donors.
8. How are indole, skatole, phenol and cresol detoxified?
9. What is the diagnostic importance of urinary indican?
10. How are hormones inactivated in the liver? What consequences are seen in cirrhosis?
11. Which factors affect drug biotransformation?
12. Give examples of P450 inducers and inhibitors and explain their clinical importance.
13. Give examples of pharmacogenetic polymorphisms.
14. How is ethanol metabolized? What is the importance of the ALDH2 polymorphism?

15. Explain the metabolic consequences of alcohol in relation to the NADH/NAD^+ ratio.

Test questions

1. What is the electron donor in microsomal oxidation?
   A) NADPH
   B) NADH
   C) $FADH_2$
   D) Glutathione
   E) Ascorbate

2. Which P450 isoform metabolizes about half of all drugs?
   A) CYP2D6
   B) CYP1A2
   C) CYP2E1
   D) CYP2C9
   E) CYP3A4

3. What is the donor in the glucuronidation reaction?
   A) PAPS
   B) S-adenosylmethionine
   C) UDP-glucuronic acid
   D) Acetyl-CoA
   E) Glutathione

4. What is the sulfate donor in the sulfation reaction?
   A) Cysteine
   B) Taurine
   C) Methionine
   D) Sodium sulfate
   E) PAPS

5. Which substance is formed in the urine from the detoxification of indole?
   A) Hippuric acid
   B) Mercapturic acid
   C) Indican (indoxyl sulfate)
   D) Vanillylmandelic acid
   E) Urobilin

6. Which substance is a P450 inducer?
   A) Ketoconazole
   B) Rifampicin
   C) Grapefruit juice
   D) Clarithromycin
   E) Cimetidine

7. With which substance is NAPQI, the toxic metabolite of paracetamol, detoxified?
   A) Glutathione
   B) Glycine
   C) Acetyl-CoA
   D) Glucuronic acid
   E) Taurine

8. Which substance is formed by the conjugation of benzoic acid with glycine?
    A) Indican
    B) Mercapturic acid
    C) Phenylacetylglutamine
    D) Hippuric acid
    E) Creatine

9. Which enzyme provides the main pathway of ethanol oxidation?
    A) Catalase
    B) Alcohol dehydrogenase
    C) CYP2E1
    D) Aldehyde dehydrogenase
    E) Monoamine oxidase

10. Which enzyme does disulfiram inhibit?
    A) Alcohol dehydrogenase
    B) Aldehyde dehydrogenase
    C) CYP2E1
    D) Catalase
    E) Glutathione S-transferase

11. What causes hypoglycemia after drinking alcohol on an empty stomach?
    A) NADH/NAD^+ rises and gluconeogenesis stops
    B) Insulin secretion increases
    C) Glycogen synthesis is enhanced
    D) Glucose is not absorbed in the intestine
    E) Glucokinase is activated

12. Which isoform converts codeine into morphine and has an important polymorphism?
    A) CYP3A4
    B) CYP2C19
    C) CYP1A2
    D) CYP2E1
    E) CYP2D6

13. Activation of clopidogrel depends on which isoform?
    A) CYP2D6
    B) CYP2E1
    C) CYP2C19
    D) CYP1A2
    E) TPMT

14. The toxicity of 6-mercaptopurine is associated with polymorphism of which enzyme?
    A) NAT2
    B) UGT1A1
    C) COMT
    D) TPMT
    E) ALDH2

15. Which is a laboratory marker of chronic alcoholism?
    A) Amylase
    B) CK-MB
    C) Troponin
    D) Carbohydrate-deficient transferrin (CDT)
    E) Lipase

Clinical cases

Case 1

A 19-year-old girl in a depressed state took 20 tablets of paracetamol (10 g). On the first day she had no complaints other than nausea. On day 3 jaundice and pain under the right costal margin appeared: ALT 6800 U/l, AST 5200 U/l, bilirubin 85 µmol/l, INR 3.8.

Questions: What is the biochemical mechanism of the liver damage? Why did the symptoms appear late? How does the antidote act?

Answer: At a therapeutic dose paracetamol is detoxified mainly by glucuronidation and sulfation; a small part is converted by CYP2E1 into reactive NAPQI, which is immediately bound by glutathione. At a high dose the conjugation pathways become saturated, a lot of NAPQI is formed and the glutathione in hepatocytes is exhausted; NAPQI binds covalently to the cysteine residues of proteins, damages the mitochondria and causes oxidative stress — necrosis of the cells in the center of the liver lobules (where CYP2E1 is abundant). The necrosis and its signs (a sharp rise in ALT, impaired synthesis — INR) develop within 24–72 hours; the early period is almost asymptomatic, which makes it dangerous. The antidote is N-acetylcysteine: a source of cysteine for glutathione synthesis, and it binds NAPQI directly; it is most effective when given early (within 8–10 hours). The patient also needs psychiatric care.

Case 2

A 28-year-old woman with tuberculosis who is taking rifampicin became pregnant even though she was taking an oral contraceptive. In addition, she takes warfarin, and her INR has fallen to 1.3 (target 2–3).

Questions: How do you explain these situations? What is the mechanism? What measures should be taken?

Answer: Rifampicin is a potent P450 inducer: through the nuclear receptor PXR it enhances the synthesis of CYP3A4, CYP2C9 and other enzymes, as well as UGT and P-glycoprotein. As a result, the estrogens and progestins of the contraceptive (CYP3A4) are broken down rapidly and their blood concentration falls below the effective level — contraception fails. Warfarin (its S-enantiomer via CYP2C9) is metabolized rapidly — the anticoagulant effect decreases (INR fell, risk of thrombosis). Measures: barrier methods of contraception during treatment and for 4 weeks after it ends; increasing the warfarin dose under INR control (and reducing the dose again after rifampicin is stopped, as the induction wears off over 2 weeks — otherwise there is a risk of bleeding).

Case 3

A 50-year-old man has been drinking large amounts of alcohol for 20 years. He was brought in confused after heavy drinking on an empty stomach in the morning. Blood glucose 2.1 mmol/l, lactate 5.8 mmol/l, β-hydroxybutyrate 4.5 mmol/l, uric acid 560 µmol/l, triacylglycerols 4.8 mmol/l, GGT 420 U/l, MCV 104 fl.

Questions: Explain all the biochemical changes in relation to alcohol metabolism.

Answer: When ethanol is oxidized by alcohol dehydrogenase and aldehyde dehydrogenase, a lot of NADH is formed and the NADH/NAD^+ ratio rises. As a result, pyruvate is reduced to lactate (lactic acidosis) and oxaloacetate to malate — substrates for gluconeogenesis are lacking; when the glycogen store is exhausted on an empty stomach, hypoglycemia develops (confusion). Lactate competes with urate secretion in the kidney — hyperuricemia. β-Oxidation slows because of the excess NADH, glycerol-3-phosphate increases and TAG synthesis is enhanced — hypertriglyceridemia and fatty liver. Acetyl-CoA formed from acetate is converted to ketone bodies when oxaloacetate is lacking, and NADH reduces acetoacetate to β-hydroxybutyrate — alcoholic ketoacidosis. The increase in GGT is due to the induction of CYP2E1 and microsomal enzymes and to liver damage; macrocytosis is due to the direct effect of alcohol on the bone marrow and folate deficiency. Treatment: thiamine before giving glucose (to prevent Wernicke encephalopathy).

Quick review

Table 26.4. *Key concepts of topic 26*

Concept	Brief definition
Biotransformation	Lipophilic → hydrophilic; phase I (P450), phase II (conjugation), phase III (excretion)
Monooxygenase	$RH + O_2 + NADPH + H^+ \rightarrow ROH + H_2O + NADP^+$
P450 chain	NADPH → P450 reductase (FAD, FMN) → cytochrome P450
Main isoforms	CYP3A4 (≈ 50%), 2D6, 2C9, 2C19, 1A2, 2E1
Conjugation donors	UDP-glucuronic acid, PAPS, GSH, acetyl-CoA, SAM, glycine
Putrefaction products	Indole → indican; phenol, cresol → sulfates
Inducers	Rifampicin, phenobarbital, carbamazepine, St John's wort, alcohol
Inhibitors	Ketoconazole, macrolides, ritonavir, grapefruit
Polymorphisms	CYP2D6 (codeine), CYP2C19 (clopidogrel), CYP2C9 (warfarin), NAT2, TPMT
Paracetamol	NAPQI → glutathione; antidote — N-acetylcysteine
Ethanol	ADH → acetaldehyde → ALDH2 → acetate; MEOS (CYP2E1)
Consequences of alcohol	NADH ↑: lactic acidosis, hypoglycemia, fatty liver, hyperuricemia

TOPIC 27

Physiologically active substances and biochemistry of the endocrine system

Learning objectives

To give the student knowledge of the classification of physiologically active substances, substances of peptide nature (cytokines, growth factors), amino acid derivatives, the hierarchy of regulatory systems and the hypothalamic–pituitary system, the classification of hormones by chemical nature, function and mechanism of action, the interrelation of the endocrine and nervous systems, the synthesis and catabolism of steroid hormones, the mechanism of action of sex hormones, the detoxification of hormones in the liver, and the determination of chorionic gonadotropin in urine.

By the end of the class the student should know:

- the classification of physiologically active substances (hormones, neurotransmitters, cytokines, growth factors, eicosanoids);
- the classification of hormones by chemical nature;
- the mechanisms of action through membrane and intracellular receptors;
- the hormones of the hypothalamus and pituitary and the feedback principle;
- the stages and enzymes of steroid hormone synthesis;
- the synthesis, transport and mechanism of action of sex hormones;
- the structure, function and diagnostic importance of β-hCG.

The student should be able to:

- determine chorionic gonadotropin in urine with an immunochromatographic test and interpret the result;
- explain the mechanism of hormone action through second messengers;
- determine the type of hormonal disorder (primary, secondary) on the basis of feedback.

1. Classification of physiologically active substances

Physiologically active substances are signaling molecules that regulate cell activity at very low concentrations (10^{-6}–10^{-12} M).

Table 27.1. *Classification of signaling molecules by range of action*

Type	Definition	Examples
Endocrine	Through the blood to distant target cells	Insulin, thyroxine, cortisol
Paracrine	To neighboring cells	Somatostatin (in the islets), histamine, NO, growth factors
Autocrine	To the cell that produced it	IL-2 (T lymphocytes), tumor growth factors
Neurocrine (neurotransmitters)	Through a synapse	Acetylcholine, norepinephrine, GABA
Neuroendocrine	Neurons secrete into the blood	Vasopressin, oxytocin, releasing hormones

Type	Definition	Examples
Juxtacrine	A ligand on the cell surface with a receptor on a neighboring cell	Notch-Delta

Locally acting substances (tissue hormones):

- **eicosanoids** (from arachidonic acid): prostaglandins, prostacyclin, thromboxanes (cyclooxygenase pathway), leukotrienes (lipoxygenase pathway) — inflammation, pain, fever, hemostasis, bronchial tone (topic 9);
- **biogenic amines:** histamine, serotonin;
- **kinins:** bradykinin, kallidin — vasodilation, permeability, pain;
- **nitric oxide (NO)**. **Substances of peptide nature:**
- **cytokines** — interleukins (IL-1, IL-6 — acute phase, fever), interferons (antiviral), tumor necrosis factor (TNF-α), chemokines, colony-stimulating factors; they act mainly through the JAK-STAT pathway;
- **growth factors** — epidermal growth factor (EGF), platelet-derived (PDGF), insulin-like (IGF-1), fibroblast (FGF), vascular endothelial (VEGF), transforming (TGF-β), nerve growth factor (NGF), erythropoietin; mainly through tyrosine kinase receptors;
- **neuropeptides** — endorphins, enkephalins (opioid peptides), substance P, neuropeptide Y. **Amino acid derivatives:** from tyrosine — catecholamines (dopamine, norepinephrine, epinephrine) and **thyroid hormones**; from tryptophan — **serotonin** and **melatonin**; from histidine — **histamine**; from glutamate — **GABA**; from arginine — NO.

2. Classification of hormones

Table 27.2. *Classification of hormones by chemical nature*

Group	Examples	Features
Protein-peptide	Insulin, glucagon, somatotropin, prolactin, PTH, calcitonin, hypothalamic hormones, vasopressin, oxytocin	Synthesized as prohormones, stored in granules, water-soluble, membrane receptors
Glycoproteins	TSH, FSH, LH, hCG	Common α subunit, specific β subunit
Amino acid derivatives	Epinephrine, norepinephrine (hydrophilic); thyroxine (T_4), triiodothyronine (T_3) (lipophilic)	Catecholamines — membrane receptors; thyroid hormones — nuclear receptors
Steroids	Cortisol, aldosterone, testosterone, estradiol, progesterone, calcitriol	From cholesterol, not stored, transported by carrier proteins, nuclear receptors
Fatty acid derivatives	Eicosanoids	Local action

By biological function: regulators of carbohydrate, lipid and protein metabolism (insulin, glucagon, cortisol, somatotropin), of water–salt metabolism (aldosterone, vasopressin, natriuretic peptides), of calcium–phosphate metabolism (PTH, calcitonin, calcitriol), of reproductive function (sex hormones, gonadotropins), and of other endocrine glands (tropic hormones).

3. Mechanisms of hormone action

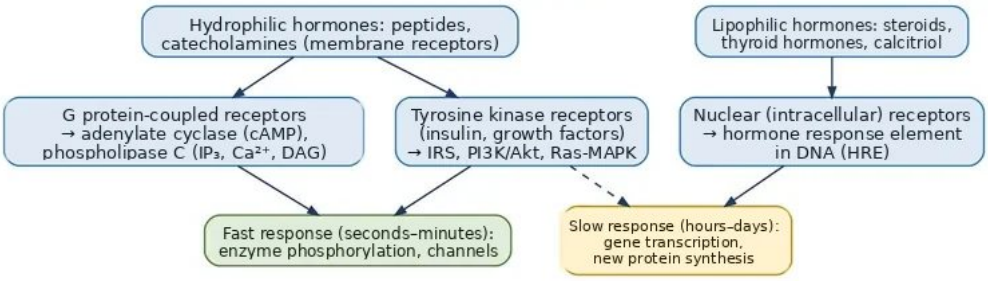


Figure 27.1. *Mechanisms of hormone action through receptors (drawn after Lippincott's Illustrated Reviews: Biochemistry)*

Through membrane receptors (hydrophilic hormones):

1. **G protein-coupled receptors** (7 transmembrane domains):
2. **G_s → adenylate cyclase → cAMP → protein kinase A** — glucagon, epinephrine (β), ACTH, TSH, PTH, vasopressin (V_2), LH, FSH;
3. **G_i** — inhibits adenylate cyclase (epinephrine α_2, somatostatin);
4. **G_q → phospholipase Cβ** → phosphatidylinositol 4,5-bisphosphate → **IP_3** (releases Ca^{2+} from the endoplasmic reticulum; Ca^{2+}–calmodulin) + **DAG** (protein kinase C) — epinephrine (α_1), vasopressin (V_1), angiotensin II, oxytocin, releasing hormones. cAMP is broken down by phosphodiesterase (inhibited by caffeine and theophylline).

2. **Tyrosine kinase receptors** — insulin, IGF-1, EGF, PDGF: autophosphorylation → IRS, PI3K/Akt, Ras-MAPK pathways.
3. **Tyrosine kinase-associated receptors (JAK-STAT)** — somatotropin, prolactin, erythropoietin, leptin, cytokines.
4. **Guanylate cyclase receptors** — natriuretic peptides (membrane), NO (soluble guanylate cyclase) → cGMP → PKG. **Through intracellular (nuclear) receptors (lipophilic hormones):** steroids, T_3, calcitriol and retinoids cross the membrane by diffusion and bind to cytoplasmic or nuclear receptors; the hormone–receptor complex (released from Hsp90) forms a dimer, binds to **hormone response elements (HRE)** in DNA and changes the transcription of target genes. The action is slow (hours) and long-lasting.

4. Hierarchy of regulatory systems. The hypothalamic–pituitary system

Hierarchy: cerebral cortex → **hypothalamus** (releasing and inhibiting hormones) → **pituitary** (tropic hormones) → **peripheral endocrine glands** → target tissues. Regulation is based on the principle of **negative feedback**: the peripheral hormone inhibits the secretion of the hypothalamus and pituitary.

Table 27.3. *Hormones of the hypothalamus and pituitary*

Hypothalamus	Pituitary (anterior lobe, adenohypophysis)	Target and action
Corticoliberin (CRH)	ACTH (from POMC; together with β-lipotropin, β-endorphin, α-MSH)	Adrenal cortex — cortisol
Thyroliberin (TRH)	TSH	Thyroid gland — T_4, T_3
Gonadoliberin (GnRH)	LH, FSH	Gonads — steroids, gametogenesis

Hypothalamus	Pituitary (anterior lobe, adenohypophysis)	Target and action
Somatoliberin / somatostatin	Somatotropin (GH)	Liver — IGF-1; growth, anabolism, lipolysis, contrainsular effect
Dopamine (prolactostatin)	Prolactin	Mammary gland — lactation
—	Posterior pituitary (neurohypophysis) — synthesized in the hypothalamus and transported along axons: vasopressin (ADH), oxytocin	Kidney (water reabsorption), vessels; uterine contraction, milk ejection

Level of disorders: primary (peripheral gland) — the peripheral hormone is low and the tropic hormone is **high** (feedback); **secondary** (pituitary) or **tertiary** (hypothalamus) — both are low. For example, in primary hypothyroidism TSH ↑, T_4 ↓; in secondary TSH ↓, T_4 ↓.

5. Interrelation of the endocrine and nervous systems. Ontogenesis

The hypothalamus is the center that links the nervous and endocrine systems: its neurons produce neurohormones (**neurosecretion**). The adrenal medulla is a modified sympathetic ganglion (acetylcholine from preganglionic fibers triggers epinephrine secretion). Hormones affect brain function (thyroid hormones — brain development, cortisol — mood and memory, sex hormones — behavior); in stress the hypothalamic–pituitary–adrenal system and the sympathoadrenal system are activated together. Circadian rhythms (cortisol high in the morning, melatonin at night) are controlled by the suprachiasmatic nucleus.

Changes in ontogenesis: in the fetal period thyroid hormones are crucial for brain and skeletal development (congenital hypothyroidism — cretinism; newborn screening — TSH); in childhood somatotropin and IGF-1 ensure growth; at puberty gonadoliberin pulsatility begins and sex hormones increase; in old age somatotropin (somatopause), sex hormones (menopause, andropause) and DHEA decrease, and insulin sensitivity declines.

6. Synthesis and catabolism of steroid hormones

All steroid hormones are synthesized **from cholesterol** (obtained mainly through LDL). The key (rate-limiting) step is the transport of cholesterol to the inner mitochondrial membrane by the **StAR protein** and its conversion to **pregnenolone** by **CYP11A1** (the side-chain cleavage enzyme); this step is stimulated through cAMP by ACTH (adrenal gland) and LH (gonads). The enzymes of synthesis belong mainly to the cytochrome P450 family.

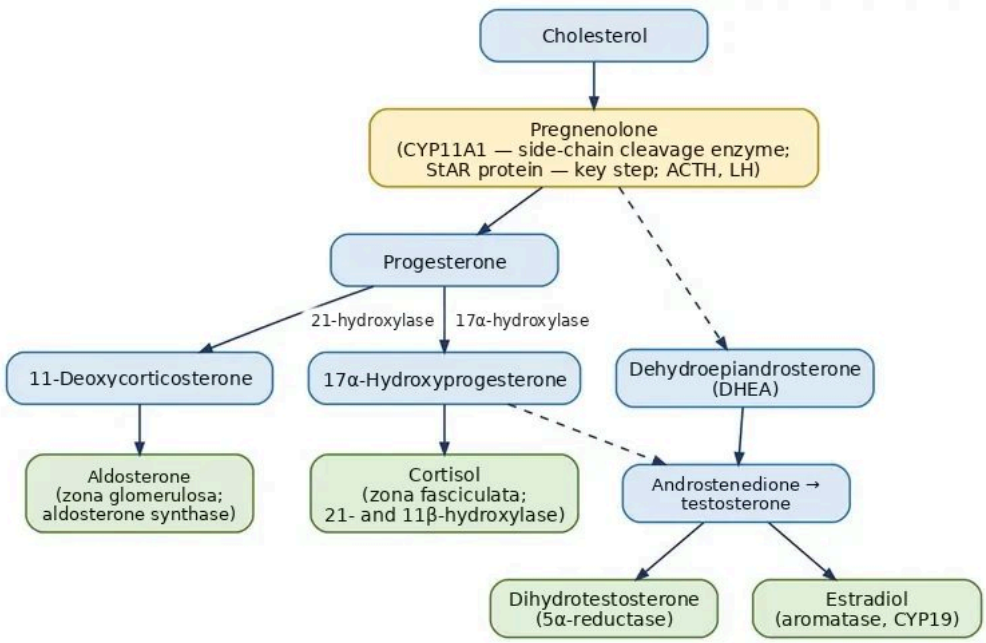


Figure 27.2. *Main pathways of steroid hormone synthesis (drawn after Severin E.S.)*

- **Zona glomerulosa: aldosterone** (aldosterone synthase — CYP11B2; stimulated by angiotensin II and K^+);
- **zona fasciculata: cortisol** (17α-, 21- and 11β-hydroxylases; ACTH);
- **zona reticularis:** androgens (DHEA, androstenedione);
- **testes (Leydig cells): testosterone** (LH); in target tissues **5α-reductase** converts it into the more active **dihydrotestosterone** (prostate, external genitalia, hair follicles);
- **ovaries:** theca cells synthesize androgens (LH), and granulosa cells convert them into **estradiol** with **aromatase** (CYP19) (FSH); the corpus luteum — **progesterone**; adipose tissue also contains aromatase (the source of estrogens after menopause).

CLINICAL SIGNIFICANCE

Congenital adrenal hyperplasia — most often **21-hydroxylase** deficiency (autosomal recessive): cortisol and aldosterone are not synthesized; by feedback ACTH increases, the gland becomes hyperplastic and the accumulated 17α-hydroxyprogesterone is converted into androgens — virilization of the external genitalia in girls, salt-wasting crisis (hyponatremia, hyperkalemia). Newborn screening — blood 17-hydroxyprogesterone. Treatment — replacement with glucocorticoids and mineralocorticoids.

Transport: in the blood most steroids are bound to proteins: cortisol — to **transcortin** (≈ 90%) and albumin; testosterone and estradiol — to **sex hormone-binding globulin (SHBG)** and albumin; only the free fraction (1–3%) is active.

Catabolism (in the liver): reduction of ring A, hydroxylation, then conjugation with **glucuronic acid** or **sulfate** — water-soluble metabolites are excreted in the urine: **17-hydroxycorticosteroids** from cortisol, **17-ketosteroids** from androgens and part of cortisol, **pregnanediol** from progesterone, and estriol glucuronides from estrogens. The daily urinary free cortisol is a screening test for Cushing syndrome. In liver cirrhosis the inactivation of estrogens decreases (topic 26).

7. Mechanism of action of sex hormones

Androgens (testosterone, dihydrotestosterone) bind to the androgen receptor (nuclear): formation of the male genitalia in the fetal period, secondary sexual characteristics at puberty, spermatogenesis

(together with FSH), **anabolic effect** (muscle protein synthesis, erythropoiesis, bone mass), libido. **Androgen receptor defect** — complete androgen insensitivity syndrome (genotype XY, female phenotype).

Estrogens (estradiol, estrone, estriol) — estrogen receptors (ERα, ERβ): development of the female genitalia and mammary glands, endometrial proliferation (the 1st phase of the menstrual cycle), reduction of bone resorption (osteoporosis after menopause), raising HDL and lowering LDL (atheroprotective), increasing the hepatic synthesis of transport proteins (SHBG, transcortin, thyroxine-binding globulin) and clotting factors (risk of thrombosis with oral contraceptives).

Progesterone — progesterone receptor: secretory phase of the endometrium, maintenance of pregnancy (reduces myometrial excitability), development of the alveoli of the mammary glands, thermogenic effect (rise in basal temperature).

Pharmacological use: oral contraceptives (estrogen + progestin — suppress gonadotropin secretion by feedback), tamoxifen (an estrogen antagonist in the breast), aromatase inhibitors (letrozole), 5α-reductase inhibitors (finasteride — in prostatic adenoma), antiandrogens, anabolic steroids (abuse — liver, heart, infertility).

8. Chorionic gonadotropin

Human chorionic gonadotropin (hCG) is a glycoprotein produced by the **syncytiotrophoblast**; its α subunit is identical to that of LH, FSH and TSH, while the **β subunit** is specific (that is why tests detect β-hCG). Its function is to act like LH and maintain the **corpus luteum** and progesterone secretion during the first 8–10 weeks of pregnancy (after that the placenta produces progesterone itself). It is detectable in the blood 8–10 days after fertilization, doubles every 48–72 hours and peaks at 8–10 weeks. At high concentrations it stimulates TSH receptors (signs of hyperthyroidism in hyperemesis of pregnancy).

Diagnostic importance: early detection of pregnancy; ectopic (extrauterine) pregnancy — hCG rises slowly; threatened miscarriage; trophoblastic diseases (**hydatidiform mole**, **choriocarcinoma** — very high hCG; a tumor marker); embryonal tumors of the testis; prenatal screening (elevated in Down syndrome in the 2nd trimester).

9. Laboratory work

Experiment 1: determination of chorionic gonadotropin in urine with an immunochromatographic test

Principle of the method: the test strip contains monoclonal antibodies against β-hCG labeled with gold particles. Urine moves along the membrane by capillary action: if hCG is present, it forms a complex with the labeled antibodies and is captured by the immobilized second antibodies in the "test" zone — a colored line appears (a "sandwich"). Antiglobulins in the "control" zone capture the excess conjugate — this shows that the test has worked correctly.

1. Collect morning urine (the highest concentration) in a clean container.
2. Dip the test strip into the urine up to the indicated mark for 10–15 seconds (do not go beyond the mark).
3. Place it on a flat horizontal surface and read the result after 3–5 minutes (a result read after 10 minutes is unreliable). **Interpretation:** two lines — positive (hCG ≥ 10–25 IU/l); only the control line — negative; no control line — the test is invalid. The test is reliable from the first

day of a missed period. False negative — too early, dilute urine; false positive — treatment with hCG preparations, trophoblastic tumors, a recent abortion.

Experiment 2: analysis of hormone mechanisms of action using a scheme (case-based)

1. For the given hormone (glucagon, epinephrine, cortisol, insulin, vasopressin), determine its chemical nature and receptor type.
2. Write down the second messenger and the final effector protein.
3. Show the speed of the response (fast or slow) and the metabolic result. **Results and analysis:** hydrophilic hormones act quickly through membrane receptors and second messengers, and lipophilic hormones act slowly through nuclear receptors by changing gene expression.

10. Practical skills

Skill: determining the level of an endocrine disorder

1. Assess the peripheral hormone (T_4, cortisol, testosterone) and the corresponding tropic hormone (TSH, ACTH, LH).
2. Peripheral hormone ↓ + tropic hormone ↑ — primary (glandular) insufficiency.
3. Peripheral hormone ↓ + tropic hormone ↓ — secondary (pituitary) or tertiary (hypothalamic).
4. Peripheral hormone ↑ + tropic hormone ↓ — primary hyperfunction (tumor, autoimmune stimulation).
5. Confirm with stimulation or suppression tests.

Skill: clinical interpretation of an hCG result

1. Compare the results of the qualitative test (urine) and the quantitative determination (blood).
2. Assess the dynamics: an increase of ≈ 2 times in 48 hours — normal pregnancy; a slow increase — ectopic or nonviable pregnancy.
3. Very high for the gestational age — multiple pregnancy or hydatidiform mole.
4. hCG outside pregnancy — tumor (choriocarcinoma, testicular tumor).

Review questions

1. How are physiologically active substances classified by range of action?
2. Give examples of locally acting hormones.
3. What are the functions of cytokines and growth factors?
4. Which physiologically active substances are formed from amino acids?
5. How are hormones classified by chemical nature?
6. Explain the mechanisms of action through G protein-coupled receptors.
7. What are the functions of IP_3 and DAG?
8. How do steroid hormones act?
9. How is the hypothalamic–pituitary system organized?
10. Explain the principle of negative feedback with an example.
11. How is the interrelation of the endocrine and nervous systems manifested?
12. What are the key step and the main enzymes of steroid hormone synthesis?
13. How are steroid hormones transported and catabolized?

14. Name the mechanism of action and the biological effects of sex hormones.
15. What are the structure, function and diagnostic importance of hCG?

Test questions

1. What is the type of signaling that acts on neighboring cells called?
    A) Endocrine
    B) Autocrine
    C) Paracrine
    D) Neuroendocrine
    E) Juxtacrine

2. Which hormone is a steroid?
    A) Insulin
    B) Adrenaline
    C) Thyroxine
    D) Cortisol
    E) Glucagon

3. What is the second messenger of glucagon?
    A) cAMP
    B) cGMP
    C) IP_3
    D) DAG
    E) Ca^{2+}–calmodulin

4. Which second messengers are formed when phospholipase C is activated?
    A) cAMP and cGMP
    B) IP_3 and DAG
    C) NO and cGMP
    D) Arachidonic acid and prostaglandins
    E) ATP and ADP

5. Where are steroid hormone receptors located?
    A) In the plasma membrane
    B) In the mitochondria
    C) In the cytoplasm and nucleus
    D) In the Golgi apparatus
    E) In lysosomes

6. What is the key step of steroid hormone synthesis?
    A) Hydroxylation of progesterone
    B) Aromatization of testosterone
    C) Conjugation of cortisol
    D) The 5α-reductase reaction
    E) Conversion of cholesterol to pregnenolone (StAR, CYP11A1)

7. Which enzyme converts testosterone into estradiol?
    A) 5α-Reductase
    B) 21-Hydroxylase
    C) 17α-Hydroxylase
    D) Aldosterone synthase
    E) Aromatase

8. Which enzyme is most often deficient in congenital adrenal hyperplasia?
    A) 21-Hydroxylase
    B) Aromatase
    C) 5α-Reductase
    D) CYP11A1
    E) Aldosterone synthase

9. Which hormone is released from the posterior pituitary?
    A) ACTH
    B) Vasopressin
    C) TSH
    D) Prolactin
    E) Somatotropin

10. What change is seen in primary hypothyroidism?
    A) TSH ↓, T_4 ↓
    B) TSH ↓, T_4 ↑
    C) TSH ↑, T_4 ↑
    D) TSH ↑, T_4 ↓
    E) TSH N, T_4 ↑

11. Which cells produce chorionic gonadotropin?
    A) Syncytiotrophoblast
    B) Corpus luteum
    C) Pituitary
    D) Granulosa cells
    E) Leydig cells

12. What is the main function of hCG?
    A) Inducing ovulation
    B) Ensuring milk secretion
    C) Enhancing uterine contraction
    D) Stopping endometrial proliferation
    E) Maintaining the corpus luteum and ensuring progesterone secretion

13. Which protein mainly carries cortisol in the blood?
    A) Albumin
    B) SHBG
    C) Thyroxine-binding globulin
    D) Transcortin
    E) Transferrin

14. To which type do insulin and growth factor receptors belong?
    A) G protein–coupled
    B) Ion channels
    C) Tyrosine kinase receptors
    D) Nuclear receptors
    E) Guanylate cyclase receptors

15. How are steroid hormones inactivated in the liver?
    A) By proteolysis
    B) By reduction and conjugation with glucuronic acid or sulfate
    C) By methylation and deamination
    D) By deiodination
    E) Phosphorylation

Clinical cases

Case 1

In the maternity ward a newborn (XX karyotype) has ambiguous external genitalia. In the 2nd week vomiting and dehydration began: Na^+ 124, K^+ 6.8 mmol/l, glucose 2.8 mmol/l. Blood 17-hydroxyprogesterone is sharply increased, and ACTH is high.

Questions: What is the diagnosis? Explain all the signs on the basis of the steroid synthesis scheme. What is the principle of treatment?

Answer: Congenital adrenal hyperplasia — 21-hydroxylase deficiency, salt-wasting form. Progesterone and 17α-hydroxyprogesterone are not 21-hydroxylated: cortisol (hypoglycemia) and aldosterone (Na^+ loss, K^+ retention, dehydration) are not synthesized. Because of the low cortisol, ACTH increases by feedback and the gland becomes hyperplastic; the accumulated 17-hydroxyprogesterone is diverted to the androgen synthesis pathway — the excess androgens virilize the external genitalia of the girl in the fetal period. Treatment: glucocorticoids (hydrocortisone — replaces cortisol and, by lowering ACTH, stops the overproduction of androgens), a mineralocorticoid (fludrocortisone), table salt; in a crisis — intravenous fluids, glucose, hydrocortisone.

Case 2

A 27-year-old woman's period is 2 weeks late; she has one-sided lower abdominal pain and bloody discharge. Blood β-hCG 1200 IU/l, and 1350 IU/l 48 hours later. Ultrasound did not detect a gestational sac in the uterine cavity.

Questions: What condition is suspected? Explain the hCG dynamics. What is the physiological role of hCG in pregnancy?

Answer: Extrauterine (ectopic, probably tubal) pregnancy. In a normal intrauterine pregnancy β-hCG roughly doubles every 48–72 hours in the early weeks; here it hardly rose (≈ 12%) — the trophoblast is not developing adequately; the absence of a gestational sac in the uterus at this hCG level indicates an ectopic location. hCG secreted by the syncytiotrophoblast maintains the corpus luteum through LH receptors and ensures progesterone secretion; progesterone maintains the endometrium and stops menstruation. An ectopic pregnancy requires urgent treatment because of the risk of tubal rupture and internal bleeding.

Case 3

A 40-year-old patient has extreme fatigue, weight loss, nausea, arterial hypotension and dark pigmentation of the skin and gums. Blood Na^+ 128, K^+ 5.9 mmol/l, morning cortisol 60 nmol/l (normal 140–690), ACTH sharply increased.

Questions: At what level is the disorder? What is the biochemical cause of the pigmentation? Explain the electrolyte changes.

Answer: Primary adrenocortical insufficiency (Addison disease, most often autoimmune). Low cortisol and high ACTH — the disorder is in the peripheral gland (feedback is preserved, and the pituitary produces a lot of ACTH). ACTH is formed from proopiomelanocortin (POMC); when POMC is cleaved, α-melanocyte-stimulating hormone also increases, and ACTH itself stimulates melanocortin-1 receptors — melanin synthesis is enhanced (pigmentation; absent in secondary insufficiency). Aldosterone is also deficient: renal Na^+ reabsorption and K^+ secretion decrease — hyponatremia, hyperkalemia, hypovolemia and hypotension. Treatment — lifelong replacement therapy with hydrocortisone and fludrocortisone.

Quick review

Table 27.4. *Key concepts of topic 27*

Concept	Brief definition
Types of signaling	Endocrine, paracrine, autocrine, neurocrine
Local hormones	Eicosanoids, histamine, serotonin, kinins, NO
Peptide regulators	Cytokines (JAK-STAT), growth factors (tyrosine kinase)
Chemical classification	Protein-peptide, glycoprotein, amino acid derivatives, steroids
Membrane receptors	G_s/G_i (cAMP), G_q (IP_3, DAG, Ca^{2+}), tyrosine kinase
Nuclear receptors	Steroids, T_3, calcitriol → HRE → transcription
Hierarchy	Hypothalamus (releasing hormones) → pituitary (tropic hormones) → glands; negative feedback
Steroid synthesis	Cholesterol → pregnenolone (StAR, CYP11A1 — key step)
Enzymes	21-, 17α-, 11β-hydroxylase; aromatase; 5α-reductase
Carriers	Transcortin, SHBG, albumin
Catabolism	Reduction in the liver + glucuronide/sulfate; 17-KS, 17-OHCS
hCG	Syncytiotrophoblast; maintains the corpus luteum; pregnancy test

TOPIC 28

Hormonal regulation of metabolism. Functions of the peripheral glands and their disorders

Learning objectives

To give the student knowledge of the hormonal regulation of carbohydrate, lipid and amino acid metabolism and its age-related features, thyroid hormones and the effect of their deficiency on growth and development, the regulation of calcium–phosphate metabolism and rickets, the regulation of water–salt metabolism, locally acting hormones, the pathogenesis of endemic goiter and hypothyroidism, and the determination of insulin by enzyme immunoassay.

By the end of the class the student should know:

- the metabolic effects and mechanisms of action of insulin, glucagon, epinephrine, cortisol and somatotropin;
- the types of diabetes mellitus and their biochemical diagnosis;
- the synthesis, transport, action and regulation of thyroid hormones;
- the biochemical basis of hypothyroidism, endemic goiter and hyperthyroidism;
- the role of PTH, calcitonin and calcitriol in calcium–phosphate metabolism;
- the causes and biochemical signs of rickets;
- the role of vasopressin, the renin–angiotensin–aldosterone system and natriuretic peptides in water–salt metabolism.

The student should be able to:

- explain the principle of determining serum insulin by ELISA and interpret the result;
- calculate the HOMA-IR index;
- interpret the results of thyroid hormones and TSH.

1. Hormonal regulation of carbohydrate, lipid and amino acid metabolism

Insulin — in the **β-cells** of the islets of Langerhans of the pancreas: preproinsulin → proinsulin (A and B chains + C-peptide) → insulin + **C-peptide** (released in equal amounts; C-peptide is an indicator of endogenous insulin secretion). **Mechanism of secretion:** glucose enters through GLUT2 → glucokinase → glycolysis and oxidation → ATP/ADP ↑ → ATP-sensitive K^+ channels close (sulfonylurea drugs act in the same way) → depolarization → Ca^{2+} channels open → exocytosis of granules. Incretins (GLP-1, GIP) enhance secretion; epinephrine (α_2) and somatostatin inhibit it.

Mechanism of insulin action: tyrosine kinase receptor → autophosphorylation → **IRS-1/2 → PI3K → Akt (PKB)**: translocation of GLUT4 to the membrane (muscle, adipose tissue), inhibition of glycogen synthase kinase-3 (glycogen synthesis ↑), activation of phosphodiesterase 3B (cAMP ↓ — lipolysis ↓), mTOR (protein synthesis ↑), inactivation of FOXO1 (gluconeogenesis genes ↓); **Ras-MAPK** — growth and division.

Table 28.1. *Metabolic effects of the contrainsular hormones*

Hormone	Mechanism	Main metabolic effects
Glucagon (α-cells)	G_s → cAMP → PKA (liver, adipose tissue)	Glycogenolysis and gluconeogenesis in the liver ↑, glycolysis and lipogenesis ↓, ketogenesis ↑

Hormone	Mechanism	Main metabolic effects
Adrenaline	β: cAMP; α_1: IP_3/Ca^{2+}	Glycogenolysis (liver and muscle), lipolysis, gluconeogenesis ↑; insulin secretion ↓ (α_2); “fight or flight”
Cortisol	Nuclear receptor — enzyme synthesis	Gluconeogenic enzymes (PEPCK, G-6-phosphatase) ↑, muscle proteolysis ↑, peripheral glucose uptake ↓; central lipogenesis, peripheral lipolysis (“permissive” effect); suppresses inflammation and immunity
Somatotropin	JAK-STAT; IGF-1 in the liver	Protein synthesis ↑ (via IGF-1), lipolysis ↑, glucose uptake ↓ (diabetogenic), growth
Thyroid hormones	Nuclear receptors	Basal metabolism ↑, all processes are accelerated

Diabetes mellitus (topic 15): **type 1** — autoimmune destruction of β-cells (antibodies to GAD, insulin, IA-2), absolute insulin deficiency, tendency to ketoacidosis, low C-peptide; **type 2** — insulin resistance + relative deficiency, often with obesity, C-peptide normal or high; **gestational**, MODY (mutations in glucokinase and other genes) and other types. **Diagnostic criteria:** fasting glucose ≥ 7.0 mmol/l, ≥ 11.1 at 2 hours in the GTT, random ≥ 11.1 (with symptoms), HbA_{1c} ≥ 6.5%.

Acute complications: diabetic ketoacidosis (type 1), hyperosmolar hyperglycemic state (type 2; glucose > 33 mmol/l, osmolality > 320, no ketosis — the small amount of insulin stops lipolysis), lactic acidosis (metformin, hypoxia), **hypoglycemia** (overdose of insulin or a sulfonylurea; < 3.9 mmol/l — the most dangerous for the brain).

Age-related features: in newborns the glycogen store is small and the gluconeogenic enzymes are immature — a tendency to hypoglycemia (especially in preterm infants and infants of diabetic mothers — fetal hyperinsulinism); type 1 diabetes is more common in children; in the elderly insulin sensitivity declines and muscle mass decreases — type 2 diabetes becomes more common.

2. Thyroid hormones

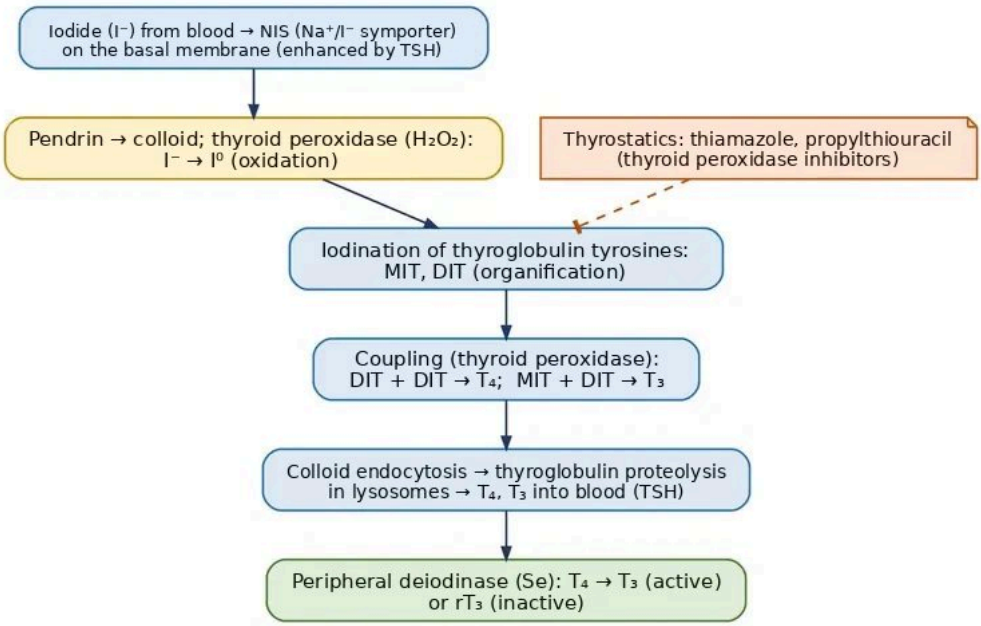


Figure 28.1. *Biosynthesis of thyroid hormones (drawn after Lippincott’s Illustrated Reviews: Biochemistry)*

Synthesis: follicular cells concentrate iodide from the blood 20–40 times through the **Na^+/I^- symporter (NIS)**; **thyroperoxidase** (heme-containing, requires H_2O_2) oxidizes iodide and iodinates the tyrosine residues within **thyroglobulin** (mono- and diiodotyrosine), then condenses them — T_4 and T_3 are stored in the colloid as part of thyroglobulin. Under the action of TSH the hormones are released into the blood by endocytosis of colloid and lysosomal proteolysis (mainly T_4, ≈ 80–90%). The daily iodine requirement is **150 µg** (250 µg in pregnancy and lactation).

Transport: > 99% is bound to proteins (thyroxine-binding globulin, transthyretin, albumin); the active forms are **free T_4** and **free T_3**. In the periphery **deiodinases** (selenoproteins) convert T_4 into the more active **T_3** (10 times higher affinity for the receptor) or into inactive reverse T_3.

Mechanism of action and effects: T_3 binds to nuclear receptors (TR, a heterodimer with the retinoid X receptor) and changes gene transcription:

- **basal metabolism and thermogenesis** ↑ — synthesis of Na^+,K^+-ATPase, mitochondrial biogenesis, uncoupling proteins;
- carbohydrates — absorption, glycogenolysis, gluconeogenesis ↑;
- lipids — lipolysis ↑, hepatic LDL receptors ↑ (cholesterol ↓);
- proteins — at physiological doses synthesis ↑ (growth), in excess catabolism ↑;
- **growth and development** — expression of the somatotropin gene, ossification, **brain development** (myelination, synaptogenesis — in the fetus and the first 2–3 years of life);
- heart — β-adrenoceptors and myosin heavy chain α ↑ (tachycardia).

Table 28.2. *Disorders of thyroid function*

Condition	Causes	Biochemical signs	Clinical signs
Hypothyroidism (primary)	Autoimmune thyroiditis (Hashimoto — antibodies to TPO), iodine deficiency, after surgery	TSH ↑, free T_4 ↓, cholesterol ↑, CK ↑	Lethargy, cold intolerance, weight gain, bradycardia, mucous edema (myxedema — accumulation of GAGs), constipation
Congenital hypothyroidism (cretinism)	Agenesis of the gland, defects of the synthetic enzymes, maternal iodine deficiency	TSH ↑ in the newborn (screening on days 3–5)	Mental and physical developmental delay — prevented by early treatment (levothyroxine)
Hyperthyroidism (thyrotoxicosis)	Graves (Basedow) disease — stimulating antibodies to the TSH receptor; toxic nodule	TSH ↓, free T_4 and T_3 ↑, antibodies to the TSH receptor	Weight loss, heat intolerance, tachycardia, tremor, exophthalmos, nervousness

Endemic goiter is enlargement of the thyroid gland in iodine-deficient regions (mountainous areas, including some regions of Central Asia). **Pathogenesis:** when iodine is lacking, T_4 synthesis decreases → TSH increases by feedback → the follicular cells undergo hypertrophy and hyperplasia (the gland enlarges); the gland adapts by producing more T_3 (which requires less iodine); if this continues for a long time, hypothyroidism develops. Iodine deficiency in pregnant women leads to impaired brain development in the fetus. Goitrogenic substances (thiocyanates and goitrin in cabbage and radish family plants) reduce iodine uptake. **Prevention** — **iodized table salt** (universal salt iodization), potassium iodide for risk groups. **Treatment** — potassium iodide, levothyroxine; surgery for nodules.

3. Regulation of calcium–phosphate metabolism

The body contains ≈ 1 kg of calcium (99% in bones as hydroxyapatite) and ≈ 0.6 kg of phosphorus (85% in bones). Total blood calcium is 2.15–2.55 mmol/l: ≈ 45% is **ionized** (active, 1.1–1.3 mmol/l), ≈ 40% is bound to albumin, and ≈ 15% forms complexes with anions. Alkalosis reduces ionized calcium (tetany).

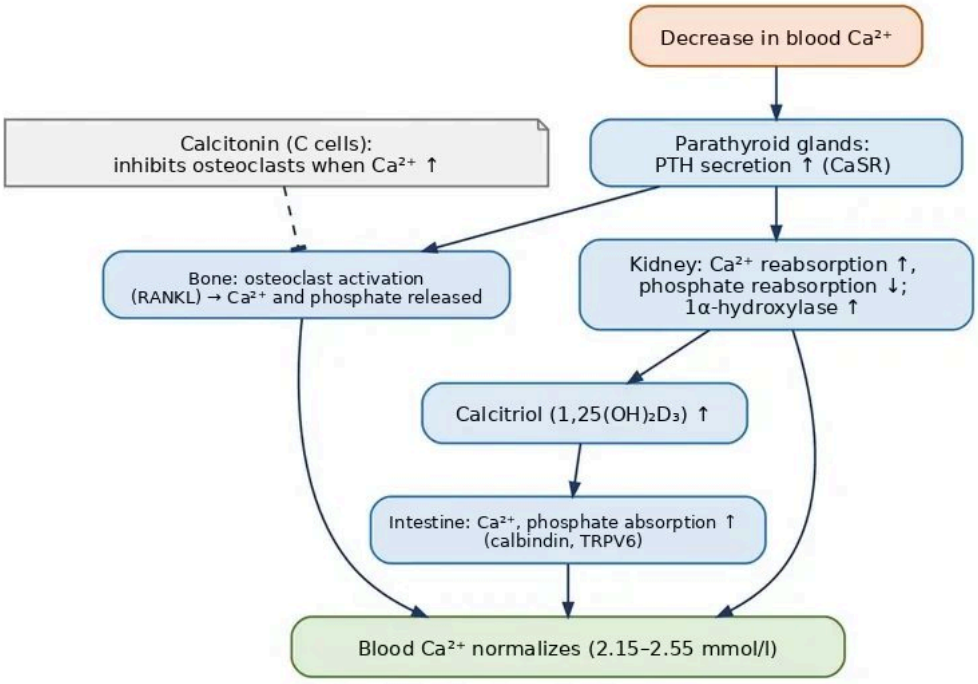


Figure 28.2. *Hormonal regulation of the blood calcium level (drawn after Severin E.S.)*

Table 28.3. *Hormones that regulate calcium–phosphate metabolism*

Hormone	Source	Blood Ca^{2+}	Blood phosphate	Main effects
Parathyroid hormone (PTH)	Parathyroid glands (when Ca^{2+} falls, via CaSR)	↑	↓	Bone resorption (via osteoblasts RANKL → osteoclasts), renal Ca^{2+} reabsorption ↑, phosphate excretion ↑ (phosphaturia), 1α-hydroxylase ↑
Calcitriol ($1,25(OH)_2D_3$)	7-Dehydrocholesterol in the skin → D_3 under UV light → 25-hydroxylation in the liver → 1α-hydroxylation in the kidney (enhanced by PTH)	↑	↑	Intestinal absorption of Ca^{2+} and phosphate (calbindin), bone mineralization and remodeling, decreases PTH secretion
Calcitonin	C cells of the thyroid gland (when Ca^{2+} rises)	↓	↓	Inhibits osteoclasts; of little physiological importance in humans
FGF-23	Osteocytes	—	↓	Phosphaturia, inhibits 1α-hydroxylase

Disorders: primary hyperparathyroidism (adenoma) — hypercalcemia, hypophosphatemia, kidney stones, osteoporosis, peptic ulcer (“bones, stones, abdominal groans, psychic moans”); **hypoparathyroidism** (after surgery) — hypocalcemia, hyperphosphatemia, tetany (Chvostek and

Trousseau signs); **chronic kidney disease** — low calcitriol, phosphate retention — secondary hyperparathyroidism, renal osteodystrophy.

Rickets in children is impaired mineralization of growing bone. **Causes:** vitamin D deficiency (little exposure to the sun, especially in those born in winter, dark skin, prolonged exclusive breastfeeding without prophylaxis, malnutrition, prematurity), malabsorption, liver and kidney diseases (impaired hydroxylation), hereditary forms (defects of 1α-hydroxylase or the vitamin D receptor — **vitamin D-dependent rickets**; **X-linked hypophosphatemic rickets** — FGF-23 excess, vitamin D-resistant). **Pathogenesis:** absorption of Ca^{2+} and phosphate decreases → hypocalcemia → secondary hyperparathyroidism → phosphaturia and hypophosphatemia, bone resorption; the osteoid is not mineralized. **Biochemical signs:** blood phosphate ↓, calcium N or ↓, **alkaline phosphatase** ↑↑ (osteoblast activity), PTH ↑, 25(OH)D ↓ (< 30 nmol/l). **Clinical signs:** sweating, restlessness, softening of the skull (craniotabes), the "rachitic rosary", bowing of the legs, delayed tooth eruption, muscle hypotonia. **Prevention and treatment** — vitamin D (prophylactic dose 400–500 IU/day), sunlight, calcium-rich food. The same condition in adults is **osteomalacia**.

4. Regulation of water–salt metabolism

Water makes up ≈ 60% of body weight in adults (≈ 75–80% in newborns); intracellular (≈ 2/3) and extracellular (interstitial and plasma) fluids. The main cation outside the cells is Na^+, and inside — K^+.

- **Vasopressin (antidiuretic hormone)** is synthesized in the hypothalamus and released from the posterior pituitary when plasma **osmolality rises** (osmoreceptors; even 1–2% is enough) and when blood volume falls sharply. In the renal collecting ducts: V_2 receptor → cAMP → **aquaporin-2** moves to the apical membrane → water is reabsorbed (concentrated urine). V_1 — vasoconstriction. **Diabetes insipidus** — vasopressin deficiency (central) or renal insensitivity to it (nephrogenic — V_2 or AQP2 mutations, lithium): 5–20 l of dilute urine per day (specific gravity < 1.005), thirst, hypernatremia. The syndrome of inappropriate vasopressin secretion — hyponatremia.
- **The renin–angiotensin–aldosterone system:** when blood pressure and Na^+ fall (and via sympathetic β_1), the juxtaglomerular cells of the kidney release **renin** → angiotensinogen (liver) → **angiotensin I** → (**angiotensin-converting enzyme**, mainly in the pulmonary endothelium) → **angiotensin II**: powerful vasoconstriction, aldosterone secretion, thirst and vasopressin, proximal Na^+ reabsorption. **Aldosterone** (a nuclear receptor) — synthesis of ENaC and Na^+,K^+-ATPase in the distal tubules and collecting ducts: Na^+ and water are retained, K^+ and H^+ are excreted. ACE inhibitors (enalapril), angiotensin receptor blockers (losartan), aldosterone antagonists (spironolactone) — in hypertension and heart failure.
- **Natriuretic peptides** (ANP, BNP) oppose the RAAS: natriuresis, vasodilation (topic 23). **Age-related features:** in newborns the concentrating ability of the kidney is low and the relative water content and turnover are high — very prone to dehydration (with diarrhea, vomiting); in the elderly thirst and concentrating ability decline.

5. Locally acting hormones

Eicosanoids (prostaglandins, thromboxanes, leukotrienes), histamine, serotonin, kinins, NO and gastrointestinal hormones (gastrin, secretin, cholecystokinin, GIP, GLP-1, somatostatin — topic 24) act at the site of formation or at a short distance and are rapidly inactivated. **Prostaglandin E_2** — fever (hypothalamus), pain, gastric protection, maintenance of renal blood flow; NSAIDs reduce it and can cause gastric ulcer and kidney damage.

6. Laboratory work

Experiment 1: determination of serum insulin by enzyme immunoassay (based on a video)

Principle of the method ("sandwich" ELISA): the wells of the plate are coated with monoclonal antibodies against one epitope of insulin. Insulin in the sample binds to them; a second, **peroxidase-labeled** antibody against another epitope is added. After washing, the substrate (tetramethylbenzidine + H_2O_2) gives a blue color, which turns yellow when the stop acid is added (450 nm). The color intensity is **directly proportional** to the amount of insulin.

1. Pipette 25 µl of the calibrators (0–300 µIU/ml), the control and the patient's serum (taken fasting) into the wells.
2. Add 100 µl of conjugate (peroxidase–anti-insulin) and incubate with shaking for 60 minutes at room temperature.
3. Wash the wells 5 times with wash buffer.
4. Add 100 µl of substrate and keep in the dark for 15 minutes; add 50 µl of stop solution.
5. Measure the absorbance at 450 nm; find the concentration from the calibration curve. **Interpretation:** fasting insulin 2–25 µIU/ml (≈ 14–174 pmol/l). **HOMA-IR** = [glucose (mmol/l) × insulin (µIU/ml)] / 22.5; > 2.5–2.7 — insulin resistance. Insulin ↓ (C-peptide ↓) — type 1 diabetes; ↑ — insulin resistance (type 2 diabetes, metabolic syndrome, polycystic ovary syndrome), insulinoma (together with hypoglycemia). In patients treated with insulin preparations, endogenous secretion must be assessed by **C-peptide**.

Experiment 2: interpretation of thyroid hormone and TSH results (case-based)

1. Assess TSH first (the most sensitive screening test; normal 0.4–4.0 mIU/l).
2. Assess free T_4 (≈ 10–22 pmol/l) and, if necessary, free T_3.
3. Take antibodies (to TPO, to the TSH receptor) into account. **Results and analysis:** TSH ↑ + free T_4 ↓ — primary hypothyroidism; TSH ↑ + free T_4 N — subclinical hypothyroidism; TSH ↓ + free T_4 ↑ — thyrotoxicosis; TSH ↓ + free T_4 ↓ — secondary hypothyroidism (pituitary).

7. Practical skills

Skill: assessing the type of diabetes and insulin secretion

1. Confirm the diagnosis of diabetes by glucose and HbA_{1c}.
2. Assess C-peptide and insulin: low — type 1 (insulin deficiency), high or normal — type 2.
3. Check for antibodies against β-cells (GAD).
4. Calculate HOMA-IR.
5. Assess ketosis and acid–base status (risk of decompensation).

Skill: identifying a disorder of calcium–phosphate metabolism

1. Assess total and ionized calcium, phosphate and albumin (albumin-corrected calcium).
2. Determine PTH, 25(OH)D, alkaline phosphatase and creatinine.
3. Ca ↑ + PTH ↑ — primary hyperparathyroidism; Ca ↓ + PTH ↓ — hypoparathyroidism; Ca ↓/N + phosphate ↓ + PTH ↑ + 25(OH)D ↓ + ALP ↑ — vitamin D deficiency (rickets, osteomalacia); phosphate ↑ + creatinine ↑ + PTH ↑ — kidney failure.

Review questions

1. What is the mechanism of insulin secretion?
2. Through which intracellular signaling pathways does insulin act?
3. How does insulin affect carbohydrate, lipid and protein metabolism?
4. Compare the metabolic effects of glucagon, epinephrine and cortisol.
5. Name the types of diabetes mellitus and the diagnostic criteria.
6. What are the age-related features of hormonal regulation?
7. How are thyroid hormones synthesized?
8. List the biological effects of thyroid hormones.
9. Name the biochemical signs of hypothyroidism and hyperthyroidism.
10. How does endemic goiter develop and how is it prevented?
11. Which hormones regulate calcium metabolism?
12. How is the active form of vitamin D formed?
13. Name the causes and biochemical signs of rickets.
14. What is the role of vasopressin and the renin–angiotensin–aldosterone system?
15. Explain the principle of determining insulin by ELISA.

Test questions

1. Which event directly triggers insulin secretion from the β-cell?
   A) A decrease in cAMP
   B) Opening of K^+ channels
   C) Hyperpolarization
   D) Closure of ATP-sensitive K^+ channels and entry of Ca^{2+}
   E) The action of glucagon

2. Which is an indicator of endogenous insulin secretion?
   A) Proinsulin
   B) C-peptide
   C) HbA_{1c}
   D) Fructosamine
   E) Glucagon

3. Through which transporter does insulin enhance glucose entry into muscle cells?
   A) GLUT1
   B) GLUT2
   C) GLUT5
   D) SGLT1
   E) GLUT4

4. Which hormone induces the synthesis of gluconeogenic enzymes?
   A) Cortisol
   B) Insulin
   C) Calcitonin
   D) Oxytocin
   E) Prolactin

5. Which enzyme oxidizes iodine and iodinates tyrosines in thyroid hormone synthesis?
   A) Deiodinase
   B) Tyrosine hydroxylase
   C) Aromatase
   D) Thyroperoxidase
   E) Monoamine oxidase

6. Which trace element is contained in the enzyme that converts T_4 into active T_3?
   A) Zinc
   B) Copper
   C) Iron
   D) Manganese
   E) Selenium

7. What is the cause of the increase in TSH in endemic goiter?
   A) Antibodies to the TSH receptor
   B) A pituitary tumor
   C) Excess thyroglobulin
   D) Feedback due to decreased T_4
   E) Calcitonin deficiency

8. What change is seen in Graves (Basedow) disease?
   A) TSH ↑, T_4 ↓
   B) TSH ↓, free T_4 ↑
   C) TSH ↑, T_4 ↑
   D) TSH ↓, T_4 ↓
   E) TSH N, T_4 N

9. How does parathyroid hormone act on the kidney?
   A) It decreases the reabsorption of Ca^{2+} and phosphate
   B) It inhibits 1α-hydroxylase
   C) It brings aquaporin-2 to the membrane
   D) It enhances Na^{+} reabsorption
   E) It increases Ca^{2+} reabsorption and enhances phosphate excretion

10. Where does the 1α-hydroxylation of vitamin D take place?
    A) In the kidney
    B) In the liver
    C) In the skin
    D) In the intestine
    E) In bone

11. The activity of which enzyme increases sharply in rickets?
    A) Alkaline phosphatase
    B) Acid phosphatase
    C) ALT
    D) Amylase
    E) CK

12. How does vasopressin act in the collecting ducts?
    A) It closes Na^+ channels
    B) It enhances K^+ secretion
    C) It brings aquaporin-2 to the apical membrane
    D) It releases renin
    E) It increases calcium reabsorption

13. Which enzyme converts angiotensin I into angiotensin II?
    A) Renin
    B) Kallikrein
    C) Angiotensin-converting enzyme
    D) Neprilysin
    E) Aldosterone synthase

14. In the determination of insulin by “sandwich” ELISA, what is the color intensity?
    A) Inversely proportional to the amount of insulin
    B) Directly proportional to the amount of insulin
    C) Independent of the amount of insulin
    D) Dependent only on the amount of glucose
    E) Dependent only on the amount of C-peptide

15. What does the HOMA-IR index assess?
    A) The number of β-cells
    B) The degree of glycation
    C) Insulin resistance
    D) The degree of ketosis
    E) Renal filtration

Clinical cases

Case 1

A 35-year-old woman living in a mountainous region was found to have an enlarged thyroid gland in the neck; weakness, cold intolerance and dry skin. TSH 9.8 mIU/l, free T_4 9 pmol/l, antibodies to TPO negative, urinary iodine 35 µg/l (normal > 100). She does not use iodized salt.

Questions: What is the diagnosis? Explain the mechanism of development of goiter and hypothyroidism. What are the measures for prevention and treatment?

Answer: Endemic (iodine deficiency) goiter with hypothyroidism. When iodine is lacking, thyroperoxidase cannot iodinate thyroglobulin sufficiently — T_4 synthesis decreases; by negative feedback the pituitary secretes a lot of TSH; TSH stimulates the growth and division of the follicular cells — the gland enlarges (goiter). As an adaptation the gland synthesizes more T_3, but if the iodine deficiency persists, hypothyroidism develops: basal metabolism and thermogenesis decrease (cold intolerance, weakness), and GAGs accumulate in the skin. The negative antibodies exclude autoimmune thyroiditis. Prevention — iodized table salt; treatment — potassium iodide, levothyroxine if necessary; iodine intake is especially important when planning a pregnancy.

Case 2

A 10-month-old child (born in winter, exclusively breastfed, no vitamin D prophylaxis) sweats, is restless, has a bald patch on the back of the head, a “rosary” on the ribs and bowed legs, and has no

teeth yet. Blood calcium 2.0 mmol/l, phosphate 0.8 mmol/l (normal 1.3–2.1), alkaline phosphatase 1100 U/l, PTH increased, 25(OH)D 15 nmol/l.

Questions: What is the diagnosis? Explain the mechanism of the biochemical changes. What are the treatment and prevention?

Answer: Vitamin D-deficiency rickets. Because of little sunlight (winter) and insufficient vitamin D in the diet (it is low in breast milk), not enough D_3 was obtained from the skin and food — 25(OH)D and calcitriol are low. Intestinal absorption of Ca^{2+} and phosphate decreases — hypocalcemia stimulates the parathyroid glands (secondary hyperparathyroidism): PTH mobilizes calcium from bone (calcium is kept close to normal) and enhances renal phosphate excretion — hypophosphatemia. Because the calcium × phosphate product is low, the osteoid is not mineralized; the osteoblasts are activated compensatorily and release a lot of alkaline phosphatase; the bones soften and become deformed. Treatment — therapeutic doses of vitamin D, calcium, sunbathing; prevention — vitamin D for all infants in the first year of life (400–500 IU/day).

Case 3

After a head injury a patient is passing 9 l of colorless urine per day and has intense thirst. Urine specific gravity 1.002, osmolality 90 mOsm/kg; blood Na^+ 152 mmol/l, osmolality 305 mOsm/kg, glucose 5.1 mmol/l. After desmopressin was given, the urine volume fell sharply and the specific gravity rose to 1.018.

Questions: What is the diagnosis? Why is the urine dilute? What does the desmopressin test show?

Answer: Central (hypothalamic–pituitary) diabetes insipidus — the synthesis or secretion of vasopressin is impaired because of the injury. Without vasopressin the V_2 receptor → cAMP → aquaporin-2 pathway in the collecting ducts does not work and water is not reabsorbed — a large amount of low-osmolality urine is excreted; the water loss raises plasma osmolality and Na^+ and stimulates the thirst center. Glucose is normal (not diabetes mellitus). The response to the synthetic vasopressin analog desmopressin (concentration of the urine) shows that the renal receptors are preserved and the defect is central; in nephrogenic diabetes insipidus there is no response. Treatment — desmopressin.

Quick review

Table 28.4. *Key concepts of topic 28*

Concept	Brief definition
Insulin secretion	Glucose → ATP ↑ → KATP closes → Ca^{2+} → exocytosis; C-peptide
Insulin signaling	IRS → PI3K/Akt (GLUT4, glycogen, lipolysis ↓); Ras-MAPK
Contrainsular hormones	Glucagon, epinephrine, cortisol, somatotropin, thyroid hormones
Diagnosis of diabetes	Fasting ≥ 7.0; GTT at 2 hours ≥ 11.1; HbA_{1c} ≥ 6.5%
Thyroid hormones	NIS, thyroperoxidase, thyroglobulin; T_4 → T_3 (deiodinase, Se)
Effects	Basal metabolism ↑, growth, brain development
Endemic goiter	Iodine ↓ → T_4 ↓ → TSH ↑ → hyperplasia; iodized salt
Calcium regulation	PTH (Ca ↑, P ↓), calcitriol (Ca ↑, P ↑), calcitonin (Ca ↓)
Rickets	Vit. D ↓ → phosphate ↓, ALP ↑↑, PTH ↑; prophylaxis 400–500 IU/day
Water–salt balance	Vasopressin (AQP2), RAAS (angiotensin II, aldosterone), ANP

Concept	Brief definition
Insulin ELISA	“Sandwich”, peroxidase, 450 nm; HOMA-IR > 2.7 — resistance

TOPIC 29

Biochemistry of the kidney. Normal and pathological components of urine

Learning objectives

To give the student knowledge of the mechanism of urine formation (filtration, reabsorption, secretion), the role of the kidneys in maintaining acid–base balance, the specific features of metabolism and energy supply in the kidney, the endocrine functions of the kidney, the general properties of urine and their age-related features, the normal and pathological chemical composition of urine, and the formation and prevention of kidney stones.

By the end of the class the student should know:

- the mechanisms of glomerular filtration and tubular reabsorption and secretion;
- the pathways by which the kidney excretes H^+ and regenerates bicarbonate (ammoniagenesis);
- the features of metabolism and energy supply in renal tissue;
- the endocrine functions of the kidney (renin, erythropoietin, calcitriol);
- the physical properties and normal chemical composition of urine;
- pathological components (protein, glucose, ketone bodies, bilirubin, blood) and their diagnostic importance;
- the types of kidney stones and the biochemical basis of their prevention.

The student should be able to:

- detect normal and pathological components of urine with test strips;
- calculate the GFR from creatinine;
- interpret the results of urinalysis.

1. Mechanism of urine formation

≈ 1.2 l of blood passes through the kidneys per minute (≈ 20–25% of the cardiac output). Each kidney has ≈ 1 million **nephrons**.

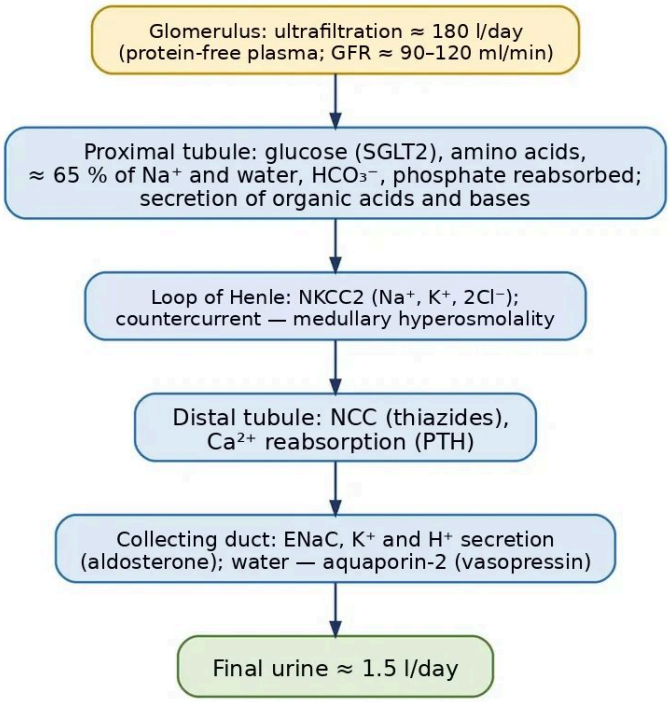


Figure 29.1. *The main functions of the parts of the nephron (drawn after Lippincott's Illustrated Reviews: Biochemistry)*

1. Glomerular filtration — under hydrostatic pressure, plasma passes through the glomerular filter (fenestrated endothelium, basement membrane — type IV collagen, negatively charged heparan sulfate, podocytes). Substances with a molecular mass < 60–70 kDa that are not negatively charged pass through the filter; albumin (66 kDa, negative charge) hardly passes. **Primary urine** is ≈ 180 l per day — its composition is close to that of protein-free plasma. The **glomerular filtration rate (GFR)** is 90–120 ml/min/1.73 m^2; it is the main indicator of kidney function.

2. Tubular reabsorption — ≈ 99% of the filtrate is reabsorbed:

- **proximal tubule** — glucose (**SGLT2**, SGLT1; completely), amino acids, ≈ 65% of Na^+ and water, HCO_3^- (≈ 80%), K^+, phosphate, urate, small proteins (endocytosis — megalin, cubilin), vitamins; the energy comes from Na^+,K^+-ATPase;
- **loop of Henle** — in the descending limb water, in the ascending limb Na^+, K^+ and $2Cl^-$ via **NKCC2** (the target of furosemide); the countercurrent mechanism creates an osmotic gradient in the renal medulla (urea also contributes);
- **distal tubule** — NaCl (**NCC**, the target of thiazides), Ca^{2+} (PTH);
- **collecting ducts** — Na^+ (ENaC, aldosterone), water (**aquaporin-2**, vasopressin). **Renal threshold:** when the maximum capacity of the transporters is exceeded, the substance passes into the urine — for glucose ≈ 8.9–10 mmol/l in the blood. **SGLT2 inhibitors** (dapagliflozin, empagliflozin) deliberately increase glucosuria, treating type 2 diabetes and protecting the heart and kidneys.

3. Secretion — active transport from the blood into the tubular lumen: H^+, K^+, NH_3, organic anions (urate, para-aminohippurate, penicillin, diuretics — OAT transporters) and cations (creatinine partly, metformin — OCT2).

Clearance is the volume of plasma completely cleared of a substance in 1 minute: C = (U × V) / P (U — concentration in urine, P — concentration in plasma, V — urine flow per minute). Inulin clearance

= GFR; para-aminohippurate clearance — renal plasma flow. In practice GFR is estimated from **creatinine** (the CKD-EPI formula; age and sex are taken into account) or from **cystatin C**.

2. The role of the kidneys in acid–base balance

≈ 50–100 mmol of nonvolatile acids (sulfuric acid from sulfur-containing amino acids, phosphoric acid, organic acids) are formed in the body per day; only the kidneys can excrete them (the lungs excrete only CO_2).

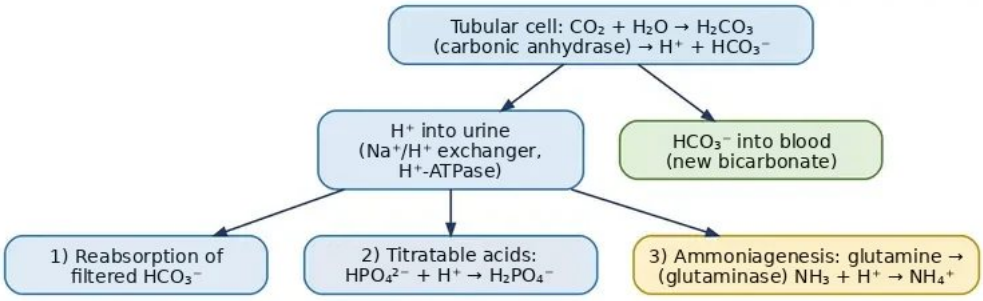


Figure 29.2. *H^+ excretion and bicarbonate regeneration in the kidney (drawn after Severin E.S.)*

In the tubular cells **carbonic anhydrase** forms H_2CO_3; H^+ is secreted into the urine (Na^+/H^+ exchanger NHE3 — proximal; H^+-ATPase and H^+,K^+-ATPase — intercalated cells of the collecting ducts), and HCO_3^- passes into the blood. In the urine H^+ is bound in three ways:

1. **Reabsorption of filtered bicarbonate** — $H^+ + HCO_3^- \rightarrow H_2CO_3 \rightarrow CO_2 + H_2O$ (brush-border carbonic anhydrase); CO_2 enters the cell.
2. **Titratable acids** — H^+ is bound mainly by the phosphate buffer: $HPO_4^{2-} + H^+ \rightarrow H_2PO_4^-$ (≈ 10–30 mmol per day).
3. **Ammoniagenesis** (the main adaptive mechanism) — in the proximal tubules **glutamine → glutaminase** → glutamate + NH_3 → (glutamate dehydrogenase) α-ketoglutarate + NH_3; NH_3 binds H^+ and is excreted as **NH_4^+** (≈ 30–50 mmol per day, up to 300 mmol in acidosis). α-Ketoglutarate is used in gluconeogenesis, and **new HCO_3^-** is formed. Urine pH is normally 5.0–7.0 (on average ≈ 6.0); acidic with a meat diet, alkaline with a plant and dairy diet.

Renal tubular acidosis: type I (distal) — H^+ secretion is impaired, urine pH > 5.5, nephrocalcinosis and stones; type II (proximal) — HCO_3^- reabsorption is impaired (together with Fanconi syndrome: glucosuria, aminoaciduria, phosphaturia); type IV — aldosterone deficiency or resistance, hyperkalemia.

3. Specific features of metabolism and energy supply in the kidney

- The kidneys make up ≈ 0.5% of body weight but consume ≈ 10% of the body's oxygen; ≈ 70–80% of the energy is spent on the work of **Na^+,K^+-ATPase** (which drives secondary active transport).
- **Cortex** (proximal tubules) — aerobic; energy sources: **fatty acids**, ketone bodies, lactate, glutamine, citrate; many mitochondria; little glycolysis.
- **Medulla** — little oxygen, **anaerobic glycolysis** predominates (glucose is the main source) — very sensitive to ischemia (papillary necrosis).
- The kidney is an organ of **gluconeogenesis** (second after the liver; in prolonged fasting and acidosis up to 40% of blood glucose); substrates — glutamine, lactate, glycerol.

- Amino acid metabolism: **the first step of creatine synthesis** (guanidinoacetate), synthesis of **arginine** from citrulline, metabolism of glycine and serine.
- Small proteins and peptide hormones (insulin, PTH, β_2-microglobulin) are reabsorbed in the tubules and degraded in lysosomes — in kidney failure the insulin requirement decreases. **Endocrine functions of the kidney: renin** (RAAS, topic 28), **erythropoietin** (in hypoxia — HIF; anemia in kidney failure), **calcitriol** (1α-hydroxylase), prostaglandins (vasodilation), kinins (kallikrein), urodilatin.

The role of the kidneys in calcium, phosphorus and bicarbonate metabolism: ≈ 98% of Ca^{2+} is reabsorbed (in the distal part it is regulated by PTH and calcitriol); phosphate — in the proximal tubule via NaPi-IIa (PTH and FGF-23 reduce it); bicarbonate — reabsorption and formation of new bicarbonate. **In chronic kidney disease:** phosphate is retained and calcitriol decreases → hypocalcemia → secondary hyperparathyroidism → renal osteodystrophy; metabolic acidosis (excretion of NH_4^+ decreases); anemia (low erythropoietin); hyperkalemia; uremia — urea, creatinine and "uremic toxins" (indoxyl sulfate, p-cresyl sulfate, guanidines) accumulate.

4. General properties of urine and their age-related features

Table 29.1. *Physical properties of urine (in adults)*

Index	Normal value	Changes
Daily volume (diuresis)	1.0–2.0 l (≈ 75% of the fluid drunk)	Polyuria > 2 l (diabetes mellitus and insipidus); oliguria < 0.5 l; anuria < 0.1 l
Color	Straw-yellow (urochrome, urobilin)	Dark brown — bilirubin; red — blood, hemoglobin, myoglobin, beetroot; colorless — polyuria
Clarity	Clear	Turbid — salts, leukocytes, bacteria
Specific gravity	1,010–1,025	High — glucosuria, proteinuria, dehydration; low (< 1.005) — diabetes insipidus; fixed at 1.010 (isosthenuria) — kidney failure
Osmolality	50–1200 mOsm/kg	—
pH	5,0–7,0	Alkaline — plant diet, urinary tract infection (urease), type I tubular acidosis
Odor	Characteristic, faint	Acetone (ketosis), ammonia (infection), "mousy" (phenylketonuria), "maple syrup" (leucinosis)

Age-related features: in newborns there is oliguria in the first days, a low specific gravity (1.002–1.006 — the concentrating ability is immature), physiological proteinuria and the "uric acid infarct" (topic 14); the daily diuresis is approximately: up to 1 year 300–600 ml, ≈ 900 ml at 5 years, ≈ 1200 ml at 10 years; creatinine excretion increases with muscle mass. In the elderly GFR decreases by ≈ 1 ml/min per year, the concentrating ability declines, and nocturia appears.

5. Chemical composition of urine

≈ 50–70 g of dry matter is excreted in the urine per day.

Table 29.2. *The main organic and inorganic components of urine (daily)*

Component	Amount	Comment
Urea	330–580 mmol (20–35 g)	≈ 85–90% of nitrogen; depends on protein intake and catabolism

Component	Amount	Comment
Creatinine	Men 8.8–17.7, women 7.1–15.9 mmol	Proportional to muscle mass, constant
Uric acid	1.5–4.5 mmol	Purine metabolism
Ammonia (NH_4^+)	30–50 mmol	Depends on acid–base status
Amino acids	0.4–1.0 g	Rich in glycine and histidine
Hippuric acid, indican	Small amounts	Detoxification products (topic 26)
Urobilin(ogen)	1–4 mg	Bilirubin metabolism
17-Ketosteroids, catecholamine metabolites	In mg amounts	Hormone metabolism
Na^+	100–260 mmol	Depends on dietary salt
K^+	35–90 mmol	—
Cl^-	110–250 mmol	—
Ca^{2+}	2.5–7.5 mmol	—
Phosphates	13–42 mmol	—
Sulfates	≈ 20 mmol	From sulfur-containing amino acids

6. Pathological components of urine

Proteinuria — urine protein > 150 mg/day (in a healthy person albumin < 30 mg/day).

- **Glomerular** — damage to the filter (glomerulonephritis, **diabetic nephropathy**, amyloidosis): selective (albumin) or nonselective (large proteins as well); **nephrotic syndrome** — > 3.5 g/day, hypoalbuminemia, edema, hyperlipidemia (topic 20).
- **Tubular** — small proteins (β_2-microglobulin, α_1-microglobulin, retinol-binding protein) are not reabsorbed: pyelonephritis, heavy metals, nephrotoxic drugs.
- **Overflow (prerenal)** — many small proteins in the blood: **Bence Jones protein** (myeloma), myoglobin (rhabdomyolysis), hemoglobin (intravascular hemolysis).
- **Postrenal** — inflammation of the urinary tract, blood.
- **Functional** — physical exertion, fever, orthostatic (in adolescents).

CLINICAL SIGNIFICANCE

Microalbuminuria (albumin 30–300 mg/day or an albumin/creatinine ratio of 3–30 mg/mmol) is the earliest sign of **diabetic nephropathy** and an indicator of cardiovascular risk; it is checked every year in diabetes mellitus and hypertension. Ordinary test strips do not detect it.

Glucosuria — when blood glucose exceeds the renal threshold (diabetes mellitus, steroids, stress) or when the threshold is lowered (**renal glucosuria** — SGLT2 gene mutation, pregnancy, Fanconi syndrome, SGLT2 inhibitors).

Ketonuria — acetoacetate and acetone: decompensated diabetes mellitus, starvation, repeated vomiting (hyperemesis of pregnancy), ketogenic diet, acetonemic vomiting in children, alcoholic ketoacidosis.

Bilirubinuria (conjugated only) — hepatocellular and obstructive jaundice; **urobilinogenuria** — hemolytic and hepatocellular jaundice (topic 25).

Hematuria (erythrocytes — glomerulonephritis, stones, tumors, infection), **hemoglobinuria** (intravascular hemolysis: G6PD deficiency, incompatible blood transfusion, burns), **myoglobinuria** (rhabdomyolysis) — the “blood” zone of the test strip is positive in all three (peroxidase activity of heme); microscopy is needed to distinguish them.

Leukocyturia and nitrites — urinary tract infection: gram-negative bacteria (E. coli) reduce nitrates to nitrites.

Other pathological components: phenylpyruvate (phenylketonuria — green color with $FeCl_3$), homogentisic acid (alkaptonuria — the urine darkens in air), porphyrins and porphobilinogen (porphyrias), cystine, aminoaciduria (Fanconi, Hartnup disease), oxalates, orotic acid (topic 14), GAGs (mucopolysaccharidoses), 5-hydroxyindoleacetic acid (carcinoid tumor), vanillylmandelic acid (pheochromocytoma), hCG (pregnancy).

7. Kidney stones and the biochemical basis of their prevention

Stone formation results from **supersaturation** of the urine with salts, a lack of crystallization inhibitors (citrate, Mg^{2+}, Tamm–Horsfall protein, nephrocalcin, GAGs) and impaired urine flow.

Table 29.3. *Types of kidney stones*

Stone	Proportion	Causes and risk factors	Prevention
Calcium oxalate	≈ 70–80 %	Hypercalciuria, hyperoxaluria (spinach, rhubarb, chocolate, tea; increased oxalate absorption in intestinal diseases; hereditary — glyoxylate metabolism), hypocitraturia, low water intake	Plenty of fluids (> 2–2.5 l of urine), oxalate restriction, normal calcium intake (binds oxalate in the intestine), salt restriction, potassium citrate, thiazides (in hypercalciuria)
Calcium phosphate	≈ 10 %	Alkaline urine, hyperparathyroidism, type I tubular acidosis	Elimination of the cause
Urate (uric acid)	≈ 5–10 %	Hyperuricosuria (gout, chemotherapy of tumors), persistently acidic urine (pH < 5.5), metabolic syndrome	Alkalinization of the urine (potassium citrate, pH 6.5–7), allopurinol, purine restriction — urate stones dissolve
Struvite (magnesium ammonium phosphate)	≈ 5–10 %	Infection with urease-producing bacteria (Proteus): urea → NH_3 → alkaline urine; “staghorn” stones	Treatment of the infection, removal of the stone
Cystine	≈ 1–2 %	Cystinuria — hereditary; reabsorption of the dibasic amino acids (cystine, ornithine, arginine, lysine — “COAL”) is impaired; cystine is poorly soluble (hexagonal crystals)	Plenty of fluids, alkalinization, thiol drugs (tiopronin)

8. Laboratory work

Experiment 1: detection of the normal and pathological components of urine with test strips

Principle of the method: on multiparameter test strips each zone works through its own specific chemical reaction.

Table 29.4. *Reactions of the test strip zones*

Zone	Basis of the reaction	Comment
Glucose	Glucose oxidase → H_2O_2 → peroxidase + chromogen	Specific; ascorbate gives a false negative result
Protein	"Protein error" of the indicator (tetrabromophenol blue): the color changes when protein binds	Detects mainly albumin; does not detect Bence Jones protein or microalbuminuria
Ketone bodies	Sodium nitroprusside (Legal) — acetoacetate, acetone	Does not detect β-hydroxybutyrate
Bilirubin	Diazo reaction	Conjugated only
Urobilinogen	Ehrlich's reagent or diazo reaction	—
Blood	Peroxidase activity of heme (H_2O_2 + tetramethylbenzidine)	Erythrocytes, Hb, myoglobin
Nitrites	Griess reaction	Bacteriuria
Leukocytes	Leukocyte esterase	Pyuria
pH	Mixture of indicators	5–9
Specific gravity	Indicator dependent on ion concentration	Does not take glucose and protein into account

1. Take fresh, well-mixed morning urine (test it within 2 hours).
2. Dip the test strip into the urine for 1–2 seconds until all the zones are immersed, and remove the excess by touching the edge of the container.
3. Holding the strip horizontally, compare each zone with the color scale on the container at the indicated time (30–120 seconds) (or read it on a urine analyzer).
4. Record the results in a table and compare them with the normal values. **Results and analysis:** normally glucose, protein (traces possible), ketone bodies, bilirubin, blood, nitrites and leukocytes are negative; urobilinogen — traces; pH 5–7; specific gravity 1.010–1.025. The causes of positive results are given in section 6; pathological results are confirmed by laboratory methods and microscopy.

Experiment 2: estimation of GFR from creatinine (calculation)

1. Determine creatinine in blood (the Jaffe reaction — an orange color with picric acid in an alkaline medium, or an enzymatic method) and creatinine and diuresis in daily urine.
2. Calculate the clearance: C = (U × V) / P, V — ml/min.
3. The Cockcroft–Gault formula: CrCl (ml/min) = [(140 − age) × weight (kg)] / [72 × creatinine (mg/dl)]; × 0.85 for women. (If creatinine is in μmol/l: mg/dl = μmol/l / 88.4.) **Results and analysis:** GFR ≥ 90 — normal; 60–89 — mildly decreased; < 60 ml/min/1.73 m^2 for more than 3 months — chronic kidney disease; < 15 — kidney failure (dialysis). Drug doses must be adjusted to the GFR.

9. Practical skills

Skill: interpreting a urinalysis

1. Assess the physical properties: volume, color, clarity, specific gravity, pH.
2. Assess the chemical components: protein, glucose, ketones, bilirubin, urobilinogen, blood, nitrites, leukocytes.

3. Group the results into syndromes: diabetes (glucose + ketones + high specific gravity), nephrotic (a lot of protein), nephritic (protein + blood), infection (leukocytes + nitrites + alkaline pH), jaundice (bilirubin, urobilinogen).
4. Compare with the blood tests (glucose, creatinine, bilirubin).

Skill: determining the type of kidney stone and its prevention

1. Assess the urine pH and the crystals in the sediment.
2. Determine calcium, oxalate, urate, citrate and cystine in daily urine.
3. Assess blood calcium, PTH and uric acid.
4. Recommend a diet and drug prophylaxis appropriate to the type of stone.

Review questions

1. How does glomerular filtration take place? What is the structure of the filter?
2. What is GFR and how is it estimated?
3. Which substances are reabsorbed in the proximal tubule?
4. What is the renal threshold for glucose? How do SGLT2 inhibitors act?
5. Which substances are secreted in the tubules?
6. By what pathways do the kidneys excrete H^+?
7. What is the importance of ammoniagenesis?
8. What is the energy supply of renal tissue?
9. List the endocrine functions of the kidney.
10. What biochemical changes are seen in chronic kidney disease?
11. Name the physical properties of urine and their age-related features.
12. List the main organic components of urine.
13. Explain the types of proteinuria.
14. What is the diagnostic importance of glucosuria, ketonuria and bilirubinuria?
15. Name the types of kidney stones and the biochemical basis of their prevention.

Test questions

1. What is the daily volume of primary urine in a healthy person?
   A) 1.5 l
   B) 15 l
   C) ≈ 180 l
   D) 500 l
   E) 50 l

2. What is the main transporter that reabsorbs glucose in the proximal tubule?
   A) SGLT2
   B) GLUT4
   C) NKCC2
   D) ENaC
   E) Aquaporin-2

3. Which transporter does furosemide inhibit?
   A) NKCC2 (loop of Henle)
   B) NCC
   C) ENaC
   D) SGLT2
   E) NHE3

4. What is the main adaptive mechanism of renal H^+ excretion in acidosis?
   A) Titratable acids
   B) Ammoniagenesis (glutaminase)
   C) Glucose secretion
   D) Urea synthesis
   E) Creatinine secretion

5. What is the main energy source in the renal medulla?
   A) β-Oxidation of fatty acids
   B) Ketone bodies
   C) Glutamine
   D) Anaerobic glycolysis
   E) Oxidation of lactate

6. What is the approximate renal threshold for glucose?
   A) 3.3–5.5 mmol/l
   B) 5.5–6.1 mmol/l
   C) 8.9–10 mmol/l
   D) 15–20 mmol/l
   E) 25 mmol/l

7. Which hormone is not produced by the kidney?
   A) Renin
   B) Erythropoietin
   C) Calcitriol
   D) Prostaglandins
   E) Aldosterone

8. What is the earliest laboratory sign of diabetic nephropathy?
   A) Microalbuminuria
   B) Glucosuria
   C) Ketonuria
   D) Hematuria
   E) Isosthenuria

9. To which type of proteinuria does Bence Jones protein belong?
   A) Glomerular
   B) Tubular
   C) Overflow (prerenal)
   D) Postrenal
   E) Functional

10. Which substance is not detected by the ketone zone of the test strip?
    A) Acetoacetate
    B) Acetone
    C) It detects both
    D) β-Hydroxybutyrate
    E) None of them

11. What does a positive nitrite result in the urine indicate?
    A) Ketosis
    B) Proteinuria
    C) Hemolysis
    D) Jaundice
    E) Bacterial infection

12. On which reaction is the “blood” zone of the test strip based?
    A) Diazo reaction
    B) Peroxidase activity of heme
    C) Legal's reaction
    D) Griess reaction
    E) The biuret reaction

13. Which is the most common kidney stone?
    A) Urate
    B) Cystine
    C) Struvite
    D) Xanthine
    E) Calcium oxalate

14. To what state must the urine be brought to dissolve urate stones?
    A) Acidification (pH < 5)
    B) Alkalinization (pH 6.5–7)
    C) Concentration
    D) Increasing oxalate
    E) Increasing calcium

15. The reabsorption of which amino acids is impaired in cystinuria?
    A) Phenylalanine, tyrosine
    B) Glycine, proline
    C) Valine, leucine, isoleucine
    D) Cystine, ornithine, arginine, lysine
    E) Glutamate, aspartate

Clinical cases

Case 1

A 58-year-old patient has had type 2 diabetes mellitus for 15 years. Edema of the face and legs. Blood glucose 11.2 mmol/l, HbA_{1c} 9.1%, creatinine 210 µmol/l, albumin 28 g/l, potassium 5.6 mmol/l, Hb 98 g/l. Urine protein 4.2 g/day, glucose positive.

Questions: What complication has developed? What is the biochemical mechanism of the kidney damage? Explain the cause of the anemia and hyperkalemia.

Answer: Diabetic nephropathy with nephrotic-range proteinuria and chronic kidney disease. Prolonged hyperglycemia nonenzymatically glycates the proteins of the glomerular basement membrane (advanced glycation end products), the mesangial matrix increases via TGF-β, and negatively charged heparan sulfate decreases; hyperfiltration and hypertension damage the glomeruli — first microalbuminuria, then heavy proteinuria (hypoalbuminemia, edema), then GFR decreases (creatinine increased). Anemia is due to decreased renal production of erythropoietin; K^+ secretion is impaired because of the decreased GFR and the hyporeninemic hypoaldosteronism that is common in diabetes — hyperkalemia. Treatment: control of glycemia and BP, ACE inhibitors or angiotensin receptor blockers, SGLT2 inhibitors (depending on GFR).

Case 2

A 40-year-old man presented with sudden severe colicky pain radiating from the loin to the groin. The urine contains many erythrocytes, pH 5.0, uric acid crystals. Blood uric acid 520 µmol/l. No stone was visible on X-ray; CT showed an 8 mm stone in the renal pelvis.

Questions: What type of stone is this? Why is it not visible on X-ray? What are the mechanism of stone formation and its treatment?

Answer: A urate (uric acid) stone. Urate stones absorb little X-ray radiation (radiolucent) and are visible on CT. Hyperuricemia and high urate excretion together with persistently acidic urine (at pH < 5.5 the solubility of uric acid is low, as it is un-ionized) lead to crystallization; in the metabolic syndrome ammoniagenesis is reduced and the urine is more acidic. Treatment: plenty of fluids, alkalinization of the urine with potassium citrate to pH 6.5–7 (urate stones dissolve — unlike other stones), allopurinol, restriction of purines and alcohol; pain relief.

Case 3

A 6-year-old child has fever, vomiting and abdominal pain. The urine smells of acetone. On the test strip ketone bodies are +++, glucose negative. Blood glucose 3.8 mmol/l. The child has eaten almost nothing for 2 days.

Questions: What is the cause of the ketonuria? Why is there no glucosuria? How is it distinguished from diabetes mellitus?

Answer: An acetonemic state (ketosis) against the background of fasting and vomiting. Children have a small glycogen store that is quickly exhausted; insulin decreases and glucagon and cortisol increase — lipolysis is enhanced, and β-oxidation of fatty acids in the liver forms a lot of acetyl-CoA, which, because oxaloacetate is used for gluconeogenesis, is converted into ketone bodies; acetoacetate and acetone are detected in the urine. Blood glucose is normal or low — because it does not exceed the renal threshold, there is no glucosuria. In diabetes mellitus, by contrast, ketonuria is accompanied by hyperglycemia (> 11 mmol/l) and glucosuria. Treatment: drinking and glucose-containing fluids (also intravenously), antiemetic measures — glucose restores insulin secretion and stops ketogenesis.

Quick review

Table 29.5. *Key concepts of topic 29*

Concept	Brief definition
Filtration	≈ 180 l/day; GFR 90–120 ml/min; albumin does not pass
Reabsorption	Proximal: glucose (SGLT2), amino acids, 65% of Na^+; ≈ 99% of the filtrate

Concept	Brief definition
Targets of diuretics	Furosemide — NKCC2; thiazides — NCC; spironolactone — aldosterone
H^+ excretion	HCO_3^- reabsorption, titratable acids, ammoniagenesis (glutaminase)
Energy	Cortex — aerobic (fatty acids); medulla — anaerobic glycolysis
Endocrine	Renin, erythropoietin, calcitriol, prostaglandins
Urine	1–2 l; specific gravity 1.010–1.025; pH 5–7
Main components	Urea 20–35 g, creatinine, uric acid, NH_4^+
Proteinuria	> 150 mg/day; microalbuminuria — diabetic nephropathy
Test strips	Glucose oxidase, protein error, nitroprusside, diazo, heme peroxidase, Griess
Stones	Ca oxalate (≈ 75%), urate (alkalinization), struvite (urease), cystine

TOPIC 30

Biochemistry of the central and peripheral nervous system

Learning objectives

To give the student knowledge of the specific features of nervous system function, the structural components of the central and peripheral nervous system, the protein and lipid composition of nervous tissue, the features of metabolism and energy exchange in nerve cells, the biochemical differences between neurons and neuroglia, signal transmission across synapses and neurotransmitters, the biochemistry of sensory systems, and the biochemical basis of mental and neurological diseases.

By the end of the class the student should know:

- the chemical composition of nervous tissue and the properties of myelin;
- the specific features of energy metabolism in the brain (dependence on glucose, ketone bodies);
- the structure and function of the blood–brain barrier;
- the metabolic cooperation between neurons and glial cells;
- the synthesis, inactivation and receptors of the main neurotransmitters;
- the biochemical basis of vision, hearing, smell and taste;
- the biochemical basis of Alzheimer and Parkinson diseases, depression, schizophrenia and epilepsy and the mechanism of action of drugs.

The student should be able to:

- isolate lipids from brain tissue and perform qualitative reactions for them;
- interpret the biochemical analysis of cerebrospinal fluid;
- explain the mechanism of drugs that act on neurotransmitters.

1. Specific features and structural components of the nervous system

The **central nervous system** is the brain and spinal cord; the **peripheral** one consists of nerves, ganglia and nerve endings. Cellular composition: **neurons** (≈ 86 billion) and **neuroglia** — astrocytes, oligodendrocytes (myelin in the CNS), Schwann cells (myelin in the periphery), microglia (immune defense), ependyma.

The blood–brain barrier (BBB) — **tight junctions** between capillary endothelial cells (claudin, occludin), basement membrane, pericytes and astrocyte end-feet. Glucose (GLUT1), amino acids and ketone bodies (MCT1) cross via specific transporters; small lipophilic substances by diffusion (ethanol, anesthetics, nicotine); proteins, most drugs and conjugated bilirubin do not cross. **P-glycoprotein** pumps many drugs back out. In newborns the BBB is immature (risk of kernicterus). Dopamine does not cross the BBB — in Parkinson disease its precursor **L-DOPA** is given.

Cerebrospinal fluid — ≈ 500 ml is formed per day in the choroid plexuses; protein 0.15–0.45 g/l (≈ 200 times less than in plasma), glucose 2.2–3.9 mmol/l (≈ 60% of blood glucose), chlorides 120–130 mmol/l.

2. Chemical composition of nervous tissue

The brain contains ≈ 78% water (≈ 84% in gray matter, ≈ 70% in white matter); half of the dry residue is **lipids** (≈ 55% in white matter), and the rest is proteins.

Lipids: phospholipids (phosphatidylethanolamine, plasmalogens, phosphatidylserine, phosphatidylcholine), **sphingolipids** — sphingomyelin, **cerebrosides** (galactosylceramide — specific to myelin), sulfatides, **gangliosides** (contain sialic acid; in gray matter, in neuronal membranes), **cholesterol** (in free form; the brain contains ≈ 25% of the body's cholesterol, synthesized locally). The brain contains **almost no triacylglycerols** — fatty acids are not an energy source. Membranes are rich in ω-3 **docosahexaenoic acid** (retina, synapses).

Myelin is the multilayered wrapping of the membrane of an oligodendrocyte or Schwann cell around an axon; ≈ 70–80% lipid (cholesterol, cerebrosides, plasmalogens), ≈ 20–30% protein (**myelin basic protein**, proteolipid protein); electrical insulation — saltatory conduction (speeds up to 100 m/s). **Demyelination:** multiple sclerosis (autoimmune response to myelin proteins), Guillain–Barré syndrome, B_{12} deficiency (spinal cord), leukodystrophies; **sphingolipidoses** (defects of lysosomal enzymes): Tay–Sachs (hexosaminidase A — GM2 ganglioside), Gaucher (glucocerebrosidase), Niemann–Pick (sphingomyelinase), Krabbe (galactocerebrosidase).

Proteins: neurospecific proteins — **S100** (astrocytes; a marker of brain injury), neuron-specific enolase, glial fibrillary acidic protein (GFAP), tubulin and **tau protein** (microtubules), neurofilaments, synaptic proteins (synapsins, SNARE), ion channels and pumps, receptors, neuropeptides. Axonal transport (kinesin — anterograde, dynein — retrograde) requires ATP.

3. Specific features of energy metabolism in nervous tissue

- The brain makes up ≈ 2% of body weight but at rest consumes **≈ 20% of the oxygen** and ≈ 25% of the glucose (≈ 120 g per day).
- ≈ 50–60% of the energy is spent on the work of **Na^+,K^+-ATPase** — to restore the membrane potential; the rest goes to the synthesis and reuptake of transmitters and to axonal transport.
- The main substrate is **glucose** (GLUT1 — BBB, GLUT3 — neurons; insulin-independent), which is completely oxidized aerobically. **The glycogen store is very small** (in astrocytes), and there is no oxygen store: if the circulation stops, consciousness is lost in 10 seconds and irreversible damage occurs in 4–5 minutes.
- **In prolonged starvation** the brain obtains ≈ 2/3 of its energy from **ketone bodies** (topic 15); in newborns (breast milk fats) ketone bodies are also an important source.
- Fatty acids cross the BBB poorly and are hardly used for energy.
- **Hypoglycemia** (< 2.8–3.0 mmol/l) — headache, confusion, seizures, coma; if prolonged, neurons die. **Hypoxia and ischemia** (stroke) — ATP is depleted, depolarization, a lot of glutamate is released — **excitotoxicity** (Ca^{2+} influx through NMDA receptors, proteases, free radicals) — necrosis and apoptosis of neurons.
- Thiamine (B_1) deficiency (the coenzyme of pyruvate dehydrogenase and α-ketoglutarate dehydrogenase) — **Wernicke–Korsakoff encephalopathy** (in alcoholism).

4. Biochemical cooperation between neurons and neuroglia

- **Astrocytes:** take up K^+ from the synaptic cleft; take up glutamate and GABA and convert them into glutamine (the **glutamate–glutamine cycle**); **detoxify ammonia** (glutamine synthetase — topics 13 and 25); store glycogen and produce **lactate** by glycolysis, which they give to neurons (the “astrocyte–neuron lactate shuttle”); produce the antioxidant glutathione; form the BBB.
- **Oligodendrocytes and Schwann cells** — myelin synthesis, nourishment of axons.
- **Microglia** — phagocytosis, cytokines, synaptic “pruning”.

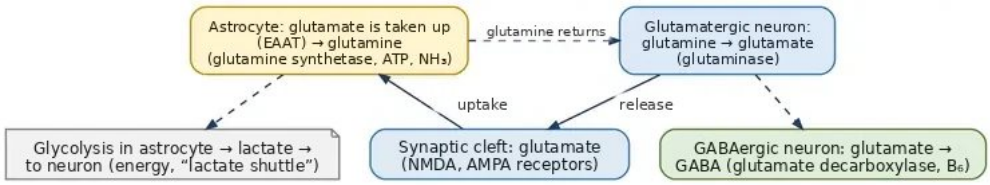


Figure 30.1. *The glutamate–glutamine cycle between neurons and astrocytes (drawn after Lippincott's Illustrated Reviews: Biochemistry)*

5. Signal transmission across synapses and neurotransmitters

The nerve impulse: resting potential (≈ −70 mV; K^+ permeability and Na^+,K^+-ATPase) → depolarization (voltage-gated Na^+ channels) → repolarization (K^+ channels). Local anesthetics (lidocaine) block Na^+ channels, and tetrodotoxin blocks them from the outside.

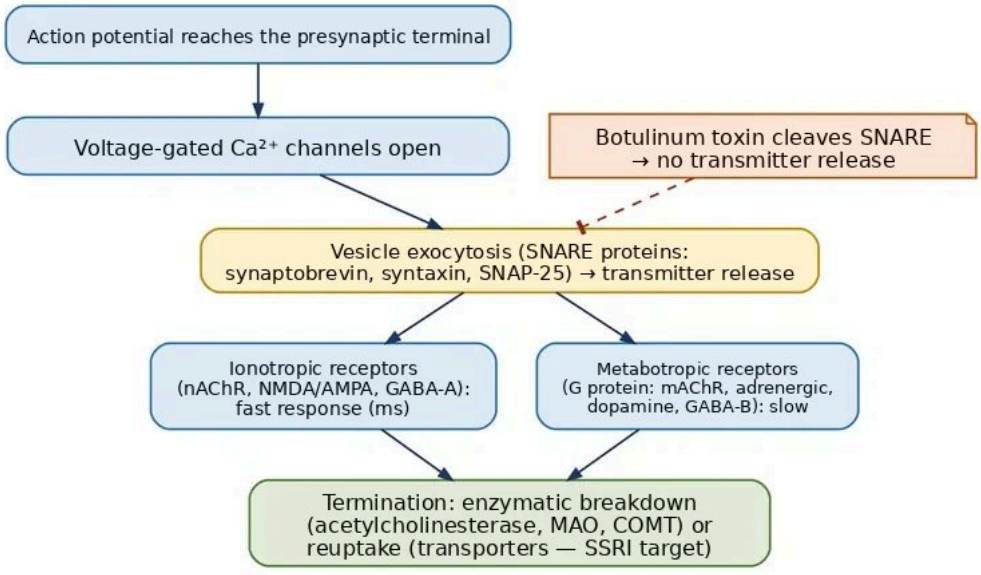


Figure 30.2. *Signal transmission in a chemical synapse (drawn after Severin E.S.)*

Table 30.1. *Main neurotransmitters*

Transmitter	Synthesis	Inactivation	Main receptors and function
Acetylcholine	Choline + acetyl-CoA (choline acetyltransferase)	Acetylcholinesterase (in the synaptic cleft)	N (ionotropic — neuromuscular junction, autonomic ganglia), M (metabotropic — parasympathetic); memory
Norepinephrine, epinephrine	Tyrosine → DOPA (tyrosine hydroxylase — key enzyme, BH_4) → dopamine → norepinephrine (β-hydroxylase, ascorbate) → epinephrine (SAM)	Reuptake (NET), MAO, COMT → VMA	α- and β-adrenoceptors; sympathetic system, attention, mood
Dopamine	DOPA decarboxylase (B_6)	Reuptake (DAT), MAO-B, COMT → homovanillic acid	D_1–D_5; movement (substantia nigra — striatum), motivation and "reward", inhibition of prolactin

Transmitter	Synthesis	Inactivation	Main receptors and function
Serotonin	Tryptophan → 5-hydroxytryptophan (tryptophan hydroxylase, BH_4) → serotonin (B_6)	Reuptake (SERT), MAO-A → 5-HIAA	5-HT_1–7 (5-HT_3 — ionotropic); mood, sleep, appetite, pain; converted to melatonin in the pineal gland
Glutamate	Glutamine (glutaminase), α-ketoglutarate	Uptake into astrocytes (EAAT)	The main excitatory transmitter: NMDA, AMPA, kainate (ionotropic), metabotropic; memory (long-term potentiation)
GABA	Glutamate → glutamate decarboxylase (B_6)	Uptake, GABA transaminase → succinate (the GABA shunt)	The main inhibitory transmitter: GABA-A (Cl^- channel — enhanced by benzodiazepines, barbiturates, ethanol), GABA-B
Glycine	From serine	Uptake	Inhibitory in the spinal cord (blocked by strychnine)
Histamine	Histidine decarboxylase	Methylation, DAO	H_1 — wakefulness (first-generation antihistamines cause drowsiness)
Neuropeptides	Proteolysis of precursor proteins (POMC → β-endorphin; proenkephalin)	Peptidases	Opioid peptides (μ, δ, κ — analgesia), substance P (pain), neuropeptide Y
NO, endocannabinoids	From arginine (nNOS); from membrane lipids	—	Retrograde transmitters

CLINICAL SIGNIFICANCE

Organophosphorus compounds (pesticides, sarin) irreversibly phosphorylate acetylcholinesterase: acetylcholine accumulates, causing muscarinic (miosis, salivary and bronchial secretion, bradycardia, diarrhea) and nicotinic (muscle twitching, paralysis) signs and CNS disturbance; death — respiratory failure. Antidotes: **atropine** (an M-cholinergic blocker) and **reactivators** (pralidoxime — restores the enzyme before "aging"). Reversible inhibitors (neostigmine) — in myasthenia, donepezil — in Alzheimer disease.

6. Biochemistry of sensory systems

Vision: in the rods **rhodopsin** = opsin + **11-cis-retinal** (vitamin A). A quantum of light isomerizes 11-cis-retinal into **all-trans-retinal** → metarhodopsin II → the G protein **transducin** → **cGMP phosphodiesterase** is activated → cGMP decreases → cGMP-gated Na^+/Ca^{2+} channels close → **hyperpolarization** → glutamate release decreases → signal. One photon activates hundreds of transducin molecules (amplification). The visual cycle: all-trans-retinal → retinol → converted back to 11-cis-retinal in the pigment epithelium. **Vitamin A deficiency** — night blindness (hemeralopia), xerophthalmia. Cones contain three kinds of iodopsin (color vision); the genes are on the X chromosome — color blindness.

Hearing: sound waves bend the stereocilia of the hair cells → mechanosensitive channels open and K^+ enters (the endolymph is rich in K^+ — 150 mmol/l, produced by the stria vascularis) →

depolarization → glutamate is released. Aminoglycosides and furosemide are ototoxic.

Smell: ≈ 400 kinds of odorant receptors (G protein-coupled; the largest gene family) → G_{olf} → adenylate cyclase → cAMP → cation channels open.

Taste — topic 24: sweet, umami, bitter (G protein — gustducin), salty and sour (ion channels).

Pain (nociception): TRPV1 (capsaicin, heat), TRPA1, acid-sensing channels; mediators — prostaglandin E_2 (increases sensitivity — the basis of NSAID action), bradykinin, substance P, glutamate; endogenous analgesia — enkephalins, endorphins.

7. Biochemical basis of mental and nervous system diseases

Table 30.2. *Biochemical basis of neurological and mental diseases*

Disease	Biochemical basis	Drug action
Alzheimer disease	Plaques of β-amyloid (cleavage of APP by β- and γ-secretases — $A\beta_{42}$); hyperphosphorylated tau — neurofibrillary tangles; death of cholinergic neurons; APOE ε4 — a risk factor	Cholinesterase inhibitors (donepezil), NMDA antagonist (memantine), anti-amyloid antibodies
Parkinson disease	Death of the dopaminergic neurons of the substantia nigra; α-synuclein aggregates (Lewy bodies); mitochondrial dysfunction, oxidative stress	L-DOPA + a peripheral decarboxylase inhibitor (carbidopa), MAO-B inhibitors, dopamine agonists
Depression	The monoamine (serotonin, norepinephrine) deficiency hypothesis; cortisol ↑, neurotrophic factor (BDNF) ↓	SSRIs (fluoxetine — selective serotonin reuptake inhibitors), tricyclic antidepressants, MAO inhibitors (the "cheese reaction" with tyramine)
Schizophrenia	Increased dopamine activity in the mesolimbic pathway (positive symptoms), hypofunction of NMDA receptors	Antipsychotics — D_2 blockers (haloperidol), atypical (5-HT_{2A} + D_2)
Epilepsy	Disturbed balance of excitation (glutamate) and inhibition (GABA); mutations of ion channels	Na^+ channel blockers (carbamazepine), GABA enhancers (valproate, benzodiazepines)
Huntington disease	(CAG)n expansion in the huntingtin gene — polyglutamine aggregates	Symptomatic
Myasthenia gravis	Autoantibodies to N-acetylcholine receptors	Cholinesterase inhibitors
Prion diseases	Conversion of PrP^C into β-sheet-rich PrP^{Sc} (self-propagating)	No cure

Routes of biochemical diagnosis: cerebrospinal fluid analysis (protein, glucose, lactate, cells; $A\beta_{42}$, tau — Alzheimer disease; oligoclonal IgG — multiple sclerosis), neurospecific proteins in the blood (S100, NSE, neurofilament light chain — brain injury), transmitter metabolites in the urine (VMA, homovanillic acid — neuroblastoma, pheochromocytoma; 5-HIAA — carcinoid), therapeutic drug monitoring (lithium, antiepileptics), genetic tests, PET (glucose metabolism, amyloid).

8. Laboratory work

Experiment 1: isolation of lipids from brain tissue and qualitative reactions for them

Principle of the method: lipids dissolve in organic solvents; cholesterol and phospholipids give characteristic reactions.

1. Grind 1 g of brain tissue with 10 ml of chloroform–methanol mixture (2:1) and filter (extract).
2. **Cholesterol — the Salkowski reaction:** carefully add 1 ml of concentrated H_2SO_4 to 1 ml of the (chloroform) extract — the chloroform layer turns red and the acid layer gives a yellow-green fluorescence. **The Liebermann–Burchard reaction:** with acetic anhydride and H_2SO_4 — a blue-green color (the basis of cholesterol determination in serum).
3. **Phosphate (phospholipids):** evaporate the extract, heat the residue with HNO_3 and add ammonium molybdate — a yellow precipitate (phosphomolybdate).
4. **The acrolein test** (glycerol): heat the residue with $KHSO_4$ — a pungent odor (glycerol of the glycerophospholipids). **Results and analysis:** the brain extract contains a lot of cholesterol and phospholipids (the Salkowski and molybdate reactions are clearly positive); this confirms the high lipid content of the brain and the composition of its membranes (myelin).

Experiment 2: interpretation of cerebrospinal fluid analysis (case-based)

Table 30.3. *Cerebrospinal fluid parameters in various conditions*

Condition	Color	Cells	Protein	Glucose	Lactate
Normal value	Clear, colorless	0–5 lymphocytes/µl	0.15–0.45 g/l	2.2–3.9 mmol/l (60% of blood glucose)	< 2.1 mmol/l
Bacterial meningitis	Turbid, purulent	Hundreds to thousands, neutrophils	↑↑	↓↓ (consumed by bacteria and neutrophils)	↑↑
Viral meningitis	Clear	Tens to hundreds, lymphocytes	N or slightly ↑	N	N
Tuberculous meningitis	Opalescent, fibrin web	Lymphocytes	↑↑	↓	↑
Subarachnoid hemorrhage	Bloody, then xanthochromic (bilirubin)	Erythrocytes	↑	N	↑

1. Compare the given cerebrospinal fluid results with the table.
2. Compare the glucose with the blood glucose taken at the same time.
3. Determine the probable diagnosis.

9. Practical skills

Skill: explaining the mechanism of a drug that acts on a neurotransmitter

1. Determine which transmitter system the drug acts on.
2. Identify the step of action: synthesis (L-DOPA), storage (reserpine), release (botulinum toxin), receptor (agonist, antagonist), reuptake (SSRIs, cocaine), breakdown (MAO and cholinesterase inhibitors).
3. State how the amount of transmitter in the synaptic cleft changes and the clinical result.
4. Explain the adverse effects by the action on other transmitter systems.

Skill: biochemical assessment of brain injury and hypoglycemia

1. Determine blood glucose immediately (< 3.0 mmol/l — give glucose immediately).
2. Assess the signs of hypoxia (blood gases, lactate).

3. Assess neurospecific markers (S100, NSE) for the degree of injury and prognosis.
4. Remember that in alcohol-related conditions thiamine is given before glucose.

Review questions

1. What cells make up the nervous system and what are their functions?
2. How is the blood–brain barrier built and which substances does it let through?
3. What is the lipid composition of nervous tissue?
4. What are the composition and function of myelin? Name the demyelinating diseases.
5. Give examples of neurospecific proteins.
6. What is special about energy metabolism in the brain?
7. What energy source does the brain use in prolonged starvation?
8. What is excitotoxicity?
9. What role do astrocytes play in neuronal metabolism?
10. How are acetylcholine, catecholamines and serotonin synthesized and inactivated?
11. What are the synthesis and functions of glutamate and GABA?
12. Explain the biochemical mechanism of vision.
13. What is the biochemical basis of hearing and smell?
14. What is the biochemical basis of Alzheimer and Parkinson diseases?
15. Which transmitter systems are disturbed in depression and schizophrenia? How do drugs act?

Test questions

1. What proportion of the body's oxygen does the brain consume at rest?
   A) 2 %
   B) 5 %
   C) 50 %
   D) ≈ 20 %
   E) 80 %

2. Which lipid is specific to myelin?
   A) Triacylglycerol
   B) Galactocerebroside
   C) Cardiolipin
   D) Free fatty acids
   E) Cholesterol esters

3. What is the main energy source in the brain?
   A) Fatty acids
   B) Amino acids
   C) Glucogenic
   D) Lactate
   E) Glucose

4. Which substance accumulates in Tay–Sachs disease?
   A) Glucocerebroside
   B) Sphingomyelin
   C) GM2 ganglioside
   D) Galactocerebroside
   E) Dermatan sulfate

5. Which is the main inhibitory neurotransmitter?
   A) Glutamate
   B) Acetylcholine
   C) GABA
   D) Norepinephrine
   E) Dopamine

6. Which enzyme catalyzes the synthesis of GABA?
   A) Glutamate decarboxylase
   B) Glutaminase
   C) Glutamine synthetase
   D) Tyrosine hydroxylase
   E) DOPA decarboxylase

7. Which enzyme breaks down acetylcholine in the synaptic cleft?
   A) Acetylcholinesterase
   B) Monoamine oxidase
   C) COMT
   D) Choline acetyltransferase
   E) Pseudocholinesterase

8. What is the key enzyme of catecholamine synthesis?
   A) DOPA decarboxylase
   B) Tyrosine hydroxylase
   C) Dopamine β-hydroxylase
   D) Phenylalanine hydroxylase
   E) COMT

9. What is the mechanism of action of fluoxetine?
   A) It inhibits MAO
   B) It blocks D_2 receptors
   C) It enhances the GABA-A receptor
   D) It inhibits serotonin reuptake
   E) It inhibits acetylcholinesterase

10. Which neurons die in Parkinson disease?
    A) The dopaminergic neurons of the substantia nigra
    B) Cholinergic neurons
    C) Serotonergic neurons
    D) GABAergic neurons
    E) Glutamatergic neurons

11. What change occurs under the action of light in vision?
    A) All-trans-retinal is converted into 11-cis-retinal
    B) 11-cis-retinal is isomerized into all-trans-retinal
    C) cGMP increases
    D) Na^+ channels open
    E) The photoreceptor depolarizes

12. What is the antidote in poisoning with organophosphorus compounds?
    A) N-acetylcysteine
    B) Naloxone
    C) Atropine and pralidoxime
    D) Flumazenil
    E) Methylene blue

13. What are the plaques in Alzheimer disease made of?
    A) α-Synuclein
    B) Prion protein
    C) Huntingtin
    D) Hyperphosphorylated tau
    E) β-Amyloid

14. Which change in the cerebrospinal fluid is characteristic of bacterial meningitis?
    A) Glucose increased, protein decreased
    B) Lymphocytes, glucose normal
    C) Clear, colorless, unchanged
    D) Many neutrophils, glucose sharply decreased, protein increased
    E) Only erythrocytes

15. What is the main enzyme that detoxifies ammonia in the brain?
    A) Carbamoyl phosphate synthetase I
    B) Glutaminase
    C) Arginase
    D) Glutamate decarboxylase
    E) Glutamine synthetase

Clinical cases

Case 1

A 19-year-old man with type 1 diabetes mellitus went to training after injecting insulin without eating. 1 hour later he developed sweating, tremor and confusion, then seizures and loss of consciousness. Blood glucose 1.9 mmol/l.

Questions: What condition is this? Why is the brain especially sensitive to hypoglycemia? How should help be provided?

Answer: Hypoglycemic coma. The brain obtains its energy almost exclusively from the aerobic oxidation of glucose; its glycogen store is very small, it cannot use fatty acids, and the use of ketone bodies develops only as an adaptation to prolonged starvation (here insulin has also stopped ketogenesis). When glucose falls, ATP formation decreases, Na^+,K^+-ATPase stops working and the neurons depolarize — confusion, seizures (a lot of glutamate is released), coma; if prolonged, irreversible damage. Sweating and tremor are a contrainsular adrenergic reaction. Help: if conscious,

a sweet drink; if unconscious, 40% glucose intravenously or glucagon intramuscularly (hepatic glycogenolysis); then a carbohydrate meal and analysis of the cause.

Case 2

A 65-year-old man has a resting hand tremor, slowness of movement, muscle rigidity and short steps when walking. Parkinson disease was diagnosed and levodopa + carbidopa was prescribed.

Questions: What is the biochemical basis of the disease? Why is levodopa given rather than dopamine itself? Why is carbidopa added?

Answer: The dopaminergic neurons of the substantia nigra die (because of α-synuclein aggregates, mitochondrial dysfunction and oxidative stress), and dopamine decreases in the striatum; the balance with acetylcholine is disturbed, and movement disorders appear. Dopamine does not cross the blood–brain barrier, whereas its precursor L-DOPA crosses through an amino acid transporter and is converted to dopamine in the brain by DOPA decarboxylase (B_6). Carbidopa is a peripheral DOPA decarboxylase inhibitor (it does not cross the BBB): it stops the conversion of L-DOPA into dopamine in the blood, ensuring that more of it reaches the brain, and reduces the peripheral adverse effects (nausea, arrhythmia). Protein-rich food competes with L-DOPA for absorption and for crossing the BBB.

Case 3

A farmer working in the field was brought in after spraying pesticide with blurred vision, sharply constricted pupils, profuse salivation, shortness of breath, diarrhea and muscle twitching. Blood cholinesterase activity is sharply decreased.

Questions: What is the diagnosis? What is the biochemical mechanism of the symptoms? What are the principles of treatment?

Answer: Acute poisoning with an organophosphorus compound. The poison phosphorylates the serine in the active site of acetylcholinesterase and inhibits it irreversibly; acetylcholine accumulates in the synapses. Through M-cholinergic receptors — miosis, blurred vision, increased salivary and bronchial secretion, bronchospasm, bradycardia, diarrhea; through N-receptors — muscle fibrillation, then paralysis (respiratory muscles); in the CNS — agitation, seizures, coma. A decrease in blood cholinesterase activity confirms the diagnosis. Treatment: washing the poison off the skin, high-dose atropine (stops the muscarinic signs), oxime reactivators (pralidoxime — detaches the phosphate group from the enzyme, effective if given early), benzodiazepines (seizures), artificial ventilation.

Quick review

Table 30.4. *Key concepts of topic 30*

Concept	Brief definition
BBB	Tight junctions, astrocytes; glucose (GLUT1) and ketone bodies cross; dopamine does not
Brain lipids	Phospholipids, cerebrosides, gangliosides, cholesterol; no TAG
Myelin	≈ 75% lipid; multiple sclerosis; sphingolipidoses
Energy	20% of O_2; glucose; ketone bodies in starvation
Excitotoxicity	Ischemia → glutamate → NMDA → Ca^{2+} → neuronal death
Astrocytes	Glutamate–glutamine cycle, NH_3 detoxification, lactate

Concept	Brief definition
Transmitters	Acetylcholine (AChE), catecholamines (MAO, COMT), serotonin, glutamate, GABA
Excitatory / inhibitory	Glutamate / GABA (glutamate decarboxylase, B_6)
Vision	11-cis → all-trans-retinal → transducin → cGMP ↓ → hyperpolarization
Alzheimer	β-Amyloid, tau; cholinesterase inhibitors
Parkinson	Dopamine ↓; L-DOPA + carbidopa
Depression, schizophrenia	Serotonin ↓ (SSRIs); dopamine ↑ (D_2 blockers)

Answers to the test questions

Topic 1: 1-B; 2-C; 3-A; 4-D; 5-E; 6-C; 7-E; 8-D; 9-B; 10-A; 11-B; 12-D; 13-A; 14-E; 15-C.

Topic 2: 1-E; 2-B; 3-E; 4-C; 5-A; 6-A; 7-B; 8-B; 9-D; 10-D; 11-C; 12-C; 13-A; 14-E; 15-D.

Topic 3: 1-E; 2-C; 3-B; 4-A; 5-D; 6-E; 7-A; 8-A; 9-E; 10-B; 11-B; 12-C; 13-D; 14-C; 15-D.

Topic 4: 1-B; 2-D; 3-A; 4-A; 5-E; 6-D; 7-C; 8-D; 9-E; 10-C; 11-E; 12-C; 13-B; 14-A; 15-B.

Topic 5: 1-B; 2-A; 3-C; 4-A; 5-C; 6-E; 7-B; 8-D; 9-D; 10-A; 11-E; 12-B; 13-C; 14-E; 15-D.

Topic 6: 1-E; 2-D; 3-B; 4-D; 5-D; 6-C; 7-A; 8-E; 9-C; 10-A; 11-B; 12-A; 13-B; 14-E; 15-C.

Topic 7: 1-B; 2-D; 3-E; 4-D; 5-A; 6-E; 7-C; 8-E; 9-C; 10-B; 11-D; 12-B; 13-C; 14-A; 15-A.

Topic 8: 1-C; 2-A; 3-E; 4-A; 5-A; 6-B; 7-D; 8-C; 9-E; 10-B; 11-E; 12-D; 13-C; 14-D; 15-B.

Topic 9: 1-A; 2-C; 3-B; 4-A; 5-D; 6-A; 7-C; 8-E; 9-D; 10-C; 11-B; 12-E; 13-D; 14-B; 15-E.

Topic 10: 1-E; 2-C; 3-C; 4-B; 5-D; 6-A; 7-A; 8-B; 9-B; 10-D; 11-E; 12-A; 13-D; 14-C; 15-E.

Topic 11: 1-C; 2-E; 3-A; 4-E; 5-A; 6-D; 7-E; 8-C; 9-B; 10-C; 11-B; 12-B; 13-A; 14-D; 15-D.

Topic 12: 1-A; 2-B; 3-C; 4-C; 5-A; 6-C; 7-B; 8-D; 9-E; 10-A; 11-D; 12-B; 13-E; 14-E; 15-D.

Topic 13: 1-A; 2-D; 3-E; 4-C; 5-A; 6-A; 7-C; 8-B; 9-B; 10-D; 11-C; 12-E; 13-B; 14-E; 15-D.

Topic 14: 1-B; 2-A; 3-C; 4-A; 5-B; 6-C; 7-E; 8-D; 9-D; 10-E; 11-D; 12-A; 13-B; 14-C; 15-E.

Topic 15: 1-A; 2-B; 3-A; 4-E; 5-C; 6-E; 7-D; 8-A; 9-E; 10-D; 11-C; 12-C; 13-B; 14-B; 15-D.

Topic 16: 1-E; 2-C; 3-B; 4-B; 5-C; 6-A; 7-A; 8-D; 9-B; 10-A; 11-E; 12-C; 13-D; 14-E; 15-D.

Topic 17: 1-C; 2-B; 3-E; 4-C; 5-D; 6-D; 7-A; 8-B; 9-D; 10-A; 11-B; 12-E; 13-E; 14-A; 15-C.

Topic 18: 1-D; 2-D; 3-B; 4-A; 5-A; 6-B; 7-E; 8-E; 9-D; 10-A; 11-C; 12-C; 13-B; 14-E; 15-C.

Topic 19: 1-D; 2-E; 3-E; 4-B; 5-A; 6-C; 7-C; 8-B; 9-B; 10-A; 11-E; 12-A; 13-C; 14-D; 15-D.

Topic 20: 1-B; 2-B; 3-A; 4-D; 5-E; 6-D; 7-E; 8-C; 9-A; 10-B; 11-A; 12-D; 13-C; 14-E; 15-C.

Topic 21: 1-E; 2-D; 3-C; 4-C; 5-A; 6-D; 7-D; 8-E; 9-B; 10-B; 11-C; 12-A; 13-E; 14-A; 15-B.

Topic 22: 1-C; 2-C; 3-D; 4-B; 5-A; 6-E; 7-E; 8-A; 9-A; 10-C; 11-D; 12-B; 13-D; 14-B; 15-E.

Topic 23: 1-E; 2-B; 3-B; 4-C; 5-C; 6-D; 7-C; 8-D; 9-B; 10-A; 11-D; 12-A; 13-E; 14-A; 15-E.

Topic 24: 1-E; 2-A; 3-E; 4-D; 5-A; 6-E; 7-C; 8-B; 9-C; 10-A; 11-B; 12-B; 13-D; 14-C; 15-D.

Topic 25: 1-A; 2-B; 3-D; 4-A; 5-B; 6-C; 7-D; 8-E; 9-C; 10-B; 11-E; 12-A; 13-D; 14-E; 15-C.

Topic 26: 1-A; 2-E; 3-C; 4-E; 5-C; 6-B; 7-A; 8-D; 9-B; 10-B; 11-A; 12-E; 13-C; 14-D; 15-D.

Topic 27: 1-C; 2-D; 3-A; 4-B; 5-C; 6-E; 7-E; 8-A; 9-B; 10-D; 11-A; 12-E; 13-D; 14-C; 15-B.

Topic 28: 1-D; 2-B; 3-E; 4-A; 5-D; 6-E; 7-D; 8-B; 9-E; 10-A; 11-A; 12-C; 13-C; 14-B; 15-C.

Topic 29: 1-C; 2-A; 3-A; 4-B; 5-D; 6-C; 7-E; 8-A; 9-C; 10-D; 11-E; 12-B; 13-E; 14-B; 15-D.

Topic 30: 1-D; 2-B; 3-E; 4-C; 5-C; 6-A; 7-A; 8-B; 9-D; 10-A; 11-B; 12-C; 13-E; 14-D; 15-E.

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